GrappStar build guide

One guide from parts to first mission: print, wire, flash, calibrate, test. Every hands-on step is in order. Every wiring step starts with the battery unplugged.

Built 2026-11-05 01:54 Sydney time from commit `fd24cdb`.

How to use it: read section 1 and 2 first, then work through the "Order of work" list in section 1. Where a section says "From `some/file.md`", that text comes straight from the repo, so it is always the latest version. Re-build the guide after any change, from the repo folder: `kedge/.venv/bin/python docs/build/make_guide.py`.

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From `docs/build/FIRST_DAY.md`*:*

First session back: what to do, in order

Everything else waits until these are done. The measuring takes an afternoon; the coupons wait for the 608s to arrive.

1. Read (15 minutes)

- Build guide sections 1 (what the system is) and 2 (safety).
- The rule for every wiring step: battery unplugged first.

2. Measure (1 to 2 hours; caliper, kitchen scale, water bottles, phone camera)

Take a photo of every caliper reading and put them in a new folder `originals/measure_2026-11/` (create it). Write each number into `docs/build/MEASURE_RESULTS.md` (a fill-in sheet; print it).

1. Wiper motor (MEASURE_LIST A1 to A5, A8): output shaft diameter and flats, end thread and length, the mount hole pattern (press the motor face onto paper), the housing size, the boss and hub.
2. Launch springs (B1 to B3), safety glasses on while a loaded spring hangs: free length with a steel rule or tape (the spring is longer than a caliper), then hang about 1, 2 and 3 kg of water bottles (weigh each load: W1, W2, W3 in kg) from ONE spring and measure hook to hook each time (L1, L2, L3 in mm). Both springs. `python3 base/physics/calib.py spring L0 W1:L1 W2:L2 W3:L3` gives the rate, the initial tension and the force at full cock, for each spring. Enter the mean of the two springs' rate and initial tension; the STRONGER spring's force at full cock decides the spring anchor holes (MEASURE_LIST B4: over 38.4 N, hole set F).
3. Small parts (H1, H3, H4): drum N20 shaft, CSK20PP keyway depths, BTS7960 heatsink height and hole spacing.
4. Count the parts box (group C): worm N20s (4 needed), the spare MP1584, 8 mm rod or M8 bolts, 693ZZ / F693ZZ left after the dart (the base needs 6 x 693ZZ), resistors (2 x 10 k, 1 k, 680 R, 100 k, 22 k), a small magnet for the AS5600, spare XT60s, and the coupon extras (MEASURE_LIST C21).
5. Update the design with the numbers (build guide section 14, "After you measure"): the easy way is to open Claude Code in the repo, give it MEASURE_RESULTS.md and ask it to enter the numbers in params.py and re-run the base pipeline and the physics. By hand: the commands are in that section.

3. Buy

- 608 bearings, 10 (search "608ZZ bearing 8x22x7").
- Only if not in the box: a small neodymium disc magnet for the AS5600 (6 to 10 mm across, 2 to 3 mm thick, or a 6 x 2.5 mm diametric disc).

4. Print first (test pieces only)

1. The dart fit coupon (`kedge/pack_v30/fit_coupon.stl`, 20 minutes). Check it against run sheet section 4.
2. The base fit coupon (`base/pack/fit_coupon/coupon_plate_1.3mf`, PLA; checks in its COUPON.md; tile A uses three 608s) and the two base risk coupon plates (`base/pack/coupons/risk_plate_1_PLA.3mf` PLA, `risk_plate_2_PPA-CF.3mf` PPA-CF: hardened

nozzle, dry spool) and the test sheet `COUPON_TEST.pdf` (test stage 0). Every check must pass before the big base parts. The coupons need a few bought parts (C21) and one each of 608, F693ZZ, 693ZZ and a CSK20PP.

3. The dart plates A and B do not wait for the base coupon or the measurements: start them once the dart fit coupon passes.

5. Do NOT do yet

- Do not print the big base parts until the coupons pass AND the measurements are in params.py and the pipeline has been re-run (the motor adapter, can stay and wiper shaft coupler depend on them).
- Do not connect the LiPo to anything new. No wiring with a battery plugged in, ever.
- Do not fit the launch springs (they go on at test stage 5).
- Do not fire anything. When the time comes: never without the tether tied on.
- Do not plug the dart into the Mac with its power loop plug in.

1. What the system is

GrappStar is a rover that can pull itself out of trouble. A launcher on the rover fires a small dart (KEDGE) on a Dyneema tether. The dart lands, screws itself into the soil, and the rover winches against it. Then the dart lets go and is reeled home.

Three parts:

- **Rover.** Jetson Orin Nano Super on the rover's 3S LiPo. It runs the mission, the operator web page, and the vision (rover camera Razer Kiyo X, and the dart's video through an RC832 receiver and a DIGITNOW USB capture).
- **Base station** (on the rover). ONE 12 V wiper motor drives everything. A sliding dog collar on the main shaft couples the wiper to the LAUNCH train, to nothing (NEUTRAL), or to the REEL train. A small N20 motor turns a selector drum (like a motorcycle gear drum) through four zones in a fixed order: LAUNCH, NEUTRAL, WRAP, REEL.
  - LAUNCH, wiper forward: draws the spring carriage back (241 mm), fires at the end of the stroke, resets.
  - LAUNCH, wiper backward: tilts the rail between 6 and 47 degrees to aim. A self-locking worm holds the angle.
  - WRAP zone: the collar stays in NEUTRAL while the drum turns a wrap cup 7 times round the capstan.
  - REEL, wiper forward: the capstan (REEL gear 50T to C1 49T on the rail bridge, a 2:1 bevel, a shaft along the rail and three gears at the reel head: 2.307 times wiper speed) winds in and tows. A one-way bearing stops it ever turning out. Tow limit 50 N by default.
  - Rail reel (decision D49): the storage spool (below the line) and the capstan with its wrap cup (above it) ride the rail behind the carriage and tilt with it, so the line runs straight from the spool through the cup eyelet to the dart tail at every pitch, with no pulleys.
  - Take-up: every forward wiper turn also winds the spool through a slip clutch, so slack never builds up.
  - Payout at launch: the line skips the capstan and pulls off the free-spinning spool against a fixed light payout brake (0.25 N; it also stops the spool overrunning after landing).
  - Range: a few metres with the two springs (6 to 47 degrees of pitch). Two springs only in v1 (decision D38).
  - Protection: current trips in the firmware and a printed torque limiter (about 5 N m) in the wiper coupler. Any fault: wiper off, drum to NEUTRAL (except when the wraps are on, the spring is drawn or the drum itself faulted: then the drum holds, decision D27). With the N20 swung out of mesh the drum turns by hand by its gear rim.
  - Brain: an ESP32-S3 DevKitC on USB from the Jetson, with an AS5600 angle sensor on the main shaft and printed contact switches (an M3 screw head touching an M3 nut).
- **Dart (KEDGE v30).** 42 mm x 152 mm, its own XIAO ESP32S3, 2S 18350 pack, WiFi access point. After landing a kick leg swings over the nose and plants a foot auger, then two drill pods swing down and screw in. A nose camera sends analog video at 5.8 GHz.

How they talk: the Jetson talks to the base over USB serial (JSON lines, `firmware/base/PROTOCOL.md`) and to the dart over WiFi (HTTP at 192.168.4.1, `firmware/dart/API.md`). Video is analog 5.8 GHz into the USB capture.

A mission (the fixed sequence in BASE_SPEC section 10.1): pick a target on the rover camera view; the Jetson looks up the pitch; the base aims, cocks and fires on your confirmation; the dart video is checked for the landing zone; the base lowers the rail to 6 degrees, then wraps the capstan at once while the line is still slack (the drum parks at WRAPPED); the dart anchors; the base tidies the slack, shifts into REEL and tows; the base stops, then the dart releases; the base reels in to about 1.5 m, unwraps, and brings the dart home on the take-up until it seats; back to LAUNCH.

Range today (from `base/out/ballistics.json`, 0.25 N payout brake, measured from the ground point under the rail pivot): best about 5.0 m at 41 degrees, shortest about 2.7 m at 6 degrees, leaving the rail at about 7.0 m/s. These numbers change after the spring and wiper measurements; rebuild the guide.

Order of work

0. Get the repo onto the Mac (Terminal; if macOS asks to install the command line developer tools for `git`, say Install and run the command again): `cd ~ && GIT_LFS_SKIP_SMUDGE=1 git clone https://github.com/HarryHYu/GrappStar.git` (skips the 145 MB photo PDF). If `~/GrappStar` then has no `docs/build` folder, run `cd ~/GrappStar && git checkout base-v1`. `git pull` updates it later.

1. Measure the wiper motor shaft, mount and housing (section 14, A1 to A5 and A8), the springs (group B), and the parts in H1, H3, H4. Put the numbers in `base/spec/params.py` and re-run the base pipeline and the physics (section 14, "After you measure"). The base parts that touch the motor depend on them. A6 (speed, current) needs the base electronics: do it at step 7. A7 is optional. They change no printed part.
2. Confirm the parts box (section 14, group C). The only purchase is 608 bearings (and an AS5600 magnet if none is in the box).
3. Print the dart fit coupon, the base fit coupon and the two base risk coupon plates, and test them (section 4, test stage 0). The dart plates (section 4) do not wait for the base coupons or the measurements: start them any time after the dart fit coupon passes.
4. Flash the dart and build it (sections 6 and 8, and the dart run sheet `kedge/pack_v30/KEDGE_v30_run_sheet.pdf`: print it separately).
5. Set up the Jetson on its own wall supply (section 9, steps 0 to 3 only; its step 4 comes at the end of step 7).
6. Flash the base ESP32 from the Mac (section 8.1). Battery out is enough: any wiring already on the board can stay.
7. Wire the base electronics on the bench, wiper motor clamped in a vice and the drum N20 loose (section 7.2, firmware/base/WIRING.md Parts A to D), then bench test them (section 11.1), then calibration parts (a) and (b) (section 10). Measure A6 here. At the end: battery out, Jetson still on its wall supply, plug the ESP32 into the Jetson and do Jetson section 9 step 4 (it creates `/dev/grappstar-base`, which every later `base` command uses).
8. Print the base (section 4, after the coupons pass and the measurements are in) and assemble it with the motor and drum N20 (section 5).
9. Put the base on the rover and wire the rover (section 7.1), then do the first power-on of everything together (7.1, its part 4). Then finish the Jetson setup (section 9, steps 5 to 7: cameras, dart WiFi, rover camera calibration).
10. Calibrate (section 10). Commands go to the base from the Jetson from now on (see section 10).
11. Run the staged tests from stage 4 on (section 12; stages 0 to 3 were done in steps 3, 7 and 9).

2. Safety

Read this before any hands-on step.

Electrical

- **Battery unplugged before you touch, add or move any wire.** On the dart that means the power loop plug out. The first XIAO died from wiring with the battery live.
- Order for every power-up: USB cables in first, battery last. Power-down: battery out first. The base ESP32 may stay on the Mac's USB while the battery goes in (firmware/base/WIRING.md Part D does this), but never plug the USB in while the battery is in. The dart: never USB and the power loop plug in at the same time.
- Check for shorts with the meter on beep before the first power-on of anything you wired: red to black must not beep.
- The 3S LiPo can deliver over 100 A into a short. Keep the 20 A and 10 A fuses in their leads. Never bypass them.
- LiPo care: charge on the iMax B6 in balance mode at 1C (5.2 A) or less, on a non-flammable surface (tile, concrete, a LiPo bag), never unattended and never overnight. Recharge when the pack reads 3.6 V per cell at rest (10.8 V); never run it below 3.3 V per cell under load (9.9 V).
- LiPo storage: if it will sit for more than a few days, use the B6 "storage" mode (3.8 V per cell), keep it in a LiPo bag or metal box, away from heat (never in a hot car). A puffed, dented or hot pack: stop, put it outside on concrete, then to battery recycling. Never puncture or short one.
- 18350 cells: never let both cells touch metal at the same time. Charge out of the dart in the XTAR MC2, or through the tail balance socket with the loop plug out: on the B6 choose Li-ion (LiIo) 2S balance, 1.0 A or less (not the LiPo 3S program). Check each cell's direction against the marks in the cell bay and meter the pack (7.2 to 8.4 V) before the loop plug goes in. Look the cells over after a hard landing; a dented cell is scrapped.
- Setting a buck on the live LiPo (dart step 3, base WIRING Part A): use clip leads or insulate every free lead end first, so nothing loose can touch.
- Never connect the DRV8833 motor supply (VM) to the LiPo directly: 10.8 V maximum. It runs from the 9.0 V buck.
- The dart VTX must have its antenna on before it is powered.

Mechanical

- The wiper motor is strong (about 15 N m) and self-locking. Hands out of the gears, drum and rail while the battery is plugged in.
- **Stopping (the same everywhere):** 1. send `estop` (the operator page STOP button sends a plain `stop`: wiper off, drum stays). 2. If anything still moves, pull lead #2's XT60: both motors lose power, the Jetson stays up. Keep that plug within arm's reach whenever the base is powered; it is the physical kill switch. 3. Only then the main battery plug.
- **Cutting power freezes stored energy, it does not release it.** Both worms self-lock, so a drawn spring and a line under tension stay loaded with the power off. The dart has its own battery.
- **Before you walk up to the base or the dart** (to measure a range, fetch the dart, or touch anything): `estop`; lead #2 out; telemetry shows `cock` 0 and `H` (SW_HOME) closed; the line is slack (no tension); the dart reads IDLE on the operator page, or its loop plug is out. If the carriage is drawn and `decock` fails: (1) `clear` the fault and send `{"cmd":"mode","to":"LAUNCH"}` if the drum is not in LAUNCH, then `decock` again; (2) if it replies `SWITCH_DISAGREE` and you can SEE the carriage is drawn back, `{"cmd":"decock","force":true}` (it runs back by the travel the saved angle needs; it trusts you); (3) if the firmware cannot run the wiper at all, unload by hand: battery out (the worm holds the spring where it is), safety glasses on, spring guard on, stand to the side, and unhook ONE spring at a time from its anchor pin with a hook tool (a bent coat-hanger wire or a long screwdriver through the hook), never with fingers in the hook. Then the carriage is slack and safe. Afterwards, with the carriage home: `{"cmd":"cal","what":"cock_rest"}`, then `aim` again before any shot.
- Never turn the drum by hand while the carriage is drawn (unless `cock` is 0 AND `SW_HOME` is closed) or while the line is under tension: shifting under load can dry fire the launcher or drop the line. De-cock and slacken the line first.
- The launch springs store energy. Cocked carriage = loaded. Never put a hand in front of the rail or the carriage. To unload a cocked base without firing, see troubleshooting. The spring hooks have little margin at full draw (about 40 N each at full cock against the 38.4 N rating of the Century C-243, decision D30, until the

springs are measured): during shots stand beside the base, not behind or in front of the springs, and fit the printed spring guard if the pack has one. After B2, if the stronger spring's force at full cock is over the rating, use the fallback hole set (MEASURE_LIST B4).

- **Line under tension:** the capstan never turns back (one-way bearing), so a towing line cannot be let off at the base. During any pull (hand test or tow) stand beside or behind the rover, never in line with the dart, glasses on. To slacken it, release the dart (`release`) or drive the rover toward the dart. Calibrate the current-to-tension number (MEASURE_LIST G9) before any tow: with uncalibrated values a simulated tow reached 61 N against a 40 N limit.
- Chock or brake the rover during every shot and every tow. Strap the LiPo down under a cover on the rover.
- **Picking up a landed dart:** send dart STOP, check it reads IDLE (an armed dart can start anchoring when moved), then pull its loop plug before you touch it.
- The dart is 279 g leaving the rail at about 7 m/s: about 7 J, like a hard-thrown cricket ball, with sharp augers. From the first spring test on: safety glasses on, nobody in front of the rail or within 2 m to either side, and the area in front clear for 30 m (four times the longest expected range, for a misfire at a wrong pitch).
- Before every shot: look along the rail, call "firing" out loud, keep hands off the rail and carriage.
- A cocked carriage stays loaded until it fires or is de-cocked (`decock`). Treat the base as LOADED unless telemetry shows `cock 0` AND `H` (SW_HOME) closed in `sw`. Unknown (`cock` null) counts as loaded.
- **Never fire a dart, dummy or real, without the tether tied on.** The tether keeps it nose first; without it the dart can tumble or come down tail first.
- Augers and drills are sharp and turn with force. Hold the dart by the hull, never near a pod or the foot.
- Soldering: iron on its stand, fan on for fumes, 3 s at most per joint near PLA. Heat gun on low, no lighter near PLA.

3. Parts

Everything below is already in hand except the 608 bearings (and possibly the AS5600 magnet). Section 14 group C lists the items to confirm by looking in the box.

The only purchase: 608 bearings

- Search: "608ZZ bearing 8x22x7" (or "608-2RS", sealed). Buy 10. Skateboard bearings are 608s and fine.
- Specs: 8 mm bore, 22 mm outside, 7 mm wide, rated well over 10,000 rpm.
- Uses: the base CAD uses 4 (2 in the aim crank, 2 in the spool on the rail head); the coupons borrow some for testing; the rest are spares. Axles: M8 bolt or 8 mm rod (check the box).

Possible extra purchases

- **AS5600 magnet** (optional): the Adafruit board does not include one. The magnet cap takes any small neodymium disc (6 to 10 mm across, 2 to 3 mm thick) mounted edge-on, or a 6 x 2.5 mm diametric disc. Look in the box first (old speakers, fridge magnets with a neodymium disc). Without a magnet the firmware has a less precise fallback (decision D12): a printed 6-bump index cam read by the contact SW_INDEX on GPIO35, switched on in firmware/base config.h.
- Not a purchase: a second pair of launch springs. The CAD has holes for them, but the drive train is not rated for four springs (decision D38). v1 uses the two springs you have.

Base station and rover

Part	Qty	Note
12 V wiper motor, self-locking worm	1	Rebuilt to 2 wires, park linkage removed. Measure it first (section 14 A)
BTS7960 motor driver module	1	Drives the wiper
ESP32-S3 DevKitC-1-N8	1	Base brain, USB to the Jetson
Adafruit AS5600 (STEMMA QT)	1	Main shaft angle. Needs a small magnet (see above)
DRV8833 carrier	1	DRV8833 #3, freed from the dart by the new dart pin map
MP1584 buck	1	MP1584 #2 from the dart build, set to 9.0 V
N20 right-angle worm motor	1	Selector drum (count your N20s: section 14 C3)
693ZZ bearings (3 x 8 x 4)	1	Lip roller (no tether pulleys since D49; the dart uses 5 F693ZZ; count what is left, section 14 C20)
CSK20PP one-way bearings (20 x 47 x 14)	2	#1 spring cock and fire, #2 capstan backstop
Century 16 x 165.1 x 1.4 mm extension springs	2	Launch springs (Bunnings 0487961). Measure them (section 14 B)
608 bearings	10	The purchase
Dyneema braid 150 lb, 1 mm	20 m	Tether
100 kg load cell + HX711	1	Optional line tension
3S LiPo 5200 mAh (Zeee), iMax B6 charger	1	Rover battery
XT60 lead #1 (10 A fuse), lead #2 (20 A fuse), spare XT60s, fuse holder		Lead #2 feeds the wiper; build the Y harness from spares
Barrel plug lead 5.5 x 2.5 mm	1	Jetson from the LiPo
Jetson Orin Nano Super	1	
Lockmaner RC832 receiver, DIGITNOW USB capture, RCA lead	1 each	Dart video in
Razer Kiyo X	1	Rover camera
Resistors 10 k x 2, 1 k, 680 R, 100 k, 22 k		Pull-downs and dividers (firmware/base/WIRING.md)
Perfboard about 55 x 31 mm	1, optional	Two bay bosses take one (2 x M3 x 10) for the dividers and the 3V3 / GND buses; WIRING.md works without it (section 14 C22)
M3 screws, M3 nuts, M3 brass heat-set inserts		Frame joints and printed contacts
Dupont jumpers, 14 / 18 / 22 / 26 AWG wire, heat shrink		

Dart (KEDGE v30)

The full dart list with quantities is in the dart run sheet, section 2 (`kedge/pack_v30/KEDGE_v30_run_sheet.pdf`). In short: XIAO ESP32S3, 2 x DRV8833 (the third goes to

the base), INA219, BNO085, MP1584 #1 (5.04 V), OTOTEC 2S BMS, 2 x 18350 cells, 3 worm N20s (kick, deploy, foot), 2 drill gearmotors, 5 x F693ZZ, Caddx Ant, Zeus VTX and antenna, JST-XH sockets, Schottky diode, capacitor, M3 hardware, 15 g nose ballast.

Tools

Drill with 3 mm and 3.4 mm bits and a deburring tool (rover deck holes), an M3 heat-set insert tip for the soldering iron, a 13 mm socket with a 1/4 inch driver (coupler M8 nut). Soldering iron and solder, flux, multimeter, wire strippers, flush cutters, small flat screwdriver (MP1584 trimmer), heat shrink and heat gun, caliper, hex keys, tweezers, kitchen scale, luggage scale, tape measure, safety glasses, 5-minute epoxy, threadlocker, zip ties, foam tape.

Base bill of materials (from the model)

From `base/out/base/BOM.md`*:*

Bill of materials

Printed parts

part	subsystem	material	qty	mass g each	mass g total	notes
drum_bracket_b	drum	PLA	1	25.6	25.6	Drum bracket B: axle, guide rod, wrap idler journal; foot screwed to the reel head spine
drum_dog_post	drum	PLA	1	7.1	7.1	Dog contact post on the shift rail: SW_DOG_R and SW_DOG_L screws (collar fully engaged)
drum_drum_a	drum	PLA	1	103.2	103.2	Drum A: N20 gear 120T m0.8, groove 1 barrel (follower), hex spigot
drum_drum_b	drum	PLA	1	82.4	82.4	Drum B: brake cam (spool clutch lever), Gray code bands A B C, wrap segment gear + locking arc
drum_follower	drum	PLA	1	6.2	6.2	Follower slider: pin in groove 1, slides along the guide rod; the link takes its motion to the bridge
drum_fork	drum	PLA	1	19.9	19.9	Compliant shift fork: thin yoke is the spring (about 1.5 N/mm), lug for the bell crank pin, contact tab
drum_index_block	drum	PLA	1	33.2	33.2	SW_INDEX block: column on the motor plate front face (2 x M3 x 10 into the frame inserts), M3 nut contact
drum_index_lever	drum	PLA	1	0.9	0.9	SW_INDEX lever on the 6-bump coupler cam (D12 fallback encoder)
drum_link_eye_1	drum	PLA	1	3.0	3.0	Link eye pressed on the rod end, M3 screw through the follower lug
drum_link_eye_2	drum	PLA	1	3.0	3.0	Link eye pressed on the rod end, M3 screw through the bell crank input arm
drum_magnet_cap	drum	PLA	1	1.9	1.9	AS5600 magnet cap on the spigot end (6 x 2.5 diametric or edge-on disc), M3 x 6 onto the flat
drum_n20_cradle	drum	PLA	1	12.1	12.1	N20 cradle on the reel head front plate: pivot + swing screw (hand fallback), SW_N20 contact
drum_n20_strap	drum	PLA	1	2.7	2.7	End plate closing the N20 cradle, 2 x M3 x 8
drum_shear_pin	drum	PLA	6	0.2	1.1	Torque limiter shear pin d4 (about 5 N m), prints lying; print 6, 5 are spares
drum_sw_lever_0	drum	PLA	1	0.9	0.9	Code switch lever A: flexure, round nub, M3 x 6 contact screw
drum_sw_lever_1	drum	PLA	1	0.9	0.9	Code switch lever B: flexure, round nub, M3 x 6 contact screw
drum_sw_lever_2	drum	PLA	1	0.9	0.9	Code switch lever C: flexure, round nub, M3 x 6 contact screw
drum_switch_block	drum	PLA	1	18.3	18.3	Code switch block on top of the drum: 3 lever feet, 3 M3 nut contacts (SW_DRUM_A, B, C)

part	subsystem	material	qty	mass g each	mass g total	notes
frame_cable_anchor_1	frame	PLA	1	4.0	4.0	cable-tie anchor (strain relief), 1 x M3 x 12 into the deck
frame_cable_anchor_2	frame	PLA	1	4.0	4.0	cable-tie anchor (strain relief), 1 x M3 x 12 into the deck
frame_camera_mount	frame	PLA	1	166.5	166.5	Kiyo X mount: its own clip grips the 16 mm edge plate (or a 1/4-20 screw), 4 x M3 x 10 into the deck
frame_can_stay	frame	PLA	1	28.4	28.4	can stay: post on the motor plate (2 x M3 x 12 into plate inserts), cable tie or hose clamp round the can through the cradle slots; reprint with the adapter
frame_cheek_r	frame	PLA	1	205.2	205.2	motor plate (y -10..0, front face = the spec's y 0): window for the puck adapter, 8 rim holes (M3 x 12 from the front into the puck lip), bridge to the motor leg (2 x M3 x 12), can stay and hood inserts; carved round the other subsystems
frame_deck_1	frame	PLA	1	233.6	233.6	deck: M3 inserts for every frame foot; 10 x M3 rover holes (it is its own drill template)
frame_deck_2	frame	PLA	1	304.3	304.3	deck: M3 inserts for every frame foot; 12 x M3 rover holes (it is its own drill template)
frame_deck_3	frame	PLA	1	231.4	231.4	deck: M3 inserts for every frame foot; 9 x M3 rover holes (it is its own drill template)
frame_deck_4	frame	PLA	1	301.6	301.6	deck: M3 inserts for every frame foot; 14 x M3 rover holes (it is its own drill template)
frame_ebay	frame	PLA	1	175.3	175.3	electronics bay: ESP32-S3 cradle (USB slot), BTS7960 on standoffs with vents, DRV8833, MP1584, HX711 pockets, fuse cradle, wire exits with tie anchors
frame_ebay_esp_clamp	frame	PLA	1	1.6	1.6	ESP32 hold-down over the module end, 2 x M3 x 8 into the cradle posts
frame_ebay_lid	frame	PLA	1	71.4	71.4	bay lid: grille over the BTS7960, 4 x M3 into the corner bosses
frame_guard_front	frame	PLA	1	56.8	56.8	front gear guard between the drum gears and the bay, M3 x 10 into the deck
frame_guard_hood	frame	PLA	1	14.1	14.1	gear hood over the reel and bus gears, M3 x 10 through its lugs into the motor plate
frame_motor_leg	frame	PLA	1	117.3	117.3	motor support leg (prints upright): 4 x M3 x 10 into the deck, 2 x M3 x 12 along y into the bridge behind the motor plate
frame_rc832_holder	frame	PLA	1	22.8	22.8	RC832 holder: receiver stands antenna-end up in the slot, leads out low, 2 x M3 x 10
launch_aim_cap	launch	PLA	1	5.2	5.2	Worm shaft saddle cap, 4 x M3x30 through the legs into 13 deep pilots in the saddles
launch_aim_crankpin_clip	launch	PLA	1	0.1	0.1	Printed C clip on the crank pin, inboard of the conrod
launch_aim_drag_pad	launch	PLA	1	0.4	0.4	Flexure drag pad: cantilever finger t 2, tip bump 0.45 into the sleeve d20 (strain 0.43 %, about 2.3 N), root block pushed axially into the arm hub pocket
launch_aim_drag_shim	launch	PLA	1	0.1	0.1	Preload shim behind the drag pad root block (nominal; print up to 0.25 thicker for more drag, strain stays under 0.7 %)
launch_aim_housing	launch	PLA	1	185.4	185.4	Aim housing on the deck (4 x M3 x 10 from the top through its counterbored foot flanges into the deck inserts): sleeve end bushing, countershaft bushings, worm saddles, M8 wheel bolt (head on the outer boss, nut in the inner face hex pocket), engaged and swung-out swing arm stop posts
launch_aim_idler_clip	launch	PLA	1	0.1	0.1	Printed C clip on the arm pin, holds the idler

part	subsystem	material	qty	mass g each	mass g total	notes
launch_aim_lugpin_clip	launch	PLA	1	0.1	0.1	Printed C clip on the lug pin, outboard of the conrod
launch_carriage	launch	PLA	1	59.6	59.6	Carriage: rack, shoes, dart cradle, pusher with tether hole, spring arms and pegs, SW feet
launch_carriage_yoke	launch	PLA	1	23.5	23.5	Bolted yoke: both spring arms (hit the stop, d8.1 holes for the 8 mm steel rod spring pegs, D5 holes at y +-56 empty), cross bar, keys, SW_SEATED block, rigid SW_HOME foot. Prints on the peg boss bottoms and key plates (w 62)
launch_drive_washer	launch	PLA	1	0.2	0.2	Spacer washer: drive gear hub to the left cheek
launch_gear_clip	launch	PLA	1	0.3	0.3	C clip on the big gear pin groove
launch_gear_washer	launch	PLA	1	0.5	0.5	Spacer washer: big gear hub to the left cheek
launch_guard_l_a	launch	PLA	1	65.0	65.0	Spring guard left rear (end wall, top cover) (D33): 2 mm side plate outside both spring pairs, 4 wall brackets (M3x12 self tap into the rail wall); fit by moving it inward
launch_guard_l_b1	launch	PLA	1	4.5	4.5	Spring guard left mid front of the stop frame (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall); fit by moving it inward
launch_guard_l_b2	launch	PLA	1	17.9	17.9	Spring guard left mid under the stop spring (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall) and a foot on the front plate (M3x10 self tap down); fit by moving it inward
launch_guard_l_c	launch	PLA	1	4.8	4.8	Spring guard left front (top cover) (D33): 2 mm side plate outside both spring pairs, 2 bosses on the front plate edge (M3x12 self tap along y); fit by moving it inward
launch_guard_r_a	launch	PLA	1	65.0	65.0	Spring guard right rear (end wall, top cover) (D33): 2 mm side plate outside both spring pairs, 4 wall brackets (M3x12 self tap into the rail wall); fit by moving it inward
launch_guard_r_b1	launch	PLA	1	4.5	4.5	Spring guard right mid front of the stop frame (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall); fit by moving it inward
launch_guard_r_b2	launch	PLA	1	21.8	21.8	Spring guard right mid under the stop spring (D33): 2 mm side plate outside both spring pairs, 2 wall brackets (M3x12 self tap into the rail wall) and a foot on the front plate (M3x10 self tap down); fit by moving it inward
launch_guard_r_c	launch	PLA	1	4.8	4.8	Spring guard right front (top cover) (D33): 2 mm side plate outside both spring pairs, 2 bosses on the front plate edge (M3x12 self tap along y); fit by moving it inward
launch_muzzle_roller	launch	PLA	1	0.5	0.5	Lip roller at the carriage front: the tether bends down over it in tow and retrieval (groove r 6 for the 1 mm line), 693ZZ inside
launch_muzzle_washer_l	launch	PLA	1	0.0	0.0	Lip roller thrust washer (0.9 thick, flange on the cheek side) between the bearing inner race and the cheek: put both in the roller bore, then drop roller and washers into the fork slot
launch_muzzle_washer_r	launch	PLA	1	0.0	0.0	Lip roller thrust washer (0.9 thick, flange on the cheek side) between the bearing inner race and the cheek: put both in the roller bore, then drop roller and washers into the fork slot
launch_rail_front	launch	PLA	1	137.7	137.7	Rail front piece: shoe track end, spring anchor posts with straps, muzzle tie

part	subsystem	material	qty	mass g each	mass g total	notes
launch_rail_mid	launch	PLA	1	178.7	178.7	Rail mid piece: shoe track, cheeks on the sleeve, big gear pin bosses, lever seats, SW_GAP bracket pilots
launch_rail_rear	launch	PLA	1	115.9	115.9	Rail rear piece: shoe track, walls, flange at the rear end for the reel head (D49)
launch_rail_splice_l	launch	PLA	1	0.3	0.3	Top splice strap over the front rail joint (4 x M3 x 10 self tap)
launch_rail_splice_r	launch	PLA	1	0.3	0.3	Top splice strap over the front rail joint (4 x M3 x 10 self tap)
launch_seat_clip	launch	PLA	1	0.1	0.1	Head cap on the plunger pin (rests on the yoke block)
launch_stop_frame_l	launch	PLA	1	32.7	32.7	Stop frame (print outboard face down): outboard preload ledge, anchor seat, rear foot (2 x M3x12 into the rail wall), front foot (2 x M3x16 into the rail front plate, hex key through the d6.5 holes in the front wall). Fit before the spring guards and the main springs
launch_stop_frame_r	launch	PLA	1	32.7	32.7	Stop frame (print outboard face down): outboard preload ledge, anchor seat, rear foot (2 x M3x12 into the rail wall), front foot (2 x M3x16 into the rail front plate, hex key through the d6.5 holes in the front wall). Fit before the spring guards and the main springs
launch_stop_spring_l	launch	PLA	1	81.6	81.6	Carriage stop flexure: 16 loops (31 tapered leaves t 3.15/1.0 x 14), print upside down on the flat w 83.5 face, 100% infill / all perimeters; preload 6.6 mm by 2 x M3x20 draw screws through the frame front wall
launch_stop_spring_r	launch	PLA	1	81.6	81.6	Carriage stop flexure: 16 loops (31 tapered leaves t 3.15/1.0 x 14), print upside down on the flat w 83.5 face, 100% infill / all perimeters; preload 6.6 mm by 2 x M3x20 draw screws through the frame front wall
launch_sw_home	launch	PLA	1	0.8	0.8	SW_HOME printed flex lever on the right rail lip, at rest (prints on its side): beam 0.8 thick, contact gap 0.50 (one screw turn), pad pretravel 0.70 + overtravel 0.25, peak strain 0.10 % (limit 0.8 %)
launch_sw_seated	launch	PLA	1	0.8	0.8	SW_SEATED printed flex lever on the left rail lip, at rest (prints on its side): beam 0.8 thick, contact gap 0.50 (one screw turn), pad pretravel 0.70 + overtravel 0.25, peak strain 0.10 % (limit 0.8 %)
reel_bridge	reel	PLA	1	118.5	118.5	Rail bridge: plate inboard of the main-axis stack with countershafts C and T, the c1' stub, pillars to the u-shaft bushings, cad-drum's shift rail; box arm bolted to rail cheek A. Print on its inboard face
reel_cup	reel	PLA	1	12.7	12.7	Wrap cup (you v7, D49 axial eyelet at r 26): the line runs straight through the eyelet. Print open end down
reel_cup_clip	reel	PLA	1	0.2	0.2	your v7 C-clip on the cup shaft neck in front of the cup
reel_head_front	reel	PLA	1	108.4	108.4	Head front plate: bolts the reel head to the rail rear flange, u-shaft T rear bushing, link window. Print flat
reel_head_mid	reel	PLA	1	19.3	19.3	Head mid plate: spool shaft inboard bushing. Print flat
reel_head_p1	reel	PLA	1	55.0	55.0	Head plate P1: capstan tube, u-shaft C and G_b bushings, clutch lever post, spool shaft leg. Print flat

part	subsystem	material	qty	mass g each	mass g total	notes
reel_head_p1b	reel	PLA	1	10.7	10.7	Head plate P1b: capstan tube rear bushing, SW_ARM. Print flat
reel_head_p2	reel	PLA	1	94.1	94.1	Head plate P2: cup shaft and wrap idler rear journals. Print flat
reel_head_spine	reel	PLA	1	104.2	104.2	Head spine: inboard wall from the front plate to P2; the drum brackets and plates screw to it. Print flat
reel_shroud	reel	PLA	1	7.6	7.6	Spool shroud: keeps loose loops on the spool, open at the top front where the line leaves
reel_spool	reel	PLA	1	30.8	30.8	Spool d40 core, d62 flanges, 20 wide: 20 m of 1 mm Dyneema, turns free on two 608s. Print flange down
reel_sw_base	reel	PLA	1	7.0	7.0	SW_ARM contact base on P2's front face
reel_sw_lever	reel	PLA	1	1.3	1.3	SW_ARM lever: rides the cup gear's raised track (contact closed), drops into its home notch (open)
sensors_as5600_carrier	sensors	PLA	1	12.7	12.7	AS5600 carrier: plate with 4 pegs through the board holes, column screwed to the aim housing spine
sensors_as5600_strip_bot	sensors	PLA	1	1.0	1.0	AS5600 clamp strip on the board edge over the pegs, M3 from the strip front into a nut in the carrier back
sensors_as5600_strip_top	sensors	PLA	1	1.0	1.0	AS5600 clamp strip on the board edge over the pegs, M3 from the strip front into a nut in the carrier back
sensors_sw_gap	sensors	PLA	1	9.9	9.9	SW_GAP printed contact base (cad.lib.switch, coupon 2 geometry): with its arm screwed to the right rail cheek outer face below the big gear, closed by the big gear cam bump (gear in its gap)
sensors_sw_gap_lever	sensors	PLA	1	1.4	1.4	SW_GAP lever, t 1.4 mm, print on its side so layers run along the beam
sensors_sw_rail_low	sensors	PLA	1	7.3	7.3	SW_RAIL_LOW printed contact base (cad.lib.switch, coupon 2 geometry): on its bracket on the aim housing foot, closed by the crank cam at the LOW dead point
sensors_sw_rail_low_bracket	sensors	PLA	1	17.8	17.8	SW_RAIL_LOW bracket on the aim housing foot extension: an M3x10 into the extension and its own deck screw, an M3x16 from the top through the counterbored foot and the extension into a deck insert
sensors_sw_rail_low_lever	sensors	PLA	1	1.3	1.3	SW_RAIL_LOW lever, t 1.4 mm, print on its side so layers run along the beam
drum_axle	drum	PPA-CF	1	8.9	8.9	Fixed drum axle d10, pressed into the reel head front plate and bracket B
drum_collar	drum	PPA-CF	1	44.0	44.0	Dog collar: hex bore, 4 back-cut roofed dogs each face, fork groove (bridges over the lower dogs)
drum_coupler	drum	PPA-CF	1	89.3	89.3	Coupler: 5 N m shear pin limiter (D8), bus gear 50T m1.5 (D25), SW_INDEX cam, REEL journal, hex socket; prints hub end down, tree support under the bus gear ring
drum_coupler_hub	drum	PPA-CF	1	14.6	14.6	Coupler hub on the wiper shaft (bore parametric: WIPER), M8 nut inside, shear pin hole
drum_crank	drum	PPA-CF	1	6.9	6.9	Bell crank 1:1 on the bridge shelf: turns the link's motion along the rail into the fork's shift
drum_dogring_launch	drum	PPA-CF	1	19.0	19.0	LAUNCH dog ring: 3 slots for the launch sleeve lugs, 3 x M3 from the sleeve side

part	subsystem	material	qty	mass g each	mass g total	notes
drum_gear_reel	drum	PPA-CF	1	95.9	95.9	REEL output gear m2 50T face 12 (D20), free on its sleeve, dog face toward the collar; meshes C1 on the rail bridge (D49)
drum_idler	drum	PPA-CF	1	8.7	8.7	Wrap idler shaft: pinion 10T m1.25 (locked at home by the drum lock arc), lock cam, journals in bracket B and P2
drum_idler_gear	drum	PPA-CF	1	35.8	35.8	Wrap idler output gear 49T m1.5 (D bore) to the cad-reel cup gear 28T: 7 cup turns
drum_n20_pinion	drum	PPA-CF	1	0.5	0.5	N20 pinion 10T m0.8, D bore pressed on the N20 shaft
drum_reel_washer	drum	PPA-CF	1	0.9	0.9	REEL gear retaining washer on the coupler journal end, 2 x M3 x 6
drum_rod	drum	PPA-CF	1	8.9	8.9	Follower guide rod d10, pressed into the reel head front plate and bracket B
drum_shaft	drum	PPA-CF	1	45.2	45.2	Main shaft: hex 22 AF, d12 spigot into the launch sleeve; prints on a hex flat (support under the spigot)
drum_shift_rail	drum	PPA-CF	1	8.2	8.2	Shift rail d10 on the rail bridge (pressed into the bridge plate boss)
frame_motor_adapter	frame	PPA-CF	1	302.8	302.8	puck adapter: slotted M6 holes (+-4 mm) for the motor bosses, rear lip with 8 M3 inserts for the rim screws, 2 inserts for cad-drum's SW_INDEX block; reprint when the motor is measured
launch_aim_arm	launch	PPA-CF	1	6.6	6.6	Swing arm on the sleeve, dragged by the printed PLA flexure pad in its hub; printed idler pin with a printed C clip groove
launch_aim_bevel30	launch	PPA-CF	1	8.8	8.8	Bevel m1.5 z30 on the countershaft D flat, drives the z15 2:1
launch_aim_conrod	launch	PPA-CF	1	14.2	14.2	Conrod 156 between pin centres (PPA-CF), printed C clips on both pins
launch_aim_crank	launch	PPA-CF	1	35.3	35.3	Worm wheel m2 z20 face 16 = crank gear, crank arm r30 and d8 crank pin in one part, two 608 inside
launch_aim_idler	launch	PPA-CF	1	7.5	7.5	Splitter idler m1.5 z18 face 12 on the arm pin
launch_aim_lug	launch	PPA-CF	1	16.9	16.9	Lug bracket (PPA-CF), 2 x M3 into the rail wall, d8 lug pin at (u 115.2, w 0)
launch_aim_output	launch	PPA-CF	1	26.2	26.2	Splitter output m1.5 z30 face 12 with its d8 countershaft (D flat)
launch_aim_sun	launch	PPA-CF	1	18.3	18.3	Splitter sun m1.5 z30 face 12, keyed on the launch sleeve
launch_aim_worm	launch	PPA-CF	1	21.3	21.3	Worm m2 pd24 single start (lead angle 4.8 deg, self-locking) + d8 shaft + bevel z15
launch_big_gear	launch	PPA-CF	1	177.6	177.6	Big gear m2.5: sector layer 27 of 40 teeth (rack, v10 release teeth) + full 40T layer (drive), SW_GAP cam bump, on the rail pin
launch_drive_gear	launch	PPA-CF	1	29.9	29.9	Drive gear m2.5 26T, sprag #1 inside on an integral key rib
launch_gear_pin	launch	PPA-CF	1	5.4	5.4	Snap pin d10 for the big gear, C clip; prints lying on its flat
launch_seat_plunger	launch	PPA-CF	1	1.0	1.0	SW_SEATED plunger: foot pad + stem with a 45 deg chamfer, clip groove on top
launch_seat_pushrod	launch	PPA-CF	1	0.2	0.2	SW_SEATED pushrod d3.2 with the stem slot
launch_sleeve	launch	PPA-CF	1	49.3	49.3	Launch sleeve d20: dog ring flange, integral key ribs (sprag #1, splitter sun), journals

part	subsystem	material	qty	mass g each	mass g total	notes
launch_yoke_peg_l	launch	PPA-CF	1	1.2	1.2	Spring peg on the yoke arm (printed PPA-CF lying down, or 8 mm steel rod 20 long): pressed into the blind d8.1 hole from below up to its end (a drop of CA if it is loose), the spring hook sits on the 6 mm below the boss (held at w 60 by the spring line); ends deburred
launch_yoke_peg_r	launch	PPA-CF	1	1.2	1.2	Spring peg on the yoke arm (printed PPA-CF lying down, or 8 mm steel rod 20 long): pressed into the blind d8.1 hole from below up to its end (a drop of CA if it is loose), the spring hook sits on the 6 mm below the boss (held at w 60 by the spring line); ends deburred
reel_b1g	reel	PPA-CF	1	2.6	2.6	Bevel B1 gear m1.5 24T (2:1 up to u-shaft C)
reel_b1p	reel	PPA-CF	1	1.2	1.2	Bevel B1 pinion m1.5 12T on the front end of u-shaft C
reel_b2g	reel	PPA-CF	1	0.9	0.9	Bevel B2 m1 20T on countershaft T
reel_b2p	reel	PPA-CF	1	1.3	1.3	Bevel B2 m1 20T on the front end of u-shaft T
reel_b3a	reel	PPA-CF	1	1.3	1.3	Bevel B3 m1 20T on the rear end of u-shaft T
reel_b3b	reel	PPA-CF	1	1.4	1.4	Bevel B3 m1 20T on the spool shaft
reel_c	reel	PPA-CF	1	17.4	17.4	Capstan tube C (PPA-CF d20): key rib in the drum, G_c on its flat, journals in P1 and P1b. Print standing on its rear end
reel_c1	reel	PPA-CF	1	93.7	93.7	C1 m2 49T: meshes the REEL output gear 50T across the pivot
reel_c1p	reel	PPA-CF	1	27.8	27.8	c1' m1.5 46T, free on its stub: driven by the bus gear 50T, drives the swing idler
reel_cs_c	reel	PPA-CF	1	11.4	11.4	Countershaft C (PPA-CF): sprag #2, C1 and the B1 bevel. Print standing
reel_cs_t	reel	PPA-CF	1	2.8	2.8	Countershaft T (PPA-CF): T1 and the B2 bevel
reel_d	reel	PPA-CF	1	7.6	7.6	Clutch disk D on the spool shaft flat: its friction ring presses the spool's outboard face. Print ring up
reel_drum	reel	PPA-CF	1	45.0	45.0	Capstan drum dia 40 x 40 (you v7), keyed on the tube. Print flange down
reel_ga	reel	PPA-CF	1	9.6	9.6	G_a m1.5 26T on the rear end of u-shaft C
reel_gb	reel	PPA-CF	1	15.1	15.1	G_b m1.5 32T idler on its stub in P1
reel_gc	reel	PPA-CF	1	5.3	5.3	Capstan gear G_c m1.5 23T on the tube flat
reel_gi	reel	PPA-CF	1	11.4	11.4	Swing idler m1.5 30T
reel_gk	reel	PPA-CF	1	15.0	15.0	Cup gear G_k m1.5 28T: driven by the drum's wrap idler 49T (7 cup turns per wrap); the raised track on its rear face holds SW_ARM closed, the notch at home lets it open. Print track face down
reel_k	reel	PPA-CF	1	10.2	10.2	Cup shaft K (PPA-CF d10): runs inside the tube, cup in front, G_k behind P1b, journal in P2. Print standing on its rear end
reel_lever	reel	PPA-CF	1	6.9	6.9	Clutch lever (PPA-CF): the drum's brake cam presses its upper arm; the lower arm is the clutch spring leaf, its FIRM screw squeezes the spool. Print flat
reel_s	reel	PPA-CF	1	5.7	5.7	Spool shaft S (PPA-CF d8): B3 bevel and the clutch disk on its flats, journals in the mid plate and the P1 leg. Print lying on its 1 mm flat
reel_sprag_hub	reel	PPA-CF	1	12.2	12.2	Sprag #2 hub on the bridge: holds the CSK20PP capstan backstop
reel_stub	reel	PPA-CF	1	1.1	1.1	c1' stub axle (PPA-CF d8, pressed into the bridge)

part	subsystem	material	qty	mass g each	mass g total	notes
reel_swing_arm	reel	PPA-CF	1	3.2	3.2	Swing arm on c1's hub: carries the idler into mesh with T1 on forward wiper turns
reel_t	reel	PPA-CF	1	1.1	1.1	Thrust ring: the lever's FIRM screw pushes it onto the inboard 608 inner race (clutch squeeze)
reel_t1	reel	PPA-CF	1	4.6	4.6	T1 m1.5 20T: take-up at 2.5 x wiper
reel_uc	reel	PPA-CF	1	23.7	23.7	u-shaft C (PPA-CF d10, 244 long): along the rail from the B1 bevel to G_a. Print standing (244 of the 260 mm height); an 8 mm steel rod with printed sleeves is the stiff fallback
reel_ut	reel	PPA-CF	1	6.5	6.5	u-shaft T (PPA-CF d8): along the rail from the B2 bevel to the B3 bevel. Print standing

Printed totals

material	parts	mass g	filament m (1.75 mm)
PLA	95	4011	1344.9
PPA-CF	60	1456	484.4

Bought parts

Bearings

item	count	used by
608ZZ 8x22x7	4	launch_aim_brg_1, launch_aim_brg_2, reel_brg_1, reel_brg_2
693ZZ 3x8x4	1	launch_muzzle_roller_brg
Adafruit AS5600 STEMMA QT PID 6357	1	sensors_as5600
CSK20PP 20x47x14 sprag #1	1	launch_sprag
CSK20PP one-way bearing 20x47x14	1	reel_sprag

Fasteners

item	count	used by
M3 nut	6	launch_sw_seated_nut, launch_sw_home_nut, sensors_as5600_strip_top, sensors_as5600_strip_bot, sensors_sw_rail_low_nut, sensors_sw_gap_nut
M3 nut (jam)	2	sensors_sw_rail_low_screw, sensors_sw_gap_screw
M3 nyloc nut	1	launch_muzzle_roller_axle
M3x10 self tap	2	launch_guard_l_b2, launch_guard_r_b2
M3x10 self tap (lever clamp)	6	reel_sw_lever, sensors_sw_rail_low_lever, sensors_sw_gap_lever
M3x10 SHCS	4	drum_index_block, sensors_sw_rail_low_screw, sensors_sw_gap_screw
M3x10 SHCS (base to P2)	4	reel_sw_base
M3x10 SHCS (foot into the housing)	1	sensors_sw_rail_low_bracket
M3x10 SHCS + 2 M3 nuts	1	reel_sw_hw
M3x10 SHCS + M3 nut	1	reel_firm_screw
M3x12 self tap	17	launch_guard_l_a, launch_guard_l_b1, launch_guard_l_b2, launch_guard_l_c, launch_guard_r_a, launch_guard_r_b1, launch_guard_r_b2, launch_guard_r_c
M3x12 SHCS	4	reel_head_mid, sensors_as5600_strip_top, sensors_as5600_strip_bot
M3x12 SHCS (arm into the right cheek)	2	sensors_sw_gap
M3x12 SHCS (posts)	2	reel_head_p1b
M3x12 SHCS (spine)	7	reel_head_front, reel_head_p1, reel_head_p2
M3x12 SHCS (through the spine)	4	drum_bracket_b, drum_switch_block
M3x16 SHCS	5	launch_muzzle_roller_axle, reel_sprag_hub
M3x16 SHCS (arm to cheek A)	2	reel_bridge
M3x16 SHCS (deck screw)	1	sensors_sw_rail_low_bracket
M3x16 SHCS (into the housing spine)	2	sensors_as5600_carrier
M3x16 SHCS (into the rail flange inserts)	4	reel_head_front
M3x16 SHCS (pivot, tapped into the shelf)	1	drum_crank

item	count	used by
M3x16 SHCS + M3 nut	1	reel_light_screw
M3x16 SHCS + nyloc nut	2	drum_link_eye_1, drum_link_eye_2
M3x16 SHCS + washer	1	reel_lever_pin
M3x6 SHCS	2	launch_sw_seated_screw, launch_sw_home_screw
M3x8 self tap (base to bracket)	4	sensors_sw_rail_low
M3x8 SHCS (tab to the P1 leg)	1	reel_shroud
M8 hex nut ISO 4032	1	launch_aim_nut
M8x25 hex bolt ISO 4017	1	launch_aim_bolt
STEMMA QT JST SH 4 pin cable	1	sensors_as5600

Other

item	count	used by
(part of the wiper motor)	1	frame_wiper_shaft
12 V car wiper motor (self-locking worm), 2-wire rebuild	1	frame_wiper_motor
8 mm steel rod, 26 mm long	1	reel_gb_axle
8 mm steel rod, cut to length (see note), ends deburred	2	launch_anchor_pin_l, launch_anchor_pin_r
8 mm steel rod, cut to the length on the print sheet	1	drum_link
BTS7960 43 A module (IBT-2)	1	frame_pcb_bts7960
Century C-243 (5/8 x 6-1/2 in x 0.054 in wire) extension spring	2	launch_spring_l_hook_a, launch_spring_r_hook_a
Dyneema braid 1 mm (LINE)	21	launch_line_a0, launch_line_a1, launch_line_a2, launch_line_a3, launch_line_a4, launch_line_a5, launch_line_a6, launch_line_a7, launch_line_a8, launch_line_a9, launch_line_a10, launch_line_a11, launch_line_a12, launch_line_a13, launch_line_a14, launch_line_a15, launch_line_a16, launch_line_a17, launch_line_a18, launch_line_a19, launch_line_a20
Espressif ESP32-S3-DevKitC-1-N8	1	frame_pcb_esp32
HX711 load cell amplifier (optional)	1	frame_pcb_hx711
inline ATO fuse holder + 20 A fuse	1	frame_fuse_20a
KEDGE v30 dart (reference, built separately)	1	launch_dart
launch_line_p (bought, no bought_ref)	1	launch_line_p
launch_spring_l_0 (bought, no bought_ref)	1	launch_spring_l_0
launch_spring_l_1 (bought, no bought_ref)	1	launch_spring_l_1
launch_spring_l_2 (bought, no bought_ref)	1	launch_spring_l_2
launch_spring_l_3 (bought, no bought_ref)	1	launch_spring_l_3
launch_spring_l_4 (bought, no bought_ref)	1	launch_spring_l_4
launch_spring_l_hook_c (bought, no bought_ref)	1	launch_spring_l_hook_c
launch_spring_r_0 (bought, no bought_ref)	1	launch_spring_r_0
launch_spring_r_1 (bought, no bought_ref)	1	launch_spring_r_1
launch_spring_r_2 (bought, no bought_ref)	1	launch_spring_r_2
launch_spring_r_3 (bought, no bought_ref)	1	launch_spring_r_3
launch_spring_r_4 (bought, no bought_ref)	1	launch_spring_r_4
launch_spring_r_hook_c (bought, no bought_ref)	1	launch_spring_r_hook_c
MP1584EN buck module, set to 9.0 V	1	frame_pcb_mp1584
N20 right-angle self-locking worm gearmotor 12 V 27 rpm (have)	1	drum_n20
perfboard 55 x 31 (dividers, pull-downs, 3V3 and GND buses), 2 x M3 x 10 on wall bosses	1	frame_perfboard
Pololu DRV8833 2130 (headers off)	1	frame_pcb_drv8833
Razer Kiyo X webcam	1	frame_kiyo_x
RC832 5.8 GHz receiver (80 x 65 x 15)	1	frame_rc832
sensors_as5600_col_screws (bought, no bought_ref)	1	sensors_as5600_col_screws
sensors_as5600_nut_bot (bought, no bought_ref)	1	sensors_as5600_nut_bot
sensors_as5600_nut_top (bought, no bought_ref)	1	sensors_as5600_nut_top

item	count	used by
sensors_as5600_screw_bot (bought, no bought_ref)	1	sensors_as5600_screw_bot
sensors_as5600_screw_top (bought, no bought_ref)	1	sensors_as5600_screw_top
sensors_sw_gap_jam (bought, no bought_ref)	1	sensors_sw_gap_jam
sensors_sw_rail_low_jam (bought, no bought_ref)	1	sensors_sw_rail_low_jam

Parts that are not printed and have no bought_ref: launch_spring_l_hook_c, launch_spring_l_0, launch_spring_l_1, launch_spring_l_2, launch_spring_l_3, launch_spring_l_4, launch_spring_r_hook_c, launch_spring_r_0, launch_spring_r_1, launch_spring_r_2, launch_spring_r_3, launch_spring_r_4, launch_line_p, sensors_as5600_nut_top, sensors_as5600_screw_top, sensors_as5600_nut_bot, sensors_as5600_screw_bot, sensors_as5600_col_screws, sensors_sw_rail_low_jam, sensors_sw_gap_jam.

4. Printing

Printer: Bambu X2D (bed 256 x 256 x 260 mm). PLA for everything except loaded gears and keyed shafts, which are PPA-CF (the print list says which).

Magnet for the bench: the AS5600 magnet cap is coupon 10 on the PLA risk plate (`base/pack/coupons/rc10_magnet_cap.stl`; it fits the main shaft spigot, so it is for the base, not the bare motor); glue the magnet in it. For the bench test on the clamped motor, tape the magnet centred on the shaft end and clamp the AS5600 board 1 to 2 mm from it. `status` must show `"magnet":"OK"` (WEAK: move the board closer; STRONG: further; NONE: no magnet seen).

Print settings (the 3MF files carry geometry only: set material and settings in Bambu Studio for each plate):

Plate	Material	Layer	Walls	Infill	Top / bottom	Hole compensation
Dart fit coupon, dart plates A and B	PLA	0.16 mm	4	40% gyroid	5 / 5	start at 0; adjust from the dart fit coupon (run sheet section 4)
Base fit coupon, risk plate 1	PLA	0.2 mm	4	40% gyroid	5 / 5	0 ("print as exported")
Risk plate 2	PPA-CF	0.2 mm	6	60% gyroid	6 / 6	0
Base plates, PLA	PLA	0.2 mm	4	40% gyroid	5 / 5	0, unless the coupons showed a fit off
Base plates, PPA-CF	PPA-CF	0.2 mm	6	60% gyroid	6 / 6	0, unless the coupons showed a fit off

Supports: none on the coupon plates and the dart plates. The base plates do need them: the base print sheet (below) has a supports column, and on every plate that holds a part marked "yes, tree supports from the plate", turn supports on in Bambu Studio (type tree, "On build plate only"). The two carriage stop flexures (`launch_stop_spring_l` and `launch_stop_spring_r`) also need 100 % infill (all perimeters), set on that object. The base print sheet has the final word for each part.

Hole compensation is a process setting in Bambu Studio, "X-Y hole compensation" (Process, Quality tab; the name may move between versions). It is saved with each project, so check it on every plate.

PPA-CF (risk plate 2 and the loaded base parts): hardened steel nozzle only (CF wears a brass nozzle quickly). Dry the spool first (about 80 C for 8 hours in a filament dryer, or the printer's own drying if it has one; it absorbs water and prints weak and stringy otherwise). Use the Bambu PPA-CF profile for temperatures. Print with the room ventilated, and wear a dust mask when sanding or trimming CF parts.

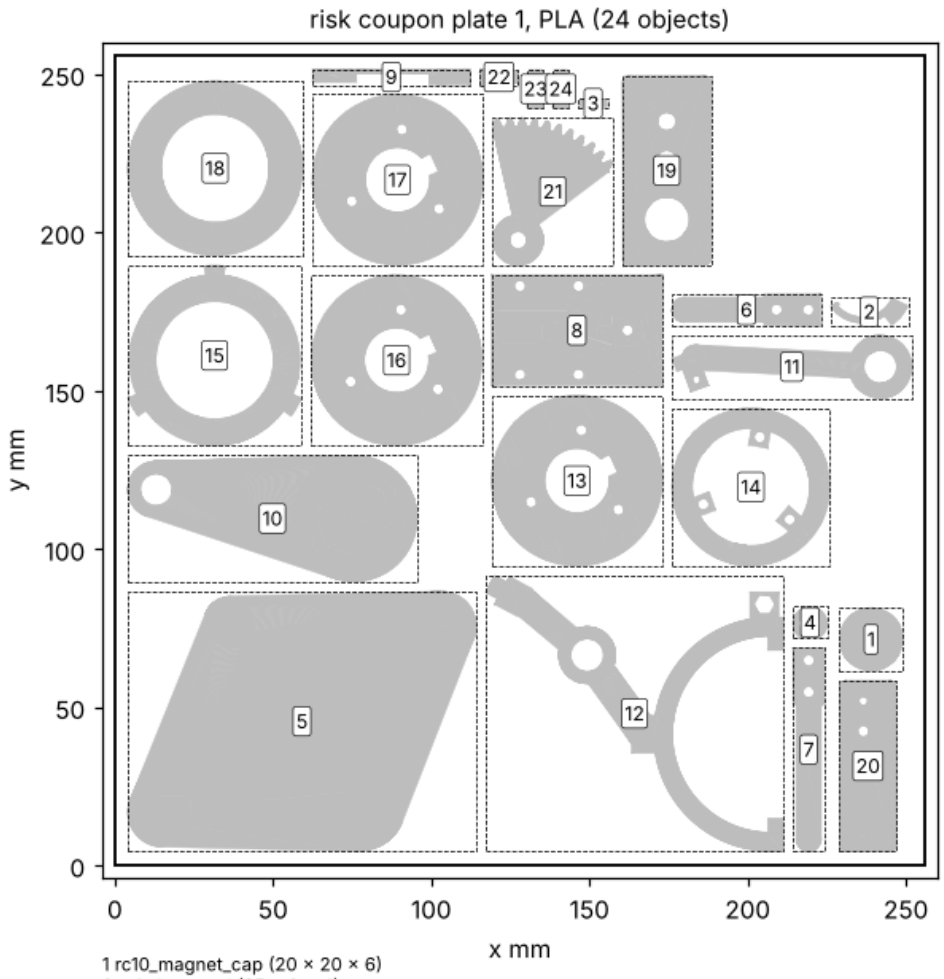
Order: dart fit coupon first (20 minutes, checks your printer's hole sizes), then the base fit coupon (`base/pack/fit_coupon/coupon_plate_1.3mf`, PLA) and the two base risk coupon plates (below, test stage 0), then the dart plates, then the base once the coupons pass and the wiper and spring numbers are in (section 1, order of work).

Day-one risk coupons (print before any big part)

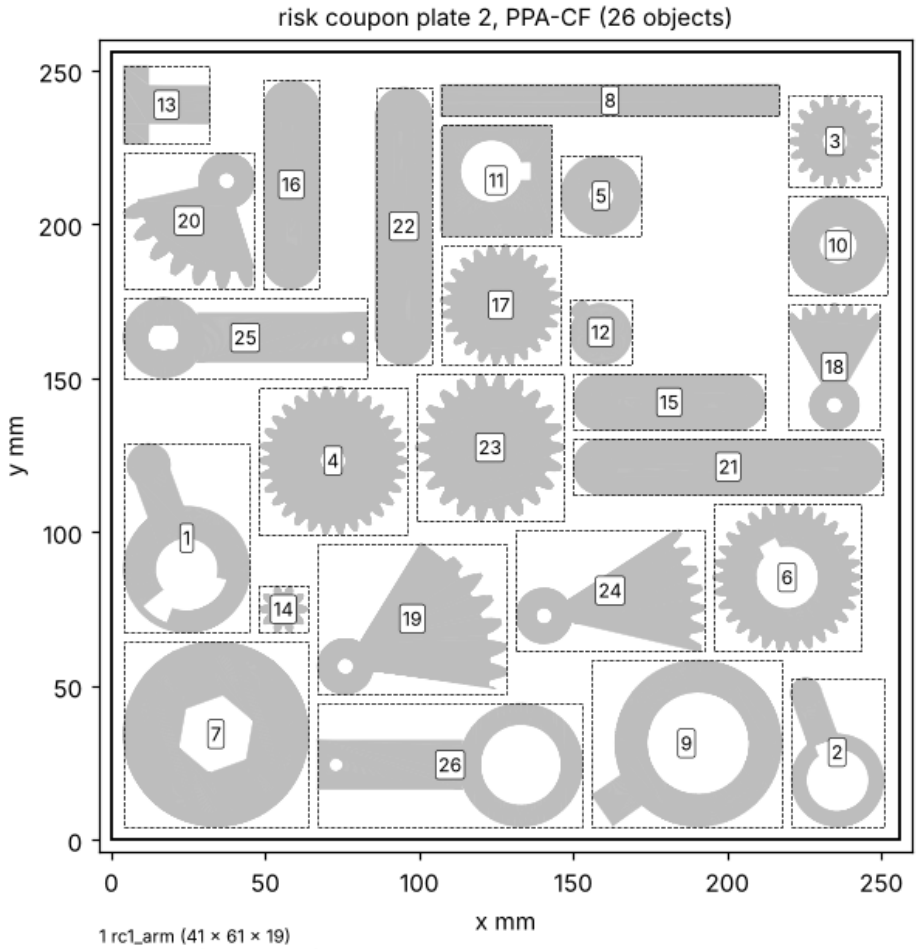
Two plates from the base print pack with small test pieces of the riskiest features: swing splitter, printed contact, dog clutch, slip clutch, bearing seats, heat-set inserts and nut traps, key rib, gear meshes, shear pin. Print them, then work through the test sheet. Do not print the big base parts until every coupon passes.

Plates: `base/pack/coupons/risk_plate_1_PLA.3mf` (PLA, 24 pieces) and `risk_plate_2_PPA-CF.3mf` (PPA-CF, 26 pieces). Each piece is also there on its own (`base/pack/coupons/rcN_*.stl`) for a reprint. A printable copy of the test sheet is `base/pack/coupons/COUPON_TEST.pdf`. The coupons need a few bought parts (MEASURE_LIST C21): M3 screws and a jam nut, heat-set inserts, one each of 608ZZ, F693ZZ and 693ZZ, a CSK20PP. Take the CSK20PP and bearings back out of the coupons after testing: they go into the base.

If a coupon fails: write down which test and what you saw, and do not change the CAD yourself. The sheet names the design numbers to change; give the result to a Claude Code session in the repo and ask it to change them and re-run the coupons (in `base/`: `../kedge/.venv/bin/python -m tools.risk_coupon`, after `bash kedge/setup_mac.sh` once from the repo root). Then reprint only that piece from its `rcN_*.stl`.



Risk coupon plate (risk_plate_1_PLA.png)



Risk coupon plate (risk_plate_2_PPA-CF.png)

From `base/pack/coupons/COUPON_TEST.md`*:*

Day-one risk coupons: test sheet

Print these before the big parts. Each coupon is made from the same geometry functions and numbers as the real part, so a pass here is a pass there. Let parts cool 10 minutes before testing. The fit coupon (tolerance tiles A to H, base/pack/fit_coupon/COUPON.md) is printed alongside.

Settings

PLA: 0.4 mm nozzle, 0.2 mm layers, 4 walls, 40 % gyroid infill unless noted, 5 top and 5 bottom layers, hole and XY compensation 0, no supports. PPA-CF: hardened steel nozzle (never brass), filament dried (80 C, 8 h) and fed from a dry box, 0.2 mm layers, 6 walls, 60 % gyroid infill, 6 top and 6 bottom layers, hole and XY compensation 0, no supports. The 3MF files carry geometry only: set the material and these settings in Bambu Studio.

If a coupon fails

To change a number: on the Mac, after bash kedge/setup_mac.sh (from the repo root), cd base and run ../kedge/.venv/bin/python -m tools.risk_coupon (it rewrites base/pack/coupons and base/pack/fit_coupon). Easier: give the failing result to a Claude session and let it change the numbers. Change one number at a time and reprint only that coupon.

Plates

plate	material	objects
risk_plate_1_PLA.3mf	PLA	24
risk_plate_2_PPA-CF.3mf	PPA-CF	26

Tests

1. Swing-arm direction splitter

- Parts: rc1_plate, rc1_sleeve, rc1_sun, rc1_arm, rc1_drag_pad, rc1_drag_shim, rc1_idler_clip, rc1_idler, rc1_output, rc1_crank, rc1_stop, rc1_stop_e
- Bought: 4 x M3x16 (two stop fingers); everything else printed
- Do: Push the drag pad root block and the shim into the arm hub pocket (finger in its recess). On the plate: sleeve on the centre pin, sun on its rib, arm on the sleeve above the sun with its pin down through the idler, printed C clip into the pin groove, output on its pin, crank on top, the two stop fingers on their columns (2 x M3x16 each). Turn the crank 10 turns each way.
- Pass: Clockwise seen from above: the arm swings the idler into the output at once and comes to rest on the engaged stop finger, the output turns, no skipped teeth. Brake the output lightly with a finger and keep turning: the arm stays on that finger (the idler is held at full depth, never pulled in deeper) and the crank still turns smoothly, no binding. Anticlockwise: the arm swings out to the other finger within about 20 deg of crank, the idler runs free and the output stays still. No jam at engagement. The pad drags the arm reliably both ways and the clip stays on.
- If it fails, change: launch_aim.py: FIT (pin and bore play), DRAG_INT and DRAG_SHIM_ADJ (drag: or print a thicker shim), clips.py for the C clip, A() alpha_d = alpha_e + 20 (swing out), _stop_post_xy (swung-out stop), _stop_post_e_xy (engaged stop: binding with the output braked means it lets the idler in too deep); launch_gears.py center_distance extra / spur backlash (mesh).
- Made from: launch_aim.sun_solid, arm_solid, idler_solid (m1.5 z30/18/30, face 12, FIT 0.35), launch_gears.spur for the output, launch_aim.A: centre distances 36.10 / 36.10, output at (11.7, 68.4), engaged arm 96.2 deg on the engaged stop (launch_aim._stop_post_e_xy), swung out 116.2 deg on the other stop (launch_aim._stop_post_xy), sleeve rib launch.RIB_W 5.8 / RIB_R 12.6; drag_pad_solid, drag_shim_solid (DRAG_INT 0.45, shim up to +0.25), pin_clip_solid

2. Printed contact switch

- Parts: rc2_switch_base, rc2_switch_lever
- Bought: M3x10 SHCS ISO 4762 + M3 jam nut, M3 nut, 2x M3x10 self tapping (pan or SHCS) through the lever root into the pad, at most M3x11.7, 2 wires, multimeter
- Do: Nut into the pillar trap (drop of glue), lever clamped with 2 M3 self tappers, contact screw head down through the lever, set the head 0.9 mm above the nut (0.5 mm per turn), lock the jam nut. Meter on beep between screw and nut. Press the lever tip 100 times.
- Pass: Measure the tip travel with calipers (depth rod from the lever tip down to the base top, note the reading at rest, press until it beeps, read again): beep within 0.8 to 1.4 mm of tip travel (design 1.14), silent when let go, tip stops on the hard stop without cracking, lever back flat, same after 100 presses.
- If it fails, change: cad/lib/switch.py contact_switch args travel, overtravel, t, b; STRAIN_LIMIT, NUT_PRESS (loose nut), SCREW_LENGTHS.
- Made from: cad.lib.switch.contact_switch(lever_len=36) with the library defaults (travel 0.8, overtravel 0.6, beam t 1.6): gap 0.91, pretravel at the tip 1.14, close force 0.83 N, strain 0.0061

3. Dog clutch and compliant fork

- Parts: rc3_base, rc3_reel_face, rc3_collar, rc3_fork, rc3_dog_post, rc3_rail
- Bought: M3 nut (yoke tab), 2 x M3x12 (SW_DOG screws in the post), 1 x M3 (post clamp), multimeter
- Do: Press the rail into the base. REEL face on the journal (dogs up), collar on the hex (dogs down), fork pads in the collar groove and its sleeve on the rail, dog post clamped on the rail below the fork, SW_DOG screws in the post set 0.5 mm short of the tab nut at full engagement. Push the fork 7 mm down with the dogs tip to tip, then turn the REEL face until they drop in. Turn the face hard both ways, then pull the fork back. Meter on beep between the lower screw and the tab nut.

- Pass: Yoke flexes when tip to tip (no jam, roofs slide off), dogs drop in within a quarter turn, under hand torque the dogs pull in, never walk out, a light pull on the fork releases them, the yoke springs back flat; the meter beeps only with the dogs fully in, never tip to tip.
- If it fails, change: _drum_kin.py DEFAULTS dog_backcut, dog_roof, dog_deg, dog_gap, fork_k; drum.py GEOM fork_t, yoke_r, groove, dog_touch, dogpost_s; fits.py SHAFT_IN_BORE (collar on the hex, face on the journal), HOLE_SLIDE (fork on the rail).
- Made from: drum.dog_teeth (dog_n 4, dog_h 5, dog_deg 36, backcut 4, roof 1.2, r 18..26), drum.fork, drum.dog_post (dog_touch 0.5), drum.rail, drum.hexprism, GEOM journal d33, groove (82.0, 88.0), stroke 7, neutral gap 2.5

4. Face slip clutch with the flexure spring

- Parts: rc4_hub_plate, rc4_friction_disc, rc4_pressure_plate, rc4_flexure, rc4_adjuster_plate
- Bought: 3 x M3x20 SHCS, 3 x M3 nut, string, coins in a small bag, kitchen scale
- Do: Stack on the key stub: hub plate (counterbores down), friction disc, pressure plate (pads up), flexure (beam down, tabs on the screws), adjuster plate (traps up); 3 screws from below, nuts on top. Run the nuts down until the flexure just touches (zero), then turn all 3 nuts the same, 1/4 turn (0.125 mm) at a time. Clamp the stub level in a vice, tie a string to a disc lug (r 28.5 mm), lay it over the rim and hang a bag of coins weighed on a kitchen scale. Add or take coins to find the hold and slip weights at each setting.
- Pass: At each setting the lower weight must not pull the disc round and the higher one must (smooth steady slip, no judder); same result 3 times; the flexure never bottoms before 1 turn. Bands: 1/4 turn: 35 g holds, 60 g slips (design 48 g, 0.013 N m); 1/2 turn: 70 g holds, 120 g slips (design 95 g, 0.027 N m); 3/4 turn: 105 g holds, 180 g slips (design 143 g, 0.040 N m); 1 turn: 140 g holds, 240 g slips (design 190 g, 0.053 N m).
- If it fails, change: cad/lib/clutch.py: slip_clutch args flex_t, flex_gap, flex_ro, r_in, r_out, n_grooves; MU_PLA_PLA, E_PLA, SIGMA_FLEX_PLA_SUSTAINED (creep: if the torque falls overnight).
- Made from: cad.lib.clutch.slip_clutch() defaults (reel does not use it: its clutch is the spool face plus the PETG rocker): shaft d20 key 6, friction r 18.5..27, mu 0.3, flexure t 0.8 gap 0.5, 0.106 N m per mm, travel 0.50; on the shared keyed stub of coupon 7

5. Bearing seats

- Parts: rc5_seats_608_693, rc5_csk20_seat
- Bought: 608ZZ, F693ZZ, 693ZZ, CSK20PP
- Do: Press each bearing in square with a flat block or a vice (tile left to right: 608, F693ZZ, 693ZZ); CSK20PP into the round boss with its outer keyway on the printed key.
- Pass: Firm press, seated flat on the lip, stays in when tapped upside down, spins as freely as before; the CSK20PP key slides in without force, it still freewheels one way and locks the other.
- If it fails, change: fits.py BRG_RIB_INTERF, BRG_RIB_BORE (all seats); bearings.py CSK20_KEY_RCLEAR, CSK20_OKEY_W (key), KEY_CLEAR in fits.py. If the 608 is off, print tile A of the fit coupon (base/pack/fit_coupon, COUPON.md).
- Made from: cad.lib.bearings.seat_608(bottom lip, lip_d 14) as reel, seat_f693(), seat_693(), seat_csk20(key_deg 90) as the reel capstan block (wall 4); fits BRG_RIB_INTERF 0.15, BRG_RIB_BORE 0.3, CSK20_KEY_RCLEAR 0.2

6. M3 holes

- Parts: rc6_m3_tile
- Bought: M3x4x5 heat-set insert, M3 nut, 2 x M3x10
- Do: Left to right: heat-set the insert (220 C), press the nut in, push a screw through, self-tap a screw in.
- Pass: Insert flush and does not spin when the screw is tightened; nut presses in and holds; screw drops through the clearance hole; the tap hole cuts a thread and holds a hard pull without splitting.
- If it fails, change: fits.py M3_INSERT_D, M3_NUT_AF, M3_CLEAR_D, M3_TAP_D (tiles C, D, E of the fit coupon in base/pack/fit_coupon bracket them).
- Made from: cad.lib.fasteners.m3_insert (d 4.0), m3_nut_trap (AF 5.8), m3_clear (d 3.4), m3_tap (d 2.6)

7. Key rib and spigot

- Parts: rc7_key_stub, rc7_key_bore, rc7_spigot, rc7_bushing
- Bought: CSK20PP from coupon 5 (inner keyway)
- Do: Slide the key stub into the keyed tile and into the CSK20PP inner race (the real interface). Push the spigot into the sleeve slice and turn it.

- Pass: Stub slides in by hand, key in the keyway, no visible rock. Play: clamp the tile, tape a ruler to the stub end, mark the ruler 60 mm from the axis, rock the stub both ways: the mark moves under 1 mm (about 1 deg). The spigot turns freely; rocked at its free end it moves under 0.5 mm.
- If it fails, change: fits.py KEY_CLEAR, SHAFT_IN_BORE; shafts.py KEY20_H, SHAFT_PRINT_UNDERSIZE; launch.py SPIGOT_BORE (now 12.35); drum.py GEOM spig_d.
- Made from: cad.lib.shafts.keyed_shaft(d 20, key_w 6, key_h 2.6) and keyed_bore (clear 0.3); drum.main_shaft (spig_d 12); launch.sleeve_solid (SPIGOT_BORE 12.35)

8. Gear meshes m1.25, m1.5, m2, m2.5

- Parts: rc8_m125_plate, rc8_m125_seg, rc8_m125_pinion, rc8_m15_plate, rc8_m15_z42, rc8_m15_z24, rc8_m2_plate, rc8_m2_z50, rc8_m2_z22, rc8_m25_plate, rc8_m25_big, rc8_m25_drive
- Bought: none
- Do: Each pair on its plate (label = module), flat faces down; turn or rock through every tooth.
- Pass: Smooth, no tight spot, a little play when rocked; m2.5: the tall tooth and the cut-down last tooth pass without catching, then the mesh lets go at the gap.
- If it fails, change: fits.py GEAR_BACKLASH, GEAR_CENTER_EXTRA (m1.25, m1.5, m2); launch_gears.py center_distance extra, spur backlash, sector_gear_2d tall_factor (m2.5); _drum_kin.py pin_x (m1.25 pinion).
- Made from: m1.25: gears.spur_gear z60 sector / z10 shift 0.4 tip_short 0.08 (drum.tip_short_for), cd 44.33 (drum.a_seg); m1.5: gears.spur_gear z42 sector / z24, cd 49.60; m2: gears.spur_gear z50 sector (drum GEOM out_m, out_z) / z22 (reel Z_CAPG), cd 72.10; m2.5: launch_gears.sector_gear_2d z40 (n_present 27, last 6 + gap) / launch_gears.spur z26, cd 82.60 (launch L); coupon face 6, fits GEAR_BACKLASH 0.15, GEAR_CENTER_EXTRA 0.1

9. Shear-pin torque limiter

- Parts: rc9_hub_lever, rc9_outer_lever, rc9_shear_pin x3
- Bought: string or a strap, luggage scale
- Do: Print the pins lying down on their flat, as exported (D44: shear plane across the layers; never stand them up). Hub slice into the outer slice, pin through both holes (head outside). Clamp the hub lever, hang the luggage scale on the outer lever hole (r 60 mm) at right angles and pull slowly until the pin shears. Break all 3 pins. Torque = kg x 9.81 x 0.06 N m.
- Pass: Each pin shears cleanly between 5.9 and 10.2 kg on the scale (3.5 to 6 N m, drum_NOTES; design 4.9 N m = 8.3 kg), the 3 pins within 20 percent of each other, both slices undamaged (holes still round); a fresh pin goes in by hand.
- If it fails, change: drum.py GEOM shear_d (torque goes with d squared), hub_od (radius), shear_s; fits for the hole: drum.coupler shear hole (shear_d + 0.1) and shear_pin (shear_d - 0.04). Spec D8 / D26 limit 5 N m.
- Made from: drum.coupler_hub and drum.coupler (sliced s 4.5..20), drum.shear_pin: d4 PLA in single shear at r 13 (shear_s 9); 4.9 N m at 30 MPa

10. AS5600 magnet cap (bench test)

- Parts: rc10_magnet_cap
- Bought: magnet: not included with the board, see MEASURE_LIST C13 (any small neodymium disc edge-on, or 6 x 2.5 diametric), M3x6, superglue
- Do: Glue the magnet into the end pocket (flush, any way round), push the cap on the shaft end (flat to the flat), M3x6 onto the flat. It fits the d12 main shaft spigot of the base (for the bare-motor bench test, tape the magnet to the wiper shaft end instead). AS5600 chip on the shaft axis, 1 to 2 mm from the magnet face.
- Pass: Cap does not wobble or slip on the shaft; the AS5600 angle follows the shaft smoothly through full turns (no jumps), magnet status OK (not too weak, not too strong) at 1 to 2 mm.
- If it fails, change: drum.py GEOM cap_s, spig_d; magnet_holder pocket d6.2 x 2.6 (magnet loose or tight); fits.py HOLE_SLIDE (cap on the spigot).
- Made from: drum.magnet_holder (spig_d 12 with its flat, cap_s (195.0, 201.5), magnet pocket d6.2 x 2.6), printed as drum prints it

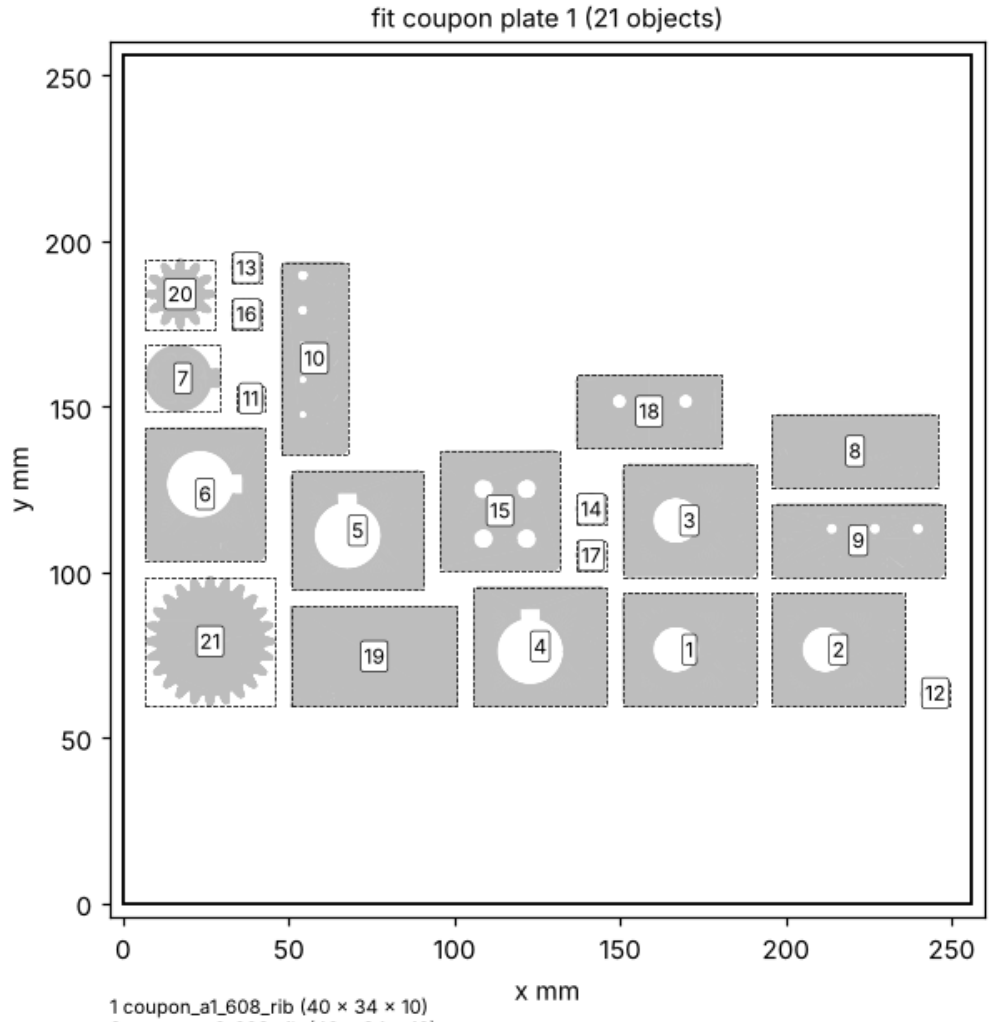
Parts

part	material	qty	print bbox mm	what
rc1_plate	PLA	1	110.1 x 82.2 x 33.1	Splitter base: centre pin d8, output pin d8, two stop finger columns (2 x M3 tap each)
rc1_sleeve	PPA-CF	1	26.0 x 26.0 x 29.3	Sleeve stub d20 with the integral key rib (launch.RIB_W, RIB_R) under the sun and the crank

part	material	qty	print bbox mm	what
rc1_sun	PPA-CF	1	47.8 x 48.0 x 12.0	launch_aim.sun_solid: sun m1.5 z30 as built
rc1_arm	PPA-CF	1	61.4 x 41.0 x 19.3	launch_aim.arm_solid: swing arm as built (drag pad pocket, d8 pin, clip groove)
rc1_drag_pad	PLA	1	24.6 x 8.7 x 4.0	launch_aim.drag_pad_solid: printed flexure drag pad as built
rc1_drag_shim	PLA	1	9.4 x 3.0 x 4.0	launch_aim.drag_shim_solid: nominal preload shim as built
rc1_idler_clip	PLA	1	10.8 x 10.0 x 1.5	launch_aim.pin_clip_solid: printed C clip for the d8 idler pin
rc1_idler	PPA-CF	1	30.0 x 29.7 x 12.0	launch_aim.idler_solid: idler m1.5 z18 as built
rc1_output	PPA-CF	1	47.8 x 48.0 x 12.0	Output m1.5 z30 (launch_gears.spur, face 12) on a d8 pin
rc1_crank	PPA-CF	1	48.2 x 30.3 x 20.0	Hand crank, keyed on the sleeve rib like the sun
rc1_stop	PLA	1	47.1 x 10.0 x 6.4	Stop finger: rounded end at the as-built swung-out stop post position
rc1_stop_e	PLA	1	64.4 x 10.0 x 6.4	Engaged stop finger: rounded end at the as-built engaged stop post position (launch_aim._stop_post_e_xy)
rc2_switch_base	PLA	1	54.0 x 35.0 x 11.5	contact_switch base: pad, nut pillar, hard stop, wire channel
rc2_switch_lever	PLA	1	50.0 x 5.0 x 8.0	contact_switch lever, printed on its side (Y up)
rc3_base	PLA	1	40.0 x 91.5 x 97.5	Base: d30 journal (REEL sleeve) and hex 22 AF (main shaft), rail hole (press)
rc3_reel_face	PPA-CF	1	61.6 x 54.0 x 9.5	REEL gear dog face (drum.dog_teeth, phase 45) with a handle
rc3_collar	PPA-CF	1	60.0 x 60.0 x 18.0	Collar lower half: drum.dog_teeth facing down, fork groove, hex bore
rc3_fork	PLA	1	87.1 x 94.1 x 83.0	drum.fork as built: compliant yoke, rail sleeve, follower pin, M3 nut in the tab
rc3_dog_post	PLA	1	20.0 x 75.9 x 41.7	drum.dog_post as built: SW_DOG_L / SW_DOG_R screw bosses, clamps on the rail
rc3_rail	PPA-CF	1	110.0 x 10.0 x 9.0	drum.rail profile d10 with the print flat, cut to 110 long
rc4_hub_plate	PLA	1	54.0 x 54.0 x 6.0	clutch.slip_clutch hub_plate (library defaults)
rc4_friction_disc	PLA	1	57.0 x 55.0 x 3.0	clutch.slip_clutch friction_disc (library defaults)
rc4_pressure_plate	PLA	1	54.0 x 54.0 x 3.5	clutch.slip_clutch pressure_plate (library defaults)
rc4_flexure	PLA	1	50.0 x 50.0 x 1.7	clutch.slip_clutch flexure (library defaults)
rc4_adjuster_plate	PLA	1	54.0 x 54.0 x 4.0	clutch.slip_clutch adjuster_plate (library defaults)
rc5_seats_608_693	PLA	1	60.0 x 28.0 x 9.6	608 (reel spool seat), F693ZZ and 693ZZ crush-rib seats
rc5_csk20_seat	PLA	1	55.3 x 55.3 x 16.0	CSK20PP seat with the printed key (reel capstan block, key_deg 90)
rc6_m3_tile	PLA	1	54.0 x 18.0 x 8.6	M3 heat-set hole, nut trap, clearance and tap holes (fits.py values)
rc7_key_stub	PPA-CF	1	21.2 x 20.0 x 31.0	shafts.keyed_shaft d20 key 6 (reel capstan tube numbers), 31 long, also the coupon 4 clutch shaft
rc7_key_bore	PPA-CF	1	36.0 x 36.0 x 8.6	shafts.keyed_bore d20 key 6, 8 long (keyed hubs on the sleeve rib are PPA-CF)
rc7_spigot	PPA-CF	1	28.0 x 25.4 x 22.0	drum.main_shaft slice: hex 8 long + d12 spigot 20 long, on a hex flat
rc7_bushing	PPA-CF	1	32.0 x 32.0 x 16.0	launch.sleeve_solid slice 16 long: spigot bore d12.35
rc8_m125_plate	PPA-CF	1	62.3 x 18.0 x 11.0	Pins at 44.33
rc8_m125_seg	PLA	1	38.1 x 46.7 x 6.0	drum segment gear m1.25 z60 sector
rc8_m125_pinion	PPA-CF	1	15.8 x 15.2 x 6.0	drum wrap pinion m1.25 z10 x0.4 tip_short 0.08
rc8_m15_plate	PPA-CF	1	67.6 x 18.0 x 11.0	Pins at 49.60
rc8_m15_z42	PPA-CF	1	41.0 x 29.6 x 6.0	drum wrap idler output m1.5 z42 sector
rc8_m15_z24	PPA-CF	1	38.9 x 38.9 x 6.0	reel cup ring gear m1.5 z24
rc8_m2_plate	PPA-CF	1	90.1 x 18.0 x 11.0	Pins at 72.10
rc8_m2_z50	PPA-CF	1	61.0 x 39.3 x 6.0	drum REEL output m2 z50 sector
rc8_m2_z22	PPA-CF	1	48.0 x 47.7 x 6.0	reel capstan gear m2 z22
rc8_m25_plate	PPA-CF	1	100.6 x 18.0 x 11.0	Pins at 82.60
rc8_m25_big	PPA-CF	1	61.5 x 49.0 x 6.0	launch big gear m2.5 z40 sector: teeth 21..26 (tall, relieved)
rc8_m25_drive	PPA-CF	1	42.5 x 44.0 x 6.0	launch drive gear m2.5 z26 sector
rc9_hub_lever	PPA-CF	1	79.0 x 26.0 x 15.5	drum.coupler_hub slice (shear pin hole) with a lever, string hole at r 60
rc9_outer_lever	PPA-CF	1	86.0 x 40.0 x 9.5	drum.coupler slice (shear pin hole) with a lever, string hole at r 60
rc9_shear_pin	PLA	3	12.1 x 5.0 x 4.1	drum.shear_pin as built (d4 PLA), printed LYING on its 0.4 mm flat as drum prints it (D44): the shear plane runs across the layers

part	material	qty	print bbox mm	what
rc10_magnet_cap	PLA	1	20.0 x 20.0 x 6.5	drum.magnet_holder as built: AS5600 magnet cap on the d12 spigot end

Base fit coupon



Base fit coupon plate (coupon_plate_1.png)

From `base/pack/fit_coupon/COUPON.md`*:*

Fit coupon

Print this set FIRST, before any real part. Each tile tells you one number in `base/cad/lib/fits.py`. Print everything in PLA with your normal profile (0.4 nozzle, 0.2 layers, 4 walls), flat as exported, no supports. Let the parts cool fully (10 minutes) before testing. The first character of every label is the tile letter below; the rest is the value that tile was made with.

How to change a fit: open `base/cad/lib/fits.py`, change the one number named in the step, save, then rerun `../kedge/.venv/bin/python -m tools.risk_coupon` from `base/` (on the Mac after `bash kedge/setup_mac.sh`; it rewrites this fit coupon in base/pack/fit_coupon and the risk coupons in base/pack/coupons), then the audit of your subsystem. Change one number at a time. If none of the variants is right, pick the closest one, move the number one more step in that direction and print only that tile again.

Parts

part	qty	print bbox mm	what
coupon_a1_608_rib	1	34.0 x 40.0 x 9.6	608 crush-rib seat, rib interference 0.10 mm radial
coupon_a2_608_rib	1	34.0 x 40.0 x 9.6	608 crush-rib seat, rib interference 0.15 mm radial
coupon_a3_608_rib	1	34.0 x 40.0 x 9.6	608 crush-rib seat, rib interference 0.20 mm radial
coupon_b1_key_bore	1	36.0 x 40.0 x 8.6	20 mm keyed bore, clearance 0.20 mm diametral
coupon_b2_key_bore	1	36.0 x 40.0 x 8.6	20 mm keyed bore, clearance 0.30 mm diametral
coupon_b3_key_bore	1	36.0 x 40.0 x 8.6	20 mm keyed bore, clearance 0.40 mm diametral
coupon_b_key_shaft	1	22.6 x 19.9 x 16.0	Keyed shaft stub d 20, 16 long, prints upright
coupon_c_m3_insert	1	50.0 x 22.0 x 8.6	M3 heat-set insert holes 3.8 / 4.0 / 4.2
coupon_d_m3_nut	1	52.0 x 22.0 x 6.6	M3 nut traps AF 5.6 / 5.8 / 6.0
coupon_e_m3_holes	1	58.0 x 20.0 x 6.6	M3 tap and clearance holes 2.5 / 2.6 / 2.7 / 3.2 / 3.4
coupon_f_pin_host	1	36.0 x 36.0 x 4.6	Host plate for the d 6 pins: snap row on top, clip row below
coupon_f_snap_pin_1	1	9.0 x 9.0 x 9.2	Snap pin d 6, fit 0.25 (1 notch)
coupon_f_clip_pin_1	1	9.0 x 9.0 x 8.4	Clip pin d 6, fit 0.25 (1 notch)
coupon_f_snap_pin_2	1	9.0 x 9.0 x 9.2	Snap pin d 6, fit 0.35 (2 notches)
coupon_f_clip_pin_2	1	9.0 x 9.0 x 8.4	Clip pin d 6, fit 0.35 (2 notches)
coupon_f_c_clip	2	8.4 x 7.8 x 1.5	C clip for the d 6 clip pin
coupon_g_eye	1	44.0 x 22.0 x 5.6	Dyneema eye (v7) in 3 mm and 5 mm plates
coupon_h_gear_plate	1	50.1 x 30.0 x 9.0	Gear plate: pins d 5 at centre distance 27.10
coupon_h_gear_z12	1	21.0 x 21.0 x 5.0	Spur gear m1.5 z12, backlash 0.15
coupon_h_gear_z24	1	39.0 x 39.0 x 5.0	Spur gear m1.5 z24, backlash 0.15

A. 608 bearing seat (labels A .10 / .15 / .20: rib interference, mm)

You need: three 608 bearings (or one, pressed in and out).

1. Press a 608 into each tile with your thumb or a vice. Use a flat block, keep it square.
2. Good: it goes in with a firm push, sits flat on the lip, and does not fall out when you turn the tile over and tap it. The outer race must not be pinched: spin the bearing, it should spin as freely as before.
3. Too tight (needs a hammer, tile cracks, bearing feels rough): pick a smaller value. Too loose (falls out, wobbles): pick a bigger value.
4. In fits.py set `BRG_RIB_INTERF` to the value printed on the best tile (now 0.15).

B. 20 mm keyed bore (labels B .20 / .30 / .40: bore clearance, mm diametral)

1. Push the printed keyed shaft stub (coupon_b_key_shaft) into each tile, key in the keyway.
2. Good for a running fit: it slides in by hand with no force, turns by the key only, and has no visible wobble when you rock it.
3. Too tight: needs force or files: bigger value. Too loose: rocks or clicks: smaller value.
4. In fits.py set `SHAFT_IN_BORE` to the best value (now 0.3).
5. If the key binds but the round part is fine, raise `KEY_CLEAR` by 0.05 (now 0.2); if the key rattles while the bore is good, lower it by 0.05.

C. M3 heat-set inserts (labels 3.8 / 4.0 / 4.2: hole diameter)

You need: three M3x4x5 heat-set inserts, soldering iron with an insert tip.

1. Set the iron to about 220 C. Press an insert into each hole, straight, until flush.
2. Good: it goes in with light pressure, a small ring of plastic rises round it, and an M3 screw tightened firmly does not spin the insert or pull it out.
3. Spins or pulls out: smaller hole. Needs a lot of force, or plastic bulges a lot: bigger hole.
4. In fits.py set `M3_INSERT_D` to the best hole (now 4.0).

D. M3 nut traps (labels 5.6 / 5.8 / 6.0: across flats)

1. Drop an M3 nut into each hex pocket.
2. Good: it presses in with a finger, stays put when you turn the tile over, and does not turn when you drive a screw in from below.
3. Will not go in: bigger. Falls out: smaller (a nut that just drops in is fine if a screw holds it).
4. In fits.py set `M3_NUT_AF` to the best value (now 5.80).

E. M3 holes (labels 2.5 / 2.6 / 2.7 / 3.2 / 3.4: hole diameter)

1. Drive an M3 screw into each of 2.5, 2.6, 2.7 (self tapping into plastic) and push one through 3.2 and 3.4.
2. Tap hole good: the screw cuts its thread with moderate force and holds when you pull hard, the plastic does not split. Set `M3_TAP_D` to the best one (now 2.6).
3. Close hole good: the screw passes with a little drag, no play. Set `M3_CLOSE_D` to it (now 3.2).
4. Clearance hole good: the screw drops through freely. Set `M3_CLEAR_D` to it (now 3.4). If 3.2 is already loose, it can come down too.
5. These holes print vertical. Holes printed lying down come out smaller: that extra is `HOLE_HORIZ_EXTRA` (now 0.15), leave it unless a horizontal hole is tight.

F. d 6 snap pin and clip pin (1 notch = fit .25, 2 notches = fit .35)

The fit is taken off the pin shank: shank d = 6 minus the fit. The host plate has 6 mm holes: top row for the snap pins, bottom row for the clip pins (the labels 1 and 2 only say which pin to try there). Print two C clips.

1. Push each snap pin through a top hole until the barbs click past the far face.
2. Put each clip pin through a bottom hole and push a C clip onto the groove from the side.
3. Good: the pin turns freely in the hole with very little side play, the snap pin needs a firm push and does not fall out, the clip snaps on with a click and stays.
4. If the 1 notch pins (fit .25) are best for sliding, keep `HOLE_SLIDE` = 0.25. If they bind and the 2 notch pins (fit .35) are right, set `HOLE_SLIDE` to about .30 to .35. If the 2 notch pins rattle, `HOLE_CLEAR` (now 0.35) can come down by 0.05 to 0.1.
5. If a prong snaps off or the clip cracks, tell the CAD side (clips.py proportions), not fits.py.

G. Dyneema eye (labels G 3 and 5: plate thickness, mm)

1. Thread 1 mm Dyneema through both eyes and saw it back and forth under load for a minute.
2. Good: the line runs smoothly, no fuzz on the line, no sharp edge you can feel with a fingernail at either face.
3. If the eye is rough, sand the cones; if the throat is too tight to thread, raise `line.EYE_THROAT_D` (now 4.4); this one is in line.py, not fits.py.

H. Gear mesh m1.5 z12 / z24 (label H 1.5), centre distance 27.10 mm

1. Drop both gears onto the pins of the gear plate and turn the big gear by hand through several turns.
2. Good: smooth, no tight spot, a tiny bit of play you can just feel when you rock one gear while holding the other.
3. Tight or notchy: raise `GEAR_CENTER_EXTRA` (now 0.1) by 0.05 or `GEAR_BACKLASH` (now 0.15) by 0.05. Lots of play (clacks when reversing): lower one of them by 0.05.
4. If a gear is tight on its pin but the mesh is fine, raise `SHAFT_IN_BORE` (now 0.3); check tile B first, it tests the same number.

Record

Write the numbers you chose and the date at the top of fits.py as a comment, for example `# coupon 2026-11-10: BRG_RIB_INTERF 0.15, SHAFT_IN_BORE 0.30, ...`, then commit.

Dart prints

Dart plates (in `kedge/pack_v30/`, also inside `kedge/dist_v30/KEDGE_v30_build_pack.zip` under `print/`):

- `kedge/pack_v30/plate_A_hull.3mf`
- `kedge/pack_v30/plate_B_legs_tray.3mf`
- `kedge/pack_v30/fit_coupon.stl` (print this first, 20 minutes)

Dart part	Print pose	Size on bed mm	Volume cm3
hull_top	split face down (on the bed), saddle up	121 x 50 x 35	17.41
hull_bottom	split face down, belly up	121 x 42 x 21	11.5
nose	tip down on the bed (flat Ø22), spigot up	42 x 42 x 27	8.25

Dart part	Print pose	Size on bed mm	Volume cm3
tail_cap	end face down, spigot up (no supports)	42 x 42 x 15	7.13
tray	keel down on the bed; discs stand up, decks bridge 26 mm between discs	109 x 39 x 36	15.98
kick_leg	on its side (pole axis horizontal, hub face down)	119 x 15 x 15	8.7
drill_leg_L	upright, motor pocket open to the top (as installed folded)	46 x 15 x 13	3.37
drill_leg_R	upright, motor pocket open to the top (as installed folded)	46 x 15 x 13	3.37
drill_leg_L_auger	base down, nut slot sideways	17 x 17 x 73	5.5
drill_leg_R_auger	base down, nut slot sideways	17 x 17 x 73	5.5
foot_auger	base down	17 x 17 x 40	3.06
coupler_L	standing on its socket end	7 x 6 x 36	1.24
coupler_R	standing on its socket end	7 x 6 x 27	0.9

Settings: PLA, 0.4 nozzle, 0.16 mm layers, 4 walls, 40% gyroid infill, 5 top and bottom layers, supports off, brim on for the augers and couplers. Hole compensation from the fit coupon (run sheet section 4).

Base prints

The base print pack comes from the base pipeline (`base/out/base/pack`: PRINT_SHEET.pdf, the plates in `plates/plate_<n>_<material>.3mf`, print_report.json). It holds every part already in print orientation, the plates as 3MF, and the material for each part.

Print sheet

Generated 2026-11-04 13:39. Bed 256 x 256 x 260 mm (Bambu X2D). Masses and times are rough estimates: walls and infill per material from the settings below, volume at a typical print rate plus 5 s per 0.2 mm layer, the layer time counted once per plate.

Warnings

- launch_guard_l_a: print bbox 250.0 x 49.0 fits the bed but not inside the 5 mm margin; printed alone, centred on its own plate
- launch_guard_r_a: print bbox 250.0 x 49.0 fits the bed but not inside the 5 mm margin; printed alone, centred on its own plate

Parts

part	material	qty	print bbox mm	volume cm3	mass g (each)	time h (each, alone)	overhang mm2	supports
frame_deck_2	PLA	1	222.4 x 211.8 x 8.0	360.4	304.3	7.63	0	no
frame_deck_4	PLA	1	214.4 x 211.8 x 8.0	358.9	301.6	7.56	0	no
frame_deck_1	PLA	1	222.4 x 168.8 x 8.0	276.2	233.6	5.87	0	no
frame_deck_3	PLA	1	214.4 x 168.8 x 8.0	274.2	231.4	5.81	0	no
frame_cheek_r	PLA	1	215.0 x 220.0 x 30.0	248.2	205.2	5.32	1084	yes, tree supports from the plate (1084 mm2 over 45 deg)
frame_camera_mount	PLA	1	58.0 x 72.0 x 236.4	236.8	166.5	5.79	672	yes, tree supports from the plate (672 mm2 over 45 deg)
launch_aim_housing	PLA	1	176.5 x 59.0 x 183.0	212.8	185.4	5.89	761	yes, tree supports from the plate (761 mm2 over 45 deg)
launch_rail_mid	PLA	1	92.5 x 68.0 x 170.0	201.0	178.7	5.63	2356	yes, tree supports from the plate (2356 mm2 over 45 deg)

part	material	qty	print bbox mm	volume cm3	mass g (each)	time h (each, alone)	overhang mm2	supports
frame_motor_leg	PLA	1	49.0 x 69.0 x 187.0	159.9	117.3	4.22	222	yes, tree supports from the plate (222 mm2 over 45 deg)
launch_rail_front	PLA	1	131.5 x 128.0 x 48.5	146.5	137.7	3.76	7890	yes, tree supports from the plate (7890 mm2 over 45 deg)
frame_ebay	PLA	1	124.0 x 124.0 x 58.0	141.3	175.3	4.77	879	yes, tree supports from the plate (879 mm2 over 45 deg)
drum_drum_a	PLA	1	97.1 x 97.1 x 43.0	138.6	103.2	2.87	719	yes, tree supports from the plate (719 mm2 over 45 deg)
launch_rail_rear	PLA	1	151.2 x 68.0 x 40.1	132.0	115.9	3.16	3676	yes, tree supports from the plate (3676 mm2 over 45 deg)
reel_bridge	PLA	1	220.0 x 192.1 x 88.0	114.1	118.5	3.56	686	yes, tree supports from the plate (686 mm2 over 45 deg)
reel_head_front	PLA	1	188.0 x 186.0 x 6.0	109.3	108.4	2.74	50	no
reel_head_spine	PLA	1	130.8 x 188.0 x 6.0	106.9	104.2	2.64	0	no
drum_drum_b	PLA	1	76.7 x 76.7 x 41.0	105.9	82.4	2.34	3126	yes, tree supports from the plate (3126 mm2 over 45 deg)
reel_head_p2	PLA	1	104.3 x 182.0 x 6.0	96.0	94.1	2.38	20	no
launch_stop_spring_l	PLA	1	154.9 x 51.5 x 14.0	65.8	81.6	2.13	57	no
launch_stop_spring_r	PLA	1	154.9 x 51.5 x 14.0	65.8	81.6	2.13	57	no
frame_ebay_lid	PLA	1	124.0 x 124.0 x 7.0	57.6	71.4	1.83	80	no
reel_head_p1	PLA	1	100.0 x 177.0 x 37.0	55.4	55.0	1.63	98	no
launch_guard_l_a	PLA	1	250.0 x 49.0 x 40.0	52.4	65.0	1.90	93	no
launch_guard_r_a	PLA	1	250.0 x 49.0 x 40.0	52.4	65.0	1.90	93	no
launch_carriage	PLA	1	205.2 x 240.2 x 43.0	48.0	59.6	1.78	6453	yes, tree supports from the plate (6453 mm2 over 45 deg)
frame_guard_front	PLA	1	155.0 x 104.0 x 16.4	45.8	56.8	1.53	40	no
drum_index_block	PLA	1	25.0 x 68.0 x 39.5	41.0	33.2	1.10	61	no
frame_can_stay	PLA	1	55.0 x 55.0 x 71.0	31.0	28.4	1.20	906	yes, tree supports from the plate (906 mm2 over 45 deg)

part	material	qty	print bbox mm	volume cm3	mass g (each)	time h (each, alone)	overhang mm2	supports
reel_spool	PLA	1	62.0 x 62.0 x 22.4	28.4	30.8	0.92	1791	yes, tree supports from the plate (1791 mm2 over 45 deg)
launch_stop_frame_l	PLA	1	174.6 x 53.0 x 55.1	26.3	32.7	1.20	541	yes, tree supports from the plate (541 mm2 over 45 deg)
launch_stop_frame_r	PLA	1	174.6 x 53.0 x 55.1	26.3	32.7	1.20	541	yes, tree supports from the plate (541 mm2 over 45 deg)
drum_bracket_b	PLA	1	76.1 x 108.6 x 6.0	24.4	25.6	0.68	24	no
drum_switch_block	PLA	1	26.0 x 66.0 x 21.0	20.1	18.3	0.60	216	yes, tree supports from the plate (216 mm2 over 45 deg)
launch_carriage_yoke	PLA	1	43.3 x 126.6 x 28.0	18.9	23.5	0.78	1406	yes, tree supports from the plate (1406 mm2 over 45 deg)
frame_rc832_holder	PLA	1	97.0 x 21.6 x 40.0	18.4	22.8	0.84	0	no
reel_head_mid	PLA	1	53.8 x 58.5 x 6.0	18.1	19.3	0.52	143	yes, tree supports from the plate (143 mm2 over 45 deg)
launch_guard_r_b2	PLA	1	145.3 x 32.5 x 40.0	17.6	21.8	0.82	56	no
drum_fork	PLA	1	86.0 x 68.5 x 51.0	16.6	19.9	0.85	1227	yes, tree supports from the plate (1227 mm2 over 45 deg)
sensors_sw_rail_low_bracket	PLA	1	31.0 x 48.0 x 74.0	16.1	17.8	0.96	65	no
launch_guard_l_b2	PLA	1	145.3 x 32.5 x 40.0	14.4	17.9	0.72	33	no
sensors_as5600_carrier	PLA	1	34.0 x 67.5 x 11.0	12.4	12.7	0.39	90	no
frame_guard_hood	PLA	1	122.0 x 56.1 x 14.4	11.4	14.1	0.45	84	no
reel_cup	PLA	1	48.0 x 53.4 x 18.0	10.2	12.7	0.44	1210	yes, tree supports from the plate (1210 mm2 over 45 deg)
drum_n20_cradle	PLA	1	53.5 x 32.0 x 19.5	9.7	12.1	0.44	114	yes, tree supports from the plate (114 mm2 over 45 deg)
reel_head_p1b	PLA	1	38.0 x 44.5 x 16.0	9.0	10.7	0.38	41	no
sensors_sw_gap	PLA	1	63.5 x 43.3 x 11.5	8.0	9.9	0.33	64	no
reel_shroud	PLA	1	70.0 x 65.9 x 29.5	6.1	7.6	0.39	0	no
sensors_sw_rail_low	PLA	1	48.0 x 35.0 x 11.5	5.9	7.3	0.26	22	no

part	material	qty	print bbox mm	volume cm3	mass g (each)	time h (each, alone)	overhang mm2	supports
drum_dog_post	PLA	1	21.8 x 65.0 x 48.2	5.7	7.1	0.51	90	no
reel_sw_base	PLA	1	46.0 x 35.0 x 11.6	5.7	7.0	0.26	22	no
drum_follower	PLA	1	25.7 x 38.6 x 17.0	5.2	6.2	0.27	260	yes, tree supports from the plate (260 mm2 over 45 deg)
launch_aim_cap	PLA	1	45.5 x 14.0 x 19.4	4.2	5.2	0.27	0	no
launch_guard_l_c	PLA	1	32.0 x 54.5 x 26.0	3.8	4.8	0.30	3	no
launch_guard_r_c	PLA	1	32.0 x 54.5 x 26.0	3.8	4.8	0.30	3	no
launch_guard_l_b1	PLA	1	11.7 x 49.0 x 40.0	3.6	4.5	0.39	22	no
launch_guard_r_b1	PLA	1	11.7 x 49.0 x 40.0	3.6	4.5	0.39	22	no
frame_cable_anchor_1	PLA	1	16.0 x 24.0 x 10.0	3.2	4.0	0.17	141	yes, tree supports from the plate (141 mm2 over 45 deg)
frame_cable_anchor_2	PLA	1	24.0 x 16.0 x 10.0	3.2	4.0	0.17	141	yes, tree supports from the plate (141 mm2 over 45 deg)
drum_link_eye_1	PLA	1	13.0 x 13.0 x 23.0	2.5	3.0	0.24	29	no
drum_link_eye_2	PLA	1	13.0 x 13.0 x 23.0	2.5	3.0	0.24	29	no
drum_n20_strap	PLA	1	53.5 x 17.9 x 3.0	2.2	2.7	0.09	0	no
drum_magnet_cap	PLA	1	20.0 x 20.0 x 6.5	1.5	1.9	0.09	40	no
frame_ebay_esp_clamp	PLA	1	43.4 x 8.0 x 4.0	1.3	1.6	0.07	0	no
sensors_sw_gap_lever	PLA	1	40.0 x 9.0 x 8.0	1.1	1.4	0.09	32	no
sensors_sw_rail_low_lever	PLA	1	44.0 x 5.0 x 8.0	1.1	1.3	0.09	32	no
reel_sw_lever	PLA	1	42.0 x 5.0 x 8.0	1.0	1.3	0.09	32	no
sensors_as5600_strip_top	PLA	1	25.4 x 11.8 x 3.1	0.8	1.0	0.05	0	no
sensors_as5600_strip_bot	PLA	1	25.4 x 11.8 x 3.1	0.8	1.0	0.05	0	no
drum_index_lever	PLA	1	12.9 x 47.5 x 6.0	0.7	0.9	0.06	94	no
drum_sw_lever_0	PLA	1	13.7 x 47.5 x 6.0	0.7	0.9	0.06	97	no
drum_sw_lever_1	PLA	1	13.7 x 47.5 x 6.0	0.7	0.9	0.06	97	no
drum_sw_lever_2	PLA	1	13.7 x 47.5 x 6.0	0.7	0.9	0.06	97	no

part	material	qty	print bbox mm	volume cm3	mass g (each)	time h (each, alone)	overhang mm2	supports
launch_sw_seated	PLA	1	41.5 x 8.4 x 6.0	0.7	0.8	0.06	24	no
launch_sw_home	PLA	1	41.5 x 8.4 x 6.0	0.7	0.8	0.06	24	no
launch_muzzle_roller	PLA	1	14.0 x 14.0 x 5.0	0.4	0.5	0.05	64	no
launch_gear_washer	PLA	1	18.0 x 18.0 x 2.3	0.4	0.5	0.03	0	no
launch_aim_drag_pad	PLA	1	8.8 x 24.5 x 4.0	0.3	0.4	0.04	0	no
launch_gear_clip	PLA	1	17.3 x 18.0 x 1.4	0.2	0.3	0.02	0	no
launch_rail_splice_l	PLA	1	22.0 x 5.0 x 3.0	0.2	0.3	0.03	0	no
launch_rail_splice_r	PLA	1	22.0 x 5.0 x 3.0	0.2	0.3	0.03	0	no
launch_drive_washer	PLA	1	32.0 x 32.0 x 0.7	0.2	0.2	0.01	0	no
drum_shear_pin	PLA	6	12.1 x 5.0 x 4.1	0.1	0.2	0.03	0	no
reel_cup_clip	PLA	1	10.0 x 9.7 x 2.4	0.1	0.2	0.02	0	no
launch_aim_drag_shim	PLA	1	8.6 x 7.0 x 4.0	0.1	0.1	0.03	0	no
launch_aim_idler_clip	PLA	1	10.8 x 10.0 x 1.4	0.1	0.1	0.01	0	no
launch_aim_crankpin_clip	PLA	1	10.8 x 10.0 x 1.4	0.1	0.1	0.01	0	no
launch_aim_lugpin_clip	PLA	1	10.8 x 10.0 x 1.4	0.1	0.1	0.01	0	no
launch_seat_clip	PLA	1	5.6 x 5.6 x 2.4	0.1	0.1	0.02	0	no
launch_muzzle_washer_l	PLA	1	6.2 x 6.2 x 0.9	0.0	0.0	0.01	0	no
launch_muzzle_washer_r	PLA	1	6.2 x 6.2 x 0.9	0.0	0.0	0.01	0	no
frame_motor_adapter	PPA-CF	1	194.0 x 194.0 x 16.0	282.2	302.8	13.57	5484	yes, tree supports from the plate (5484 mm2 over 45 deg)
launch_big_gear	PPA-CF	1	105.0 x 105.0 x 27.3	184.2	177.6	8.09	639	yes, tree supports from the plate (639 mm2 over 45 deg)
drum_gear_reel	PPA-CF	1	104.0 x 103.9 x 19.0	87.9	95.9	4.39	0	no
reel_c1	PPA-CF	1	101.9 x 102.0 x 12.0	87.4	93.7	4.25	0	no
drum_coupler	PPA-CF	1	78.0 x 77.9 x 61.5	79.6	89.3	4.40	1998	yes, tree supports from the plate (1998 mm2 over 45 deg)
drum_shaft	PPA-CF	1	170.3 x 25.4 x 22.0	42.7	45.2	2.16	922	yes, tree supports from the plate (922 mm2 over 45 deg)
reel_drum	PPA-CF	1	46.0 x 46.0 x 43.0	41.5	45.0	2.30	251	yes, tree supports from the plate (251 mm2 over 45 deg)
launch_sleeve	PPA-CF	1	64.0 x 64.0 x 85.0	39.4	49.3	2.78	2907	yes, tree supports from the plate (2907 mm2 over 45 deg)

part	material	qty	print bbox mm	volume cm3	mass g (each)	time h (each, alone)	overhang mm2	supports
drum_collar	PPA-CF	1	60.0 x 60.0 x 26.0	38.5	44.0	2.14	2694	yes, tree supports from the plate (2694 mm2 over 45 deg)
launch_aim_crank	PPA-CF	1	62.3 x 59.6 x 38.0	28.9	35.3	1.83	985	yes, tree supports from the plate (985 mm2 over 45 deg)
drum_idler_gear	PPA-CF	1	76.5 x 76.5 x 7.0	28.7	35.8	1.64	0	no
launch_drive_gear	PPA-CF	1	70.0 x 69.7 x 28.8	23.9	29.9	1.53	2457	yes, tree supports from the plate (2457 mm2 over 45 deg)
reel_c1p	PPA-CF	1	72.0 x 71.9 x 9.6	22.3	27.8	1.30	0	no
launch_aim_output	PPA-CF	1	48.0 x 47.8 x 59.5	21.8	26.2	1.58	0	no
reel_uc	PPA-CF	1	10.0 x 10.0 x 243.5	19.0	23.7	2.75	4	no
launch_aim_worm	PPA-CF	1	28.0 x 28.0 x 54.5	17.1	21.3	1.33	2779	yes, tree supports from the plate (2779 mm2 over 45 deg)
drum_dogring_launch	PPA-CF	1	62.0 x 62.0 x 10.5	15.2	19.0	0.92	123	yes, tree supports from the plate (123 mm2 over 45 deg)
launch_aim_sun	PPA-CF	1	48.0 x 47.8 x 12.0	14.6	18.3	0.89	0	no
reel_c	PPA-CF	1	22.6 x 20.0 x 58.5	13.9	17.4	1.18	33	no
launch_aim_lug	PPA-CF	1	30.0 x 58.0 x 27.8	13.6	16.9	0.95	995	yes, tree supports from the plate (995 mm2 over 45 deg)
reel_gb	PPA-CF	1	51.0 x 51.0 x 7.0	12.1	15.1	0.72	0	no
reel_gk	PPA-CF	1	45.0 x 45.0 x 12.4	12.0	15.0	0.75	1031	yes, tree supports from the plate (1031 mm2 over 45 deg)
drum_coupler_hub	PPA-CF	1	26.0 x 26.0 x 32.5	11.7	14.6	0.88	37	no
launch_aim_conrod	PPA-CF	1	173.0 x 17.0 x 6.0	11.3	14.2	0.67	0	no
reel_cs_c	PPA-CF	1	22.6 x 20.0 x 40.5	10.2	11.4	0.79	182	yes, tree supports from the plate (182 mm2 over 45 deg)
reel_sprag_hub	PPA-CF	1	56.0 x 56.0 x 14.0	9.8	12.2	0.64	33	no
reel_gi	PPA-CF	1	48.0 x 47.8 x 6.0	9.1	11.4	0.55	0	no
reel_k	PPA-CF	1	10.0 x 10.0 x 108.6	8.1	10.2	1.21	14	no
reel_ga	PPA-CF	1	42.0 x 41.8 x 7.0	7.7	9.6	0.47	0	no
drum_axle	PPA-CF	1	95.3 x 10.0 x 9.0	7.1	8.9	0.46	0	no
drum_rod	PPA-CF	1	95.3 x 10.0 x 9.0	7.1	8.9	0.46	0	no

part	material	qty	print bbox mm	volume cm3	mass g (each)	time h (each, alone)	overhang mm2	supports
launch_aim_bevel30	PPA-CF	1	52.6 x 52.4 x 9.7	7.0	8.8	0.46	590	yes, tree supports from the plate (590 mm2 over 45 deg)
drum_idler	PPA-CF	1	22.0 x 18.6 x 54.5	7.0	8.7	0.77	331	yes, tree supports from the plate (331 mm2 over 45 deg)
drum_shift_rail	PPA-CF	1	88.0 x 10.0 x 9.0	6.5	8.2	0.43	0	no
reel_d	PPA-CF	1	53.0 x 53.0 x 3.0	6.0	7.6	0.36	1050	yes, tree supports from the plate (1050 mm2 over 45 deg)
launch_aim_idler	PPA-CF	1	30.0 x 29.7 x 12.0	6.0	7.5	0.42	0	no
drum_crank	PPA-CF	1	56.5 x 57.5 x 9.6	5.5	6.9	0.37	0	no
reel_lever	PPA-CF	1	67.0 x 66.0 x 8.0	5.5	6.9	0.36	20	no
launch_aim_arm	PPA-CF	1	63.6 x 41.0 x 19.3	5.3	6.6	0.43	14	no
reel_ut	PPA-CF	1	8.0 x 8.0 x 104.5	5.2	6.5	1.01	4	no
reel_s	PPA-CF	1	99.0 x 8.0 x 7.0	4.5	5.7	0.30	0	no
launch_gear_pin	PPA-CF	1	53.0 x 16.0 x 12.5	4.3	5.4	0.33	68	no
reel_gc	PPA-CF	1	37.4 x 37.5 x 7.0	4.3	5.3	0.29	0	no
reel_t1	PPA-CF	1	33.0 x 33.0 x 6.0	3.7	4.6	0.25	0	no
reel_swing_arm	PPA-CF	1	72.6 x 19.0 x 9.6	2.5	3.2	0.21	0	no
reel_cs_t	PPA-CF	1	12.0 x 12.0 x 24.5	2.2	2.8	0.29	39	no
reel_b1g	PPA-CF	1	39.0 x 39.0 x 6.2	2.1	2.6	0.16	679	yes, tree supports from the plate (679 mm2 over 45 deg)
reel_b3b	PPA-CF	1	22.0 x 22.0 x 8.8	1.1	1.4	0.12	0	no
reel_b2p	PPA-CF	1	22.0 x 22.0 x 7.8	1.0	1.3	0.11	149	yes, tree supports from the plate (149 mm2 over 45 deg)
reel_b3a	PPA-CF	1	22.0 x 22.0 x 7.8	1.0	1.3	0.11	0	no
launch_yoke_peg_l	PPA-CF	1	20.0 x 8.0 x 7.4	1.0	1.2	0.11	0	no
launch_yoke_peg_r	PPA-CF	1	20.0 x 8.0 x 7.4	1.0	1.2	0.11	0	no
reel_b1p	PPA-CF	1	21.0 x 21.0 x 11.4	1.0	1.2	0.13	87	no
reel_stub	PPA-CF	1	20.0 x 8.0 x 6.8	0.9	1.1	0.10	0	no
reel_t	PPA-CF	1	18.0 x 18.0 x 5.6	0.9	1.1	0.09	0	no

part	material	qty	print bbox mm	volume cm3	mass g (each)	time h (each, alone)	overhang mm2	supports
launch_seat_plunger	PPA-CF	1	29.5 x 15.0 x 19.6	0.8	1.0	0.18	0	no
drum_reel_washer	PPA-CF	1	35.1 x 35.1 x 1.4	0.8	0.9	0.05	0	no
reel_b2g	PPA-CF	1	22.0 x 22.0 x 6.8	0.7	0.9	0.09	149	yes, tree supports from the plate (149 mm2 over 45 deg)
drum_n20_pinion	PPA-CF	1	9.8 x 9.5 x 8.0	0.4	0.5	0.08	0	no
launch_seat_pushrod	PPA-CF	1	16.0 x 4.8 x 3.6	0.1	0.2	0.03	0	no

- **frame_deck_2**: As modelled, bottom face on the bed. deck: M3 inserts for every frame foot; 12 x M3 rover holes (it is its own drill template)
- **frame_deck_4**: As modelled, bottom face on the bed. deck: M3 inserts for every frame foot; 14 x M3 rover holes (it is its own drill template)
- **frame_deck_1**: As modelled, bottom face on the bed. deck: M3 inserts for every frame foot; 10 x M3 rover holes (it is its own drill template)
- **frame_deck_3**: As modelled, bottom face on the bed. deck: M3 inserts for every frame foot; 9 x M3 rover holes (it is its own drill template)
- **frame_cheek_r**: Rotated: the part's -y axis points up. motor plate (y -10..0, front face = the spec's y 0): window for the puck adapter, 8 rim holes (M3 x 12 from the front into the puck lip), bridge to the motor leg (2 x M3 x 12), can stay and hood inserts; carved round the other subsystems
- **frame_camera_mount**: As modelled, bottom face on the bed. Kiyo X mount: its own clip grips the 16 mm edge plate (or a 1/4-20 screw), 4 x M3 x 10 into the deck
- **launch_aim_housing**: As modelled, bottom face on the bed. Aim housing on the deck (4 x M3 x 10 from the top through its counterbored foot flanges into the deck inserts): sleeve end bushing, countershaft bushings, worm saddles, M8 wheel bolt (head on the outer boss, nut in the inner face hex pocket), engaged and swung-out swing arm stop posts
- **launch_rail_mid**: Rotated: the part's -x axis points up. Rail mid piece: shoe track, cheeks on the sleeve, big gear pin bosses, lever seats, SW_GAP bracket pilots
- **frame_motor_leg**: As modelled, bottom face on the bed. motor support leg (prints upright): 4 x M3 x 10 into the deck, 2 x M3 x 12 along y into the bridge behind the motor plate
- **launch_rail_front**: As modelled, bottom face on the bed. Rail front piece: shoe track end, spring anchor posts with straps, muzzle tie
- **frame_ebay**: As modelled, bottom face on the bed. electronics bay: ESP32-S3 cradle (USB slot), BTS7960 on standoffs with vents, DRV8833, MP1584, HX711 pockets, fuse cradle, wire exits with tie anchors
- **drum_drum_a**: As modelled, bottom face on the bed. Drum A: N20 gear 120T m0.8, groove 1 barrel (follower), hex spigot
- **launch_rail_rear**: Rotated: the part's -z axis points up. Rail rear piece: shoe track, walls, flange at the rear end for the reel head (D49)
- **reel_bridge**: Rotated: the part's +y axis points up. Rail bridge: plate inboard of the main-axis stack with countershafts C and T, the c1' stub, pillars to the u-shaft bushings, cad-drum's shift rail; box arm bolted to rail cheek A. Print on its inboard face
- **reel_head_front**: Rotated: the part's -x axis points up. Head front plate: bolts the reel head to the rail rear flange, u-shaft T rear bushing, link window. Print flat
- **reel_head_spine**: Rotated: the part's -y axis points up. Head spine: inboard wall from the front plate to P2; the drum brackets and plates screw to it. Print flat
- **drum_drum_b**: Rotated: the part's -z axis points up. Drum B: brake cam (spool clutch lever), Gray code bands A B C, wrap segment gear + locking arc
- **reel_head_p2**: Rotated: the part's -x axis points up. Head plate P2: cup shaft and wrap idler rear journals. Print flat

- **launch_stop_spring_l**: Rotated: the part's -z axis points up. Carriage stop flexure: 16 loops (31 tapered leaves t 3.15/1.0 x 14), print upside down on the flat w 83.5 face, 100% infill / all perimeters; preload 6.6 mm by 2 x M3x20 draw screws through the frame front wall
- **launch_stop_spring_r**: Rotated: the part's -z axis points up. Carriage stop flexure: 16 loops (31 tapered leaves t 3.15/1.0 x 14), print upside down on the flat w 83.5 face, 100% infill / all perimeters; preload 6.6 mm by 2 x M3x20 draw screws through the frame front wall
- **frame_ebay_lid**: Rotated: the part's -z axis points up. bay lid: grille over the BTS7960, 4 x M3 into the corner bosses
- **reel_head_p1**: Rotated: the part's +x axis points up. Head plate P1: capstan tube, u-shaft C and G_b bushings, clutch lever post, spool shaft leg. Print flat
- **launch_guard_l_a**: Rotated: the part's -y axis points up. Spring guard left rear (end wall, top cover) (D33): 2 mm side plate outside both spring pairs, 4 wall brackets (M3x12 self tap into the rail wall); fit by moving it inward
- **launch_guard_r_a**: Rotated: the part's +y axis points up. Spring guard right rear (end wall, top cover) (D33): 2 mm side plate outside both spring pairs, 4 wall brackets (M3x12 self tap into the rail wall); fit by moving it inward
- **launch_carriage**: Rotated: the part's +y axis points up. Carriage: rack, shoes, dart cradle, pusher with tether hole, spring arms and pegs, SW feet
- **frame_guard_front**: Rotated: the part's -x axis points up. front gear guard between the drum gears and the bay, M3 x 10 into the deck
- **drum_index_block**: As modelled, bottom face on the bed. SW_INDEX block: column on the motor plate front face (2 x M3 x 10 into the frame inserts), M3 nut contact
- **frame_can_stay**: Rotated: the part's -y axis points up. can stay: post on the motor plate (2 x M3 x 12 into plate inserts), cable tie or hose clamp round the can through the cradle slots; reprint with the adapter
- **reel_spool**: Rotated: the part's -z axis points up. Spool d40 core, d62 flanges, 20 wide: 20 m of 1 mm Dyneema, turns free on two 608s. Print flange down
- **launch_stop_frame_l**: Rotated: the part's -y axis points up. Stop frame (print outboard face down): outboard preload ledge, anchor seat, rear foot (2 x M3x12 into the rail wall), front foot (2 x M3x16 into the rail front plate, hex key through the d6.5 holes in the front wall). Fit before the spring guards and the main springs
- **launch_stop_frame_r**: Rotated: the part's +y axis points up. Stop frame (print outboard face down): outboard preload ledge, anchor seat, rear foot (2 x M3x12 into the rail wall), front foot (2 x M3x16 into the rail front plate, hex key through the d6.5 holes in the front wall). Fit before the spring guards and the main springs
- **drum_bracket_b**: As modelled, bottom face on the bed. Drum bracket B: axle, guide rod, wrap idler journal; foot screwed to the reel head spine
- **drum_switch_block**: As modelled, bottom face on the bed. Code switch block on top of the drum: 3 lever feet, 3 M3 nut contacts (SW_DRUM_A, B, C)
- **launch_carriage_yoke**: As modelled, bottom face on the bed. Bolted yoke: both spring arms (hit the stop, d8.1 holes for the 8 mm steel rod spring pegs, D5 holes at y +-56 empty), cross bar, keys, SW_SEATED block, rigid SW_HOME foot. Prints on the peg boss bottoms and key plates (w 62)
- **frame_rc832_holder**: As modelled, bottom face on the bed. RC832 holder: receiver stands antenna-end up in the slot, leads out low, 2 x M3 x 10
- **reel_head_mid**: Rotated: the part's +y axis points up. Head mid plate: spool shaft inboard bushing. Print flat
- **launch_guard_r_b2**: Rotated: the part's +y axis points up. Spring guard right mid under the stop spring (D33): 2 mm side plate outside both spring pairs, 2 wall brackets (M3x12 self tap into the rail wall) and a foot on the front plate (M3x10 self tap down); fit by moving it inward
- **drum_fork**: As modelled, bottom face on the bed. Compliant shift fork: thin yoke is the spring (about 1.5 N/mm), lug for the bell crank pin, contact tab
- **sensors_sw_rail_low_bracket**: As modelled, bottom face on the bed. SW_RAIL_LOW bracket on the aim housing foot extension: an M3x10 into the extension and its own deck screw, an M3x16 from the top through the counterbored foot and the extension into a deck insert
- **launch_guard_l_b2**: Rotated: the part's -y axis points up. Spring guard left mid under the stop spring (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall) and a foot on the front plate (M3x10 self tap down); fit by moving it inward
- **sensors_as5600_carrier**: Rotated: the part's -y axis points up. AS5600 carrier: plate with 4 pegs through the board holes, column screwed to the aim housing spine

- **frame_guard_hood**: Rotated: the part's -z axis points up. gear hood over the reel and bus gears, M3 x 10 through its lugs into the motor plate
- **reel_cup**: As modelled, bottom face on the bed. Wrap cup (you v7, D49 axial eyelet at r 26): the line runs straight through the eyelet. Print open end down
- **drum_n20_cradle**: As modelled, bottom face on the bed. N20 cradle on the reel head front plate: pivot + swing screw (hand fallback), SW_N20 contact
- **reel_head_p1b**: Rotated: the part's +x axis points up. Head plate P1b: capstan tube rear bushing, SW_ARM. Print flat
- **sensors_sw_gap**: As modelled, bottom face on the bed. SW_GAP printed contact base (cad.lib.switch, coupon 2 geometry): with its arm screwed to the right rail cheek outer face below the big gear, closed by the big gear cam bump (gear in its gap)
- **reel_shroud**: Rotated: the part's +y axis points up. Spool shroud: keeps loose loops on the spool, open at the top front where the line leaves
- **sensors_sw_rail_low**: As modelled, bottom face on the bed. SW_RAIL_LOW printed contact base (cad.lib.switch, coupon 2 geometry): on its bracket on the aim housing foot, closed by the crank cam at the LOW dead point
- **drum_dog_post**: Rotated: the part's -z axis points up. Dog contact post on the shift rail: SW_DOG_R and SW_DOG_L screws (collar fully engaged)
- **reel_sw_base**: As modelled, bottom face on the bed. SW_ARM contact base on P2's front face
- **drum_follower**: As modelled, bottom face on the bed. Follower slider: pin in groove 1, slides along the guide rod; the link takes its motion to the bridge
- **launch_aim_cap**: Rotated: the part's -z axis points up. Worm shaft saddle cap, 4 x M3x30 through the legs into 13 deep pilots in the saddles
- **launch_guard_l_c**: Rotated: the part's -y axis points up. Spring guard left front (top cover) (D33): 2 mm side plate outside both spring pairs, 2 bosses on the front plate edge (M3x12 self tap along y); fit by moving it inward
- **launch_guard_r_c**: Rotated: the part's +y axis points up. Spring guard right front (top cover) (D33): 2 mm side plate outside both spring pairs, 2 bosses on the front plate edge (M3x12 self tap along y); fit by moving it inward
- **launch_guard_l_b1**: Rotated: the part's -y axis points up. Spring guard left mid front of the stop frame (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall); fit by moving it inward
- **launch_guard_r_b1**: Rotated: the part's +y axis points up. Spring guard right mid front of the stop frame (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall); fit by moving it inward
- **frame_cable_anchor_1**: As modelled, bottom face on the bed. cable-tie anchor (strain relief), 1 x M3 x 12 into the deck
- **frame_cable_anchor_2**: As modelled, bottom face on the bed. cable-tie anchor (strain relief), 1 x M3 x 12 into the deck
- **drum_link_eye_1**: Rotated: the part's -z axis points up. Link eye pressed on the rod end, M3 screw through the follower lug
- **drum_link_eye_2**: Rotated: the part's +z axis points up. Link eye pressed on the rod end, M3 screw through the bell crank input arm
- **drum_n20_strap**: As modelled, bottom face on the bed. End plate closing the N20 cradle, 2 x M3 x 8
- **drum_magnet_cap**: Rotated: the part's -z axis points up. AS5600 magnet cap on the spigot end (6 x 2.5 diametric or edge-on disc), M3 x 6 onto the flat
- **frame_ebay_esp_clamp**: As modelled, bottom face on the bed. ESP32 hold-down over the module end, 2 x M3 x 8 into the cradle posts
- **sensors_sw_gap_lever**: Rotated: the part's +y axis points up. SW_GAP lever, t 1.4 mm, print on its side so layers run along the beam
- **sensors_sw_rail_low_lever**: Rotated: the part's +y axis points up. SW_RAIL_LOW lever, t 1.4 mm, print on its side so layers run along the beam
- **reel_sw_lever**: Rotated: the part's +y axis points up. SW_ARM lever: rides the cup gear's raised track (contact closed), drops into its home notch (open)
- **sensors_as5600_strip_top**: Rotated: the part's +y axis points up. AS5600 clamp strip on the board edge over the pegs, M3 from the strip front into a nut in the carrier back

- **sensors_as5600_strip_bot**: Rotated: the part's +y axis points up. AS5600 clamp strip on the board edge over the pegs, M3 from the strip front into a nut in the carrier back
- **drum_index_lever**: As modelled, bottom face on the bed. SW_INDEX lever on the 6-bump coupler cam (D12 fallback encoder)
- **drum_sw_lever_0**: As modelled, bottom face on the bed. Code switch lever A: flexure, round nub, M3 x 6 contact screw
- **drum_sw_lever_1**: As modelled, bottom face on the bed. Code switch lever B: flexure, round nub, M3 x 6 contact screw
- **drum_sw_lever_2**: As modelled, bottom face on the bed. Code switch lever C: flexure, round nub, M3 x 6 contact screw
- **launch_sw_seated**: Rotated: the part's +y axis points up. SW_SEATED printed flex lever on the left rail lip, at rest (prints on its side): beam 0.8 thick, contact gap 0.50 (one screw turn), pad pretravel 0.70 + overtravel 0.25, peak strain 0.10 % (limit 0.8 %)
- **launch_sw_home**: Rotated: the part's +y axis points up. SW_HOME printed flex lever on the right rail lip, at rest (prints on its side): beam 0.8 thick, contact gap 0.50 (one screw turn), pad pretravel 0.70 + overtravel 0.25, peak strain 0.10 % (limit 0.8 %)
- **launch_muzzle_roller**: As modelled, bottom face on the bed. Lip roller at the carriage front: the tether bends down over it in tow and retrieval (groove r 6 for the 1 mm line), 693ZZ inside
- **launch_gear_washer**: As modelled, bottom face on the bed. Spacer washer: big gear hub to the left cheek
- **launch_aim_drag_pad**: As modelled, bottom face on the bed. Flexure drag pad: cantilever finger t 2, tip bump 0.45 into the sleeve d20 (strain 0.43 %, about 2.3 N), root block pushed axially into the arm hub pocket
- **launch_gear_clip**: As modelled, bottom face on the bed. C clip on the big gear pin groove
- **launch_rail_splice_l**: As modelled, bottom face on the bed. Top splice strap over the front rail joint (4 x M3 x 10 self tap)
- **launch_rail_splice_r**: As modelled, bottom face on the bed. Top splice strap over the front rail joint (4 x M3 x 10 self tap)
- **launch_drive_washer**: As modelled, bottom face on the bed. Spacer washer: drive gear hub to the left cheek
- **drum_shear_pin**: Rotated: the part's -z axis points up. Torque limiter shear pin d4 (about 5 N m), prints lying; print 6, 5 are spares
- **reel_cup_clip**: As modelled, bottom face on the bed. your v7 C-clip on the cup shaft neck in front of the cup
- **launch_aim_drag_shim**: As modelled, bottom face on the bed. Preload shim behind the drag pad root block (nominal; print up to 0.25 thicker for more drag, strain stays under 0.7 %)
- **launch_aim_idler_clip**: As modelled, bottom face on the bed. Printed C clip on the arm pin, holds the idler
- **launch_aim_crankpin_clip**: As modelled, bottom face on the bed. Printed C clip on the crank pin, inboard of the conrod
- **launch_aim_lugpin_clip**: As modelled, bottom face on the bed. Printed C clip on the lug pin, outboard of the conrod
- **launch_seat_clip**: Rotated: the part's -z axis points up. Head cap on the plunger pin (rests on the yoke block)
- **launch_muzzle_washer_l**: Rotated: the part's -z axis points up. Lip roller thrust washer (0.9 thick, flange on the cheek side) between the bearing inner race and the cheek: put both in the roller bore, then drop roller and washers into the fork slot
- **launch_muzzle_washer_r**: Rotated: the part's -z axis points up. Lip roller thrust washer (0.9 thick, flange on the cheek side) between the bearing inner race and the cheek: put both in the roller bore, then drop roller and washers into the fork slot
- **frame_motor_adapter**: Rotated: the part's -z axis points up. puck adapter: slotted M6 holes (+-4 mm) for the motor bosses, rear lip with 8 M3 inserts for the rim screws, 2 inserts for cad-drum's SW_INDEX block; reprint when the motor is measured
- **launch_big_gear**: Rotated: the part's -z axis points up. Big gear m2.5: sector layer 27 of 40 teeth (rack, v10 release teeth) + full 40T layer (drive), SW_GAP cam bump, on the rail pin
- **drum_gear_reel**: As modelled, bottom face on the bed. REEL output gear m2 50T face 12 (D20), free on its sleeve, dog face toward the collar; meshes C1 on the rail bridge (D49)
- **reel_c1**: As modelled, bottom face on the bed. C1 m2 49T: meshes the REEL output gear 50T across the pivot
- **drum_coupler**: As modelled, bottom face on the bed. Coupler: 5 N m shear pin limiter (D8), bus gear 50T m1.5 (D25), SW_INDEX cam, REEL journal, hex socket; prints hub end down, tree support under the bus gear

ring

- **drum_shaft**: Rotated: the part's +y axis points up. Main shaft: hex 22 AF, d12 spigot into the launch sleeve; prints on a hex flat (support under the spigot)
- **reel_drum**: As modelled, bottom face on the bed. Capstan drum dia 40 x 40 (you v7), keyed on the tube. Print flange down
- **launch_sleeve**: Rotated: the part's -z axis points up. Launch sleeve d20: dog ring flange, integral key ribs (sprag #1, splitter sun), journals
- **drum_collar**: As modelled, bottom face on the bed. Dog collar: hex bore, 4 back-cut roofed dogs each face, fork groove (bridges over the lower dogs)
- **launch_aim_crank**: As modelled, bottom face on the bed. Worm wheel m2 z20 face 16 = crank gear, crank arm r30 and d8 crank pin in one part, two 608 inside
- **drum_idler_gear**: As modelled, bottom face on the bed. Wrap idler output gear 49T m1.5 (D bore) to the cad-reel cup gear 28T: 7 cup turns
- **launch_drive_gear**: As modelled, bottom face on the bed. Drive gear m2.5 26T, sprag #1 inside on an integral key rib
- **reel_c1p**: As modelled, bottom face on the bed. c1' m1.5 46T, free on its stub: driven by the bus gear 50T, drives the swing idler
- **launch_aim_output**: As modelled, bottom face on the bed. Splitter output m1.5 z30 face 12 with its d8 countershaft (D flat)
- **reel_uc**: As modelled, bottom face on the bed. u-shaft C (PPA-CF d10, 244 long): along the rail from the B1 bevel to G_a. Print standing (244 of the 260 mm height); an 8 mm steel rod with printed sleeves is the stiff fallback
- **launch_aim_worm**: As modelled, bottom face on the bed. Worm m2 pd24 single start (lead angle 4.8 deg, self-locking) + d8 shaft + bevel z15
- **drum_dogring_launch**: Rotated: the part's -z axis points up. LAUNCH dog ring: 3 slots for the launch sleeve lugs, 3 x M3 from the sleeve side
- **launch_aim_sun**: As modelled, bottom face on the bed. Splitter sun m1.5 z30 face 12, keyed on the launch sleeve
- **reel_c**: As modelled, bottom face on the bed. Capstan tube C (PPA-CF d20): key rib in the drum, G_c on its flat, journals in P1 and P1b. Print standing on its rear end
- **launch_aim_lug**: Rotated: the part's +y axis points up. Lug bracket (PPA-CF), 2 x M3 into the rail wall, d8 lug pin at (u 115.2, w 0)
- **reel_gb**: As modelled, bottom face on the bed. G_b m1.5 32T idler on its stub in P1
- **reel_gk**: As modelled, bottom face on the bed. Cup gear G_k m1.5 28T: driven by the drum's wrap idler 49T (7 cup turns per wrap); the raised track on its rear face holds SW_ARM closed, the notch at home lets it open. Print track face down
- **drum_coupler_hub**: As modelled, bottom face on the bed. Coupler hub on the wiper shaft (bore parametric: WIPER), M8 nut inside, shear pin hole
- **launch_aim_conrod**: Rotated: the part's +y axis points up. Conrod 156 between pin centres (PPA-CF), printed C clips on both pins
- **reel_cs_c**: Rotated: the part's -z axis points up. Countershaft C (PPA-CF): sprag #2, C1 and the B1 bevel. Print standing
- **reel_sprag_hub**: Rotated: the part's -z axis points up. Sprag #2 hub on the bridge: holds the CSK20PP capstan backstop
- **reel_gi**: As modelled, bottom face on the bed. Swing idler m1.5 30T
- **reel_k**: As modelled, bottom face on the bed. Cup shaft K (PPA-CF d10): runs inside the tube, cup in front, G_k behind P1b, journal in P2. Print standing on its rear end
- **reel_ga**: As modelled, bottom face on the bed. G_a m1.5 26T on the rear end of u-shaft C
- **drum_axle**: Rotated: the part's -x axis points up. Fixed drum axle d10, pressed into the reel head front plate and bracket B
- **drum_rod**: Rotated: the part's +x axis points up. Follower guide rod d10, pressed into the reel head front plate and bracket B
- **launch_aim_bevel30**: As modelled, bottom face on the bed. Bevel m1.5 z30 on the countershaft D flat, drives the z15 2:1

- **drum_idler**: As modelled, bottom face on the bed. Wrap idler shaft: pinion 10T m1.25 (locked at home by the drum lock arc), lock cam, journals in bracket B and P2
- **drum_shift_rail**: Rotated: the part's +x axis points up. Shift rail d10 on the rail bridge (pressed into the bridge plate boss)
- **reel_d**: As modelled, bottom face on the bed. Clutch disk D on the spool shaft flat: its friction ring presses the spool's outboard face. Print ring up
- **launch_aim_idler**: As modelled, bottom face on the bed. Splitter idler m1.5 z18 face 12 on the arm pin
- **drum_crank**: Rotated: the part's -y axis points up. Bell crank 1:1 on the bridge shelf: turns the link's motion along the rail into the fork's shift
- **reel_lever**: Rotated: the part's -x axis points up. Clutch lever (PPA-CF): the drum's brake cam presses its upper arm; the lower arm is the clutch spring leaf, its FIRM screw squeezes the spool. Print flat
- **launch_aim_arm**: Rotated: the part's -z axis points up. Swing arm on the sleeve, dragged by the printed PLA flexure pad in its hub; printed idler pin with a printed C clip groove
- **reel_ut**: As modelled, bottom face on the bed. u-shaft T (PPA-CF d8): along the rail from the B2 bevel to the B3 bevel. Print standing
- **reel_s**: Rotated: the part's +x axis points up. Spool shaft S (PPA-CF d8): B3 bevel and the clutch disk on its flats, journals in the mid plate and the P1 leg. Print lying on its 1 mm flat
- **launch_gear_pin**: Rotated: the part's -x axis points up. Snap pin d10 for the big gear, C clip; prints lying on its flat
- **reel_gc**: As modelled, bottom face on the bed. Capstan gear G_c m1.5 23T on the tube flat
- **reel_t1**: As modelled, bottom face on the bed. T1 m1.5 20T: take-up at 2.5 x wiper
- **reel_swing_arm**: Rotated: the part's -z axis points up. Swing arm on c1's hub: carries the idler into mesh with T1 on forward wiper turns
- **reel_cs_t**: Rotated: the part's -z axis points up. Countershaft T (PPA-CF): T1 and the B2 bevel
- **reel_b1g**: As modelled, bottom face on the bed. Bevel B1 gear m1.5 24T (2:1 up to u-shaft C)
- **reel_b3b**: As modelled, bottom face on the bed. Bevel B3 m1 20T on the spool shaft
- **reel_b2p**: As modelled, bottom face on the bed. Bevel B2 m1 20T on the front end of u-shaft T
- **reel_b3a**: As modelled, bottom face on the bed. Bevel B3 m1 20T on the rear end of u-shaft T
- **launch_yoke_peg_l**: Rotated: the part's +x axis points up. Spring peg on the yoke arm (printed PPA-CF lying down, or 8 mm steel rod 20 long): pressed into the blind d8.1 hole from below up to its end (a drop of CA if it is loose), the spring hook sits on the 6 mm below the boss (held at w 60 by the spring line); ends deburred
- **launch_yoke_peg_r**: Rotated: the part's +x axis points up. Spring peg on the yoke arm (printed PPA-CF lying down, or 8 mm steel rod 20 long): pressed into the blind d8.1 hole from below up to its end (a drop of CA if it is loose), the spring hook sits on the 6 mm below the boss (held at w 60 by the spring line); ends deburred
- **reel_b1p**: As modelled, bottom face on the bed. Bevel B1 pinion m1.5 12T on the front end of u-shaft C
- **reel_stub**: Rotated: the part's -x axis points up. c1' stub axle (PPA-CF d8, pressed into the bridge)
- **reel_t**: As modelled, bottom face on the bed. Thrust ring: the lever's FIRM screw pushes it onto the inboard 608 inner race (clutch squeeze)
- **launch_seat_plunger**: As modelled, bottom face on the bed. SW_SEATED plunger: foot pad + stem with a 45 deg chamfer, clip groove on top
- **drum_reel_washer**: As modelled, bottom face on the bed. REEL gear retaining washer on the coupler journal end, 2 x M3 x 6
- **reel_b2g**: As modelled, bottom face on the bed. Bevel B2 m1 20T on countershaft T
- **drum_n20_pinion**: As modelled, bottom face on the bed. N20 pinion 10T m0.8, D bore pressed on the N20 shaft
- **launch_seat_pushrod**: Rotated: the part's -z axis points up. SW_SEATED pushrod d3.2 with the stem slot

Plates

plate	material	objects	mass g	time h
plate_1_PLA.3mf	PLA	7: drum_link_eye_1, drum_link_eye_2, drum_n20_cradle, launch_carriage, launch_guard_l_c, launch_guard_r_c, sensors_sw_rail_low	95	2.6
plate_2_PLA.3mf	PLA	5: drum_index_block, drum_magnet_cap, drum_switch_block, frame_cheek_r, frame_rc832_holder	281	7.3

plate	material	objects	mass g	time h
plate_3_PLA.3mf	PLA	6: drum_index_lever, drum_n20_strap, drum_sw_lever_2, frame_cable_anchor_1, frame_deck_2, launch_aim_cap	318	8.1
plate_4_PLA.3mf	PLA	6: drum_follower, frame_cable_anchor_2, frame_deck_4, launch_aim_drag_pad, sensors_as5600_strip_bot, sensors_as5600_strip_top	314	7.9
plate_5_PLA.3mf	PLA	6: drum_dog_post, reel_bridge, reel_cup, reel_head_p1b, reel_sw_base, sensors_sw_rail_low_bracket	174	4.9
plate_6_PLA.3mf	PLA	16: drum_shear_pin__c1, drum_shear_pin__c2, drum_shear_pin__c3, drum_shear_pin__c4, drum_shear_pin__c5, frame_deck_1, launch_aim_crankpin_clip, launch_aim_lugpin_clip, launch_gear_clip, launch_gear_washer, launch_guard_r_b1, launch_rail_rear, launch_sw_home, launch_sw_seated, reel_cup_clip, reel_shroud	365	9.4
plate_7_PLA.3mf	PLA	4: frame_deck_3, launch_guard_l_b2, launch_guard_r_b2, launch_rail_mid	450	12.4
plate_8_PLA.3mf	PLA	9: drum_shear_pin__c6, frame_motor_leg, launch_aim_idler_clip, launch_guard_l_b1, launch_muzzle_washer_l, launch_muzzle_washer_r, launch_stop_frame_l, launch_stop_spring_l, reel_head_front	345	9.9
plate_9_PLA.3mf	PLA	4: launch_stop_frame_r, reel_head_mid, reel_head_p2, reel_head_spine	250	6.6
plate_10_PLA.3mf	PLA	6: launch_aim_housing, launch_carriage_yoke, launch_rail_front, reel_head_p1, reel_spool, sensors_sw_gap_lever	434	12.1
plate_11_PLA.3mf	PLA	12: drum_drum_a, drum_fork, drum_sw_lever_0, drum_sw_lever_1, frame_ebay, frame_ebay_esp_clamp, frame_guard_front, launch_aim_drag_shim, launch_drive_washer, launch_muzzle_roller, launch_seat_clip, sensors_as5600_carrier	372	9.7
plate_12_PLA.3mf	PLA	11: drum_bracket_b, frame_camera_mount, frame_can_stay, frame_ebay_lid, frame_guard_hood, launch_rail_splice_l, launch_rail_splice_r, launch_stop_spring_r, reel_sw_lever, sensors_sw_gap, sensors_sw_rail_low_lever	401	11.6
plate_13_PLA.3mf	PLA	1: drum_drum_b	82	2.3
plate_14_PLA.3mf	PLA	1: launch_guard_l_a	65	1.9
plate_15_PLA.3mf	PLA	1: launch_guard_r_a	65	1.9
plate_16_PPA-CF.3mf	PPA-CF	13: drum_n20_pinion, drum_reel_washer, drum_rod, frame_motor_adapter, launch_aim_arm, reel_b1g, reel_b1p, reel_b2g, reel_b3a, reel_drum, reel_ga, reel_gc, reel_gk	401	18.1
plate_17_PPA-CF.3mf	PPA-CF	18: drum_coupler, drum_coupler_hub, drum_gear_reel, drum_idler, drum_shift_rail, launch_aim_bevel30, launch_big_gear, launch_gear_pin, launch_seat_pushrod, reel_b2p, reel_b3b, reel_c, reel_c1, reel_cs_c, reel_d, reel_gb, reel_s, reel_ut	569	26.0
plate_18_PPA-CF.3mf	PPA-CF	22: drum_axle, drum_crank, drum_dogring_launch, drum_idler_gear, drum_shaft, launch_aim_crank, launch_aim_idler, launch_aim_lug, launch_aim_worm, launch_drive_gear, launch_seat_plunger, launch_sleeve, launch_yoke_peg_l, launch_yoke_peg_r, reel_c1p, reel_cs_t, reel_k, reel_lever, reel_sprag_hub, reel_stub, reel_t1, reel_uc	369	18.1
plate_19_PPA-CF.3mf	PPA-CF	7: drum_collar, launch_aim_conrod, launch_aim_output, launch_aim_sun, reel_gi, reel_swing_arm, reel_t	118	5.7

Totals on the plates

- PLA: 95 objects on 15 plates, 4011 g, about 1345.0 m of 1.75 mm filament, 108.6 h
- PPA-CF: 60 objects on 4 plates, 1456 g, about 484.4 m of 1.75 mm filament, 67.9 h

Recommended settings

PLA

- 0.4 mm nozzle, 0.2 mm layers
- 4 walls, 5 top and 5 bottom layers
- 40 % gyroid infill
- Nozzle 220 C, bed 55 to 60 C, textured PEI plate
- Door open or top vented

PPA-CF

- Hardened steel nozzle (0.4 or 0.6 mm), never a brass nozzle
- 0.2 mm layers
- 6 walls, 6 top and 6 bottom layers

- 60 % gyroid infill
- Dry the filament first (80 C for 8 h or more) and print from a dry box
- Enclosure closed, chamber heated if the printer can
- Nozzle 290 to 320 C, bed 100 to 110 C (check the spool label)

Base plates:

- `base/out/base/pack/plates/plate_10_PLA.3mf`
- `base/out/base/pack/plates/plate_11_PLA.3mf`
- `base/out/base/pack/plates/plate_12_PLA.3mf`
- `base/out/base/pack/plates/plate_13_PLA.3mf`
- `base/out/base/pack/plates/plate_14_PLA.3mf`
- `base/out/base/pack/plates/plate_15_PLA.3mf`
- `base/out/base/pack/plates/plate_16_PPA-CF.3mf`
- `base/out/base/pack/plates/plate_17_PPA-CF.3mf`
- `base/out/base/pack/plates/plate_18_PPA-CF.3mf`
- `base/out/base/pack/plates/plate_19_PPA-CF.3mf`
- `base/out/base/pack/plates/plate_1_PLA.3mf`
- `base/out/base/pack/plates/plate_2_PLA.3mf`
- `base/out/base/pack/plates/plate_3_PLA.3mf`
- `base/out/base/pack/plates/plate_4_PLA.3mf`
- `base/out/base/pack/plates/plate_5_PLA.3mf`
- `base/out/base/pack/plates/plate_6_PLA.3mf`
- `base/out/base/pack/plates/plate_7_PLA.3mf`
- `base/out/base/pack/plates/plate_8_PLA.3mf`
- `base/out/base/pack/plates/plate_9_PLA.3mf`

Before printing the base:

1. The wiper and spring numbers are in params.py, and the base pipeline and the physics were re-run (section 14, "After you measure"); the audit listed only the known items (README.md, "Open items").
2. Print one test part from the pack in PPA-CF and check a 608 drops into its seat with a firm thumb press, and an M3 heat-set insert sets straight. Keep hole compensation at 0 for base parts: base fits are tuned in the CAD through the coupons, not in the slicer.
3. PPA-CF: as in the PPA-CF paragraph and the settings table at the top of this section.

5. Base assembly

The assembly steps come from the base pipeline: every step is generated from the same model that passed the collision audit, with a picture per step. They are in `base/out/base/guide.pdf` (attached as an appendix at the end of this guide).

Present: `base/out/base/guide.pdf` (attached at the end of this guide).

General rules for the base:

- Heat-set inserts: soldering iron with the insert tip, about 220 C in PLA and about 280 C in the PPA-CF motor puck; press straight in, let it cool before loading.
- Bearings: press in square with a flat block, never hammer on the inner race.
- Keyed shafts: check the key slides in the sprag (CSK20PP) keyway by hand before final assembly.
- Turn every train by hand (battery unplugged) after each step: it must turn without binding before the next step.
- Leave the springs off until test stage 5 (section 12).
- Motor puck: bolt the wiper motor to the puck adapter on the bench first (M6 bolts through the puck slots from the front, step "Wiper motor on the puck"), then feed the puck into the motor plate window from behind and fit M3 x 12 from the front through the plate into the lip inserts (step "Puck into the plate" says which rim holes), before any shaft goes in.
- Drum coupler: coupler hub onto the wiper shaft (flats in line), the M8 nut on the thread, then the outer coupler over the hub and a printed shear pin through both (step "Main shaft: coupler and limiter").
- AS5600: push the board onto the 4 carrier pegs and clamp it with the two printed strips (M3 x 12 into nuts in the carrier back); no heat staking. Glue the magnet in its cap.
- Wires: leave a 150 mm loop at the rail pivot for SW_SEATED and SW_HOME. Never route a wire across a moving part.
- Contacts: set each gap by turning the screw (MEASURE_LIST G12).
- Manual drum fallback: loosen the cradle swing screw (M3, in the slot), swing the N20 out of mesh until n disappears from `sw`, turn the drum by hand by its gear rim, then swing the N20 back and tighten. Battery unplugged. Never turn the drum by hand while the carriage is drawn (`cock` above 0 or `SW_HOME` open) or while the line is under tension: shifting under load can dry fire the launcher or drop the line. De-cock and slacken the line first.
- Aim only with the carriage at HOME (springs slack). Fire low-pitch shots with the rover braked or chocked.

6. Dart build

The dart is built from three documents. This section says which to use for what, and in what order.

- **KEDGE v30 run sheet** (`kedge/pack_v30/KEDGE_v30_run_sheet.pdf`, a separate 10-page PDF from the dart project: print it alongside this guide): printing, board prep, mechanical assembly, function test. Its own section numbers are used below.
- `firmware/dart/WIRING.md` (below): the new wiring. It replaces the run sheet's wiring (its sections 7 and 9 and its wiring picture).
- `firmware/dart/BENCH_TEST.md` (below): the bench checks for the new firmware.

What changed from the run sheet

- The dart now uses two DRV8833 boards. DRV#3 goes to the base selector drum; its pocket on the keel stays empty.
- DRV#2 channel A drives BOTH drill motors (right drill leads still swapped), channel B drives the foot.
- No pin bridge on DRV#2 (ignore the bridge in run sheet 7.3).
- To reverse a motor, swap its two thin input wires at the driver instead of the motor leads (run sheet 7.6 and 10 swap motor leads by v29 number): drills = D8 / D9 at DRV#2 AIN1 / AIN2, foot = D6 / D7 at DRV#2 BIN1 / BIN2, deploy = D0 / D1 at DRV#1 AIN1 / AIN2, kick = D2 / D3 at DRV#1 BIN1 / BIN2. Exception: if only ONE drill turns the wrong way, swap that drill's motor leads.
- **Wire numbers changed.** The tape flags carry the NEW numbers from the new cut list. Wherever the run sheet names a wire number (sections 2 (parts table, "in wire 18"), 6.4, 6.5, 7.1, 7.6, 8b, 8d, 8e, 8f and 10), look it up in the table "Run sheet wire numbers" below. The two that matter most: run sheet 6.4 "3V3 wires (25 to 27) go only to the SLP pad" means NEW wires 23 and 24; run sheet 6.5 "wire 18 has a Schottky diode" means NEW wire 16.
- The old cut list, `wires.json` and `wiring.png` in `kedge/pack_v30/` are the 53-wire v29 files: replaced by `firmware/dart/wiring/` (wire_cut_list.csv, wiring.png).
- Use the firmware in `firmware/dart/`, not the copy inside the v30 pack. Ignore the run sheet's `firmware/reel` sketch: `firmware/base/` replaces it. The Adafruit INA219 library is no longer needed (BNO08x still is).
- Run sheet section 10, "Put them in kedge_logic.h (kickContactMa, kickStallMa, deployStallMa)": kickContactMa is still there, but kickStallMa and deployStallMa are gone (now `stallMa[]`). Enter the measured currents as BENCH_TEST.md says.
- Foot direction: the foot auger is right-hand thread, so it screws in when it turns **clockwise seen from the motor end, looking toward the auger tip**. The run sheet's "clockwise, seen from the shaft end" in 7.6 means the same only if you stand behind the motor; use the motor-end wording.

Dart build order (follow this; it merges the three documents)

1. Run sheet section 4: print and check the fit coupon.
2. Run sheet section 5: start plate A, flash the firmware from `firmware/dart/` (section 8.2 of this guide).
3. Run sheet section 6 steps 1 to 3: clip headers, set the buck, cut the wires (cut and flag them from the NEW cut list below, not the run sheet's section 9). Set MP1584 #1 to 5.04 V and MP1584 #2 to 8.0 V from the 3S LiPo on XT60 lead #1 (run sheet 6.2), before either buck is soldered to anything. (This replaces WIRING.md step B2.) MP1584 #2 later becomes the base drum supply at 9.0 V. Then fit the one capacitor (470 to 1000 uF, 10 V or more) across MP1584 #1's output: long leg OUT+, striped leg OUT- (run sheet 7.1; this is the capacitor WIRING.md B2 fits).
4. Run sheet section 6 step 4: find the SLP pad on DRV#1 and DRV#2 (WIRING.md step B6). Meter on ohms (not beep: beep only catches a dead short): SLP to VM must read over 50 ohms (a Pololu pull-up of tens of kilohms is fine). Under 50 ohms means the pad is the motor supply: do not put 3V3 on it.
5. Battery unplugged. Solder the bench harness with WIRING.md Path B, steps B3 to B10 (B2 is replaced by step 3 above), on the bench, outside the hull, motor leads left free at their far ends. B1 (cells and BMS) and every wire end that goes to the BMS (P+, P-) wait for step 9. Heat-shrink or tape every free wire end (wire 9 at its socket end, wire 13 at its BMS end): they are live during the bench power-up.
6. Battery unplugged. **Click the u.FL antenna onto the VTX** (never power the VTX without it). Bench power-up: the 8.0 V MP1584 #2 on the LiPo feeds the INA219 VIN+ pad and main ground, exactly as run sheet 7.5. Check the readings there (about 4.7 V at the XIAO 5V pad, VTX LED on; on the KEDGE WiFi page key z finds 0x40 and 0x4a). USB unplugged whenever the bench supply is connected. BENCH_TEST.md section "Power:

bench supply or battery" says how the bench supply stands in for the battery.

7. Motor directions on the bench (BENCH_TEST.md, the steps that need only a loose motor; run sheet 7.6 with the new wire numbers). Mark each motor lead. Disconnect the bench supply.
8. Start plate B. Run sheet 8a: nose ballast.
9. Battery unplugged. Run sheet 8b: battery contacts and BMS, using the new wire numbers.
10. Run sheet 8c, 8d, 8e, 8f in order (pods, tray, close the hull, kick leg and foot). The run sheet closes the hull with the kick and drill leads threaded out through their holes, then solders those motors outside: that is intended. Antenna still on before every loop-plug-in.
11. BENCH_TEST.md for everything that needs the legs fitted, then run sheet section 10 (function test in damp soil). Steps that use the Jetson (`kedge_client.py`, the RC832) wait until sections 7 and 9 are done: from the Jetson use `python3 -m grappstar dart status`.

Dart wiring (new pin map)

From `firmware/dart/WIRING.md`*:*

KEDGE dart wiring v4 (DRV8833 #3 removed)

This replaces Appendix A of `kedge/CLAUDE.md` (the v29 list of 53 wires) and the old 53-wire files in `kedge/pack_v30/`. The new list has 50 wires: `firmware/dart/wiring/wire_cut_list.csv`. Diagram: `firmware/dart/wiring/wiring.svg` / `wiring.png`. Changes only: `firmware/dart/wiring/changes_from_v29.csv`. Regenerate everything with `python3 firmware/dart/tools/wires_v4.py` (lengths from the v30 model, same rule as before).

What changed:

- DRV8833 #3 leaves the dart and goes to the base station (selector drum). **The dart geometry does not change.** v30 is still the default build; the DRV#3 pocket on the keel in the sled simply stays empty.
- DRV#2 channel A now drives **both** drill motors (wired in parallel on AOUT1 / AOUT2). The pair stalls at about 0.7 A, inside one channel's 1.2 A. The right drill's leads stay swapped so the augers counter-rotate.
- DRV#2 channel B now drives the **foot** motor (it was on DRV#3).
- The pin bridge on DRV#2 (AIN1 to BIN1, AIN2 to BIN2) is **gone**. Channels A and B now have their own inputs.

Pin map (firmware v4):

XIAO pin	Goes to	Motor
D0 / D1	DRV#1 AIN1 / AIN2	deploy N20
D2 / D3	DRV#1 BIN1 / BIN2	kick N20
D8 / D9	DRV#2 AIN1 / AIN2	both drill motors
D6 / D7	DRV#2 BIN1 / BIN2	foot N20 (these two pins twitch at reset: harmless on the slow worm)
D4 / D5	INA219 + BNO085 SDA / SCL	I2C
3V3	DRV#1 SLP, DRV#2 SLP, INA219 VCC, BNO085 VIN	logic power

Rules for every step

1. **Battery unplugged.** Pull the power loop plug (the ON switch in the tail) before you touch any wire, and leave it out until the step says so. The first XIAO died from wiring with the battery live.
2. USB unplugged while you solder. Never have USB and the power loop plug in at the same time unless a step says so.
3. Heat shrink on before you solder, not after. Keep the iron off the PLA.
4. After each solder joint, tug the wire gently. It must not move.

Path A: you already wired the v29 harness (convert it)

Do these steps in order. Wire numbers in brackets are the v29 numbers, then the new numbers.

Step A1. Battery unplugged. Pull the power loop plug. Unplug USB. Take the cells out if the hull is open.

Step A2. Battery unplugged. Remove DRV#3. Unsolder and remove these three wires completely:

- [13] DRV#2 VM to DRV#3 VM (red, 22 AWG)
- [17] DRV#2 GND to DRV#3 GND (black, 22 AWG)
- [27] DRV#2 SLP to DRV#3 SLP (red/white, 28 AWG)

Wires [13] and [17] share the DRV#2 VM and GND pads with wires [12] and [16] (new 12 and 15): leave those in place and tug them after. Also unsolder the foot motor wires [52] [53] from DRV#3, and remove the XIAO D6 / D7 wires [42] [43] completely (desolder them at the XIAO end too and throw them away). Lift DRV#3 out of the sled and put it in a labelled bag for the base station. The pocket stays empty.

Step A3. Battery unplugged. Cut the bridge on DRV#2. In v29 the D8 wire went to AIN1 *and* BIN1 (bridged), and the D9 wire to AIN2 *and* BIN2. Desolder the jumper wire or solder blob that joins AIN1 to BIN1 and AIN2 to BIN2 (desoldering braid). The 45 mm D8 and D9 wires stay. After this:

- D8 goes to **AIN1 only** ([40], new wire 37)
- D9 goes to **AIN2 only** ([41], new wire 38)

Check with the meter on beep: AIN1 to BIN1 must NOT beep. AIN2 to BIN2 must NOT beep.

Step A4. Battery unplugged. Move the right drill to channel A. Unsolder the right drill wires from BOUT1 / BOUT2 and solder them to the channel A pads, keeping the swap:

- right drill **black** to **AOUT1** (the same pad as the left drill red) ([50], new wire 47)
- right drill **red** to **AOUT2** (the same pad as the left drill black) ([51], new wire 48)

Two wires in one pad hole, one pad at a time: desolder the left drill red from AOUT1 (braid or pump). Strip 3 mm of it and of the right drill black, twist the two bare ends together, tin them, push them through the hole, solder. Trim the ends flush on the far side. Check with a magnifier that no strand touches the next pad. Tug both. Then the same on AOUT2 with the left drill black and the right drill red.

Step A5. Battery unplugged. Foot inputs to DRV#2 B. Cut two new 45 mm 28 AWG wires (purple and grey):

- XIAO D6 to DRV#2 **BIN1** (purple, new wire 39)
- XIAO D7 to DRV#2 **BIN2** (grey, new wire 40)

Step A6. Battery unplugged. Foot motor to DRV#2 B. The foot wires are now longer, because DRV#2 sits in the board bay behind the cells, not in the sled. Cut two new 330 mm silicone 26 AWG wires (red and black). Run them forward along the top of the hull through the disc notches (with the camera wires), out the skin hole, a 60 mm loop round the kick hub, then down the pole channel:

- DRV#2 **BOUT1** to foot N20 **+ (red)** ([52], new wire 49)
- DRV#2 **BOUT2** to foot N20 **- (black)** ([53], new wire 50)

If the old 255 mm foot wires are already through the pole channel, you may instead solder an 80 mm extension at the DRV#2 end: joint inside the hull, heat shrunk, never in the hub loop.

Step A7. Battery unplugged. Meter checks (meter on beep, nothing powered):

- DRV#2 AOUT1 beeps to the left drill red AND the right drill black.
- DRV#2 AOUT2 beeps to the left drill black AND the right drill red.
- DRV#2 BOUT1 beeps to the foot red, BOUT2 to the foot black.
- No beep between any AOUT pad and any BOUT pad.
- No beep between XIAO D6, D7, D8, D9 (each pair).
- DRV#2 GND beeps to DRV#1 GND, and DRV#2 VM beeps to DRV#1 VM (the pads reheated in step A2).
- Meter on ohms (not beep): AOUT1 to AOUT2 and BOUT1 to BOUT2 read a few ohms or more (that is the motor winding), never 0.0.
- VM to GND: a short chirp is a capacitor charging and is fine; a steady beep is a short: find it before power.

Then go to "Power-on check" below.

Path B: fresh build (the whole harness)

Use `firmware/dart/wiring/wire_cut_list.csv` for colours, gauges and cut lengths. Cut every wire 10 mm longer than listed if you are unsure; trim at the end. Work in this order. Every step starts with the battery unplugged.

Step B1. Battery unplugged. Cells and BMS (wires 1 to 8). Solder the plate tabs before the plates go in their recesses. The series link (wire 2) carries the full current: 22 AWG. Charge socket pins: 1 = B-, 2 = BM, 3 = B+. Wire 8: BMS P+ to power loop socket pin 1 (red, 22 AWG).

Step B2. Battery unplugged. Set the buck before it is wired in (as in the run sheet step 6.2). With the LiPo unplugged, clip or solder the MP1584 IN+ / IN- to the fused side of XT60 lead #1 (red to IN+, black to IN-), nothing on the output, free ends insulated. Meter on beep: IN+ to IN- must not beep steadily (a short). Then plug the LiPo in, put the meter on DC volts across OUT+ and OUT-, turn the small brass screw until it reads **5.04 V**, and unplug the LiPo. Then fit the capacitor across OUT (long leg OUT+, striped leg OUT-; rated 10 V or more).

Step B3. Battery unplugged. Motor rail (wires 9 to 15). Power loop socket pin 2 to INA219 VIN+, the buck IN+ taps the same VIN+ pad, INA219 VIN- to DRV#1 VM, then DRV#1 VM to DRV#2 VM. Grounds: BMS P- to buck IN-, buck IN- to DRV#1 GND, DRV#1 GND to DRV#2 GND. **The chain ends at DRV#2.** There is no DRV#3.

Step B4. Battery unplugged. 5 V (wires 16 to 21). Click the u.FL antenna onto the VTX now: never power a VTX without its antenna. Buck OUT+ to XIAO 5V **through the Schottky diode, stripe towards the XIAO**. Buck OUT- to XIAO GND. Buck OUT to the VTX, VTX pads on to the camera.

Step B5. Battery unplugged. Camera video (wire 22): camera yellow to VTX video in, twisted with the camera ground.

Step B6. Battery unplugged. Find the SLP pad (before wires 23, 24). On each DRV8833 board find the pad named SLP, nSLEEP or EEP. Meter on **ohms** (beep mode only catches a dead short): that pad to VM. **Under 50 ohms means it is tied to the motor supply on this board: do not connect 3V3 to it.** (Pololu boards pull SLP up through a resistor of tens of kilohms, which is fine; on a generic board with a solder bridge from SLP to VCC, open the bridge first.) Then: XIAO 3V3 to DRV#1 SLP (wire 23), DRV#1 SLP to DRV#2 SLP (wire 24).

Step B7. Battery unplugged. Sensors (wires 25 to 32). 3V3 and GND from the XIAO to the INA219, then on to the BNO085. SDA (blue): D4 to INA219 SDA to BNO085 SDA. SCL (green): D5 to INA219 SCL to BNO085 SCL.

Step B8. Battery unplugged. Driver inputs (wires 33 to 40): D0, D1 to DRV#1 AIN1, AIN2 (deploy). D2, D3 to DRV#1 BIN1, BIN2 (kick). D8, D9 to DRV#2 AIN1, AIN2 (drills). D6, D7 to DRV#2 BIN1, BIN2 (foot). **No bridges between pads.**

Step B9. Battery unplugged. Motors, leave them hanging outside the hull for now (wires 41 to 50):

- DRV#1 AOUT1 / AOUT2 to deploy red / black.
- DRV#1 BOUT1 / BOUT2 to kick red / black.
- DRV#2 AOUT1 to left drill red **and** right drill black; DRV#2 AOUT2 to left drill black **and** right drill red (two wires per pad, twisted and tinned together exactly as in step A4).
- DRV#2 BOUT1 / BOUT2 to foot red / black (330 mm silicone).

Step B10. Battery unplugged. Meter checks: the list in step A7, plus 5V to GND and 3V3 to GND: no steady beep. The motors are not connected yet on this path, so AOUT1 to AOUT2 and BOUT1 to BOUT2 read open (`OL`): that is correct.

Optional, if you have them (recommended). Battery unplugged, USB unplugged before you solder them:

- 470 to 1000 uF, 10 V or more, across the MP1584 output (long leg to OUT+, short leg to OUT-), as in the build guide. It keeps the XIAO running through a short power break at impact (the JST plug is the only power path), so the flight state survives. Glue or tape the power loop plug so it cannot shake loose.
- 100 uF, 16 V, across DRV#1 VM and GND (long leg to VM): bulk capacitor for the motor rail (TI asks for 10 uF or more).

Power-on check (both paths)

1. **Battery unplugged.** Motors still outside the hull, legs free to turn, nothing touching the augers.
2. USB only, power loop plug out: flash the firmware (README.md). Serial Monitor at 115200. Type `z`: expect `0x40` and `0x4a` (or `0x4b`).
3. Antenna on the VTX. Unplug USB. Insert the power loop plug. Nothing should move, nothing should get warm. Join WiFi `KEDGE`.
4. Open `http://192.168.4.1`. Status line must show `IDLE`, `INA219 ok`, `BNO085 ok`, rail about 7 to 8.4 V.
5. Go through BENCH_TEST.md: steps 0 to 6 (directions, free currents) on the loose motors now, the rest after assembly (the table at the top of BENCH_TEST.md says which).

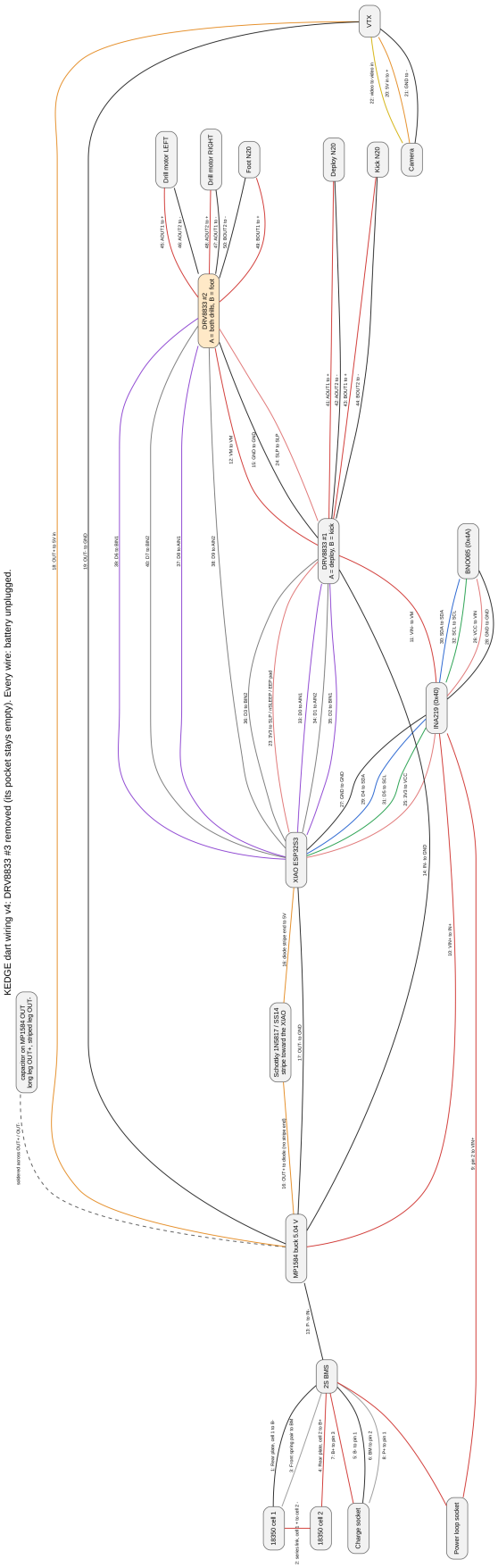
Drill direction check (BENCH_TEST.md step 4): hold "drills in" (key 3) and look at each auger from its motor end, looking toward the tip. The left auger (right-hand thread) must turn **clockwise**, the right auger (left-hand thread) **anticlockwise**: then both screw into the ground.

- Left auger wrong: battery unplugged, swap the two thin input wires at DRV#2 AIN1 and AIN2 (D8 and D9). This reverses both drills. Check again.
- Then if the right auger is wrong: battery unplugged, swap the right drill's own two wires on AOUT1 / AOUT2.
- Swapping a motor's two input wires at its driver is the easy way to reverse any motor.

Totals (cut from the reel)

Wire	mm
22 AWG red	520

Wire	mm
22 AWG black	360
24 AWG orange / black	160 / 160
26 AWG silicone red / black (foot pair)	330 / 330
26 AWG red / black (other motors)	760 / 855
26 AWG white / orange	180 / 95
28 AWG red/white, black, blue, green, purple, grey, yellow, orange	240, 225, 120, 120, 200, 200, 105, 105



Dart wiring v4 (numbers match the new cut list). To zoom in, open firmware/dart/wiring/wiring.svg on a screen.

Run sheet wire numbers

When the run sheet names a wire number, use the new number from this table.

Run sheet wire	New wire	Run sheet says
1	1	Rear plate, cell 1 (-) to BMS B-
2	2	Front spring, cell 1 (+) to Front spring, cell 2 (-)
3	3	Front spring pair (cell 1 + / cell 2 -) to BMS BM
4	4	Rear plate, cell 2 (+) to BMS B+
5	5	BMS B- to Charge socket pin 1 (tail window)
6	6	BMS BM to Charge socket pin 2
7	7	BMS B+ to Charge socket pin 3
8	8	BMS P+ to Power loop socket pin 1 (tail window)
9	9	Power loop socket pin 2 to INA219 VIN+
10	10	INA219 VIN+ (same pad) to MP1584 IN+
11	11	INA219 VIN- to DRV#1 VM
12	12	DRV#1 VM to DRV#2 VM
13	REMOVED (no DRV#3)	DRV#2 VM to DRV#3 VM
14	13	BMS P- to MP1584 IN-
15	14	MP1584 IN- (same pad) to DRV#1 GND
16	15	DRV#1 GND to DRV#2 GND
17	REMOVED (no DRV#3)	DRV#2 GND to DRV#3 GND
18	16	MP1584 OUT+ (5.04 V) to XIAO 5V via a Schottky diode (stripe toward the XIAO)
19	17	MP1584 OUT- to XIAO GND
20	18	MP1584 OUT+ to VTX 5V in
21	19	MP1584 OUT- to VTX GND
22	20	VTX 5V in (same pad) to Camera + (red)
23	21	VTX GND (same pad) to Camera - (black)
24	22	Camera video (yellow) to VTX video in
25	23	XIAO 3V3 to DRV#1 SLP / nSLEEP / EEP pad (NOT VCC or VIN)
26	24	DRV#1 SLP to DRV#2 SLP
27	REMOVED (no DRV#3)	DRV#2 SLP to DRV#3 SLP
28	25	XIAO 3V3 to INA219 VCC
29	26	INA219 VCC to BNO085 VIN
30	27	XIAO GND to INA219 GND
31	28	INA219 GND to BNO085 GND
32	29	XIAO D4 to INA219 SDA
33	30	INA219 SDA to BNO085 SDA
34	31	XIAO D5 to INA219 SCL
35	32	INA219 SCL to BNO085 SCL
36	33	XIAO D0 to DRV#1 AIN1
37	34	XIAO D1 to DRV#1 AIN2
38	35	XIAO D2 to DRV#1 BIN1
39	36	XIAO D3 to DRV#1 BIN2
40	37	XIAO D8 to DRV#2 AIN1 + BIN1 (bridge the two pads)
41	38	XIAO D9 to DRV#2 AIN2 + BIN2 (bridge the two pads)
42	39	XIAO D6 to DRV#3 AIN1
43	40	XIAO D7 to DRV#3 AIN2
44	41	DRV#1 AOUT1 to Deploy N20 + (red)
45	42	DRV#1 AOUT2 to Deploy N20 - (black)
46	43	DRV#1 BOUT1 to Kick N20 + (red)
47	44	DRV#1 BOUT2 to Kick N20 - (black)
48	45	DRV#2 AOUT1 to Drill motor LEFT + (red)
49	46	DRV#2 AOUT2 to Drill motor LEFT - (black)

Run sheet wire	New wire	Run sheet says
50	47	DRV#2 BOUT1 to Drill motor RIGHT - (black) SWAPPED
51	48	DRV#2 BOUT2 to Drill motor RIGHT + (red) SWAPPED
52	49	DRV#3 AOUT1 to Foot N20 + (red)
53	50	DRV#3 AOUT2 to Foot N20 - (black)

New dart wire cut list (replaces run sheet section 9)

From `firmware/dart/wiring/wire_cut_list.csv`*:*

#	from	to	colour	gauge	cut length mm	note	v29 #
1	Rear plate, cell 1 (-)	BMS B-	black	22 AWG	45	solder to the plate tab before it goes in its recess	1
2	Front spring, cell 1 (+)	Front spring, cell 2 (-)	red	22 AWG	45	the series link: carries the full battery current; glue both springs in the front recess	2
3	Front spring pair (cell 1 + / cell 2 -)	BMS BM	white	26 AWG	55	midpoint sense	3
4	Rear plate, cell 2 (+)	BMS B+	red	22 AWG	45		4
5	BMS B-	Charge socket pin 1 (tail window)	black	26 AWG	80	JST-XH 3-pin, balance charge	5
6	BMS BM	Charge socket pin 2	white	26 AWG	125		6
7	BMS B+	Charge socket pin 3	red	26 AWG	80		7
8	BMS P+	Power loop socket pin 1 (tail window)	red	22 AWG	80	JST-XH 2-pin; the shorting plug is the ON switch	8
9	Power loop socket pin 2	INA219 VIN+	red	22 AWG	95	main + after the loop	9
10	INA219 VIN+ (same pad)	MP1584 IN+	red	22 AWG	140	buck taps before the shunt, so INA219 reads motor current only	10
11	INA219 VIN-	DRV#1 VM	red	22 AWG	50	motor rail 6.0-8.4 V	11
12	DRV#1 VM	DRV#2 VM	red	22 AWG	65	daisy chain ends here (no DRV#3)	12
13	BMS P-	MP1584 IN-	black	22 AWG	130	main ground	14
14	MP1584 IN- (same pad)	DRV#1 GND	black	22 AWG	120		15
15	DRV#1 GND	DRV#2 GND	black	22 AWG	65		16
16	MP1584 OUT+ (5.04 V)	XIAO 5V via a Schottky diode (stripe toward the XIAO)	orange	24 AWG	160	set and check 5.04 V BEFORE connecting; 1N5817 or SS14 in heat shrink	18
17	MP1584 OUT-	XIAO GND	black	24 AWG	160		19
18	MP1584 OUT+	VTX 5V in	orange	26 AWG	95		20
19	MP1584 OUT-	VTX GND	black	26 AWG	95		21
20	VTX 5V in (same pad)	Camera + (red)	orange	28 AWG	105	runs forward along the top through the disc notches	22
21	VTX GND (same pad)	Camera - (black)	black	28 AWG	105		23
22	Camera video (yellow)	VTX video in	yellow	28 AWG	105	twist with the camera ground	24
23	XIAO 3V3	DRV#1 SLP / nSLEEP / EEP pad (NOT VCC or VIN)	red/white	28 AWG	55	wakes the driver; meter check first (WIRING.md step B6)	25

#	from	to	colour	gauge	cut length mm	note	v29 #
24	DRV#1 SLP	DRV#2 SLP	red/white	28 AWG	65		26
25	XIAO 3V3	INA219 VCC	red/white	28 AWG	60		28
26	INA219 VCC	BNO085 VIN	red/white	28 AWG	60		29
27	XIAO GND	INA219 GND	black	28 AWG	60		30
28	INA219 GND	BNO085 GND	black	28 AWG	60		31
29	XIAO D4	INA219 SDA	blue	28 AWG	60	I2C SDA	32
30	INA219 SDA	BNO085 SDA	blue	28 AWG	60		33
31	XIAO D5	INA219 SCL	green	28 AWG	60	I2C SCL	34
32	INA219 SCL	BNO085 SCL	green	28 AWG	60		35
33	XIAO D0	DRV#1 AIN1	purple	28 AWG	55	deploy	36
34	XIAO D1	DRV#1 AIN2	grey	28 AWG	55	deploy	37
35	XIAO D2	DRV#1 BIN1	purple	28 AWG	55	kick	38
36	XIAO D3	DRV#1 BIN2	grey	28 AWG	55	kick	39
37	XIAO D8	DRV#2 AIN1 (only AIN1: NO bridge to BIN1)	purple	28 AWG	45	drills	40
38	XIAO D9	DRV#2 AIN2 (only AIN2: NO bridge to BIN2)	grey	28 AWG	45	drills	41
39	XIAO D6	DRV#2 BIN1	purple	28 AWG	45	foot (slow worm, harmless if it twitches at reset)	42
40	XIAO D7	DRV#2 BIN2	grey	28 AWG	45	foot	43
41	DRV#1 AOUT1	Deploy N20 + (red)	red	26 AWG	165	motor sits in the sled cradle	44
42	DRV#1 AOUT2	Deploy N20 - (black)	black	26 AWG	165		45
43	DRV#1 BOUT1	Kick N20 + (red)	red	26 AWG	150	out through the hole in the saddle floor	46
44	DRV#1 BOUT2	Kick N20 - (black)	black	26 AWG	150		47
45	DRV#2 AOUT1	Drill motor LEFT + (red)	red	26 AWG	195	out the left flank hole, 30 mm loop for the pod swing	48
46	DRV#2 AOUT2	Drill motor LEFT - (black)	black	26 AWG	195		49
47	DRV#2 AOUT1 (same pad as the LEFT red)	Drill motor RIGHT - (black) SWAPPED	black	26 AWG	170	reversed on purpose: the pair counter-rotates	50
48	DRV#2 AOUT2 (same pad as the LEFT black)	Drill motor RIGHT + (red) SWAPPED	red	26 AWG	170		51
49	DRV#2 BOUT1	Foot N20 + (red)	red	26 AWG	330	forward along the top through the disc notches, out the skin hole, 60 mm loop round the hub, through the pole channel	52
50	DRV#2 BOUT2	Foot N20 - (black)	black	26 AWG	330	silicone wire, it flexes every deploy	53

Dart bench tests

From `firmware/dart/BENCH_TEST.md`*:*

KEDGE dart bench test

Goal: every motor turns the right way and stops on its own, both sensors work, the current thresholds are real numbers, and one full anchor + release runs on soil. About 1 hour.

When to run each part:

Steps	When
0 to 2, and the direction + free current parts of 3 to 6	loose motors on the bench (bench harness, before routing into the hull)
stall at the end stops in 3 to 6, then 7 to 10	assembled: hull closed, legs and pods fitted (run sheet 8e and 8f done)
11, and 12 step 4 (camera picture)	after the Jetson, RC832 and capture card are set up (rover section)

You need: charged 18350 cells (XTAR MC2), multimeter, phone or laptop on WiFi `KEDGE`, a stopwatch, a tray or bucket of damp soil at least 10 cm deep, a folded towel, and this sheet to write numbers on.

Directions used below (firmware convention, +1 in `/status motors`):

Key	Motor	Must do
1	deploy	pods swing **down**
2	kick	leg swings **out**, over the nose
3	drills	augers turn to **screw in**: left clockwise, right anticlockwise, seen from the motor end, looking toward the tip
4	foot	foot auger turns to **screw in**: clockwise seen from the motor end, looking toward the tip (right-hand foot auger)

If a motor turns the wrong way: **battery unplugged**, swap that motor's two thin input wires at its driver (deploy: DRV#1 AIN1/AIN2, kick: DRV#1 BIN1/BIN2, drills: DRV#2 AIN1/AIN2, foot: DRV#2 BIN1/BIN2). Then check again.

Power: bench supply or battery

The loose-motor steps (0 to 2, and the direction and free-current parts of 3 to 6) usually run before the cells and the power loop socket are fitted. Then power comes from the **bench supply**: the spare MP1584 (#2) set to **8.0 V**, fed by the 3S LiPo on XT60 lead #1 (10 A fuse), as in run sheet step 7.5.

- Connect it with the LiPo unplugged: bench MP1584 OUT+ to the dart's **INA219 VIN+** pad, bench MP1584 OUT- to the dart's **main ground** (the dart MP1584 IN- pad). Use small clip leads, or two temporary wires soldered to those pads. Check the polarity twice: red to VIN+, black to ground.
- "Insert the power loop plug" below then means: plug the LiPo into the bench MP1584. "Pull the plug" means: unplug the LiPo. "The battery voltage" means: the bench supply voltage, metered at INA219 VIN+ to ground (about 8.0 V).
- **USB out** while the bench supply is connected, exactly as with the battery.
- The assembled steps (end-stop stalls in 3 to 6, then 7 to 10) use the real cells and the power loop plug.
- **Before the cells go in: LiPo unplugged, remove the bench leads** (unclip, or desolder the two temporary wires from INA219 VIN+ and the MP1584 IN- pad). Nothing of the bench supply may stay inside the hull.

Step 0. Set up

1. **Battery unplugged.** Wiring done and meter-checked (WIRING.md step A7 or B10). Firmware flashed (README.md).
2. Loose motors on the bench, or the assembled dart (hull closed, run sheet 8e and 8f done) with the legs free to swing. Nothing near the augers. Fingers clear.
3. The u.FL antenna must be clicked onto the VTX before any power (a VTX without its antenna can burn out). Unplug USB. Then insert the power loop plug. Nothing moves, nothing gets warm in 30 s. If anything heats: pull the plug.
4. Join `KEDGE`, open `http://192.168.4.1`. The status line must show `IDLE`, `INA219 ok`, `BNO085 ok`.

Step 1. Power and current sensor

1. Meter the supply: the battery at the power loop socket, or the bench supply at INA219 VIN+ to ground. Compare with `rail` on the page: within 0.15 V (the INA219 reads after its shunt, so `rail` can be a few tens of mV lower). Write both down.
2. With nothing running, `current` reads 0 to 5 mA (the INA219 sees the motor rail only). When a motor runs (step 3) the current must read **positive**. A negative current means INA219 VIN+ and VIN- are swapped: the firmware then treats the sensor as failed (`INA219 MISSING`). Battery unplugged and swap them.
3. Press `i` on the page: it prints rail, current, and the peak since the last `i`.

Pass: INA219 ok, rail agrees with the meter. If `INA219 MISSING`: battery unplugged, check D4/D5 and 3V3/GND to the INA219. Then, with the power off (plug out or LiPo unplugged), plug in USB: Serial Monitor `z` must list 0x40.

Step 2. BNO085 and attitude

1. Press b: roll, pitch and |a| about 1.00 g with the dart still.
2. Status `IDLE`, no key pressed, hands off the legs. Tilt the dart nose up: roll or pitch changes by about 90 (which one depends on how the chip sits; step 7 maps the axes). Roll it on its side: the other one changes by about 90.
3. If `BNO085 MISSING`: battery unplugged, check wires. Then, plug still out, plug in USB: Serial `z` must list 0x4a (or 0x4b). If the Serial Monitor stays blank: right port, 115200 baud, tap R on the XIAO.

If a motor in steps 3 to 6 does not stop by itself at the end within 8 s (the hold cap), the default stall threshold is above its real stall current. So press `i` once BEFORE each hold (it resets the peak), then hold, release, and press `i` again: the peak it prints is the stall current.

Also check once, battery unplugged, meter on ohms: each DRV8833 SLP pad to VM. Under 50 ohms means the pad is tied to the motor supply on that board: do not connect 3V3 to it (WIRING step B6).

Step 3. Deploy (DRV#1 A)

1. Hold `deploy down` (key 1) for 1 s. Pods swing **down**. Wrong way: fix as above.
2. Hold it until the pods reach the end. They stop by themselves (event `stalled: deploy at ... mA, stopped`). Write down that mA: **deploy stall =** ____ **mA**. While moving freely, `current` reads **deploy free =** ____ **mA**.
3. Hold `deploy up` (`!`) until they stow and stop by themselves.

Step 4. Drills (DRV#2 A, both motors)

1. Pods down first (step 3). Hold `drills in` (3) for 2 s. Check the directions in the table (left clockwise, right anticlockwise, from motor end to tip). If both turn the same way: WIRING.md "Drill direction check".
2. **Drill pair free =** ____ **mA** (expect 60 to 70). Press a printed stick or a pencil end lightly against one auger flight (never a cloth or fingers: cloth winds in and pulls fingers with it): current rises. Do not stall both for long (two sticks, never hands). Stalling both reads **drill pair stall =** ____ **mA** (expect about 700 at 8 V).
3. Hold `drills out` (#): they reverse.

Step 5. Kick (DRV#1 B)

1. Hold `kick out` (2): the leg swings **out over the nose**. Wrong way: fix as above.
2. Assembled (kick leg fitted). **Battery unplugged.** Lay the dart on its back with the kick free to swing in air, legs stowed. Plug in (power on with the legs stowed: the firmware counts the kick's travel from here). Speed 255. Hold `kick out` (2) until the leg stops by itself at its outer end stop (a hold is capped at 8 s: if it is not there yet, release and hold again). While it swings, read `current` on the status line: **kick free =** ____ **mA**. When it stops, the event line says `kick travel from stowed NNNN ms (at 8.0 V)`: **kick swing in air =** ____ **ms**. This number goes into `kickSwingMs` (step 10). The firmware measures it, so no stopwatch is needed. It must be well under 15000 (the auto limit).
3. Hold `kick back` (@) until it stows and stops by itself: **kick stall =** ____ **mA** (the `stalled: kick at ...` event). Repeat steps 2 and 3 twice more and use the middle value of the three swings.
4. STOP, then **battery unplugged**. Turn the dart belly down on a flat board, legs stowed. Plug in. Hold 2 until the housing rests on the board and the kick stops by itself: the event gives **kick to ground =** ____ **ms**. It must be under 0.8 x the swing in air, or `/auto` can call flat ground `NO_GROUND` (tell whoever owns the firmware). Bring it back with `@`, STOP, battery unplugged.

Step 6. Foot (DRV#2 B)

1. Hold `foot in` (4): the foot auger turns the way that screws in. Wrong way: fix as above.
2. **Foot free =** ____ **mA** (expect about 40). Brake it by holding the coupler with pliers, or press a printed stick against the auger (never a cloth or fingers): **foot stall =** ____ **mA** (expect 500 to 600).
3. Hold `foot out` ($): it reverses.

Step 7. BNO085 axes (for the Jetson flight analysis)

This step needs power for the IMU while you turn the dart by hand: status `IDLE` (not armed), nobody else on the page or the Jetson, no key pressed, hands off the legs and augers.

1. Lay the dart flat, belly down. Open `http://192.168.4.1/flightlog` and look at the last sample: `ax, ay, az`. The axis reading about +9.8 points **up**.
2. Stand the dart on its tail, nose up: the axis reading about +9.8 points to the **nose**.

3. Lay it on its right side (kick side up): the axis reading +9.8 points **left**.
4. Write: **nose = ___ (e.g. +x), left = ___, up = ___**. The Jetson flight analysis needs this mapping.

Step 8. Flight log (safe, log only)

1. Open `http://192.168.4.1/arm?on=1&auto=0` (arm, log only: no motors will run). Status shows `ARMED`.
2. Hold the dart over the towel on the floor. Jerk it sharply upwards and let go about 40 cm above the towel.
3. Status should go `FLIGHT` then `LANDED` (or stay `ARMED` if the jerk was under 4 g: jerk harder).
4. Wait until the status shows `LANDED`, then 4 s more (if the towel landing ends as `timeout` that is about 19 s after the jerk, and the log is already frozen by then; `/flightlog` is refused while armed or in flight). Open `http://192.168.4.1/flightlog`: `frozen: true`, `t_launch_ms` and `t_land_ms` set, a few hundred samples. Save it. The towel drop usually reads `landing: "timeout"` or `"still"` (soft), which is correct.
5. Open `http://192.168.4.1/stop` to disarm.

Safe to touch, after any arm or auto test: press STOP, check the status is not `ARMED`, `FLIGHT`, `ANCHORING` or `RELEASING` (`IDLE`, `LANDED` or `ANCHORED` is fine), pull the power loop plug (or unplug the bench LiPo), and only then pick the dart up or touch a leg. An armed dart can start the auto sequence by itself after a knock.

Step 9. Anchor and release on soil (assembled, hull closed)

1. **Battery unplugged.** Legs stowed. Lay the dart on the damp soil, belly down, nose clear of the tray wall. Plug in.
2. Page: press `AUTO anchor`. Watch the status: kick swing, foot touches, kick + foot planting, foot finishing, deploy + drills, drills finishing, `ANCHORED`. Write the time of each event from the event lines on the web page (not the Serial Monitor: no USB while the battery is in). Note the working currents for `loadMa`.
3. Pull the tether gently by hand: the dart should hold.
4. Press `RELEASE`: drills out, pods up, foot out, kick back, `released`. The legs end stowed. Before lifting the dart out of the soil: STOP, status `IDLE`, plug out (the "safe to touch" rule in step 8).
5. If it stops with `FAULT`: write down the fault and the last event, press STOP, and fix the threshold (step 10).
6. Foot current drop (for the kick stall check). Press `AUTO anchor` again and wait for `ANCHORED`. Hands clear. Hold `foot in` (4) alone for 1 s and read `current` on the status line while holding (not `i`: its peak includes the start-up surge): **foot alone in soil = ____ mA**. Then hold `kick out` (2) and 4 together for 1 s: **kick + foot = ____ mA**. If (foot alone) is more than 80 mA below (kick + foot minus the kick's working current), the kick's end stop can be faked while planting: tell whoever owns the firmware. STOP after each.
7. No ground check. Press `RELEASE` and wait for `released` (legs stowed). STOP, **battery unplugged**. Lay the dart on its back on the bench, nothing within reach of the kick or the augers. Plug in. Press `AUTO anchor`: the kick swings through air to its end stop and the dart reports `FAULT NO_GROUND`; foot, pods and drills must not turn. Then press `RELEASE` to bring the kick back. **Release spins the drills (3 s), swings the pods and turns the foot too, with the dart on its back: hands and cloth well clear.** Then STOP, battery unplugged.

Step 10. Write the real thresholds

In `firmware/dart/kedge_dart/kedge_logic.h`, `struct Tuning`, set (all at full speed 255, battery fully charged). The four-motor arrays are in the order {deploy, kick, drills, foot}: `stallMa[0]` is deploy, `stallMa[1]` kick, `stallMa[2]` drills, `stallMa[3]` foot. `impactG` is the one exception: it is in `struct Flight`, not `Tuning`.

Setting	Rule	Default
`stallMa[DEPLOY]`	80 % of deploy stall	150
`stallMa[KICK]`	80 % of kick stall	150
`stallMa[DRILLS]`	85 % of drill pair stall	600
`stallMa[FOOT]`	80 % of foot stall	450
`kickContactMa`	about 2 x kick free, below kick stall / 2	60
`freeMa[...]`	measured free currents (steps 3 to 6)	30, 30, 70, 45
`loadMa[...]`	typical working current in soil (step 9)	40, 70, 420, 130
`podsTravelMs`	pods travel stowed to down at full speed (step 3: hold 1 from stowed until it stalls, time it, or use the default)	1400
`kickSwingMs`	kick swing in air from step 5 (the firmware's own `kick travel` number), in ms	4300
`impactG`	under the `peak_g` of a real landing (step 8 and the first shots), over 2 g	3.5
`kickMaxMs`	2 x kick swing time, at least 8000	15000

The PC tests use a motor model with the v29 stall currents (deploy and kick 185 mA, drills 700, foot 550: `Plant` in `test/test_logic.cpp`, and the `kStall`, `dStall`, `drStall`, `fStall` ranges in `test/fuzz_logic.cpp`). If a

stall you measured is higher, raise the model's number to your measured stall first, or `run_tests.sh` fails (for example `stallMa[2]` over 700 fails "drill stall threshold below it"). The drill pair stall must stay under 1200 mA (one DRV8833 channel).

Re-run `test/run_tests.sh`, upload, repeat steps 3, 5 and 6 and step 9 once to see the stops still work.

Step 11. Jetson link (after the Jetson is set up)

From the Jetson on `KEDGE`: `python3 -m grappstar dart status` (the jetson package). Phase `IDLE`, fields present. Or simply `curl http://192.168.4.1/status`.

Step 12. Re-check the wiring inside the closed hull

The run sheet (8e) closes the hull before the legs go on. This step is a re-check after all the tests.

1. **Battery unplugged.**
2. Check the routing: drill wires with a 30 mm loop at each flank hole for the pod swing; the foot pair with a 60 mm loop round the kick hub; nothing crossing a hinge; nothing pinched by the hull halves. The DRV#3 pocket stays empty.
3. Insert the plug and repeat steps 3, 5 and 6 (one swing each way) to make sure no wire catches.
4. Camera (after the RC832 is set up): switch on the RC832 and check the picture.

Bring back: the numbers from steps 1 to 7 and 9. They go into `Tuning` and the REPORT.

7. Electronics and wiring (base and rover)

Battery unplugged for every step in this section. Two documents, used in this order:

1. **Base electronics bay, on the bench** (7.2, `firmware/base/WIRING.md` section 8, Parts A to D): set the buck, wire the ESP32, BTS7960, DRV8833, sensors and contacts, run the meter checks, first power with the ESP32 on the Mac's USB. Then `firmware/base/BENCH_TEST.md`.
2. **The whole rover** (7.1, the system page): the XT60 Y harness, the fuses, the Jetson and video, and the first power-on of everything together (its part 4). Do this once the base is assembled and on the rover (order of work step 9), and again after any rewiring.

One-page printable pin sheets for the bench: `docs/build/Wiring_Cheat_Sheet_Base.pdf` (base ESP32-S3) and `docs/build/Wiring_Cheat_Sheet_Dart.pdf` (dart XIAO).

Before the buck output goes to the DRV8833 VM (WIRING.md Part B step 11), measure the buck on its own again: 9.0 V with no load, then connect it (battery unplugged).

When you reach WIRING.md Part B step 12 (buck input), put a 2 A inline fuse in the IN+ wire right at the B+ screw (system page, run P4).

Where the two differ, the system page adds the rover parts (Y harness, 2 A and 1 A tap fuses, Jetson, RC832); the ESP32 pins always come from WIRING.md.

7.1 The whole system

From `docs/build/SYSTEM_ELECTRICAL.md`*:*

GrappStar system electrical (rover, base and dart)

Diagram: `system_electrical.svg` / `system_electrical.png` (source `system_electrical.dot`). The ESP32 pin table is owned by `firmware/base/WIRING.md` and `firmware/base/grappstar_base/pins.h`; this page shows how the whole rover is powered and connected. If this page and WIRING.md ever disagree on a pin, WIRING.md wins. Decision D9: the DRV8833 #3 SLP pin is NOT wired (GPIO21 is now the SW_HOME contact, decision D27).

1. What is on the rover

Block	Part	Powered from
Battery	Zeee 3S LiPo 5200 mAh, XT60 (9.9 V empty in use, 11.1 V nominal, 12.6 V full)	
Wiper branch	XT60 lead #2 with the 20 A fuse, BTS7960 module, 12 V wiper motor (red and black only; never wire the green park wire, it may switch +12 V: tape it off)	LiPo
Drum branch	MP1584 #2 buck set to 9.0 V, DRV8833 #3 (the one freed from the dart), selector drum N20 worm	BTS7960 B+ / B- terminals, after the 20 A fuse
Base brain	ESP32-S3 DevKitC-1-N8	Jetson USB (the port marked USB, not UART)
Sensors on the ESP32	Adafruit AS5600 (STEMMA QT, I2C 0x36), printed contacts (M3 screw head on an M3 nut), optional HX711 + 100 kg load cell, battery divider	ESP32 3V3
Jetson	Orin Nano Super dev kit	XT60 lead #1 (10 A fuse) to a 5.5 x 2.5 mm barrel pigtail
Video in	Lockmaner RC832 5.8 GHz receiver, AV (yellow RCA) out to the DIGITNOW USB capture, which plugs into the Jetson	XT60 lead #1 branch (12 V class input)
Rover camera	Razer Kiyo X	Jetson USB
Dart	KEDGE: XIAO ESP32S3, 2 x DRV8833, INA219, BNO085, Caddx Ant, Zeus VTX	its own 2S 18350 pack. No wire to the rover: the tether is Dyneema and carries no power

2. Power tree

```
3S LiPo (XT60)
  -> XT60 Y harness (one male in, two female out; made from your spare XT60s, 14 AWG)
       -> lead #2, 20 A fuse, 14 AWG -> BTS7960 B+ / B-  -> wiper motor (BTS7960 M+ / M-, 14 AWG)
                                     \-> 2 A inline fuse, 22 AWG from the B+ / B- screws -> MP1584 #2 IN (
  9.0 V out)
                                                  -> DRV8833 #3 VM / GND -> drum N20 (AOUT1 / AOUT2)
       -> lead #1, 10 A fuse, 18 AWG -> barrel pigtail -> Jetson DC jack
                                     \-> 1 A inline fuse, 22 AWG -> RC832 power in
Jetson USB -> ESP32-S3 DevKitC (5 V from USB; the board makes its own 3V3)
                ESP32 3V3 -> BTS7960 VCC, AS5600, HX711 (DRV8833 #3 gets only AIN1, AIN2, GND)
Jetson USB -> DIGITNOW capture <- RC832 AV out
Jetson USB -> Kiyo X
```

Why it is laid out this way:

- The DRV8833 motor supply (VM) is 10.8 V maximum. The LiPo is 12.6 V full, so the drum N20 runs from MP1584 #2 set to 9.0 V. MP1584 #2 is the spare from the dart build (it was the 8.0 V bench supply there, 8.0 to 8.12 V): reset it to 9.0 V after the dart bench test.
- The buck taps the BTS7960 screw terminals, so both motor branches sit behind the 20 A fuse and the Jetson branch stays separate. A wiper stall that blows the 20 A fuse does not reboot the Jetson.
- The thin taps (MP1584 input, RC832) each get their own small inline fuse (2 A and 1 A) right at the tap. A 22 AWG wire behind only the 20 A or 10 A fuse would burn before that fuse blew if it shorted.
- The ESP32 takes power only from USB. It never connects to the LiPo directly. Its ground meets battery minus through the BTS7960 and DRV8833 GND wires (needed so the logic signals have a reference). There is a second ground path through USB, the Jetson and its barrel lead, so if the BTS7960 B- screw came loose the wiper current would try to return through the thin GND jumper and the USB cable. Keep the B- screw tight (tug test) and never run the wiper with B- disconnected.
- The wiper and the Jetson share the pack. A wiper stall (about 18 A) makes the pack voltage sag; with a low pack that sag can brown out the Jetson. Charge the pack when it reads 3.6 V per cell at rest (10.8 V).

3. Wires and connectors

Run	From	To	Gauge	Connector	Note
P1	LiPo	Y harness	14 AWG	XT60	Make the Y from spare XT60s; solder, heat shrink each joint. This short run is the only unfused part: keep the Y's legs under 10 cm so both fuses sit right at the split, sleeve it, and strap it so it cannot rub
P2	Y harness	BTS7960 B+ / B-	14 AWG	XT60 lead #2 (20 A fuse), screw terminals at the BTS7960	Red to B+, black to B-. 10 A blew on wiper inrush before: keep 20 A
P3	BTS7960 M+ / M-	wiper motor	14 AWG	screw terminals; spade or solder at the motor	Red wire (+) and black (-) of the rebuilt 2-wire motor
P4	BTS7960 B+ / B-	MP1584 #2 IN+ / IN-	22 AWG	under the same screws, 2 A inline fuse in the + wire right at the screw	Set 9.0 V with nothing on the output first
P5	MP1584 #2 OUT+ / OUT-	DRV8833 #3 VM / GND	22 AWG	solder	Check 9.0 V on the meter before you connect
P6	DRV8833 #3 AOUT1 / AOUT2	drum N20	26 AWG	solder or JST	Swap the pair if the drum turns the wrong way (see calibration)
P7	Y harness	Jetson DC jack	18 AWG	XT60 lead #1 (10 A fuse), 5.5 x 2.5 mm barrel plug (same as the plug on the Jetson supply), centre positive	Check polarity with the meter before the first plug-in
P8	lead #1 (after the fuse)	RC832 power in	22 AWG	the RC832 power lead, 1 A inline fuse in the + wire at the tap	Read the RC832 label for its input range first (see MEASURE_LIST)
S1	Jetson USB-A	ESP32-S3 USB port (native, marked USB)		USB-C or micro-USB cable to suit the board	Data cable, not charge-only
S2	Jetson USB-A	DIGITNOW capture		its own USB plug	
S3	RC832 AV out	DIGITNOW video in		yellow RCA	Video only; audio not used
S4	Jetson USB-A	Kiyo X		its own USB plug	
L1	ESP32 pins	BTS7960 RPWM, LPWM, R_EN, L_EN, IS, VCC, GND	26 to 28 AWG	Dupont jumpers	Pins in firmware/base/WIRING.md
L2	ESP32 pins	DRV8833 #3 AIN1, AIN2, GND	26 to 28 AWG	Dupont	SLP / nSLEEP is not wired: the board pulls it high, and its inputs are pulled low so the outputs are off until the ESP32 drives them
L3	ESP32 SDA / SCL / 3V3 / GND	AS5600		STEMMA QT to Dupont cable (400 mm or longer)	Magnet 1 to 2 mm from the chip
L4	ESP32 pins	each printed contact (names: BASE_SPEC section 4; GPIO for each: WIRING.md)	26 AWG twisted pair	Dupont at the ESP32, M3 ring or solder at the contact	One wire to the pin, the other to GND

Run	From	To	Gauge	Connector	Note
L5	ESP32 pins	HX711 DT, SCK, VCC, GND (optional)	26 AWG	Dupont	Load cell 4 wires into HX711 E+, E-, A+, A-
L6	BTS7960 B+	ESP32 battery pin through the 100 k / 22 k divider	26 AWG	solder the 100 k resistor right at the B+ screw (in heat shrink), so the long wire only carries the divided voltage; the 22 k goes from the pin to GND at the ESP32	12.6 V reads 2.27 V at the pin

Gauges assume runs under 0.5 m. 14 AWG carries the 18 A wiper stall; 18 AWG is fine for the Jetson (about 2.5 A at 10 V); 22 AWG is fine for the drum branch (N20 stall about 0.2 A) and the RC832 (about 0.2 A).

4. First power-on sequence (do it in this order, once, after any rewiring)

Before you start: the pull-down and divider resistors are fitted as in firmware/base/WIRING.md (10 k from R_EN to GND, 10 k from L_EN to GND, the IS divider, the battery divider). Both motors are disconnected from their drivers for this first power-on, and the DRV8833 VM wire (P5) is soldered to the DRV8833 but NOT yet to MP1584 #2 OUT (tape its free end). If the base electronics already passed firmware/base/WIRING.md Part D on the bench (the normal order of work), the buck is already set and the motors may stay connected (nothing moves without a command): in step 5 only check the buck reads 9.0 V (do not turn the screw), and skip steps 6 and

10. If the Jetson is still on its wall supply: in step 7 skip the barrel plug (plug only the ESP32 USB into the

Jetson, battery out), and in step 8 expect only the RC832 and ESP32 LEDs. Multimeter ready.

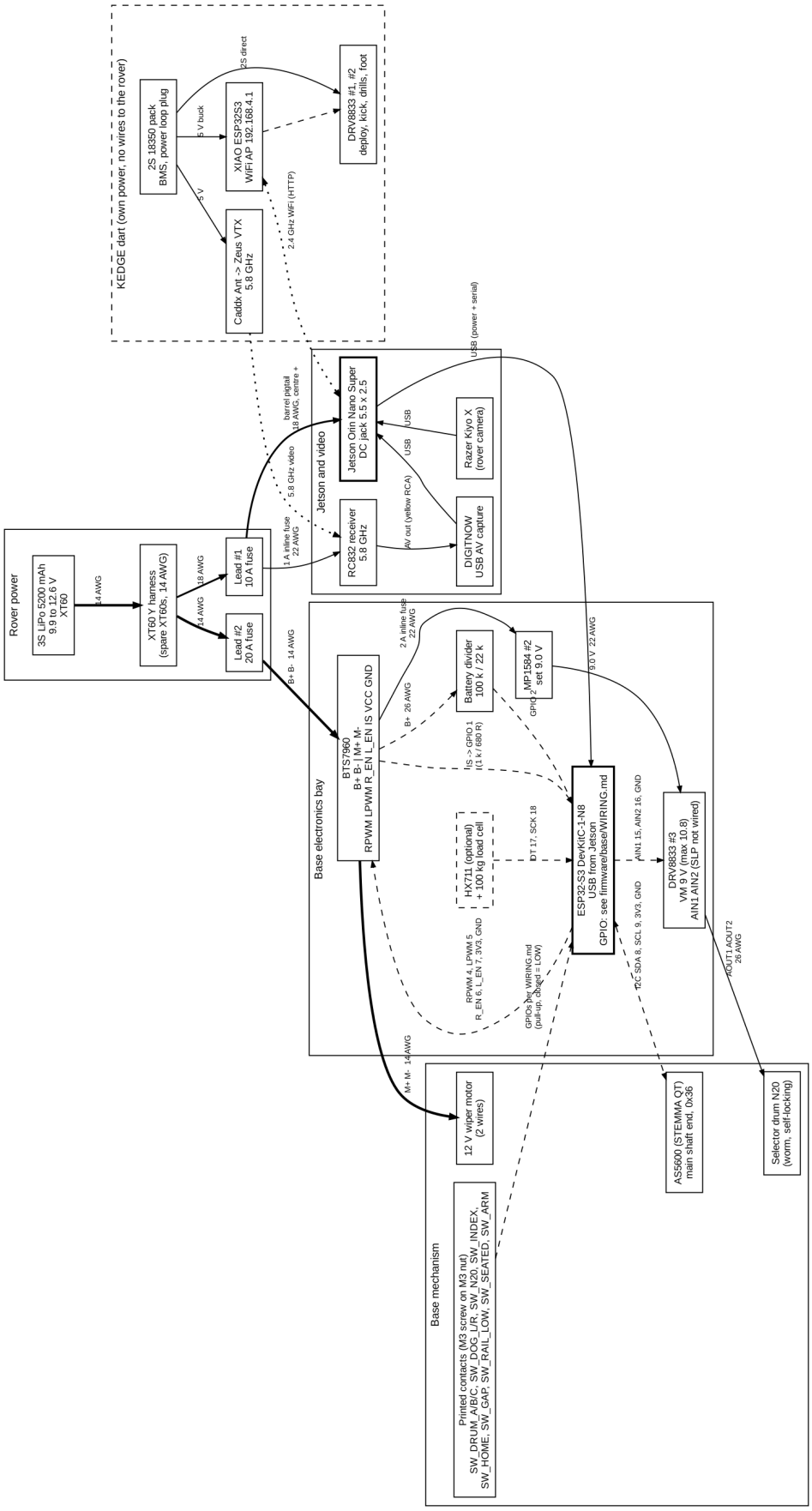
1. Battery unplugged. Jetson barrel plug and ESP32 USB cable unplugged. Pull both fuses (20 A and 10 A) and both small inline fuses.
2. Battery unplugged. Meter on beep, check for shorts: the two pins of the Y harness male XT60; red to black at the BTS7960 B+ / B-; at MP1584 #2 IN and OUT; at DRV8833 VM / GND; at the barrel plug; at the RC832 lead. Nothing may beep steadily. A short chirp that stops is a capacitor charging, not a short. A steady beep is a short: find it before going on.
3. Battery unplugged. Look at the wiring: only three DRV8833 #3 pins go to the ESP32 (AIN1, AIN2, GND). Its SLP / nSLEEP / EEP pad is NOT wired: the board holds it high by itself. Check no wire is on SLP, VM or VCC going to the ESP32.
4. Battery unplugged. Meter on DC volts on the LiPo balance lead: each cell 3.7 to 4.2 V. Then check the bare leads before anything is on them: unscrew lead #2 from the BTS7960 (or check before you first screw it on), plug the battery into the Y harness with the fuses in only for this check (free lead ends kept apart, bare ends taped over, uncover just enough to touch the probes), meter the free ends of lead #2 and lead #1: red probe on red, battery voltage, POSITIVE. Negative or zero: unplug and fix the harness. Unplug the battery, pull the fuses again, and connect lead #2 to the BTS7960 (red to B+). Then add one load at a time as below.
5. Battery unplugged. Fit the 20 A fuse and the 2 A inline fuse. Plug the battery in. Meter on DC volts:
   - BTS7960 B+ (red probe) to B- (black probe): battery voltage, positive. Negative means reversed: unplug at once.
   - MP1584 #2 IN+ to IN-: battery voltage, positive.
   - MP1584 #2 OUT: turn the brass screw until it reads 9.0 V.
6. Unplug the battery. Connect the DRV8833 VM wire (P5) to MP1584 #2 OUT.
7. Battery unplugged. Fit the 10 A fuse and the 1 A inline fuse. Plug the battery in. Meter on the barrel plug (not in the Jetson yet): centre pin positive, battery voltage. Unplug the battery. Push the barrel into the Jetson and plug the ESP32 USB cable into the Jetson (USB cables in first, battery last).
8. Plug the battery in: the Jetson boots (green LED), the RC832 lights, and the ESP32 power LED lights once the Jetson is up.
9. On the Jetson, in `~/GrappStar/jetson`: `python3 -m grappstar base ping` (expect `"ok": true`), then `python3 -m grappstar base monitor`. Check `vbat` in telemetry matches the meter within 0.3 V.
10. Unplug the battery. Connect the motors (wiper to BTS7960 M+ / M-, drum N20 to AOUT1 / AOUT2). Check the B- screw is tight. Plug the battery in. Nothing moves until a command is sent (a mode change may turn the wiper a few degrees to free the dogs: that is a command too).

Emergency stop: send `estop`. If that does not stop it, pull lead #2's XT60 (or its 20 A fuse): that kills both motors and leaves the Jetson running. Pull the main battery plug only if that is not enough.

Shut-down order (every time): `estop`, then `sudo shutdown now` on the Jetson and wait for its LED to go off, then unplug the battery.

5. Rules

- Battery unplugged before you touch, add or move any wire, or fit a fuse. This killed the first XIAO.
- Battery unplugged before you plug the ESP32 into the Mac. Flash from the Mac; battery out is enough, the wiring can stay on the board (firmware/base/README.md).
- Charge the LiPo at 3.6 V per cell at rest (10.8 V). Never run it below 3.3 V per cell under load (9.9 V); the firmware reports `vbat`.
- Never connect the DRV8833 VM to the LiPo directly (10.8 V maximum; the LiPo is 12.6 V full).
- Never put 3V3 on a DRV8833 pad called VIN, VCC or VM: on these carriers that is the motor supply. On the base, the DRV8833 SLP / nSLEEP / EEP pad is left unwired.
- Dart: power loop plug out whenever a wire moves or USB is plugged in (the dart run sheet rule).
- The VTX on the dart must have its antenna on before it is powered.



Whole system electrical diagram (bold = battery power, dashed = signal, dotted = radio)

7.2 Base ESP32 wiring (pin by pin)

From `firmware/base/WIRING.md`*:*

Base station wiring

Board: ESP32-S3 DevKitC-1-N8. Pins come from `grappstar_base/pins.h` (firmware `base-1.3.1`). Nothing here has been wired and tested on real hardware yet: check every step with the multimeter.

The rule: battery unplugged whenever a wire or plug moves. Plug the battery in last, after the checks. Never pull a plug while the battery is in. The one exception: lead #2's XT60 is the physical motor kill switch and may be pulled at any time (the build guide's motors-safe state).

1. Pin table

GPIO	Name in pins.h	Connects to	Note
4	`PIN_RPWM`	BTS7960 RPWM	wiper FORWARD (dir 1), PWM 20 kHz
5	`PIN_LPWM`	BTS7960 LPWM	wiper BACKWARD (dir -1), PWM 20 kHz
6	`PIN_REN`	BTS7960 R_EN	plus 10 k from R_EN to GND
7	`PIN_LEN`	BTS7960 L_EN	plus 10 k from L_EN to GND
1	`PIN_WIPER_IS`	BTS7960 R_IS + L_IS through 1 k	plus 680 R and a 1 uF capacitor from GPIO1 to GND (the IS divider and filter)
2	`PIN_VBAT`	battery divider middle	100 k from 3S + (at the BTS7960 B+ screw) to GPIO2, 22 k from GPIO2 to GND
8	`PIN_SDA`	AS5600 SDA	STEMMA QT blue wire
9	`PIN_SCL`	AS5600 SCL	STEMMA QT yellow wire
15	`PIN_DRUM_IN1`	DRV8833 AIN1	drum N20
16	`PIN_DRUM_IN2`	DRV8833 AIN2	drum N20
10	`PIN_SW_A`	contact `SW_DRUM_A`	drum code bit A
11	`PIN_SW_B`	contact `SW_DRUM_B`	drum code bit B
12	`PIN_SW_C`	contact `SW_DRUM_C`	drum code bit C
13	`PIN_SW_D`	contact `SW_DOG_L`	collar engaged in LAUNCH
14	`PIN_SW_E`	contact `SW_DOG_R`	collar engaged in REEL
47	`PIN_SW_F`	contact `SW_N20`	drum N20 in mesh (open = N20 swung out of mesh for the hand lever)
21	`PIN_SW_G`	contact `SW_HOME`	carriage on its front stop (uncocked)
39	`PIN_SW_H`	contact `SW_GAP`	big gear in its toothless gap (rest)
40	`PIN_SW_I`	contact `SW_RAIL_LOW`	rail at its lowest pitch (aim home)
41	`PIN_SW_J`	contact `SW_SEATED`	dart tail on the pusher
42	`PIN_SW_K`	contact `SW_ARM`	closed while the wrap cup is away, OPENS for about +-5 deg at home (one cup turn counted per opening)
35	`PIN_SW_L`	contact `SW_INDEX`	only on a build without the AS5600 magnet (decision D12). Leave open otherwise
17	`PIN_HX_DT`	HX711 DT (DOUT)	optional load cell
18	`PIN_HX_SCK`	HX711 SCK	optional load cell
3V3		AS5600 VIN, BTS7960 VCC, HX711 VCC	logic supply
GND		common ground (see section 3)	

The DRV8833 SLP pin is **not wired** to any GPIO (decision D9: the Pololu board pulls it high by itself). Its FLT pin is not used either. GPIO21 is `SW_HOME`, not SLP.

Which contact goes on which pin comes from `swBit` in `config.h`. Default: the order above, `SW_DRUM_A` on `PIN_SW_A` through `SW_ARM` on `PIN_SW_K`. `SW_INDEX` is switched off in `config.h` unless the build has no magnet.

Pins the firmware avoids on purpose: 0, 3, 45, 46 (boot strapping), 19 and 20 (the USB port), 26 to 32 (flash), 43 and 44 (UART), 38 and 48 (the RGB LED). `params.py` once suggested GPIO38 for `SW_HOME`; that is the RGB LED on newer boards, so `SW_HOME` is on GPIO21 instead.

2. Resistors and fuses: what goes where

Part	From	To	Why	If you do not have it
10 k	BTS7960 R_EN	GND (BTS7960 header)	keeps the bridge off while the ESP32 resets or is flashed	any value from 4.7 k to 47 k
10 k	BTS7960 L_EN	GND (BTS7960 header)	same	any value from 4.7 k to 47 k
1 k	R_IS + L_IS (tied)	GPIO1	series protection for the pin	any value from 1 k to 4.7 k
680 R	GPIO1	GND	turns the IS current into a voltage (sets `amps_per_volt` = 8500 / 680 = 12.5)	any value from 470 R to 1 k, then `set amps_per_volt` to 8500 / your value (1 k: 8.5)
100 k	BTS7960 B+ screw (soldered right at the screw)	the long wire to GPIO2	battery divider top: the long wire then carries only the divided voltage, never the full 12.6 V	leave the divider out (both resistors) and send `{"id":1,"cmd":"set","key":"vbat_wired","val":false}` after every start-up (it is not kept): no `vbat`, no battery checks, and no `NO_POWER` protection
1 uF capacitor (strongly recommended)	GPIO1 (at the ESP32 end of the IS wire)	GND	smooths the IS pulses. The sketch already averages 16 reads spread across the 20 kHz PWM period, so the reading is right without it; the capacitor is belt and braces against noise and timing slips	1 uF to 2.2 uF, ceramic best (an electrolytic: striped leg to GND). Without one the firmware still works; fit it when you can
22 k	GPIO2	GND	battery divider bottom (12.6 V reads 2.27 V)	as above: the firmware assumes exactly 100 k / 22 k, so do not swap in other values
2 A inline fuse	BTS7960 B+ screw	MP1584 IN+	a short in the buck or the DRV8833 blows this, not the thin wires	no small fuse: use 18 AWG wire for that lead and keep it short (under 10 cm)
20 A fuse	battery + (XT60 lead #2)	BTS7960 B+	main fuse	none: never run without it

There is NO resistor on the DRV8833 SLP pin (it is not wired at all). The 1 uF capacitor from GPIO1 to GND is strongly recommended (above). Optional: 100 nF from GPIO2 to GND (steadies `vbat`).

3. Power

```
3S LiPo + --> 20 A fuse (XT60 lead #2) --> BTS7960 B+ screw --+--> 2 A fuse --> MP1584 #2 IN+   (buck,
  set to 9.0 V first)
                                                               +--> 100 k (at the screw) --> long wire --
  > GPIO2 --> 22 k --> GND
3S LiPo - ------------------------------------------------> BTS7960 B- --> MP1584 #2 IN-

MP1584 #2 OUT+ (9.0 V) --> DRV8833 VM (Pololu labels it VIN)
MP1584 #2 OUT-         --> DRV8833 GND

BTS7960 M+ / M-  --> wiper motor red / black (never the green park switch wire)
DRV8833 AOUT1 / AOUT2 --> drum N20 motor
ESP32 power: USB from the Jetson (from the Mac on the bench)
```

- **Wiper:** 3S LiPo, then the 20 A fuse, then BTS7960 B+ and B-. Use thick wire (14 AWG) on B+, B-, M+ and M-. Never wire the wiper's green park switch wire.
- **Drum N20:** through MP1584 #2 set to **9.0 V**. The DRV8833 VM maximum is 10.8 V and a full 3S is 12.6 V, so **never connect the 3S straight to the DRV8833.** Set the buck to 9.0 V BEFORE its output touches the DRV8833 (Part A below). Take the buck input from the BTS7960 B+ and B- screws, so it sits behind the main fuse.

- **Which buck:** the dart uses ONE MP1584. MP1584 #2 is the spare from the dart build (MEASURE_LIST C4). It may still be set to some other voltage from earlier: Part A re-sets it to 9.0 V.
- **2 A fuse:** in the buck IN+ lead, right at the B+ screw.
- **ESP32:** powered over USB from the Jetson (on the bench: from the Mac). Never connect the battery to the ESP32 5V or 3V3 pins.
- **Common ground:** ESP32 GND, BTS7960 GND (logic header), DRV8833 GND and MP1584 OUT- must all be joined. Without a common ground the control signals mean nothing.
- **Battery divider:** 100 k soldered right at the BTS7960 B+ screw, then the long wire to GPIO2, 22 k from GPIO2 to GND. 12.6 V reads 2.27 V on the pin. It feeds `vbat` in telemetry and the battery checks (`NO_POWER` under 6 V, `BROWNOUT` under 9.9 V while a motor runs). With the battery unplugged `vbat` reads about 0. If you leave the divider out, send `{"id":1,"cmd":"set","key":"vbat_wired","val":false}` (then `vbat` is `null` and there is no battery check). It goes back to true at every start-up, so send it again after each power-up; without it every motor command is refused with `NO_POWER`.

4. BTS7960 details

- **10 k pull-downs on R_EN and L_EN.** While the ESP32 resets or is being flashed its pins float, and a floating enable pin can let the motor twitch. With 10 k to GND the BTS7960 stays off until the firmware drives R_EN and L_EN high. Solder each resistor between the EN pin and a GND pin on the BTS7960 header.
- **VCC** to ESP32 3V3, **GND** to ESP32 GND.
- **IS divider:** tie R_IS and L_IS together. From there a 1 k resistor to GPIO1. From GPIO1 a 680 R resistor to GND. The BTS7960 IS pin gives a small current: motor amps divided by 8500. At 20 A that is 2.35 mA, and 2.35 mA through the 680 R gives about 1.6 V at GPIO1, well under 3.3 V. So `amps_per_volt` = 8500 / 680 = 12.5 (the default).
- **1 uF capacitor from GPIO1 to GND: strongly recommended.** Solder it at the ESP32 end of the IS wire, right at GPIO1 and a GND pin. The IS pin only reports while the high side conducts (a 20 kHz pulse train); the sketch takes 16 reads 53 us apart that step through the PWM period, so their average is right even without the capacitor. The capacitor is belt and braces: it also smooths noise and covers a late read.
- **Check for an onboard IS resistor first.** With everything unpowered, tie R_IS and L_IS together and put the meter (ohms) between that point and GND. Open (no reading) means no onboard resistor: the default scale 12.5 is right. If it reads a resistance Rb, write it down. The scale changes:

`Rx = 1680 in parallel with Rb`, then `amps_per_volt = 8500 x 1680 / (Rx x 680)`.

No onboard resistor gives 12.5. Two onboard 1 k gives about 54.5. Set it with `{"id":1,"cmd":"set","key":"amps_per_volt","val":54.5}` (your number), then calibrate it (CALIBRATION.md).

5. DRV8833 details

- VM (Pololu: VIN) from the buck OUT+ at 9.0 V. GND to the buck OUT- and to the ESP32 GND.
- AIN1 to GPIO15, AIN2 to GPIO16, AOUT1 and AOUT2 to the drum N20. Only channel A is used. Leave BIN1, BIN2, BOUT1, BOUT2 open.
- **SLP: leave it unconnected** (decision D9). The board pulls it high itself, so the driver is always awake. No GPIO and no resistor on it.
- FLT: leave it unconnected.
- **If the drum does nothing: battery unplugged, wire DRV8833 SLP to the ESP32 3V3 pin (never to VM).**
- Never put 3V3 on a pad that is the motor supply. Meter check (unpowered): if a pad beeps to VM, it is the motor supply.
- If the drum turns the wrong way, swap the two N20 wires at AOUT1 and AOUT2 (battery unplugged). CALIBRATION.md section (c) says how to tell.

6. Contact switches

Each printed contact is an M3 screw head touching an M3 nut.

- One side of each contact to its GPIO, the other side to GND.
- The ESP32 uses its internal pull-up. Closed = LOW = shown as closed in telemetry.
- No resistors needed. All the GND sides may share one GND wire.

Contact	**Pin**	**GPIO**	**Telemetry `sw` key**
`SW_DRUM_A`	`PIN_SW_A`	10	a
`SW_DRUM_B`	`PIN_SW_B`	11	b

Contact	Pin	GPIO	Telemetry `sw` key
`SW_DRUM_C`	`PIN_SW_C`	12	`c`
`SW_DOG_L`	`PIN_SW_D`	13	`l`
`SW_DOG_R`	`PIN_SW_E`	14	`r`
`SW_N20`	`PIN_SW_F`	47	`n`
`SW_HOME`	`PIN_SW_G`	21	`H`
`SW_GAP`	`PIN_SW_H`	39	`G`
`SW_RAIL_LOW`	`PIN_SW_I`	40	`L`
`SW_SEATED`	`PIN_SW_J`	41	`S`
`SW_ARM`	`PIN_SW_K`	42	`W`
`SW_INDEX`	`PIN_SW_L`	35	`I` (only shows when switched on in `config.h`)

The three drum code contacts read the drum position as a code (BENCH_TEST.md step 2 has the table). All three open (code 000) never happens on a working drum: the firmware treats it as a wire off and faults (`DRUM_SENSOR`).

On the bench, before the drum is built, use two jumper wires:

- GPIO11 to GND: code "b only" = NEUTRAL.
- GPIO47 to GND: `SW_N20` closed. Without it the firmware thinks the N20 is swung out of mesh (hand mode) and refuses every drum move (`HAND_MODE`).

7. AS5600 and HX711

- **AS5600** (Adafruit STEMMA QT): use a STEMMA QT cable. Red to 3V3, black to GND, blue (SDA) to GPIO8, yellow (SCL) to GPIO9. DIR pad to GND (fixed count direction). Plug and unplug the cable only with the battery unplugged.
- **Magnet:** the Adafruit board comes without one. Any small neodymium disc mounted edge-on (decision D12), or a 6 x 2.5 mm diametric disc, in the magnet cap on the main shaft end, **1 to 2 mm from the chip, centred** on it.
- **HX711** (optional): VCC to 3V3, GND to GND, DT to GPIO17, SCK to GPIO18. Load cell wires to E+, E-, A+, A- as printed on the HX711 board. The firmware finds it at boot (`"hx711":true` in the `HELLO` line).

8. Build order, step by step

You need: multimeter, the 20 A fuse lead (XT60 lead #2), MP1584 #2, a 2 A inline fuse for the buck, 2 x 10 k, 1 k, 680 R, 100 k, 22 k, a 1 uF to 2.2 uF capacitor (strongly recommended), optionally a 100 nF capacitor (section 2 lists fallbacks), an insulated small screwdriver for the buck.

Part A: set the buck (on its own, before the build)

MP1584 #2 must be set to 9.0 V before it ever touches the DRV8833. This is done with the buck on its own: nothing else is wired to it.

1. Battery unplugged. Solder input leads to MP1584 #2, long enough to reach the BTS7960 B+ and B- screws later: red to IN+, with the 2 A inline fuse at the far (B+ screw) end of the red lead; black to IN-. Nothing on the output.
2. Battery unplugged. Connect the input leads to the fused side of the XT60 lead (red to +, black to -). Check with the meter (beep mode) that IN+ and IN- are not shorted.
3. Battery unplugged. Clip the meter probes onto OUT+ and OUT- (meter on DC volts), so you do not have to hold them.
4. Battery unplugged. Hands and tools clear of the pads. Now plug the battery in. Turn the little screw on the buck with the insulated screwdriver until the meter reads **9.0 V**. Turn slowly: many turns may be needed. Keep the screwdriver and probes off the other pads. Then unplug the battery.
5. Battery unplugged. (If it is still in, unplug it now, before anything else.) Disconnect the buck input leads from the XT60 lead. Mark the buck "9.0 V".

Part B: build

1. Battery unplugged. Solder the two 10 k pull-downs: R_EN to GND and L_EN to GND on the BTS7960 header.
2. Battery unplugged. Check the BTS7960 for an onboard IS resistor (section 4). Write the number down.
3. Battery unplugged. Wire the IS divider: R_IS and L_IS tied, 1 k to GPIO1, 680 R from GPIO1 to GND, and the 1 uF capacitor from GPIO1 to GND at the ESP32 end (strongly recommended).
4. Battery unplugged. BTS7960 logic: RPWM GPIO4, LPWM GPIO5, R_EN GPIO6, L_EN GPIO7, VCC 3V3, GND to ESP32 GND.

5. Battery unplugged. Battery divider: solder the 100 k right at the BTS7960 B+ screw, run the long wire from its other end to GPIO2, 22 k from GPIO2 to GND. Optional 100 nF from GPIO2 to GND.
6. Battery unplugged. AS5600: STEMMA QT cable to 3V3, GND, GPIO8 (SDA), GPIO9 (SCL). DIR to GND.
7. Battery unplugged. Contacts: one side to the GPIO from section 6, other side to GND. Contacts not built yet: put the two bench jumpers on (GPIO11 to GND, GPIO47 to GND, section 6).
8. Battery unplugged. HX711 if you have it: 3V3, GND, GPIO17 (DT), GPIO18 (SCK).
9. Battery unplugged. USB check with no power wiring yet: plug USB into the port labelled USB, open the Serial Monitor and do "First contact" in README.md. `status` must show `"magnet":"OK"`. Close each contact and watch telemetry `sw` (BENCH_TEST.md step 2). Unplug USB.
10. Battery unplugged. DRV8833: AIN1 GPIO15, AIN2 GPIO16, GND to ESP32 GND. SLP and FLT stay unconnected. AOUT1 and AOUT2 to the drum N20.
11. Battery unplugged. Check the buck on its own once more before it touches the DRV8833: connect only its input to the fused XT60 lead as in Part A (nothing on the output, meter on OUT+ / OUT-), plug the battery in, read 9.0 V with no load, unplug the battery and disconnect the input again. Only then, battery still unplugged: OUT+ to DRV8833 VM, OUT- to DRV8833 GND.
12. Battery unplugged. Buck input: the red lead's fuse end (the 2 A inline fuse) to the BTS7960 B+ screw, so the fuse sits right at the screw; its other end is already on the buck IN+ (If you have no 2 A fuse: 18 AWG, under 10 cm). Black lead from IN- to BTS7960 B-.
13. Battery unplugged. Power: fused XT60 lead + to BTS7960 B+, - to B-. Use 14 AWG. The 20 A fuse in the + lead.
14. Battery unplugged. Wiper motor to M+ and M- (red to M+, black to M-). On the bench the motor is clamped and has nothing attached (BENCH_TEST.md).
15. Battery unplugged. Ground: check ESP32 GND, BTS7960 GND, DRV8833 GND and buck OUT- are all joined.

Part C: checks with the multimeter (still unplugged)

16. Battery unplugged. Beep mode: **B+ to B- must NOT beep.** A very short chirp is fine (capacitors on the boards charging from the meter). A steady beep is a short: stop and find it.
17. Battery unplugged. Beep mode: buck OUT+ to OUT- must not beep. DRV8833 VM to GND must not beep.
18. Battery unplugged. Beep mode: ESP32 GND to BTS7960 GND, DRV8833 GND and buck OUT- must beep (common ground).
19. Battery unplugged. Beep mode: ESP32 3V3 to GND must not beep (a short chirp is fine). 3V3 must not beep to B+ or to DRV8833 VM. DRV8833 SLP must not beep to any GPIO.
20. Battery unplugged. Ohms: GPIO2 to GND about 22 k (a bit less is fine). R_EN to GND and L_EN to GND about 10 k.
21. Battery unplugged. Check every motor and power screw terminal is tight and no bare strand touches its neighbour.

Part D: first power

22. Battery unplugged. Plug USB in first. The Serial Monitor shows telemetry with `"state":"IDLE"`.
23. Battery unplugged. Hands clear of the motor and the drum. Now plug the battery in, last. Nothing should move. Measure the buck output again (9.0 V) and the battery at the XT60, and compare it with `vbat` in telemetry. If anything is hot, smells or a motor moves on its own: unplug the battery at once.
24. Battery unplugged. Go on with BENCH_TEST.md.

8. Flashing both ESP32s

Rules for both boards: battery unplugged (dart: power loop plug out) before the USB cable goes in. Use a data USB cable. Flash the dart XIAO before it goes into the hull (run sheet section 5).

8.1 Base: ESP32-S3 DevKitC-1-N8

From `firmware/base/README.md`*:*

GrappStar base station firmware

Firmware `base-1.3.1` for the ESP32-S3 DevKitC-1-N8 in the base station. It follows BASE_SPEC v1.2, `base/spec/params.py` 1.3.8 (generated into `params_gen.h`) and the C1 sequence of BASE_SPEC 10.1 (D22, D35, D45).

Nothing in this folder has run on real hardware yet. The logic passes the PC tests (a simulation). Go slowly the first time and follow BENCH_TEST.md before the wiper motor goes into the base.

What it does

One wiper motor (BTS7960 driver, AS5600 angle sensor) drives everything: launch, aim, take-up and reel. A small N20 motor (DRV8833 driver) turns the selector drum. The drum slides the dog collar and runs the wrap cup. Its order is LAUNCH, NEUTRAL, WRAP, then the WRAPPED park, then REEL:

- LAUNCH (drum 0 deg): the collar drives the launcher.
- NEUTRAL (41 deg): the collar drives nothing. Forward wiper turns only drive the take-up (it winds slack onto the storage reel through a slip clutch).
- WRAP window (62 to 302 deg): the collar stays in NEUTRAL while the cup lays 7 turns of line on the capstan.
- WRAPPED park (314 deg): wraps on, cup locked, collar still in NEUTRAL (it only starts to move at 318). The take-up works here too.
- REEL (347 deg): the collar drives the capstan. REEL is only reached with the wraps on.

The drum is slow (about 8 deg per second): a wrap or an unwrap takes about 40 s.

Printed contacts (12 of them) tell the firmware where everything is.

Wiper direction	LAUNCH	NEUTRAL, WRAP and the WRAPPED park	REEL
FORWARD (`dir` 1, RPWM)	cocks, fires and resets the launcher	take-up only, at most 40 % power	winds the capstan
BACKWARD (`dir` -1, LPWM)	aims (turns the rail pitch)	refused (`BACKSTOP`): it would fight the take-up bus one-way	refused (`BACKSTOP`): sprag #2 holds the capstan

The Jetson sends commands over USB as JSON lines. The base answers every command, sends telemetry 10 times a second (20 while something moves) and sends an event when an action ends. Full details: PROTOCOL.md.

A mission in short (BASE_SPEC 10.1): `mode LAUNCH`, `aim`, (`cock`), `arm`, `fire`, flight, `set line_m` (the landing distance), `aim 6` if the rail is above 15 deg (D45: the tow runs with the rail low), `wrap` right after landing (straight from LAUNCH, ends at the WRAPPED park), wait for the dart to anchor, `tidy` (take-up for a few seconds), `mode REEL` (fly-in), tow (`reel` until tension), release, reel in, `unwrap` (back to NEUTRAL with the take-up running), final approach (`retrieve`, until the dart is seated), `mode LAUNCH`.

Files

File	What
`grappstar_base/grappstar_base.ino`	the sketch: hardware glue (pins, USB, AS5600, HX711, flash). Open this one in the Arduino IDE
`grappstar_base/pins.h`	every GPIO pin
`grappstar_base/config.h`	defaults, contact list, drum sectors and parks, current trips
`grappstar_base/base_logic.h`	all the logic: commands, states, faults, telemetry
`grappstar_base/drives.h`	wiper and drum motor drive (soft start, reverse pause, current trip, stall)
`grappstar_base/sensors.h`	contact debounce, AS5600 angle tracking
`grappstar_base/persist.h`	what is saved in flash
`grappstar_base/json_lite.h`	small JSON reader and writer
`grappstar_base/params_gen.h`	GENERATED from `base/spec/params.py`. Do not edit
`tools/gen_params.py`	makes `params_gen.h`
`test/run_tests.sh`	builds and runs the PC tests

File	What
PROTOCOL.md	every command, reply, event, setting and fault code
WIRING.md	pin table, power, step by step build
BENCH_TEST.md	first checks on the bench, before the motor goes into the base
CALIBRATION.md	direction, current, drum, aim and tension calibration

Arduino IDE setup (Mac)

1. Install the Arduino IDE 2 from arduino.cc if you do not have it.
2. Arduino IDE > Settings. In "Additional boards manager URLs" add `https://espressif.github.io/arduino-esp32/package_esp32_index.json` (skip if it is already there).
3. Tools > Board > Boards Manager. Search `esp32`. Install **esp32 by Espressif Systems**, version **3.3.12**. The sketch compiles with this version (checked with arduino-cli 1.5.1). 2.0.17 should also work (the sketch handles both) but was not compiled.
4. Tools > Manage Libraries. Search `HX711`. Install **HX711 by Rob Tillaart**, version **0.6.6** (not the one by bogde). You need it even without a load cell (the sketch includes it and finds out at boot whether one is fitted).
5. File > Open: `firmware/base/grappstar_base/grappstar_base.ino`. All the `.h` files open as tabs next to it.
6. In the Tools menu set exactly these:
   - Board "ESP32S3 Dev Module"
   - "USB CDC On Boot: Enabled" (the sketch refuses to compile otherwise, on purpose)
   - "USB Mode: Hardware CDC and JTAG"
   - Flash Size: 8MB (the N8 board has 8 MB)
   - Leave the rest at the defaults.
7. Plug a USB cable into the DevKitC port labelled **USB** (the native USB port). This is the same port the Jetson uses. The other port is labelled UART: do not use it for this firmware. Note: the old ground station used the UART port. This one does not.
8. Tools > Port: pick the one called something like `/dev/cu.usbmodem...`.
9. Click Upload (the arrow). If the upload cannot connect: hold the BOOT button, tap RESET, release BOOT, then Upload again. After an upload done this way, tap RESET once.

Coming from base-1.0: the saved bench settings (current scale, free current, N per amp and so on) are stored in a new format since base-1.2. The old ones are ignored after the upload, so every bench value is back at its default. Enter them again (CALIBRATION.md).

Command line instead of the IDE (same settings): `arduino-cli compile --fqbn esp32:esp32:esp32s3:CDCOnBoot=cdc firmware/base/grappstar_base` from the repository folder. It should report about 398 kB of program (30 %) and about 35 kB of global variables (10 %). A compile only shows the code builds; it says nothing about the hardware.

First contact

1. Battery unplugged. USB only. If the contacts are not wired yet, put two jumper wires on the board first: GPIO11 to GND (makes the drum code read NEUTRAL) and GPIO47 to GND (says the drum N20 is in mesh). WIRING.md section 6 explains why. Without the GPIO11 jumper you get a harmless `FAULT DRUM_SENSOR` "drum code 000" a moment after start-up.
2. Tools > Serial Monitor. At the bottom right set the line ending to **Newline**. The baud rate does not matter (USB).
3. Tap RESET on the board. You should see a `HELLO` line, then a `BOOT` line, then telemetry lines:

```
 {"evt":"HELLO","fw":"base-1.3.1","reset":"POWERON","hx711":false,"as5600":true,"pwm":true}
 {"evt":"BOOT","t":60,"fw":"base-1.3.1","record":false,"was_moving":false,"drum_moved":false,
"drum_known":true,"wraps_known":true,"aim_known":false,"phase":"UNKNOWN","angle":"AS5600"}
 {"tel":{"state":"IDLE","mode":"NEUTRAL","drum_deg":41,"drum_pos":"NEUTRAL",...,"sw":{"b":1,"n":1},
"fault":null,"vbat":0}}
```

The board waits only 1.5 s for the Serial Monitor at start-up, so you may miss `HELLO`. That is fine. With USB only, `vbat` reads about 0: that is normal. Motor commands are refused with `NO_POWER` until the battery is in.

4. Type this in the box at the top and press Enter:

```
{"id":1,"cmd":"ping"}
```

You should get {"id":1,"ok":true,"fw":"base-1.3.1","t":...} somewhere between the telemetry lines.

5. Telemetry scrolls fast. Untick "Autoscroll" to read a line, tick it again to follow.
6. Also try {"id":2,"cmd":"status"}. It shows "magnet": OK, WEAK, STRONG or NONE.
7. Telemetry sw lists the closed contacts with one letter each: a b c (drum code), l (SW_DOG_L), r (SW_DOG_R), n (SW_N20), H (SW_HOME), G (SW_GAP), L (SW_RAIL_LOW), S (SW_SEATED), W (SW_ARM), I (SW_INDEX, only on a no-magnet build).

In the first 60 ms after a start-up motor commands, set and cal are refused with BOOTING: send again.

On the Jetson the same commands go through the GrappStar program instead of the Serial Monitor. Stop the service first (sudo systemctl stop grappstar), then run python3 -m grappstar base console: one connection, telemetry twice a second plus every event; type a JSON line, key=value words (reel duty=0.3 until=stop) or a bare stop, estop or status, and quit to end.

Every command is one line of JSON with a "cmd". The "id" is any whole number you like; the reply carries the same id (a number like 1.5 gives BAD_ID). A few examples (all in PROTOCOL.md):

```
{"id":3,"cmd":"jog","motor":"wiper","dir":1,"ms":500}
{"id":4,"cmd":"stop"}
{"id":5,"cmd":"get","key":"amps_per_volt"}
{"id":6,"cmd":"set","key":"limit_n","val":50}
{"id":7,"cmd":"clear"}
```

A refused command replies with "ok":false and an "err" code, and nothing moves.

PC tests

The logic is plain C++ and runs on a computer against a simulated base. On the Mac you need the command line tools once: xcode-select --install (gives g++ and python3). Then, from the repository folder:

```
sh firmware/base/test/run_tests.sh
```

It tests the params generator, then builds test_base (the mission and fault tests) and runs it, then the fuzz test (random commands and faults, checking that nothing unsafe ever happens). Look for ALL PASS and fuzz: no invariant violated.

How spec numbers get into the firmware

The shared numbers live in base/spec/params.py. After that file changes, run:

```
python3 firmware/base/tools/gen_params.py
```

It rewrites grappstar_base/params_gen.h. Then upload again. Never edit params_gen.h by hand. If a value is missing in params.py, the firmware uses its own default from config.h, so the build never breaks.

Bench values (current scale, free current, N per amp, tension offset, aim offset and so on) are not in params.py. You set them over USB with set or learn them with cal (see CALIBRATION.md). They are saved in flash and kept over a power cycle and over a new upload of the same firmware version.

Safety summary

- Any fault: the wiper stops at once. Then, if the wraps are off and nothing below applies, the drum goes to NEUTRAL, so the wiper turns nothing.
- The drum does NOT move, and holds where it is, when:
  - the wraps are on (wrap window, WRAPPED park, REEL): the worm and sprag #2 hold the line, and the operator unwraps when it is safe (unwrap). This applies to estop too;
  - the launch spring may be drawn: there is no sear, only the wiper worm holds the spring, and a shift under that load could dry fire. Recover with decock;
  - the drum itself faulted, its position or the wraps are unknown, or the N20 is swung out of mesh (hand mode). After a drum code that changed with the drum still near the wrap window, only unwrap is accepted until NEUTRAL.

It then sends a MODE event with "err":"DRUM_HELD" and a "why" that says which (exactly at the WRAPPED park it says "ok":true instead).

- After a power cut the base never moves by itself. Inside the wrap window it stops and reports. A cut while cocking or firing is treated as a drawn spring: cock is refused (LOADED_UNKNOWN) until a decock (PROTOCOL.md section 9).
- stop: wiper off now, drum stops. estop: the same, then the drum goes to NEUTRAL when that is safe (never with the wraps on).

- Before every shift out of LAUNCH or REEL the wiper makes a small centering jog (6 deg) so the collar is not loaded. The collar contacts `SW_DOG_L` and `SW_DOG_R` are checked during every drum move.
- Soft start: the wiper ramps up over 300 ms (inrush once blew a 10 A fuse). It always stops for 200 ms before it changes direction.
- Current trips per job (each for 30 ms, ignored for 50 ms after a start): every backward run in LAUNCH (aim, decock, backward and centering jogs) 3 A, reel in 5 A, cock and fire 6.5 A, tow 5 to 6.9 A set from `max_n` (at most 55 N, D46) (the tension estimate is the real tow limit), take-up 2.5 A, jogs and fly-in 3 A; 17 A at any time. Stall: the shaft slower than 12 deg/s for 200 ms while driving.
- AS5600 lost while the wiper runs, or its angle jumps so travel is lost: fault `ANGLE_SENSOR`. If the AS5600 is not reading, wiper actions and the drum moves that need the centering jog or the fly-in are refused (`ANGLE_SENSOR`). Exception: a build with no magnet can switch on the 6-bump index cam `SW_INDEX` in `config.h` (decision D12, off by default); then the base runs on that coarser angle (PROTOCOL.md section 9).
- Battery below 9.9 V for 300 ms while a motor runs: fault `BROWNOUT`.
- Fire needs a fresh 6-digit token from `arm`, valid 3 s, used once, plus the dart seated. `cock` also needs the dart seated. The seated contact can only read the dart with the carriage home (the carriage plunger presses it there), so `cock` checks it before the draw and that reading counts for `arm` and `fire` while cocked (until a fire, a decock, a fault or a restart).
- Both worm gears self-lock, so the mechanism stays where it stopped when the power goes. The position is saved in flash, and after a power cut mid-motion the base only trusts what its sensors can prove.
- After a fault, fix the cause, then send `{"id":1,"cmd":"clear"}`.

And the rule that matters most: **battery unplugged whenever a wire or plug moves.** Plug the battery in last, and never pull a plug while the battery is in. The one exception: lead #2's XT60 is the physical motor kill switch and may be pulled at any time (motors dead, the ESP32 and the Jetson stay up: the build guide's motors-safe state).

8.2 Dart: XIAO ESP32S3

Arduino IDE: board XIAO_ESP32S3, Tools > USB CDC On Boot: Enabled (the sketch refuses to compile otherwise). Library: Adafruit BNO08x (the INA219 library is no longer needed). Open `firmware/dart/kedge_dart/kedge_dart.ino`. After the first USB upload you can upload over WiFi (port "kedge"). On a bare XIAO the `z` scan finds nothing: 0x40 and 0x4a only appear once the INA219 and BNO085 are wired.

From `firmware/dart/README.md`*:*

KEDGE dart firmware v4

Firmware for the KEDGE dart: Seeed XIAO ESP32S3, two DRV8833 drivers (four motor channels), INA219 current sensor, BNO085 IMU. It anchors the dart after landing, releases it again, records the flight, and takes commands over WiFi.

File	What
`kedge_dart/kedge_dart.ino`	the sketch (hardware: pins, PWM, I2C, WiFi, web page, HTTP API)
`kedge_dart/kedge_logic.h`	all control logic (sequences, safety, flight detection, flight log), plain C++
`test/test_logic.cpp`, `test/run_tests.sh`	PC tests of the logic against a motor model and synthetic IMU traces
`API.md`	the HTTP API for the Jetson (contract C2)
`WIRING.md`, `wiring/`	wire cut list, change list from v29, diagram
`BENCH_TEST.md`	how to check every motor and sensor before closing the hull
`tools/wires_v4.py`	regenerates `wiring/` (run from the repo root; needs python3 with numpy, graphviz `dot`, and `kedge/out30`. Another graphviz version gives harmless image differences)
`sim/`	host simulator: the real sketch compiled on Linux with stub Arduino headers, serving the real HTTP API (`make && python3 test_sim.py`)
`REPORT.md`	build-day report

Arduino IDE setup (once)

1. Install Arduino IDE 2 (arduino.cc).
2. File > Preferences > Additional boards manager URLs, add:
   `https://espressif.github.io/arduino-esp32/package_esp32_index.json`

3. Tools > Board > Boards Manager: search `esp32`, install **esp32 by Espressif Systems** (3.3.x tested; 2.0.17 also compiles).
4. Sketch > Include Library > Manage Libraries: install **Adafruit BNO08x** (say yes to its dependencies, Adafruit BusIO). The INA219 is read directly, so the Adafruit INA219 library is no longer needed (having it installed does no harm).
5. Open `firmware/dart/kedge_dart/kedge_dart.ino` (the folder name must stay `kedge_dart`; `kedge_logic.h` sits next to it and opens as a second tab).

Tested versions (3 Nov 2026, `arduino-cli compile --fqbn esp32:esp32:XIAO_ESP32S3 --warnings all`**)**

esp32 core	Result	Flash	RAM
3.3.12 (latest)	OK, no warnings from the sketch	1,014,379 B (30 %)	70,500 B (21 %)
3.3.2	OK, no warnings from the sketch	1,018,875 B (30 %)	68,512 B (20 %)
2.0.17	OK, no warnings from the sketch	805,405 B (24 %)	66,968 B (20 %)

Libraries: Adafruit BNO08x 1.2.7, plus Adafruit BusIO 1.17.4 and Adafruit Unified Sensor 1.1.15 (installed with it). arduino-cli 1.5.2-rc.1. Note for arduino-cli users: on this board the menu value for "USB CDC On Boot: Enabled" is `CDCOnBoot=default` and `CDCOnBoot=cdc` means Disabled (the sketch then refuses to compile, on purpose).

Board settings

- Tools > Board > esp32 > **XIAO_ESP32S3**.
- Tools > **USB CDC On Boot: Enabled** (the default for this board). If it is Disabled, the sketch will not compile.
- Leave the rest at the defaults.

Upload over USB

1. **Battery unplugged** (power loop plug out). Never have the battery plug and USB in together while you work.
2. Plug the XIAO into the Mac with a USB-C data cable.
3. Tools > Port: pick the one that appears (Mac `/dev/cu.usbmodem...`, Linux `/dev/ttyACM0`, Windows `COM3` or similar).
4. Click Upload (arrow). Wait for "Hard resetting via RTS pin" / "Done uploading". If the port does not appear or the upload fails: hold the XIAO's **B** (boot) button, tap **R** (reset), release B, then upload again. After such an upload, tap R once.
5. Tools > Serial Monitor, 115200 baud, "No line ending". You should see `KEDGE v4 ready` and the key list. Type `z` and press Enter: expect `I2C device at 0x40` (INA219) and `0x4a` (BNO085; some boards answer at `0x4b`, also fine).

Upload over WiFi (after the first USB upload)

USB unplugged, power loop plug in. Join WiFi `KEDGE` (password `grappstar`). Tools > Port > network port `kedge`. Upload. Motors stop, any sequence is aborted and the dart disarms when the upload starts.

Driving it

- Web page: join `KEDGE`, open `http://192.168.4.1`. Hold a button or key to run, release to stop.
- Serial Monitor: type a key and press Enter, each press runs 1.5 s.
- Keys: `1`/`!` deploy down/up, `2`/`@` kick out/back, `3`/`#` drills in/out, `4`/`$` foot in/out, space or `s` STOP, `a` auto-anchor, `r` release, `A` arm/disarm (auto-anchor on landing), `+`/`-` speed, `i` current, `b` attitude, `p` status, `f` flight log summary, `z` I2C scan, `h` help.
- Jetson: see `API.md`.

Safety built in

- All motors are off from the first lines of `setup()`. If the chip resets, the DRV8833 inputs fall low and the motors coast.
- Every manual run has a deadman (350 ms on WiFi, 1.5 s per Serial key) and a total hold cap (8 s, 2 s without the INA219). A motor driven into an end stop stops on stall.
- WiFi client gone: manual motors stop at once. A running anchor or release sequence carries on (it is autonomous and has its own limits).
- Over-current trip 2000 mA, low rail (under 5.6 V while driving), sequence time limits, jam detection in release.
- All safety runs in its own task every 5 ms with a 2 s watchdog; a stuck web client or Serial port cannot block it.
- After a brownout or watchdog reset the dart boots with motors off and reports `FAULT` until `/stop`, `/auto`, `/release` or `/arm`.

- INA219 missing: manual keys only (on timers), auto and release refused. BNO085 missing: no arming, no flight log.
- A stuck I2C bus is recovered (9 clocks and a Wire restart). If the web loop hangs for 30 s while nothing moves, the dart is not armed or in flight, and no OTA upload runs, it restarts itself.
- **Power on with the legs stowed.** The firmware keeps track of the kick's travel from power-on to tell a jammed kick from one that is already down or stowed.

Tuning

The current thresholds in `kedge_logic.h` (`struct Tuning`) are from the v29 bench numbers at 8 V. Measure the real ones with BENCH_TEST.md steps 3 to 6 and write them in at step 10. Then run the PC tests.

PC tests

```
cd firmware/dart/test && ./run_tests.sh
```

Needs g++ and python3 (Mac: run `xcode-select --install` once for g++). Prints every check and `ALL PASS`, runs the fuzz test, then checks the JSON files the tests wrote (into `test/out/`).

If `make` is found it also runs the whole simulator suite below (about 2 minutes). That part needs Linux, make, node 18 or newer and Playwright with Chromium (`npm i playwright && npx playwright install chromium`). Without node the browser check is skipped; node without Playwright makes the run fail. `./run_tests.sh --quick` runs only the logic and fuzz checks (seconds, any OS).

Simulator (Linux, optional)

Needs Linux, g++, make, pthread, python3, and for the page check node 18+ with Playwright and Chromium (`npm i playwright && npx playwright install chromium`).

```
cd firmware/dart/sim && make && python3 test_sim.py      # 12 HTTP tests against the real sketch
python3 test_jetson_client.py                            # 9 tests: the Jetson's own dart client
  against the real sketch (C2)
node test_page.js                                        # 11 checks of the web page in headless
  Chromium (one logged 409 is expected)
./kedge_sim --port 8790                                  # then open http://localhost:8790 (the real
  web page)
```

`/sim/launch`, `/sim/state` and `/sim/set?k=v` drive the simulated hardware (see `sim/sim_main.cpp`).

Compile check without the IDE (optional)

Run from the repo root:

```
arduino-cli core install esp32:esp32 --additional-urls https://espressif.github.io/arduino-esp32/
  package_esp32_index.json
arduino-cli lib install "Adafruit BNO08x"
arduino-cli compile --fqbn esp32:esp32:XIAO_ESP32S3 --warnings all firmware/dart/kedge_dart
```

9. Jetson setup

The Jetson already runs JetPack 6.2 (Ubuntu 22.04, Python 3.10) from the SD card with WiFi and SSH. Keep numpy below 2 and do not pip install opencv-python (both broke things in August).

From `jetson/README.md`*:*

GrappStar rover software (Jetson)

This folder is the software that runs on the rover's Jetson Orin Nano. It talks to the base station (USB serial), the KEDGE dart (WiFi), and two cameras, and it serves the operator page in a web browser.

Run everything from this folder (`jetson/`) with `python3 -m grappstar <command>`. Every command has `--help`, for example `python3 -m grappstar serve --help`. If this README and `--help` disagree, `--help` is right (and please fix this README).

Marked **NOT VERIFIED** below: things nobody has checked on the real hardware yet.

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7. Calibrate the rover camera
8. Operator page tour and a full dry mission in sim
9. First real mission checklist
10. Troubleshooting
11. File map

Command summary:

Command	What it does
`python3 -m grappstar serve`	operator page at `http://<jetson>:8080`. Options: `--sim`, `--port 8080`, `--base /dev/grappstar-base`, `--dart 192.168.4.1`, `--rover-cam /dev/grappstar-rovercam`, `--dart-cam /dev/grappstar-dartcam`, `--fourcc YUYV` or `MJPG`
`python3 -m grappstar cameras`	list cameras and the formats they offer
`python3 -m grappstar base status` / `ping` / `monitor`	talk to the base. `monitor` prints telemetry live (Ctrl+C stops). Only one program can hold the base port at a time (serve, monitor, console, cmd): a second one says "port busy"
`python3 -m grappstar base cmd '<json>'`	send one raw command, e.g. `'{"cmd": "status"}'`
`python3 -m grappstar base console`	one connection that prints telemetry (twice a second) and every event, and sends what you type: a word (`stop`, `estop`, `status`), a word with `key=value` pairs (`reel duty=0.3 max_n=20 until=stop`) or a JSON line. `quit` ends. Use it for calibration (watch the tension, type `stop`)
`python3 -m grappstar dart status`	one dart status reading
`python3 -m grappstar dart key <c> --hold <s>`	hold a dart motor key for s seconds
`python3 -m grappstar dart auto` / `release` / `flightlog`	auto anchor, release sequence, download the flight log
`python3 -m grappstar mockbase --pty`	fake base for testing; prints a pty path to use as `--base`
`python3 -m grappstar calibrate intrinsics` / `mount`	rover camera calibration (step 7)
`python3 -m grappstar replay <logdir>`	re-run the vision on a recorded mission
`python3 -m grappstar selftest`	run checks that need no hardware

---

0. What you need

- Jetson Orin Nano Super with JetPack 6.2 on its SD card (already done in August: hostname `GrappStar`).

- Internet for the Jetson during install (home WiFi or Ethernet).
- A way to type on it: monitor and keyboard, or SSH from the Mac: `ssh <your user>@GrappStar.local` (or the IP address, which was 192.168.0.249 on home WiFi in August).
- For later steps: the base station (ESP32-S3 DevKitC-1) and a USB cable, the Razer Kiyo X, the RC832 receiver and the DIGITNOW USB capture dongle, the KEDGE dart, a phone or laptop for the operator page.
- A printed checkerboard for step 7.

Two things learned in August that this kit works around:

- snap is broken on JetPack 6.2. Use `apt` for everything, never `snap`.
- numpy must stay below 2. pip once upgraded it to 2.x and broke system packages. The install keeps it at 1.x.

1. Get the code

On the Jetson:

```
cd ~
GIT_LFS_SKIP_SMUDGE=1 git clone https://github.com/HarryHYu/GrappStar.git
cd GrappStar
git checkout main
cd jetson
```

- `GIT_LFS_SKIP_SMUDGE=1` skips the big caliper photo PDF (145 MB). The Jetson does not need it.
- If `main` has no `jetson/` folder yet (before the build day merge), use the integration branch instead: `git checkout base-v1`.
- To update later: `cd ~/GrappStar && git pull`.

2. Install

```
cd ~/GrappStar/jetson
bash setup/install.sh
```

It asks for your password (sudo). What it does, in order:

1. apt installs `python3-numpy python3-serial python3-pytest python3-pip v4l-utils network-manager`.
2. OpenCV: JetPack normally already has OpenCV for Python (NVIDIA's build, with GStreamer). If `import cv2` already works the script leaves it alone. Only if it is missing does it run `apt install python3-opencv`. Why not `pip install opencv-python`? The pip version has no GStreamer, carries its own JPEG and video libraries (August's JPEG mismatch was exactly this kind of clash), and can drag numpy up to 2.x.
3. `pip install --user -r requirements.txt` for anything apt did not provide. The file pins `numpy>=1.21,<2`.
4. Adds you to the groups `dialout` (serial ports) and `video` (cameras).
5. Installs `setup/99-grappstar.rules` so the devices get fixed names: `/dev/grappstar-base`, `/dev/grappstar-base-uart`, `/dev/grappstar-rovercam`, `/dev/grappstar-dartcam`. `/dev/grappstar-base` needs the base's USB serial number: step 4 sets it. Until then that name is missing.
6. Runs `setup/check_env.py` and prints a report.

Then **log out and back in** (or `sudo reboot`) so the new groups work.

Check any time with:

```
python3 setup/check_env.py
```

Look for: numpy 1.x, cv2 with `V4L2 YES` (and `GStreamer YES` on the Jetson), pyserial, groups dialout and video yes.

Optional: start the operator page automatically at boot (do this after step 3 works):

```
bash setup/install.sh --service
systemctl status grappstar          # should say "active (running)"
journalctl -u grappstar -f          # live log, Ctrl+C to leave
```

The service runs `python3 -m grappstar serve --port 8080` as your user from this folder, and restarts it if it crashes. While the service runs, do not also start `serve` by hand (both want port 8080). Stop it with `sudo systemctl stop grappstar`, remove it from boot with `sudo systemctl disable --now grappstar`.

Optional: onnxruntime-gpu for a GPU terrain model. Not needed to run anything. See the note at the bottom of `requirements.txt` (Jetson AI Lab wheel, NOT VERIFIED).

Optional: maximum performance mode (already set in August): `sudo nvpmodel -q` shows the mode. `jtop` shows load.

3. Test with no hardware

Nothing needs to be plugged in for this step. It also works on a PC.

```
cd ~/GrappStar/jetson
python3 -m pytest                  # all tests; ends with "N passed"
python3 -m grappstar selftest      # quick checks, prints PASS or FAIL per check
python3 -m grappstar serve --sim   # operator page with a simulated base, dart and cameras
```

With `serve --sim` running, open the page:

- on the Jetson itself: `http://localhost:8080`
- from the Mac or phone on the same network: `http://GrappStar.local:8080` or `http://<jetson IP>:8080` (find the IP with `hostname -I`).

You should see both camera views moving (simulated) and base and dart status updating. Ctrl+C in the terminal stops the server. Step 8 walks through a whole simulated mission.

Test the real base client against a fake base (two terminals):

```
python3 -m grappstar mockbase --pty           # terminal 1: prints a path like /dev/pts/5
python3 -m grappstar base ping --base /dev/pts/5    # terminal 2, using the path it printed
```

4. Plug in the base

Battery unplugged. For this step the base runs from USB only: the Jetson's USB powers the DevKitC. Do not connect the 3S LiPo, the BTS7960 or the motors yet (if you are following the build guide, they are already wired: battery unplugged, just check the USB port). Any time you change wiring, unplug the battery first.

Why this step matters: the base (ESP32-S3 DevKitC) and the dart (XIAO ESP32S3) show up on USB with the same ID, 303a:1001. To tell them apart, the Jetson remembers the base's own USB serial number (each board has a different one, it looks like `F4:12:FA:12:34:56`). Then `/dev/grappstar-base` only ever points at the base.

1. Battery unplugged. Unplug the dart's USB cable from the Jetson if it is plugged in. Only the base may be plugged in by USB for this step.
2. Plug a USB cable from the Jetson into the DevKitC port marked **USB** (the ESP32-S3 native port), not the one marked UART or COM. Check it is seen:

   ```
   lsusb                              # expect a line with 303a:1001 (Espressif)
   ls /dev/ttyACM*                    # expect /dev/ttyACM0
   ```

3. Save the base's serial number:

   ```
   bash setup/install.sh --no-apt
   ```

It prints `ESP32-S3 board on /dev/ttyACM0, serial ...` and asks "Is only the base plugged in?". Answer `y`. It writes the serial into the installed rule and prints `base serial ...`. If it says more than one board is plugged in, unplug the dart and run it again. By hand instead: `udevadm info -q property -n /dev/ttyACM0 | grep ID_SERIAL_SHORT`, then `bash setup/install.sh --no-apt --base-serial <that value>`. Running install.sh again later keeps the saved serial. New base board? Run it again with `--base-serial <new value>`.

4. Unplug the base's USB cable and plug it back in. Check the name appeared:

   ```
   ls -l /dev/grappstar-base
   ```

`/dev/grappstar-base -> ttyACM0` means it worked. If you used the UART port instead you get `/dev/grappstar-base-uart`; move the cable to the USB port.

5. Talk to it:

   ```
    sudo systemctl stop grappstar       # only if you installed the service: it holds the port and the
   cameras
    python3 -m grappstar base ping      # expect a reply with "ok": true
    python3 -m grappstar base status
    python3 -m grappstar base monitor   # telemetry lines about 10 per second, Ctrl+C stops
   ```

6. Only after this works, and following the base build guide: battery unplugged, wire the power side, then connect the battery last.

Now the dart may be plugged into the Jetson by USB too: it never gets the base's name. As a second guard, the software only treats a port as the base once a valid base reply (a C1 JSON line) arrives, so it never sends base commands to the dart.

The base firmware must use the native USB CDC (Arduino IDE: "USB CDC On Boot: Enabled", "USB Mode: Hardware CDC and JTAG"). The base talks JSON lines (contract C1, `firmware/base/PROTOCOL.md`).

5. Cameras

Battery unplugged before connecting the RC832 receiver's power. Always screw the antennas onto the RC832 and onto the dart's Caddx Ant VTX before powering them: a VTX powered without an antenna can burn out.

1. Plug in the Kiyo X and the DIGITNOW capture dongle (USB). Connect the RC832 video out (yellow) to the dongle's video in. Battery unplugged while you wire the RC832 power; power it as its label says (NOT VERIFIED which supply on the rover).
2. If the boot service is installed, stop it first (it holds the cameras): `sudo systemctl stop grappstar`. List the cameras:

```
python3 -m grappstar cameras
ls -l /dev/grappstar-rovercam /dev/grappstar-dartcam
```

If a name is missing, run `lsusb`, find the camera's `ID vvvv:pppp`, and fix the line in `setup/99-grappstar.rules` (the Kiyo X ID 1532:0e4f and the dongle ID 534d:2109 are NOT VERIFIED). Then rerun `bash setup/install.sh --no-apt` and replug.

3. See exactly which formats each camera offers:

```
v4l2-ctl -d /dev/grappstar-rovercam --list-formats-ext
v4l2-ctl -d /dev/grappstar-dartcam --list-formats-ext
```

Look for `YUYV` and `MJPG` and the sizes listed under each.

4. Pick the format. The software always asks for a format explicitly, because on JetPack the default path through NVIDIA's JPEG library failed in August (Cheese crashed). Default is YUYV:

```
python3 -m grappstar serve --fourcc YUYV     # safe, works at lower resolutions
python3 -m grappstar serve --fourcc MJPG     # needed for 1920x1080 on the Kiyo X
```

If the picture is black or you see JPEG errors with one, try the other.

5. Dart video: power the dart (its VTX transmits). On the RC832, press the button to step through channels until the picture is clean. The RC832 must be on the same band and channel as the Caddx Ant VTX (the VTX shows its channel with its LED or button; see its manual). Snow means wrong channel, no antenna, or too far.
6. Start `serve` and check both views on the page. The dart view is analog: some snow and dropouts are normal, and the software marks bad frames and skips them for vision.

6. Dart WiFi (and how the operator connects)

The dart is a WiFi access point: name **KEDGE**, password **grappstar**, address **192.168.4.1**. The Jetson must join it to control the dart. At the same time the operator's phone or laptop must reach the Jetson's page.

The plan

- The Jetson's built-in WiFi joins KEDGE. Always.
- The operator connects in one of three ways. **Start with A** (nothing extra to buy). Move to B or C for real missions if A is slow or drops.

Option	Operator connects to	Page address	Good	Bad
A	KEDGE too	`http://<Jetson KEDGE IP>:8080`	nothing extra, simplest	all video goes through the dart's small ESP32 radio, which also carries the dart's own commands. NOT VERIFIED how well it copes
B	Jetson hotspot GRAPPSTAR-ROVER on a USB WiFi adapter	`http://10.42.0.1:8080`	separate radio for video	needs a USB WiFi adapter with Linux AP mode support
C	travel router plugged into the Jetson's Ethernet	Jetson's address on the router	most reliable	one more box to power

The ESP32 access point allows only a few clients at once (4 by default), so with option A keep it to the Jetson and one operator device.

Join KEDGE from the Jetson

Do the install (step 2) first: once the built-in WiFi is on KEDGE, the Jetson has no internet through it.

Power the dart, then on the Jetson:

```
nmcli device                                # find the WiFi interface, usually wlP1p1s0
nmcli device wifi list                      # KEDGE should be listed
sudo nmcli device wifi connect KEDGE password grappstar ifname wlP1p1s0 name kedge
# KEDGE has no internet: never use it as the default route
# keep retrying when the dart is off, and prefer it over home WiFi
sudo nmcli connection modify kedge ipv4.never-default yes ipv6.method ignore \
    connection.autoconnect yes connection.autoconnect-priority 10 connection.autoconnect-retries 0
ping -c 3 192.168.4.1
python3 -m grappstar dart status
```

Replace `wlP1p1s0` if `nmcli device` shows a different name. Back to home WiFi for updates: `nmcli connection up <home wifi name>` (list names with `nmcli connection show`). Note: with priority 10, the Jetson picks KEDGE whenever the dart is on.

Dart commands:

```
python3 -m grappstar dart status            # phase, battery, motors
python3 -m grappstar dart key 2 --hold 1.0  # hold "kick out" for 1 s
python3 -m grappstar dart auto              # auto anchor sequence
python3 -m grappstar dart release           # release sequence
python3 -m grappstar dart flightlog         # download the last flight log
```

Dart keys: `1` deploy down, `!` deploy up, `2` kick out, `@` kick back, `3` drills in, `#` drills out, `4` foot in, `$` foot out. Quote the symbols in the shell, e.g. `dart key '!' --hold 1.0`.

Tell the Jetson how the IMU sits in the dart

The dart's flight log is in the BNO085 chip's own axes. The bench test in `firmware/dart/BENCH_TEST.md` (step 7) finds how the chip sits in the hull. Write it as three entries: which chip axis points along the nose (body x), which points left (body y) and which points up when the dart lies level (body z), with a minus sign where the chip axis points the other way. Examples: `x,y,z` (chip matches the dart), `y,-x,z` (chip turned 90 degrees). Put it in `settings.json` (below) as `"imu_axes": "y,-x,z"`, or pass `--imu-axes y,-x,z` to `serve`. A mirrored entry (for example `y,x,z`) is refused: one sign is wrong.

Settings file

Copy `settings.example.json` to `settings.json` in this folder and edit it once. `serve` (and the boot service) read it, so you do not have to type options. Options typed on the command line still win. Keys: `base`, `dart`, `rover_cam`, `dart_cam`, `fourcc`, `rover_fourcc`, `rover_size`, `dart_size`, `tow_limit`, `drag`, `imu_axes`.

Option A: the operator joins KEDGE too

1. On the Jetson, find its KEDGE address:
   ```
   ip -4 addr show wlP1p1s0         # look for "inet 192.168.4.x"
   ```
2. Phone or laptop: join WiFi KEDGE (password grappstar). It will say "no internet": choose to stay connected (Android asks; turn off "switch to mobile data" if it keeps leaving).
3. Open `http://192.168.4.x:8080` (the x from step 1). `http://GrappStar.local:8080` may also work (mDNS works on iPhone and Mac, not on every Android).

Option B: hotspot on a USB WiFi adapter

1. Plug in the USB WiFi adapter. `bash setup/hotspot.sh list` shows it (a name like `wlx...`).
2. Start the hotspot (it also starts at every boot from now on):
   ```
   bash setup/hotspot.sh up                # picks the USB WiFi adapter
   bash setup/hotspot.sh up wlx00c0ca123456 # or name it
   ```

It only ever uses a USB adapter. It refuses the built-in WiFi and the interface the `kedge` connection uses, and says "No USB WiFi adapter found" if none is plugged in.

This runs `nmcli device wifi hotspot ifname <adapter> con-name grappstar-hotspot ssid GRAPPSTAR-ROVER password grappstar`.

3. Operator: join **GRAPPSTAR-ROVER** (password grappstar), open `http://10.42.0.1:8080`.
4. Stop it: `bash setup/hotspot.sh down`. Status: `bash setup/hotspot.sh status`.

Do not run the hotspot on the built-in WiFi: that radio is on KEDGE.

Option C: travel router on Ethernet

1. Set up the travel router as a normal WiFi router (its own WiFi name and password, DHCP on). Plug a cable from one of its LAN ports to the Jetson's Ethernet port.
2. On the Jetson:
   ```
   nmcli device                                  # find the Ethernet name, usually eth0 or enP8p1s0
   sudo nmcli connection add type ethernet ifname eth0 con-name rover-lan ipv4.method auto \
       connection.autoconnect yes
   ip -4 addr show eth0                          # the Jetson's address on the router
   ```

Replace `eth0` with the name `nmcli device` showed.

3. Operator: join the router's WiFi and open `http://<that address>:8080`. In the router's admin page you can reserve that address so it never changes.

7. Calibrate the rover camera

The rover aims the base at things it sees, so it must know how its camera maps pixels to angles (intrinsics) and how the camera sits on the rover (mount: height and tilt).

You need: the rover camera plugged in (`/dev/grappstar-rovercam`, step 5) and a printed checkerboard. Run every command from this folder (`jetson/`). Files go into `calib/`. Every command has `--help`.

Print the checkerboard

- **9 by 6 inner corners** (that is 10 by 7 squares), **25 mm squares**. A generator such as calib.io ("Checkerboard", 10 columns, 7 rows of squares, 25 mm) or OpenCV's `pattern.png` works.
- Print at 100 percent ("actual size", no "fit to page"). Measure a square with calipers.
- `--square` is in **metres**: 25 mm is `--square 0.025`. If you measured 25.3 mm, type `--square 0.0253` (not 25.3: that would be 1000 times too big).
- Tape it flat on something stiff (cardboard, a clipboard). A bent board gives bad results.

Intrinsics

1. Stop the operator page if it is running (it holds the camera): `sudo systemctl stop grappstar`.
2. Take 20 pictures of the board:

```
python3 -m grappstar snap --n 20
```

Before each picture, hold the board in a new place: near and far, left, right, top, bottom corners, and tilted. Keep it still, avoid glare, then press Enter. The pictures are saved as `calib/images/snap_00.png` to `snap_19.png`.

3. Work out the intrinsics from them:

```
python3 -m grappstar calibrate intrinsics --images calib/images/*.png --square 0.025
```

4. It prints the result and saves `calib/intrinsics.json`. Check `rms` (the error in pixels): under 0.5 is good, over 1 means take the pictures again. `n_used` is how many pictures it could use.

Mount

1. Fix the camera in its final position on the rover. If it moves later, redo this part.
2. Rover on flat ground. Lay the board flat on the ground in front of it, all of it in view.
3. Take one picture, then work out the mount:

```
python3 -m grappstar snap --n 1 --out calib/ground
python3 -m grappstar calibrate mount --image calib/ground/snap_00.png --square 0.025
```

It uses `calib/intrinsics.json` from the step above automatically (it prints "using intrinsics"). If it prints a WARNING about an uncalibrated camera instead, do the Intrinsics part first.

4. It saves camera height, pitch and roll to `calib/rovercam.json`. Check the height against a tape measure.
5. Check: put an orange object at a distance you measured with a tape (for example 5.0 m) from the ground point under the rail pivot (the launcher's main axis, D29), click it on the page, and compare the range the page shows (it adds the camera offset to the pivot by itself). Within about 5 percent is good.

8. Operator page tour and a full dry mission in sim

Start the simulator and open the page (step 3):

```
python3 -m grappstar serve --sim
```

What is on the page (layout may differ a little from this list):

- **Camera views**: rover camera and dart camera, live. Tap (or click) the rover view to mark a target, or press **Auto target**.
- **Base panel**: link, state, mode, drum position, aim angle, current, battery, fault.
- **Dart panel**: link, phase (IDLE, ARMED, FLIGHT, LANDED, ANCHORING, ANCHORED, RELEASING, FAULT), battery.
- **Mission panel**: current step, history, **Preflight**, **Start mission**, **Abort**, and the **prompt** box. The mission stops and asks you before each big move. The confirmations, in order: **Base may move**, **Aim**, **Cock**, **Fire** then **FIRE NOW** (two presses), and the landing verdict (**Anchor** or **Retrieve and reshoot**). After "Base may move" the base is LIVE: drum shifts, aiming and small wiper turns happen between prompts.
- **Recover dart**: if a mission is aborted while the dart is still out, the Mission panel shows "The dart is still out" with a **Recover dart** button. It releases if needed, wraps, reels the dart in, unwraps and goes back to launch. Keep clear of the line.
- **LAUNCHER LOADED** (red banner at the top): the spring is drawn. Hands off the rail. Press **Decock** in the banner to unwind it before you touch the launcher.
- **Target**: range and bearing of the clicked or auto found target, and the ballistic solution (pitch, flight time). A typed range is measured from the ground point under the rail pivot (D29), not from the camera or the rover nose; tapped targets get the camera offset added automatically.

- **Results**: after a flight, the landing zone verdict (GO, NO_GO or UNSURE with reasons), the landing image, the aerial map built from the dart video, and the flight analysis.
- **Manual**: jog the wiper and drum motors, dart keys, record on or off; **Unwrap** brings the drum back through the wrap window to NEUTRAL with the take-up running (the way out when the drum stopped inside the window: there `mode` and drum jogs are refused).
- **STOP** (wiper off now) and **ESTOP** (wiper off and drum to neutral). Always visible at the top. Keyboard: **Esc** is STOP from anywhere on the page (Space too, when you are not typing). Holding STOP for 1 s is ESTOP. Know where they are before you press anything else.

Security note: the page needs no setup. The server puts a fresh session token into the page each time it starts, and every button press must carry it, so another web site cannot press buttons for you. After a server restart the page reloads itself to get the new token. Anyone on the same WiFi can still open the page, so for a password add `--token SOMEPASSWORD` to `serve` and open `http://<jetson>:8080/?token=SOMEPASSWORD` once on each phone or laptop (it remembers it). This is plain HTTP, not encryption. STOP and ESTOP always work from the page, with or without a token.

The mission steps, in order (the page highlights the current one):

Step	What happens	You do
PREFLIGHT	links, faults, dart seated, battery, videos, a target set	fix what it lists, then Retry; then confirm "Base may move"
LAUNCH_MODE	drum to LAUNCH (about 4 s per drum step)	
AIM	pitch from the ballistics table; the page says how far to turn the rover	turn the rover, press Aim
COCK	the base draws the spring (up to about 20 s)	confirm Cock
ARM	the dart's landing detector is armed (it only records; you decide on anchoring)	
FIRE	two confirmations, then the shot (FIRED, then the launcher RESET)	Fire, then FIRE NOW
FLIGHT	waits for the dart to say LANDED (or the dart video to go still)	
LANDING_ZONE	landing report, map, verdict; the base is told how much line is out	Anchor, or Retrieve and reshoot
RAIL_LOW	if the rail is above 15 degrees, it is lowered to its minimum pitch before the tow (skipped on a reshoot)	
BASE_NEUTRAL	launcher at rest, drum to NEUTRAL	
WRAP	the capstan wraps go on while the line is still slack; the drum parks at WRAPPED (about 35 to 40 s, drum 8.1 deg/s)	
ANCHOR	the dart digs in (kick, foot, drills)	if it reports warnings, choose Tow anyway or Release and reshoot
TIDY	slack wound onto the storage reel for a few seconds	
FLY_IN	drum on to REEL: the capstan is now driven	
CHECK	a gentle test pull must reach the check tension and hold it	
TOW	reel to the tension limit and hold	drive the rover gently, then Take up slack or Rover free
RELEASE	base stops (the worm holds the line), then the dart unscrews and folds	
RETRIEVE	capstan reels in until about 1.5 m is left	
UNWRAP	capstan wraps come off, drum back to NEUTRAL	
FINAL_APPROACH	the storage take-up pulls the last metre until the dart is seated (slow, about 0.058 m/s at duty 0.3)	
LAUNCH_READY	drum back to LAUNCH for the next shot	Shoot again, or Abort to finish
DONE	mission over; the log folder is shown	

A NO GO verdict comes before the wrap: the dart is not anchored, so it comes home on the take-up alone.

If you abort after COCK and before FIRE, the spring stays drawn and the red **LAUNCHER LOADED** banner shows: press **Decock** (in the banner or the Manual section) before you touch the launcher.

Dry mission in sim, step by step:

1. Set the target first: tap it on the rover view, or press **Auto target**. Check the range and solution.

2. Press **Preflight**. Everything should go green (base link, dart link, cameras, target). If it lists a problem, fix it and press Preflight again.
3. Press **Start mission**. Answer each prompt: Base may move, Aim, Cock, Fire, then FIRE NOW.
4. Watch the dart view: the simulated flight, impact and landing.
5. Read the landing verdict and look at the map. Press Anchor (or Retrieve and reshoot).
6. Follow the prompts through anchoring, tensioning, release and reeling in.
7. Try **STOP** (or press Esc) and **ESTOP** in the middle of a second run and check the mission stops and says why. If the dart is still out, press **Recover dart**. If the banner says LAUNCHER LOADED, press **Decock**.
8. Replay the recorded mission's vision:

```
python3 -m grappstar replay <logdir>
```

where `<logdir>` is the mission log folder (its path is printed in the terminal and shown on the page when the mission ends).

9. First real mission checklist

Before the first real launch:

- [ ] Steps 3 to 8 all worked. `python3 -m pytest` passes on the Jetson.
- [ ] Clear area: nobody in front of the launcher or anywhere along the flight path or the line. Check behind the target too. Everyone knows the shout for stop.
- [ ] Line attached at both ends (dart and reel) and laid so it cannot snag on the rover, feet or wires.
- [ ] Tow limit set, and you know what it is. The rule: the dart's hand-pull holding force (TEST_PLAN stage 8 step 2) minus 10 N, and 40 N for the first field tows until the wrap cup ring has been checked for wear (D46); firmware cap 55 N at the dart. Set it in two places: the base (`python3 -m grappstar base cmd '{"cmd": "set", "key": "limit_n", "val": 40}'`) and the Jetson (`settings.json` key `tow_limit`, or `serve --tow-limit <N>`, or the page's "Tow limit N at the dart" field). The default 40 N is only a start.
- [ ] Know the stop rule (the same in every GrappStar document):
    1. ESTOP or the page's STOP button.
    2. Pull lead #2's XT60 (motors off, the Jetson stays up).
    3. The main battery. Cutting power FREEZES a drawn spring and a taut line (the worm drives hold them); it does not release them. Treat a cocked launcher and a loaded line as live until Decock and unwrap.
- [ ] The base is live from the "Base may move" confirmation until DONE.
- [ ] Battery charged (3S full is 12.6 V), 20 A fuse fitted. Battery unplugged while you check any wiring.
- [ ] Antennas on the VTX and RC832 before power. Dart view clean on the page.
- [ ] Dart status OK (`python3 -m grappstar dart status`), base ping OK, preflight all green.
- [ ] Recording on, so you can replay it.
- [ ] HARD PREREQUISITE before any tow: `free_a` and `k_n_per_a` calibrated (firmware/base/CALIBRATION.md steps (d) free running current and (f) tension) and `limit_n` set. Reason: without a load cell the tension limit is estimated from the wiper current; with factory constants a simulated tow reached 61 N against a 40 N limit. The mission refuses the TOW step while `k_n_per_a` is still a firmware default (8, 17, 14.8 or 13.1) and there is no load cell.
- [ ] Start with a short shot at low pitch.
- [ ] Know the launcher's reachable range: it comes from `base/out/ballistics.json` (copied in at build time; `python3 -m grappstar solve <range>` answers for one range). The page says "not reachable" for anything outside it: move the rover instead.

10. Troubleshooting

Problem	Likely cause	Fix
No `/dev/grappstar-base`	base serial not saved yet, rules not installed, cable in the UART port, charge-only cable, or a different USB ID	`ls /dev/ttyACM* /dev/ttyUSB*`, `lsusb` (expect 303a:1001). Use the DevKitC port marked USB and a data cable. With only the base plugged in, rerun `bash setup/install.sh --no-apt`, answer `y`, and replug (step 4). If lsusb shows another ID, edit `setup/99-grappstar.rules`

Problem	Likely cause	Fix
Only `/dev/grappstar-base-uart`	cable in the UART/COM port	move it to the port marked USB
`/dev/ttyUSB0` appears then vanishes (CH340)	brltty grabs it	`bash setup/install.sh --remove-brltty`
`Permission denied` on the serial port or camera	not in groups dialout or video yet	`groups` should list both. If not, `bash setup/install.sh --no-apt`, then log out and in (or reboot)
Base ping times out	wrong firmware USB mode, or another program has the port	check the firmware is built with "USB CDC On Boot: Enabled". Stop the service (`sudo systemctl stop grappstar`) or other `serve` before using `base` commands
Camera black, or JPEG errors like "Corrupt JPEG data"	format mismatch with NVIDIA's JPEG library (seen in August)	set the format explicitly: `--fourcc YUYV`, or `--fourcc MJPG` if YUYV does not offer the size. Check sizes with `v4l2-ctl --list-formats-ext`
Camera "busy" or will not open	another program has it (Cheese, a second `serve`, the service, calibrate)	close it. `sudo fuser /dev/video*` shows who holds it
Dart view is snow	RC832 channel does not match the VTX, antenna missing, dart off or too far	step the RC832 channel, check both antennas, move closer
Dart view stuck or blue screen	RC832 has no signal, dongle not getting video	check the yellow video cable and RC832 power
Dart unreachable (`dart status` fails, no ping to 192.168.4.1)	Jetson not on KEDGE, dart off, or out of range	`nmcli device` (is the WiFi on `kedge`?), `nmcli connection up kedge`, power the dart, move closer
Page unreachable from the phone	phone on a different network, wrong address, or serve not running	phone must be on KEDGE (A), GRAPPSTAR-ROVER (B) or the router (C). Find the address with `hostname -I`. On the Jetson, `curl -I http://localhost:8080` checks serve is up. `systemctl status grappstar` if using the service
Phone keeps leaving KEDGE	it wants internet	tell it to stay connected; turn off mobile data switching; or use option B
`numpy.core.multiarray failed to import`, "compiled using NumPy 1.x cannot be run in NumPy 2"	numpy 2 got installed by pip	`python3 -m pip install --user 'numpy>=1.21,<2'`, then `python3 setup/check_env.py`
Two OpenCVs (`cv2` version or GStreamer NO in check_env on the Jetson)	`pip install opencv-python` was run	`python3 -m pip uninstall opencv-python opencv-python-headless`, keep the apt one
A program wants `snap install`	snap is broken on JetPack 6.2	use apt (or the project's apt or pip instructions) instead
Port 8080 already in use	the service is running, or an old `serve`	`sudo systemctl stop grappstar`, or close the other terminal. Or use `--port 8081`

11. File map

```
jetson/
  README.md              this guide
  INTERFACES.md          how the modules fit together (for programmers)
  requirements.txt       Python packages (numpy below 2, pyserial, pytest; OpenCV from apt)
  pytest.ini             test settings (tests/ folder)
  setup/
    install.sh           installs everything (step 2)
    check_env.py         prints what is installed and plugged in
    99-grappstar.rules   udev rules: /dev/grappstar-base, -base-uart, -rovercam, -dartcam
    grappstar.service    systemd unit to start the page at boot
    hotspot.sh           optional operator hotspot on a USB WiFi adapter (option B)
  grappstar/
    __main__.py, cli.py  the command line (python3 -m grappstar ...)
    app.py               owns the base, dart, cameras and mission; what the page talks to
    mission.py           the mission steps and operator prompts
    rover.py             rover side helpers
                         (defaults: mission.MissionConfig, app.AppConfig, cli.py)
    jlog.py              mission logs
    ballistics.py        aim solution from the ballistics table
    data/                placeholder ballistics table
    comms/               base.py (serial), dart.py (HTTP), mock_base.py, mock_dart.py
    video/               camera capture, discovery, frame quality, recording, replay
```

```
  vision/             flight.py, mosaic.py (aerial map), terrain.py (landing verdict), target.py (
aiming)
  sim/                simulated dart flight video and terrain textures
  ui/                 operator web page
tests/                one test file per module, run with python3 -m pytest
```

10. Calibration (base)

Where to type the commands: on the bench (ESP32 on the Mac's USB) in the Arduino Serial Monitor. Once the ESP32 is on the Jetson: `python3 -m grappstar base console` in `~/GrappStar/jetson`. It shows telemetry twice a second and every event, and sends what you type: a JSON line, or a word with key=value pairs (`reel duty=0.3 until=stop`, `set key=limit_n val=40`, `mode to=REEL`), or a bare `stop` / `estop` / `status`; `quit` ends. Only one program may hold the serial port: stop the operator page and the grappstar service first (`sudo systemctl stop grappstar`). Never move the USB plug with the battery in. On the rover, where CALIBRATION.md says "battery unplugged" only so you can turn the drum by hand or touch a contact while reading `sw`, use the motors-safe state: lead #2's XT60 out (motors dead, ESP32 and Jetson still powered).

Do it after the base is assembled and wired, springs off, before test stage 4. The bench test (section 11.1) comes first, on the bench (order of work step 7). Part (g), the cock geometry check, needs the springs: do it at test stage 5. The values are stored on the ESP32.

From `firmware/base/CALIBRATION.md`*:*

Base station calibration

Firmware `base-1.3.1`. Do BENCH_TEST.md first. Sections (a) and (b) can be done on the bench. The rest needs the base assembled. Do them in order: (d) needs REEL from (c), and (f) needs (b) and (d).

- Two ways to send the commands below:
  - on the bench: the Arduino Serial Monitor (line ending "Newline"), as in README.md;
  - on the Jetson: stop the service first (`sudo systemctl stop grappstar`), then run `python3 -m grappstar base console`. It keeps one connection, prints telemetry twice a second plus every event, and takes a typed JSON line, `key=value` words (for example `reel duty=0.3 until=stop`) or a bare `stop`, `estop` or `status`; `quit` ends it.
- Any whole `id` number works.
- Every value you `set` or learn with `cal` is **saved in flash automatically** (once everything has stopped) and is kept over a power cycle and over a new upload of base-1.3.1. Read a value back with `get`, for example `{"id":1,"cmd":"get","key":"amps_per_volt"}`.
- **Coming from an older build (base-1.0 to base-1.3.0):** the old saved bench values are ignored by base-1.3.1 (record version 3: it gained `gap_close_deg`). Everything is back at its default after the upload: redo or re-enter every value below.
- In the first 60 ms after a start-up commands are refused with `BOOTING`: just send again.
- If something faults, read the `FAULT` line (`code` and `why`), fix the cause, then send `{"id":1,"cmd":"clear"}`.
- `stop` is always there: `{"id":1,"cmd":"stop"}`. Keep it pasted in the box, ready to send.
- **Battery unplugged** before you move any wire, plug, contact or the meter leads. Never pull a plug while the battery is in. The one exception: lead #2's XT60 is the physical motor kill switch and may be pulled at any time.
- **Motors-safe state** (the build guide's term): lead #2's XT60 out. The motors are dead, but the ESP32 and the Jetson stay up (on the rover they run from the same LiPo, so unplugging the battery would turn the ESP32 off too). Use it for every step that only looks, turns something by hand or reads `sw`. On the bench, with the ESP32 on USB from the Mac, "battery unplugged" is the same thing. **Motors on** = lead #2's XT60 (and the battery) in.

Nothing here has been tried on the real base yet. Expect to adjust.

New build: tell the firmware the launcher is at rest

On a new build (or after a fresh upload with no saved state) the firmware does not know where the launcher is: the shot phase is `UNKNOWN`. Then `aim` is refused (`PHASE_UNKNOWN`) and leaving LAUNCH is refused (`LAUNCH_BUSY`), because only the wiper worm holds the spring and the firmware will not shift while the spring might be drawn.

Look at the launcher: carriage fully forward on its stop (`H` in telemetry `sw`) and the big gear in its gap (`G`). Then send `{"id":1,"cmd":"cal","what":"cock_rest"}`. Use `"force":true` only when you have SEEN the carriage on its stop and a contact is known broken: the firmware then trusts you completely, and a wrong force on a drawn spring can FIRE on the next forward jog or `cock`. `SWITCH_DISAGREE` means `SW_HOME` or `SW_GAP` is open: the carriage is not home or the big gear is not in its gap, or a contact is not touching.

(a) AS5600 magnet and direction

1. Motors-safe state. Magnet in its cap on the shaft end, 1 to 2 mm from the chip, centred. Any small neodymium disc mounted edge-on works (decision D12), or a 6 x 2.5 mm diametric disc.
2. Send `{"id":1,"cmd":"status"}`. You want `"magnet":"OK"`.
   - `NONE`: no magnet seen or no I2C. Check the STEMMA QT cable and the magnet.
   - `WEAK`: move the magnet closer. `STRONG`: move it further away.
3. The wiper output cannot be turned by hand (its worm locks it). So for this check turn the magnet instead: before it is glued, hold it over the chip at 1 to 2 mm and turn it slowly (or turn the AS5600 board round the magnet). Watch `wiper_deg` in telemetry: it must change smoothly, about 360 per turn. Then fix the magnet in its cap.
4. Direction (needs the motors on; done once in BENCH_TEST.md step 4, check it again in the base). Motors-safe state: check nobody's hands are near the shaft. Motors on. Drum in NEUTRAL. Send a FORWARD jog (a backward jog is only allowed at the LAUNCH park, `BACKSTOP` elsewhere):
   ```
   {"id":2,"cmd":"jog","motor":"wiper","dir":1,"ms":500}
   ```

Dir 1 is FORWARD. `wiper_deg` must go UP. If it goes DOWN, wait until it has stopped, then send:

```
{"id":3,"cmd":"set","key":"enc_sign","val":-1}
```

Jog again: now it must go up. (`BUSY` means it was still moving: send again.) The firmware also catches a wrong sign by itself while driving hard: `FAULT ANGLE_SENSOR` "AS5600 counts against the drive (enc_sign?)".

5. In `status`, `glitches` counts rejected AS5600 readings. A few are fine. If it climbs fast while the motor runs, route the AS5600 cable away from the motor wires.

(b) Wiper current

Current zero

1. Motors on, wiper stopped (`"amps"` should be near 0 anyway).
2. Send `{"id":1,"cmd":"cal","what":"current"}`. `amps` now reads 0.0 when stopped. `OUT_OF_RANGE` means the zero would be over 5 A: something is wrong with the IS wiring (WIRING.md section 4).

Current scale (multimeter with a 10 A range, low duty only)

Fit the 1 uF capacitor from GPIO1 to GND first if you have it (WIRING.md section 4, strongly recommended; the sketch averages 16 reads across the PWM period, so the reading is right without it too). If WIRING.md section 4 found an onboard IS resistor, set the computed `amps_per_volt` first.

The meter's 10 A range cannot take a start surge or a stall. So this is done only at low duty, with nothing loading the shaft: a FORWARD jog with the drum in NEUTRAL runs the take-up at 40 % power with the 300 ms soft start (on the bench: the bare motor with the GPIO11 jumper on). Never in LAUNCH or REEL with the meter in.

1. Battery unplugged. Put the meter in series with the motor + wire (BTS7960 M+ to the meter 10 A jack, meter COM to the motor red wire). In the MOTOR wire, not the battery wire. Meter on the 10 A DC range.
2. Battery in, motors on. Drum in NEUTRAL. Jog forward for 1.5 s:
   ```
   {"id":2,"cmd":"jog","motor":"wiper","dir":1,"ms":1500}
   ```

Sending the same line again before it ends keeps it running (copy the line, then paste and Enter a few times).

3. Read the meter once it is steady, and `amps` in telemetry at the same moment.
4. `new value = old value x meter amps / shown amps`. The old value is 12.5 unless you changed it. Example: meter 1.8 A, shown 1.5 A: 12.5 x 1.8 / 1.5 = 15.0. Send `{"id":3,"cmd":"set","key":"amps_per_volt","val":15.0}`.
5. Jog again and check they now agree. Then battery unplugged, take the meter out, reconnect the motor wire directly.
6. From now on every higher current (full-duty free current, stall, `peak_a` in `status`) comes from `amps` in telemetry, never from the meter.

(c) Drum direction, modes, wrap and unwrap

The drum position comes from three code contacts (a Gray code, BENCH_TEST.md step 2). The code is absolute, so there is no home search: the base always knows which sector the drum is in, even after a power cut. The drum turns slowly, about 8 deg per second.

1. Motors-safe state. The drum N20 is in place and in mesh (`n` in telemetry `sw`), the three drum code contacts are wired (the bench jumpers are OFF), and the line is slack.
2. On a new build send `{"id":1,"cmd":"cal","what":"cock_rest"}` first (carriage forward, `H` and `G` closed: see "New build" above). Without it every drum move out of LAUNCH or L_TO_N is refused with

LAUNCH_BUSY.

3. Still in the motors-safe state (the ESP32 stays up). `{"id":2,"cmd":"cal","what":"drum"}` replies `"ok":true` and sends a `MODE` event with the `drum_pos` the code shows. `POSITION_UNKNOWN` means the code is invalid (000, or a jump): check the code contacts (BENCH_TEST.md step 2).
4. Get the drum to NEUTRAL (`drum_pos NEUTRAL`, only b of the code closed) by hand, because the motor direction is not checked yet. Still in the motors-safe state: loosen the cradle swing screw (M3, in the slot), swing the N20 out of mesh until n disappears from `sw`, turn the hand lever until only b is closed, swing the N20 back and tighten the screw (n closed). Turning the drum while the board runs gives `FAULT DRUM_SENSOR` "drum code changed with the drum still": expected, send `{"id":3,"cmd":"clear"}` once the drum is still (one `clear` is enough). When the N20 swings back into mesh the firmware checks the code against the other contacts (at NEUTRAL: both dog contacts open and the cup at home, `W` absent); a mismatch is `SWITCH_DISAGREE` "NEUTRAL code with a dog contact closed" or "... with the wrap cup away from home". The drum angle is then a guess, so the next drum move first runs to a code edge and back (re-reference).
5. Direction check. Motors on. Drum at NEUTRAL. Send a drum jog toward REEL (the longest jog, 1.5 s, is about 12 deg):

   ```
   {"id":4,"cmd":"jog","motor":"drum","dir":1,"ms":1500}
   ```

The drum must move into the next sector toward REEL: `sw` shows b and c closed (code 0 1 1, WRAP_PREP), `drum_pos BETWEEN`. Then go back with `{"id":5,"cmd":"jog","motor":"drum","dir":-1,"ms":1500}`: only b again.

6. If instead `sw` shows a and b (code 1 1 0, L_TO_N, toward LAUNCH), the drum turned the wrong way: battery unplugged, swap the two N20 wires at AOUT1 and AOUT2. Bring the drum back to NEUTRAL with the hand lever as in step 4, battery in, motors on, and repeat step 5.
7. Now try the modes and wait for each `MODE` event with `"ok":true` (about 5 s each):

   ```
   {"id":6,"cmd":"mode","to":"LAUNCH"}
   {"id":7,"cmd":"mode","to":"NEUTRAL"}
   ```

After each one, telemetry `drum_pos` and `mode` must name that position, and the dog collar must sit where it should (look). In LAUNCH `l` must be closed; in NEUTRAL `l` and `r` open.

8. Wrap, REEL and unwrap. Line threaded as in the build, the dart off the rail, a little line pulled out and lying slack. Stand clear of the line.

   ```
   {"id":8,"cmd":"wrap"}
   ```

The drum turns through the WRAP window (the cup lays 7 turns on the capstan, `wraps` counts up in telemetry) and stops at the WRAPPED park (314 deg). Wait for the `WRAPPED` event (about 35 s; no `MODE` event comes). Telemetry: `drum_pos WRAPPED`, `mode NEUTRAL`, `wraps` 7, `l` and `r` open. Then:

```
{"id":9,"cmd":"mode","to":"REEL"}
```

Wait for `MODE` with `"mode":"REEL"` (about 5 s). `r` must be closed. Then:

```
{"id":10,"cmd":"unwrap"}
```

Wait for `UNWRAPPED` and `MODE` with `"mode":"NEUTRAL"` (about 40 s). `wraps` back to 0.

`mode REEL` is refused (`NOT_WRAPPED`) unless the drum is past the wrap window: always `wrap` first. And with the wraps on, `mode LAUNCH` or `mode NEUTRAL` is refused (`WRAPS_ON`): always use `unwrap`. A drum `jog` is refused inside the wrap window (`WRAPS_ON`) and a running drum jog stops when it enters it: only `wrap` and `unwrap` cross it.

9. The wiper turns a little during these moves. Keep clear:
   - leaving LAUNCH it first turns 6 deg BACKWARD (the centering jog, stopped early if the current rises) so the dogs carry no load;
   - leaving REEL (start of `unwrap`) it first turns forward a little (up to 16 deg), then 6 deg back;
   - entering REEL it creeps forward slowly from drum 318 deg until the dogs bite (`r` closes, or the current rises 0.8 A); if `r` is still open at the park it winds in 5 deg at a time, stopping at an estimated 20 N (the fly-in);
   - entering LAUNCH, if `l` does not close, it turns BACKWARD slowly (up to 40 deg) until the dogs drop in;
   - during `unwrap` it turns forward at 40 % while the drum is in WRAP and WRAP_PREP (the take-up winds the freed line).

These moves need the AS5600: without it `mode`, `wrap` and `unwrap` are refused with `ANGLE_SENSOR`. If the dogs never drop in: `FAULT DRUM_TIMEOUT` "dogs did not engage (collar contact open)" (or "... line tension reached the fly-in limit" when the line is taut). Check the `l` / `r` contact closes when the collar is engaged. `SWITCH_DISAGREE` "collar contact disagrees with the drum code" means `l` or `r` was closed where the collar cannot be engaged (both

are checked during every drum move).

10. If you get `SW_ARM counted n of 7 cup turns` (`DRUM_SENSOR`): the `SW_ARM` contact (`W`) is closed while the cup is away from home and must OPEN for about -6 to +8 deg at home (cad-reel); the firmware counts one cup turn per opening. Check it with the drum still: `W` must disappear from `sw` while the cup is at home.
11. The drum speed (`drum_dps`, default 8.1) is learned by itself between code edges. See it with `{"id":11,"cmd":"get","key":"drum_dps"}`. Nothing to do.
12. If you get `DRUM_TIMEOUT` or `DRUM_SENSOR`: read the `why`. Motors-safe state, check the code contacts close and open cleanly as the drum turns (N20 swung out of mesh, `n` open, turn the hand lever), then `clear` and try again.

(d) Free running current

The tension estimate uses the current above free running in REEL. Learn it with the wraps on and the line slack.

1. Line slack. Drum in REEL: `{"id":1,"cmd":"wrap"}`, wait for `WRAPPED`, then `{"id":2,"cmd":"mode","to":"REEL"}` and wait for `MODE` REEL.
2. Send `{"id":3,"cmd":"jog","motor":"wiper","dir":1,"ms":1500}`, then within the 1.5 s, after at least 0.8 s, send `{"id":4,"cmd":"cal","what":"free"}`. `NOT_RUNNING` means you were too early or too late: try again.
3. Easier way: jog, read `amps` in telemetry while it runs steady, then `{"id":5,"cmd":"set","key":"free_a","val":1.6}` (your number).

(e) Aim

You need a phone inclinometer app (any "level" app that shows degrees). Base assembled, no dart, nobody in front of the rail.

1. Motors-safe state. Fix the phone flat on the rail (tape or a rubber band), along the rail.
2. Motors on. `{"id":1,"cmd":"mode","to":"LAUNCH"}`.
3. The launcher must be at rest: carriage fully forward, `H` and `G` in telemetry `sw`. On a new build, declare it (top of this file).
4. See the allowed range: `{"id":3,"cmd":"get","key":"pitch_range"}` (it replies `min` and `max`, about 6 and 47).
5. Aim at a middle pitch: `{"id":4,"cmd":"aim","deg":30}`. After every shift into LAUNCH the aim is unknown, so it first turns backward until `SW_RAIL_LOW` closes (home), then goes to 30. That can take up to about half a minute: one aim cycle is 10 wiper turns. Wait for the `AIMED` event with `"ok":true`. At the end the wiper may turn FORWARD slowly for a moment: the backward aim run can drag the big gear back out of its gap (`G` opens), and the gap run closes it again before `AIMED`. `BAD_STATE` "SW_GAP did not close in the forward gap run" means check the `SW_GAP` contact.
6. Read the phone. Say it shows 33.4. Send `{"id":5,"cmd":"cal","what":"aim","deg":33.4}`. This sets `aim_offset` so the reported pitch matches the phone.
7. Check at 3 pitches: aim at a low, middle and high pitch inside the range (for example 15, 30, 42). At each, compare the phone with the `pitch_deg` in the `AIMED` event. Within about 1 deg is good.

If the error is different at different pitches

Then the offset alone cannot fix it. Most likely `SW_RAIL_LOW` closes at a different point of the aim train than the firmware assumes (`aim_home_deg`, the aim-train angle where `SW_RAIL_LOW` closes on the backward run).

1. Find the size of the error. First clear the offset: `{"id":1,"cmd":"set","key":"aim_offset","val":0}`. Jog backward in short steps: `{"id":1,"cmd":"jog","motor":"wiper","dir":-1,"ms":1500}`. Pick a pitch in the middle of the range where the rail moves quickly, for example 30. Write down `wiper_deg` when the PHONE reads 30, and `wiper_deg` when telemetry `pitch_deg` reads 30 (on the same up or down stroke).
2. D = the difference between the two `wiper_deg` numbers (as a positive number).
   - Telemetry reached 30 AFTER the phone did: the firmware is behind. New `aim_home_deg` = old + D.
   - Telemetry reached 30 BEFORE the phone did: new `aim_home_deg` = old minus D.

Get the old value with `{"id":2,"cmd":"get","key":"aim_home_deg"}` (default 3502: where cad-sensors' contact closes, about 10 crank deg before the low dead point), then set the new one, for example `{"id":3,"cmd":"set","key":"aim_home_deg","val":3480}`. The allowed range is -7200 to +7200.

3. Changing `aim_home_deg` does not re-home by itself. Send `{"id":4,"cmd":"cal","what":"aim_forget"}`: the next `aim` homes again on `SW_RAIL_LOW`.

4. Redo steps 5 to 7 above.

(f) Tension (must be done before the first tow)

Do not tow before `free_a` **(section d),** `k_n_per_a` **and** `tension_offset_n` **are calibrated.** Without a load cell the firmware stops a tow on its tension ESTIMATE, not on a measured force. With the defaults on an uncalibrated base the PC system test reached 61 N against a 40 N limit. The current trip (5 to 6.9 A, set from `max_n`) is only a backstop. The Jetson refuses a tow while `tension_cal` is false (systest F28; see it with `{"id":1,"cmd":"get","key":"tension_cal"}` or in `status`). Setting `k_n_per_a` or `tension_offset_n` in this section sets it to true (saved in flash).
`{"id":1,"cmd":"cal","what":"factory"}` puts every bench value back to its default and clears it again.

`k_n_per_a` **and** `tension_offset_n` **(used without a load cell)**

The estimate is: line N = `k_n_per_a` x (`amps` - `free_a`) + `tension_offset_n`. Defaults 14.1 and -2.5 (D49 rail reel: capstan 2.307, path gain 1.18). Since D43 these constants give the DART-side line tension (the tow limit is the tension at the dart); the luggage scale at the line end measures exactly that side, so fit them straight from the scale. Pulls against a luggage scale at three loads give both numbers (this matches MEASURE_LIST G9). Do not use jogs for this: a jog trips at 3 A, about 15 N.

You need a luggage scale and a solid post. Do (b) and (d) first and do not change `free_a` afterwards. If a load cell is fitted and live (`status "hx":true`), the firmware uses it instead: skip this part.

1. Motors-safe state. Tie the line end to the luggage scale. Hook the scale to the post. Line slack.
2. Motors on. Drum in REEL with the wraps on: `{"id":1,"cmd":"wrap"}`, wait for `WRAPPED` (the wrap only parks at the WRAPPED park), then `{"id":2,"cmd":"mode","to":"REEL"}` and wait for `MODE` REEL. The capstan only pulls in REEL.
3. Stand to the side of the line, never in line with it. Wind in slowly:
   ```
   {"id":3,"cmd":"reel","duty":0.3,"until":"stop"}
   ```

(reel-in trip 5 A). Watch the scale. When it shows about 10 N, read `amps` in telemetry at that moment (on the Jetson the console prints it twice a second), then send `{"id":4,"cmd":"stop"}` (in the console a bare `stop`). The worm and sprag #2 hold the line. Write down the scale reading N1 and the current I1.

4. Send the same `reel` line again and `stop` at about 20 N (N2, I2), then again at about 30 N (N3, I3). Keep every load under 40 N: the reel-in trips at 5 A (about 44 N with the default constants) and stops at `limit_n` anyway.
5. Fit the numbers (use the lowest and the highest load):
   - new `k_n_per_a` = (N3 - N1) / (I3 - I1)
   - new `tension_offset_n` = N1 - new `k_n_per_a` x (I1 - `free_a`)

Check the middle load: N2 should be close to new `k_n_per_a` x (I2 - `free_a`) + new `tension_offset_n`. If it is off by more than 3 N, repeat the pulls.

Example with `free_a` 1.5: N1 10 at I1 2.44, N3 30 at I3 3.97. k = 20 / 1.53 = 13.1. Offset = 10 - 13.1 x 0.94 = -2.3 (your own numbers replace the 14.1 / -2.5 defaults).

6. Set them: `{"id":5,"cmd":"set","key":"k_n_per_a","val":13.1}` and `{"id":6,"cmd":"set","key":"tension_offset_n","val":-2.3}` (your numbers; the offset range is -50 to 50). Both are saved in flash.
7. To release the line: unhook the scale from the post by hand, slowly (30 N is about 3 kg: hold it firmly). The capstan never turns back (sprag #2).
8. Optional check: `{"id":7,"cmd":"reel","duty":0.3,"max_n":20,"until":"tension"}` against the scale: it should hold (`LIMIT` `"holding":true`) with the scale at about 20 N. `stop`, release as in step 7.
9. `{"id":8,"cmd":"unwrap"}` when done.

A `FAULT` `OVERCURRENT` "wiper over 5.0 A" during a pull means the load went too high or the line jammed: `clear` and use lower loads. In the tow itself the trip is the current `max_n` needs plus 1 A, at least 5 A (D34; the old `tow_trip_a` setting is no longer used). These constants cannot be set while a motor runs or the reel holds (`BUSY`): `stop` first. On a very stiff line the tow now starts slow; residual overshoot in the simulation was under 10 percent of a 40 N limit at 3000 N/m.

HX711 load cell (only if fitted)

1. Check `HELLO` said `"hx711":true`, or `status` shows `"hx":true`.
2. Line slack, motor stopped: `{"id":1,"cmd":"cal","what":"tare"}`. `tension_n` reads about 0.
3. Hang a known weight on the line, for example 1 kg = 9.81 N. Read `tension_n`.

4. `new = old x 9.81 / shown`. Get the old value with `{"id":2,"cmd":"get","key":"hx_n_per_count"}` (default 0.0000476). Set it: `{"id":3,"cmd":"set","key":"hx_n_per_count","val":0.0000512}` (your number).
5. If `tension_n` goes negative when you pull: battery unplugged, swap the A+ and A- wires on the HX711.

(g) Wiper direction in the base and cock geometry check

This checks that FORWARD really cocks, that `SW_HOME` opens where the firmware expects, and that cocking stops before the release tooth. The springs are hooked, so treat it as loaded: no dart, nobody and no hands in front of the carriage or the rail. The build guide runs this as TEST_PLAN stage 5 step 0, then sets `require_seated` false again for the dry shots.

1. Motors-safe state. Springs hooked, no dart. Carriage fully forward.
2. Motors on. `{"id":1,"cmd":"mode","to":"LAUNCH"}`. Check `H` and `G` are in telemetry `sw`.
3. On a new build: `{"id":2,"cmd":"cal","what":"cock_rest"}`.
4. `cock` needs the dart seated. For this check without a dart send `{"id":3,"cmd":"set","key":"require_seated","val":false}`. It is NOT saved: it comes back true at every start-up, so send it again after every power-up in the no-dart stages. Set it back to true at the end.
5. Aim first: `{"id":4,"cmd":"aim","deg":30}` and wait for `AIMED`. (After a shift into LAUNCH only an aim run tells the firmware where the dogs sit. Cocking straight after a shift can fault with `BAD_STATE` "carriage not drawn".)
6. `{"id":5,"cmd":"cock"}`. Watch the carriage and telemetry:
   - The carriage must move BACK (toward the springs' stretched side). If it does not move and you get `BAD_STATE` "carriage not drawn": the wiper turns the wrong way for FORWARD. Battery unplugged, swap M+ and M-, then redo (a) step 4 (`enc_sign`) and this step.
   - `H` must disappear almost at once: the first tooth takes the rack a few degrees after rest. If it opens too far from there you get `FAULT SWITCH_DISAGREE` "SW_HOME opened at the wrong angle": the contact sits in the wrong place, the dog window is much wider than 16 deg (measure it), or the number in `config.h` needs changing (tell the firmware author).
   - `cock` rises to 1 and a `COCKED` event comes. `H` stays open.
7. Look: the carriage is drawn back and the last tooth still holds it, with clear margin before the relieved tooth. If it stops too early or too late, write down how many degrees and tell the firmware author (`cockEndDeg` in `config.h`; it is not a `set` key).
8. Let it down without firing: `{"id":6,"cmd":"decock"}` (the same as `cal decock`). The wiper runs BACKWARD slowly and the spring unwinds under the worm until `H` and `G` are both closed. A `RESET` event with `"how":"decock"` comes. This also turns the aim train, so the aim is unknown afterwards (the next `aim` homes again). The decock trips at 3 A (D40). `FAULT BAD_STATE` "decock: SW_GAP never closed" means `H` closed but `G` did not: check the `SW_GAP` contact.

If a decock fails: `clear`, `{"id":8,"cmd":"mode","to":"LAUNCH"}`, then `decock` again (backward at 3 A until `H` and `G` close). `SWITCH_DISAGREE` while the carriage is visibly drawn: `{"id":9,"cmd":"decock","force":true}` (decock by travel). If the decock faults again there is no powered way out (a backward jog is `NOT_AT_REST`, and `fire` needs `COCKED`, the dart seated, an aim and a token): battery unplugged, unload the springs by hand as the build guide says, then `{"id":10,"cmd":"cal","what":"cock_rest"}`, then `aim`.

A `RESET` with `"ok":false` and `"why":"SW_GAP open: check the carriage by eye, then cal cock_rest force"` means the carriage reads home but the big gear is out of its gap (for example after a `stop` during a gap run): nothing moved. Look: if the carriage is really forward on its stop and the springs are slack, send `{"id":11,"cmd":"cal","what":"cock_rest","force":true}`, then `aim` (its gap run closes the gap). A forward jog in LAUNCH is refused (`SWITCH_DISAGREE`) while `G` is open.

9. `{"id":7,"cmd":"set","key":"require_seated","val":true}`.

(h) SW_GAP closing angle (`gap_close_deg`)

The reset after every shot runs FORWARD until the big gear is back in its gap: the firmware re-zeroes the shot angle at the moment `SW_GAP` (`G`) closes and runs on to the rest point from there. So it must know where `G` closes: `gap_close_deg`, the angle in the 553.8 deg shot cycle. Default 531: cad-sensors' contact on the bump of the big gear is closed within about 15 big-gear deg (23 wiper deg) either side of rest. If the real contact closes later than the setting, the reset runs into the first tooth and draws the carriage: `FAULT BAD_STATE` "reset ended with the carriage off its stop" after a shot. If it closes earlier, the reset stops short of rest.

The bump is symmetric about rest, so the firmware measures the other side for you: during every `cock` it notes where `G` opens on the draw, counted from the first tooth like every shot angle, and reports it as `gap_deg` in the `COCKED` event. Do this once, together with (g):

1. Do (g) up to and including step 6 (springs hooked, no dart, `require_seated` false, nobody and no hands in front of the carriage or the rail).
2. Read `gap_deg` in the `COCKED` event, for example `{"evt":"COCKED","t":51230,"ok":true,"cock_deg":415.2,"gap_deg":22.6}`. Expected about 23. No `gap_deg` means `G` never opened during the cock: the contact sits wrong or is stuck closed. Fix it before any shot.
3. `gap_close_deg` = 553.8 minus `gap_deg` (here 531.2). Send it once the wiper has stopped: `{"id":4,"cmd":"set","key":"gap_close_deg","val":531.2}` (allowed 480 to 553; `BUSY` while a motor runs). It is saved like every bench value; `cal factory` puts 531 back.
4. Go on with (g) step 8 (`decock`, wait for `RESET`) and step 9 (`require_seated` true again).
5. The first dry shots of the build guide check it: after `FIRED` the `RESET` event must come, with `H` and `G` closed.

11. Bench tests

11.1 Base

This runs on the bench before the base is assembled (order of work step 7): wiper motor clamped in a vice, drum N20 loose. Where to type the commands: on the bench (ESP32 on the Mac's USB) in the Arduino Serial Monitor. Once the ESP32 is on the Jetson: `python3 -m grappstar base console` in `~/GrappStar/jetson`. It shows telemetry twice a second and every event, and sends what you type: a JSON line, or a word with key=value pairs (`reel duty=0.3 until=stop`, `set key=limit_n val=40`, `mode to=REEL`), or a bare `stop` / `estop` / `status`; `quit` ends. Only one program may hold the serial port: stop the operator page and the grappstar service first (`sudo systemctl stop grappstar`). Never move the USB plug with the battery in.

From `firmware/base/BENCH_TEST.md`*:*

Base station bench test

No-load checks BEFORE the wiper motor goes into the base, with firmware `base-1.3.1`. Nothing here has run on real hardware yet, so this is the first time the firmware meets the parts. Go one step at a time.

Each step says what to type, what you should see, and what to do if not. Keep `{"id":9,"cmd":"stop"}` ready to paste.

Two ways to send the commands:

- On the bench: the Arduino Serial Monitor (line ending "Newline"), as in README.md "First contact".
- On the Jetson: stop the service first (`sudo systemctl stop grappstar`), then run `python3 -m grappstar base console`. It keeps one connection, prints telemetry twice a second plus every event, and takes a typed JSON line, `key=value` words (for example `jog motor=wiper dir=1 ms=500`) or a bare `stop`, `estop` or `status`; `quit` ends it.

Two rules for the whole test:

- **Battery unplugged** before you touch any wire, jumper or plug. Never pull a plug (not even the small sensor cable) while the battery is in. The one exception: lead #2's XT60 is the physical motor kill switch and may be pulled at any time.
- Never hold a turning shaft, and never put a hand near a spinning shaft.

Set-up

1. Battery unplugged. Clamp the wiper motor firmly in a vice or to the bench. **Nothing attached to its shaft.** Put a strip of tape on the shaft as a flag so you can see it turn.
2. Battery unplugged. Magnet: the base does not exist yet, so print the magnet cap early, or tape the magnet centred on the clamped motor's shaft end. Fix the AS5600 board facing it, 1 to 2 mm away, centred (a temporary holder or tape is fine). In Step 1 `status` must show `"magnet":"OK"` (not `WEAK`, `STRONG` or `NONE`) before any wiper step.
3. Battery unplugged. The drum N20 is OUT of the drum. Tape or clamp it down so it can spin freely. Tape flag on its shaft.
4. Battery unplugged. Contacts on their wires, loose on the bench. Put two jumper wires on:
   - GPIO11 (`SW_DRUM_B`) to GND. That makes the drum code read NEUTRAL. Without it the three drum contacts are all open (code 000), which the firmware treats as a wire off: `FAULT DRUM_SENSOR` "drum code 000 (wire off or bump missing)" right after start-up.
   - GPIO47 (`SW_N20`) to GND. That says the drum N20 is in mesh. Without it every drum move is refused with `HAND_MODE`.

Leave both jumpers on for the whole bench test unless a step says otherwise.

5. Battery unplugged. Wired as in WIRING.md, and the Part C multimeter checks done. The 1 uF capacitor from GPIO1 to GND is strongly recommended (WIRING.md section 4).

Step 1: USB only

Battery unplugged. Plug USB into the port labelled USB. Open the Serial Monitor (or the two Jetson terminals).

- Type: `{"id":1,"cmd":"ping"}`
- You should see: `{"id":1,"ok":true,"fw":"base-1.3.1","t":...}`, and telemetry lines with `"state":"IDLE"`, `"mode":"NEUTRAL"`, `"sw":{"b":1,"n":1}` and `vbat` about 0 (no battery).
- If not: no output at all means wrong port or "USB CDC On Boot: Enabled" not set (README.md). Telemetry but no reply means the line ending is not "Newline". `BAD_JSON` means a typo (straight quotes only, no smart

quotes). BOOTING means you sent it in the first 60 ms after a start-up: send again.

- Type: {"id":2,"cmd":"status"}
- You should see: "magnet":"OK".
- If not: NONE: check the STEMMA QT cable (3V3, GND, GPIO8, GPIO9) and that the magnet is there. WEAK: closer. STRONG: further away.

Step 2: contacts, one jumper at a time

USB only, battery unplugged. Telemetry sw lists only the CLOSED contacts. With the two jumpers on you already see "b":1 and "n":1.

Use one spare jumper wire. For each row: battery still unplugged, put the jumper from the GPIO to GND (or press the screw head onto its nut), watch sw, then take it off again before the next row. Only one contact at a time.

Close	GPIO	You should see in sw
SW_DOG_L	13	"l":1
SW_DOG_R	14	"r":1
SW_HOME	21	"H":1
SW_GAP	39	"G":1
SW_RAIL_LOW	40	"L":1
SW_SEATED	41	"S":1
SW_ARM	42	"W":1 (in the base W means the cup is AWAY from home: the contact opens at home)

Then take the GPIO47 jumper off: "n":1 must disappear. Put it back.

(SW_INDEX on GPIO35, key I, is only used on a build without the magnet and does not show in sw by default.)

Never close SW_DOG_L and SW_DOG_R together: that is a fault (SWITCH_DISAGREE), because the collar cannot be in LAUNCH and REEL at once.

Drum code contacts

The drum reads its position as a 3-bit Gray code, one contact per bit. Neighbouring sectors differ by one contact only. Walk through the rows in order, changing ONE jumper per row (GPIO10 = a, GPIO11 = b, GPIO12 = c), and check drum_pos in telemetry.

The real drum cannot turn by itself (its worm locks it), so the first change you make by hand raises FAULT DRUM_SENSOR "drum code changed with the drum still". That is expected and nothing moves. Carry on: telemetry drum_pos keeps following the code while the base sits in FAULT.

Row	Closed	drum_pos	sector (params.DRUM)
1	b	NEUTRAL	NEUTRAL
2	a b	BETWEEN	L_TO_N
3	a	LAUNCH	LAUNCH
4	a b	BETWEEN	L_TO_N
5	b	NEUTRAL	NEUTRAL
6	b c	BETWEEN	WRAP_PREP
7	c	BETWEEN	WRAP (the wrap window)
8	a c	WRAPPED	N_TO_R (the WRAPPED park is in here)
9	a b c	REEL	REEL

Then walk back the same way (rows 8, 7, 6, 5) and finish with only b closed.

The mode field also needs the collar contacts: at row 3 it says NONE until you also close SW_DOG_L (GPIO13): then LAUNCH. At row 9 it says NONE until you close SW_DOG_R (GPIO14): then REEL. At rows 6, 7 and 8 it says NEUTRAL (the collar stays in NEUTRAL there); on the way back from REEL, row 8 (a c) says NONE, because the angle is then taken as 342 (the code edge), past the 318 collar point. Take any extra jumper off again before the next row.

At the end, only b (and the n jumper) closed, send {"id":3,"cmd":"clear"}. You should see "ok":true and "state":"IDLE". One clear is enough. STILL_PRESENT means the code is not valid (000, or you changed two contacts at once): check the jumpers and send clear again.

- If a contact never shows: check its wire goes to the right GPIO and the other side to GND.
- If it shows when open: the two sides touch, or the GPIO wire touches GND somewhere.
- If it flickers on and off while held closed: the screw and nut do not touch firmly. Clean or tighten them.

- Printed contacts that rattle can still trip faults in the base later: `SWITCH_DISAGREE` "drum code contact chattering" if a code contact keeps fuzzing for more than about 1 s during a drum move, and "SW_HOME opened at the wrong angle" if the real dog window (the free play of the collar dogs) is much wider than the 16 deg the firmware assumes. Measure the dog window when the base is built (turn the collar by hand between its two flanks, read `wiper_deg`) and tell the firmware author if it is far from 16 deg.

Step 3: battery in, current zero

Battery unplugged. USB plugged in. Then plug the battery in, last. Nothing should move.

- Look: telemetry `vbat` about the same as your meter reads at the XT60 (within about 0.3 V).
- If `vbat` stays about 0: the 100 k / 22 k divider is not connected to GPIO2.
- Type: `{"id":4,"cmd":"cal","what":"current"}` (motor stopped). You should see `"ok":true` and `amps` 0.0.

Step 4: jog the wiper, and its direction

Forward jogs are done with the drum code at NEUTRAL (the GPIO11 jumper): there a forward jog only counts as take-up, so it runs at 40 % power (trip 2.5 A). A backward jog is refused with `BACKSTOP` everywhere except at the LAUNCH park (BASE_SPEC 10: backward fights the take-up bus one-way), so the backward checks further down use a LAUNCH jumper set.

4a: forward, drum code NEUTRAL

- Type: `{"id":5,"cmd":"jog","motor":"wiper","dir":1,"ms":500}`
- You should see: the shaft turns for half a second, starting gently (300 ms soft start). Telemetry `state JOG`, then `IDLE`. `amps` shows a value while it runs (not calibrated yet, so only roughly right).
- Mark on the motor which way the shaft turned for dir 1 (CW or CCW, looking at the shaft end).

What FORWARD means: dir 1 (RPWM) is FORWARD, and FORWARD must be the direction that cocks. Sprag #1 is only printed at base assembly, so on the bench FORWARD is just a label: mark the shaft direction that RPWM turns. CALIBRATION.md section (g) confirms it on the built base (a short forward jog in LAUNCH moves the carriage BACK; if not, battery unplugged, swap M+ and M-).

- Check the angle sensor agrees: with dir 1, `wiper_deg` must go UP. If it went DOWN, wait until it has stopped, then `{"id":6,"cmd":"set","key":"enc_sign","val":-1}` and jog again.
- If it does not move but the reply was `"ok":true`: battery, fuse, B+/B- or the R_EN / L_EN wires. A longer jog that does not move ends in `FAULT STALL`, because the shaft did not turn.
- If the reply is `"err":"ANGLE_SENSOR"`: the AS5600 does not see the magnet. The wiper never starts without it (on this build: the no-magnet index fallback of README.md is off by default).
- If the reply is `"err":"NO_POWER"`: the firmware does not see the battery (`vbat` under 6 V). Check the divider.

Longer run, current and speed:

- Type: `{"id":8,"cmd":"jog","motor":"wiper","dir":1,"ms":1500}`, then `{"id":9,"cmd":"status"}` after it stops.
- You should see: `rpm` in telemetry steady while it runs. It is lower than the 45 rpm spec because a jog runs at part power. `peak_a` in `status` shows the highest current of that run. A free motor should draw a few amps at most. (This jog trips at 2.5 A: if you get `FAULT OVERCURRENT` "wiper over 2.5 A" on the bare motor, write down `peak_a`, `clear` and tell the firmware author.)
- If the current is very high with nothing attached, or the motor gets hot: `stop`, battery unplugged, check the wiring.

Reverse pause:

- The firmware always stops the wiper for 200 ms before it changes direction. That is too short to time by hand (the PC tests check it). What you can check: start `{"id":10,"cmd":"jog","motor":"wiper","dir":1,"ms":1500}` and, while it runs, send `{"id":11,"cmd":"jog","motor":"wiper","dir":-1,"ms":500}`.
- You should see: the second one refused with `"err":"BUSY"`. The motor keeps going until the first jog ends. You never get a direct reversal.
- With the motor stopped, the same backward jog here is refused with `"err":"BACKSTOP"` (not at the LAUNCH park).

4b: backward, LAUNCH jumper set

1. Battery unplugged, and USB unplugged too (the board must be off while the drum code changes, or it faults with "drum code jumped over a sector"). Move the GPIO11 jumper to GPIO10 (code a = LAUNCH), and add jumpers from GPIO13 (SW_DOG_L, collar in LAUNCH), GPIO21 (SW_HOME) and GPIO39 (SW_GAP) to GND. Keep GPIO47.
2. Battery unplugged. Plug USB in. The BOOT line says "drum_moved":true (the saved angle was NEUTRAL); that is expected and not a fault (without the GPIO13 jumper the boot cross-check gives SWITCH_DISAGREE "LAUNCH code without SW_DOG_L" instead). Telemetry: "mode":"LAUNCH", "drum_pos":"LAUNCH", sw with a, l, n, H, G.
3. Plug the battery in. Declare the launcher at rest: {"id":13,"cmd":"cal","what":"cock_rest"} (it needs H and G closed).
4. Type: {"id":14,"cmd":"jog","motor":"wiper","dir":-1,"ms":500}
5. You should see: the shaft turns the other way (about 60 % power). wiper_deg goes DOWN. Every backward run in LAUNCH trips at 3 A (D40): a FAULT OVERCURRENT "wiper over 3.0 A" on the bare motor means write down peak_a, clear and tell the firmware author. NOT_IN_LAUNCH: l is missing. NOT_AT_REST: send the cock_rest of step 3 again.
6. Do not jog forward in this set-up: with SW_HOME jumpered closed a forward jog in LAUNCH faults ("SW_HOME still closed past the first tooth").
7. Back to NEUTRAL: battery unplugged, USB unplugged. Take the GPIO10, GPIO13, GPIO21 and GPIO39 jumpers off and put the GPIO11 jumper back. Plug USB in (BOOT says "drum_moved":true again, no fault), then plug the battery in. Telemetry shows "mode":"NEUTRAL" again.

Step 5: jog the drum N20

- Type: {"id":12,"cmd":"jog","motor":"drum","dir":1,"ms":300}, then the same with "dir":-1.
- You should see: the N20 turns one way, then the other. Telemetry state JOG then IDLE. (The drum code does not change: it comes from the jumper.)
- If the reply is "err":"HAND_MODE": the GPIO47 jumper is missing.
- If it does not turn: buck output 9.0 V on DRV8833 VM? AIN1 on GPIO15, AIN2 on GPIO16? Common ground? SLP stays unconnected: that is correct (decision D9). If the drum does nothing: battery unplugged, wire DRV8833 SLP to the ESP32 3V3 pin (never to VM).
- Which way is right is checked later with the drum fitted (CALIBRATION.md section c).

Step 6: fault test, AS5600 (wiper stopped)

This test is done with the wiper STOPPED. Never move the sensor, or a hand, near a spinning shaft. (Losing the sensor while the wiper runs is checked by the PC tests: the wiper stops at once with FAULT ANGLE_SENSOR.)

1. Battery unplugged. Move the AS5600 board well away from the magnet (do not pull its plug with the battery in).
2. Plug the battery in. status shows "magnet":"NONE".
3. Type: {"id":13,"cmd":"jog","motor":"wiper","dir":1,"ms":500}
4. You should see: "ok":false,"err":"ANGLE_SENSOR". Nothing moves.
5. Battery unplugged. Put the board back. Plug the battery in. status shows "magnet":"OK", and the same jog works again.
6. If the motor moved in step 3: battery unplugged at once, and do not go on. Report it.

Step 7: fault test, stall (optional: skip if unsure)

Never hold the shaft with your hands or fingers. A wiper motor is strong enough to hurt you. Only do this with a proper stop. Skip it if you do not have one: the PC tests check stall detection.

1. Battery unplugged. Take the multimeter out of the motor wire if it is still there (a stall is too much for its 10 A range).
2. Battery unplugged. Fix a solid arm to the shaft (for example a wooden or printed lever on the shaft flat with the M8 nut). Clamp a padded block to the bench so the arm hits it after about a quarter turn of dir 1 (drum code NEUTRAL).
3. Plug the battery in. Stand clear. Type: {"id":16,"cmd":"jog","motor":"wiper","dir":1,"ms":1500}
4. You should see: the arm hits the block, and within a fraction of a second the motor stops with {"evt":"FAULT",...,"code":"OVERCURRENT","why":"wiper over 2.5 A"} (this jog trip is only 2.5 A) or "code":"STALL" followed by a {"evt":"STALL",...} line. Both are a correct stop.
5. {"id":17,"cmd":"status"} shows peak_a. Write it down.

6. `{"id":18,"cmd":"clear"}`. Battery unplugged before you remove the arm.

Step 8: stop and estop

- Type: `{"id":19,"cmd":"jog","motor":"wiper","dir":1,"ms":1500}`, then at once `{"id":20,"cmd":"stop"}`.
- You should see: the motor stops at once. `{"evt":"ABORTED",...,"what":"JOG","why":"stop"}`.
- Same again with `{"id":21,"cmd":"estop"}`.
- You should see: the motor stops at once. `ABORTED` with `"why":"estop"`, then a `MODE` line with `"ok":true`: the drum code says NEUTRAL already, so the drum does not move. In the base, estop drives the drum to NEUTRAL when that is safe; with the wraps on (or another reason) the drum holds and a `MODE` line with `"err":"DRUM_HELD"` and a `"why"` comes.
- If the motor keeps running: unplug the battery. Report it.

Step 9: brownout (optional)

This needs the battery below 12 V (check `vbat`). It raises the low-battery level for a moment so a normal battery looks low.

1. Type: `{"id":22,"cmd":"set","key":"vbat_low","val":12}`
2. Type: `{"id":23,"cmd":"jog","motor":"wiper","dir":1,"ms":1500}`
3. You should see: after about 0.3 s the motor stops with `FAULT BROWNOUT` "battery sagged under load".
4. **Set it back first** (it is saved in flash): `{"id":24,"cmd":"set","key":"vbat_low","val":9.9}`. Check with `{"id":25,"cmd":"get","key":"vbat_low"}`.
5. Then `{"id":26,"cmd":"clear"}`. (Clearing before you set it back gives `STILL_PRESENT`.)

Done

Battery unplugged. Write down: which way dir 1 turns (CW or CCW from the shaft end, the FORWARD label), `enc_sign`, the `peak_a` values, anything odd. Then build the base and go on with CALIBRATION.md.

11.2 Dart

The dart bench tests are in section 6 (`firmware/dart/BENCH_TEST.md` and the dart build order, steps 6, 7 and 11). Foot check: key 4 turns the foot auger clockwise seen from the motor end, looking toward the tip.

12. First tests, bench to field

From `docs/build/TEST_PLAN.md`*:*

First tests: bench to field

Do the stages in order. Do not skip ahead when a stage fails: fix it, then repeat the stage. Write the date and the result of each stage in a notebook (or `docs/build/test_log.md`).

How to fire: `python3 -m grappstar base fire` on the Jetson. It shows mode, pitch, cock and seated, asks you to type FIRE, then sends `arm` and `fire` with the token at once (the token is only valid 3 s, too short to type by hand).

How to send base commands: on the Jetson, in `~/GrappStar/jetson`, run `python3 -m grappstar base console`. It shows telemetry twice a second and every event, and sends what you type: a JSON line from firmware/base/PROTOCOL.md (for example `{"cmd":"jog","motor":"wiper","dir":1,"ms":300}`), a word with key=value pairs (`mode to=REEL`), or a bare `stop` / `estop` / `status`. `estop` stops everything and is always safe to send. Only one program may hold the base's serial port: quit the console before `base fire` or the operator page, and stop the grappstar service first if installed.

Rules that catch people out:

- Aim before every shot: every `FIRED` and every `decock` clears the aim, and `base fire` refuses `NOT_AIMED`. Send `aim` before every `cock` and every fire.
- `require_seated` goes back to `true` at every start-up. In stages 4 and 5 (no dart) re-send `{"cmd":"set","key":"require_seated","val":false}` after every power-up.
- On the rover the ESP32 and the Jetson run from the same LiPo. When a step says "battery unplugged" only so your hands can touch the mechanism while you still read `sw`, use the motors-safe state instead: lead #2's XT60 out (both motors dead, ESP32 and Jetson still up). For any WIRING change: the whole battery out.

Safety for every stage:

- Battery unplugged before you touch any wire.
- Hands out of the gears, the drum and the rail whenever the battery is plugged in.
- From stage 5 on: nobody in front of the rail, safety glasses on, the launch area clear for 30 m.
- Never fire a dart (dummy or real) without the tether tied on.
- Stopping and approaching: build guide section 2 (`estop`, then lead #2's XT60, then the battery; and the checklist before you walk up to the base or the dart). Cutting power does not release a drawn spring or a taut line.

Stage 0. Risk coupons (before any big print)

Print the base fit coupon (`base/pack/fit_coupon/coupon_plate_1.3mf`, PLA; its checks are in `base/pack/fit_coupon/COUPON.md`) and the two day-one risk coupon plates (`base/pack/coupons/risk_plate_1_PLA.3mf` PLA and `risk_plate_2_PPA-CF.3mf` PPA-CF; settings in section 4 of the build guide). They test, in a few hours of printing, the things most likely to go wrong: the swing splitter, a printed contact, the dog clutch, the slip clutch, the bearing seats, heat-set inserts and nut traps, the key rib, the gear meshes and the shear pin. Work through the test sheet (`base/pack/coupons/COUPON_TEST.md` or `.pdf`).

Pass: every check on the coupon sheet passes. If it fails: write down the test and what you saw. First check the print itself (right material, settings as in section 4, spool dry). If the print is right, the design number the sheet names must change: ask a Claude Code session in the repo to change it and re-run the coupons, then reprint only that piece (`rcN_*.stl`). Do not start the big base parts until the coupons pass.

Stage 1. Power-on (no motors attached)

Follow SYSTEM_ELECTRICAL.md section 4, the first power-on sequence, steps 1 to 9. Both motors stay disconnected from their drivers until step 10, unless the base already passed WIRING.md Part D on the bench (then see the note at the top of that sequence).

Pass: no steady beeps on the short check, MP1584 #2 reads 9.0 V, the Jetson boots, `ping` replies, `vbat` within 0.3 V of the meter, `fault` is null. If it fails: a beep means a short (check the Y harness and the BTS7960 screws first). No reply: check the USB cable is a data cable and in the ESP32 port marked USB; see troubleshooting in the build guide.

Stage 2. Each motor alone, out of the base

Battery unplugged. Connect the wiper motor to BTS7960 M+ / M- and clamp the motor to the bench. Connect the drum N20 to DRV8833 #3 AOUT1 / AOUT2, loose on the bench. Battery in.

1. `jog` wiper dir 1 for 300 ms, then dir -1 (backward jogs are refused outside LAUNCH with `BACKSTOP`: on the bench set the drum code to LAUNCH with the jumpers BENCH_TEST.md gives). Mark with a pen arrow on the motor which way the shaft turns for dir 1. That is FORWARD: on the built base it is the direction that draws the carriage back (cocks). Before the base exists it is only a label; once assembled, a short forward jog in LAUNCH must move the carriage BACK. If it moves forward, swap the wiper leads at M+ / M- (battery unplugged).
2. `jog` drum dir 1 for 500 ms, then dir -1. Mark the shaft direction.
3. Do the current-sense calibration now (MEASURE_LIST G1, CALIBRATION.md), then watch `amps` during a 1.5 s wiper jog (jogs are capped at 1.5 s; sending the same jog again extends it).

Pass: each motor turns both ways, stops when the jog ends, no fault. Wiper free current at jog power about 1 to 2 A, safely under the jog trips (2.5 A forward, 3 A backward); a jog that trips on a bare clamped motor means something binds or the current scale is off (redo G1). If it fails: no movement on the wiper: check R_EN / L_EN wiring and the 20 A fuse. Wiper turns only one way: one of RPWM / LPWM is not connected. Drum does nothing: check 9.0 V on VM and the AIN1 / AIN2 / GND jumpers. If it still does nothing, check which board it is (MEASURE_LIST D3): a board that does not pull SLP high by itself cannot be used as wired (decision D9); write it in the test log. Wrong direction against CALIBRATION.md: swap that motor's two leads (battery unplugged).

Stage 3. Sensors alone

1. Battery unplugged, ESP32 on USB only. Before the magnet holder is on the motor, turn the magnet by hand over the chip (the wiper worm locks its own shaft, so on the built base use short `jog`s instead): `wiper_deg` changes smoothly, no jumps, and goes up for the FORWARD way (the arrow you drew in stage 2).
2. Touch each printed contact closed with a finger: its entry in `sw` changes and changes back when released.
3. Load cell (if fitted): press it, `tension_n` rises.

Pass: every sensor responds and returns. If it fails: AS5600 not found: check SDA / SCL are not swapped and the STEMMA cable is pushed in; angle jumps: magnet too far (keep 1 to 2 mm) or off centre. Contact never changes: check the GND wire and that the screw head really touches the nut.

Stage 4. Base assembled, dry cycles, NO springs, no launch line

Before the first powered cock: break three spare shear pins on the torque arm (MEASURE_LIST G14, decision D44). Springs off, no tether, no dart. Run CALIBRATION.md first (zeros and drum positions; part (g), the cock geometry check, waits for stage 5). With no dart, `arm` refuses (`NOT_SEATED`): for this stage and stage 5 send `{"cmd":"set","key":"require_seated","val":false}` (again after every power-up); it is `true` again by itself at the next start-up, so power-cycle (or set it `true`) before stage 6.

1. `mode` LAUNCH, `mode` NEUTRAL, `mode` LAUNCH: 20 rounds. Each change ends with the right `drum_pos`. (REEL is only reachable with the wraps on: step 4.)
2. `aim` 15, 30, 45, then the minimum and maximum pitch: 5 rounds. Rail reaches each angle within 1 degree (`pitch_deg`) and holds with the motor off.
3. Cock and fire with stand-in springs: without springs nothing returns the carriage after the release, so the firmware would fault every cycle. Hook two strong rubber bands (or loops of bicycle inner tube) from the spring anchor pins to the carriage hooks instead: enough to pull the carriage home, far too weak to throw anything. Then `aim` 20, `cock`, `{"cmd":"decock"}` (the carriage creeps home, `SW_HOME` closes): 5 cycles. Then, with `aim` 20 before each one, fire with `python3 -m grappstar base fire` (it arms and fires in one go, inside the 3 s token life, after you type FIRE): the carriage travels back, releases, the bands pull it home, and the big gear re-engages: 10 cycles. Take the bands off afterwards.
4. Wrap and unwrap, with a spare line (not the launch tether) through the capstan path and slack: from LAUNCH, `wrap`, `tidy`, `mode` REEL, then `unwrap` and `mode` LAUNCH (the stage 7 table without the shot): 10 cycles. Once wrapped and in REEL, send `reel` with duty 0.3, wait 3 s, send `stop`.
5. Send `estop` in the middle of each kind of move once. The wiper stops at once and the drum goes to NEUTRAL, except when the spring is drawn, the wraps are on or the drum itself faulted: then the drum holds (`DRUM_HELD`), which is correct (decision D27).
6. Fault test: `estop` first, then with nothing moving, slide only the AS5600 sensor board away from its magnet (do not unplug anything, touch nothing else). `base status` shows `"magnet":"NONE"` and a wiper jog is refused with `ANGLE_SENSOR`. Put it back, `clear`.

Pass: all cycles complete with no fault, no clicking or skipping teeth, no part warm to the touch, the drum never stops BETWEEN positions. If it fails: a STALL fault means something binds: find it by turning by hand with the

battery unplugged. Skipped teeth: check the mesh spacing and the frame screws. Drum stops BETWEEN: re-run the drum calibration; check the drum contacts close cleanly.

Stage 5. Springs on, no dart

Fit both springs and the spring guard. Rail pointing at a clear area.

0. CALIBRATION.md (g), the cock geometry check, comes first: it needs the springs. It ends by setting `require_seated` true: set it false again for the rest of this stage.
1. `aim` to the lowest pitch. Fire once with `python3 -m grappstar base fire`. Look at every part of the launch train after the shot.
2. Five more shots, `aim` before each. Check the stop bumpers, the carriage, the rack teeth and the big gear's last tooth after each.
3. Watch `amps` while cocking: note the peak (the physics expects about 5 A; decision D8 trips at 6.5 A while cocking).

Pass: the carriage stops on the bumpers, nothing cracks or loosens over 6 shots, the cocking current peaks under 6.5 A (about 5 A expected), the rail holds its pitch while cocking. If it fails: cracks at the bumpers: stop and report (no more dry shots). Rail drops while cocking: the aim worm is slipping, check its mesh. STALL while cocking: the springs are stronger than planned (redo B2) or something binds.

Stage 6. Dummy dart WITH the tether, first shots

Never fire a dart (dummy or real) without the tether tied on. The tether is what keeps it flying nose first; a shot with no line can tumble or land tail first.

Make the dummy: the plate A hull top and bottom, nose and tail cap (spare prints or the dart shells before the electronics go in), taped shut with M8 nuts or sand inside to 279 g including the 15 g nose ballast (decision D10). Paint the nose bright. Tie the line through the tail cap holes. Route the line as in the base assembly steps (for launch it bypasses the capstan). Set the payout brake with hanging weights (G4): 40 g does not pull line off, 50 g does.

1. Check `require_seated` is back to `true` (`{"cmd":"get","key":"require_seated"}`). `aim` 20 degrees. Load the dummy, check `SW_SEATED` is closed. Fire. Walk out to the dummy only after the approach checklist (section 2).
2. Film it side-on with the phone in slow motion, tape measure on the ground in view (muzzle speed, G5).
3. Measure the range. Repeat at 30 and 36 degrees (G6).
4. Check the line paid out with no loops or overrun on the storage reel. Between shots, bring the dummy home: `{"cmd":"set","key":"line_m","val":<measured m>}`, `mode` NEUTRAL, `{"cmd":"retrieve","duty":0.3}` until `SEATED`, then `mode` LAUNCH and `aim` for the next shot (no wraps needed for a slack line; the full wrap sequence is stage 7).
5. Range calibration, after 2 or 3 shots on level ground: measure each range from the ground point under the rail pivot, then on the Jetson in `~/GrappStar/base/physics`: `python3 calib.py range 20:0.5:3.6 36:0.5:4.2` (pitch:brake N:measured range m, your numbers) fits a muzzle-speed factor; `python3 ballistics.py --v-scale <factor>` rewrites `base/out/ballistics.json`, which the Jetson uses to aim (base/physics/README.md has the details). On the Mac the same two commands need the repo's Python (the Mac's own python3 has no numpy): `../../kedge/.venv/bin/python calib.py range ...` and `../../kedge/.venv/bin/python ballistics.py --v-scale <factor>`; then copy the file across from `~/GrappStar`: `scp base/out/ballistics.json <user>@GrappStar.local:GrappStar/base/out/`. Also put the factor in `base/spec/params.py` as `LAUNCH v_scale`, so a later physics run keeps it.

Pass: clean exit (no rubbing marks on the dummy), it flies nose first, payout is clean, the range is within 30 percent of the ballistics table at the same pitch (measured from the ground point under the rail pivot; the table is `base/out/ballistics.json`, shown on the operator page and in section 1 of the build guide). If it fails: short range: measure the springs (B2) and the muzzle speed, then re-run the physics. Rubbing marks: check the cradle and the barrel against the dart. Loops on the reel: raise the brake a little (never over 1 N). Big range loss with clean payout: the line catches somewhere; watch the slow-motion video.

Stage 7. Reel cycle with the dummy

The base follows one fixed command sequence (BASE_SPEC section 10.1, decision D35). The operator page runs it for you; by hand it is (each line: what you send, what to wait for):

Step	Send	Wait for
after the shot	`{"cmd":"set","key":"line_m","val":4.5}` (the measured distance)	reply
rail down first if above 15 degrees (decision D45)	`{"cmd":"aim","deg":6}`	`AIMED`
wrap at once, line still slack	`{"cmd":"wrap"}` (from LAUNCH or NEUTRAL)	`WRAPPED` (allow 90 s; the drum parks at 314: cup locked, collar NEUTRAL, clutch FIRM)
(real dart only: anchor)	dart `auto`	dart ANCHORED
tidy the slack	`{"cmd":"tidy"}`	`TIDIED`
into REEL	`{"cmd":"mode","to":"REEL"}`	`MODE` ok (REEL)
tow (stage 8 only)	`{"cmd":"reel","duty":1,"until":"tension"}`	`LIMIT` holding
(real dart: release)	`{"cmd":"stop"}` FIRST (`ABORTED`), then dart `release`	dart released (operator page)
reel in	`{"cmd":"reel","until":"stop"}`, then `{"cmd":"stop"}` when `line_m` reads about 1.5	`ABORTED`
unwrap	`{"cmd":"unwrap"}`	`UNWRAPPED`, then `MODE` NEUTRAL
final approach	`{"cmd":"retrieve","duty":0.3}`	`SEATED` (never seat it on the capstan)
back to launch	`{"cmd":"mode","to":"LAUNCH"}`	`MODE` ok

1. After a stage 6 shot, run the table top to bottom, skipping the anchor, tow and release rows.
2. Check the line runs on the launch path again (not round the capstan). `aim` 30, fire again.

Pass: the wrap lays 7 turns on the capstan, reel-in works, unwrap returns the line to the launch path, the dummy seats, and the next shot pays out cleanly. Repeat 5 times. If it fails: wrap misses the capstan: check the wrap cup position at the end of `wrap`. Line still round the capstan after `unwrap`: do not fire; unwrap again and check the drum position. `DART_STUCK` when `retrieve` passes its time limit (about 50 s from 1.5 m of line at duty 0.3, longer with more line out; `max_s` sets it): the dummy snagged; clear it by hand after the approach checklist (section 2).

Stage 7b. Bench tow (before the first field tow)

Rover chocked, base wrapped and in REEL (stage 7 table), the line end tied to the luggage scale on a post at least 5 m away (a short stiff line overshoots the limit by up to half); everyone beside the line, glasses on. Free current and tension calibrated (CALIBRATION.md (d) and (f)).

1. First pull: `{"cmd":"reel","duty":0.5,"max_n":30,"until":"tension"}`. Slacken (below), then the real one: `{"cmd":"reel","duty":0.5,"max_n":40,"until":"tension"}` (40 N: the first-tows limit, decision D46).
2. Watch the 7 wraps on the capstan the whole time: they must not walk along the capstan, and the line must not get trapped in the gap at the capstan skirt.
3. Read the scale when the base holds (`LIMIT`): it should read 35 to 45 N.

Pass: the wraps stay in place, no line trapped, the base holds at 35 to 45 N. To slacken afterwards (and if it fails: `estop` first): the capstan cannot let line out, so walk the scale end slowly toward the base, standing beside the line. If it failed: wraps walking or trapped line: check the wrap cup position and the capstan skirt; wrong tension: redo CALIBRATION.md (f).

Stage 8. The anchor in soil

The real dart, built and tested to the dart run sheet section 10. Use real soil (moist loam is the target).

1. Place the dart on the soil by hand, tether tied, line slack. `wrap` now, before the dart anchors (decision D22, the stage 7 table). Then send `/auto` from the operator page (or key a).
2. When the dart reports ANCHORED: standing beside the line, not in line with it, glasses on, pull the tether by hand with the luggage scale, slowly, up to 50 N or until it moves (G7).
3. HARD PREREQUISITE for any tow: CALIBRATION.md (d) free running current (`free_a`) and (f) tension (`k_n_per_a`, MEASURE_LIST G9) are done, and stage 7b passed. Why: with the factory values a simulated tow reached 61 N against a 40 N limit. The tow limit rule: the hand result from step 2 minus 10 N, and never above 40 N for the first field tows, until the wrap cup ring has been checked for wear (decision D46; the firmware cap is 55 N). Example: the dart held 45 N by hand, so the limit is 35 N. Set it in the firmware (`{"cmd":"set","key":"limit_n","val":35}`) and as the operator page's tow limit (jetson/README.md, settings). Re-anchor (wrap first, as in step 1). Rover chocked, everyone beside the line. Then the stage 7 table from `tidy` on: `tidy`, `mode` REEL, then the tow: `{"cmd":"reel","duty":1,"max_n":35,"until":"tension"}` (your limit).

4. `stop` the tow first, then send release (operator page, or `python3 -m grappstar dart release`): drills out, pods up, foot out, kick back. Then finish the stage 7 table: reel in to about 1.5 m, `unwrap`, `retrieve`, `mode` LAUNCH.

Pass: the dart anchors on its own, holds at least 20 N in loam, the base stops at its tension limit, the release sequence frees the dart, it reels home. If it fails: augers back out: swap that motor's leads (dart run sheet section 10). Pulls out at low force: try firmer soil, and report the force (it sets the reel limit). Base overshoots the limit: redo CALIBRATION.md (f) (MEASURE_LIST G9).

Stage 9. Full mission

Before the first mission: `free_a` and `k_n_per_a` are calibrated (CALIBRATION.md (d) and (f)), the tow limit is set by the stage 8 rule in both the firmware (`limit_n`) and the operator page, `require_seated` is `true`, and the Jetson README steps 5 to 7 are done (cameras, dart WiFi, rover camera calibration: targeting needs it).

From the operator page on the Jetson (jetson/README.md): preflight, pick a target on the Kiyo view, aim, confirm the fire prompt, flight video from the dart, landing verdict, anchor, tow, release, reel home.

1. First run: a short target well inside the range the operator page offers (about two thirds of the best range shown on the operator page), soft soil, open ground.
2. Watch every prompt: the software stops and asks before each dangerous step.

Pass: the whole sequence runs with only the expected operator confirmations, the dart lands within 2 m of the target, anchors, the rover is pulled toward the dart, the dart releases and is reeled home. If it fails: note the mission step and the fault code in the log (`mission.jsonl` on the operator page), then repeat the stage where that step was first tested.

Expected numbers (physics model)

Compare your test results with these. They are recomputed from params.py, so after you enter the measured spring and wiper numbers and re-run the physics, they update.

From `base/physics/SUMMARY.md`*:*

Base station physics: summary

Params: base/spec/params.py v1.4.0. Generated by base/physics/summary.py.

Spring (each, A228, SF 1.2 on Shigley static limits)

Item	Value
Rate used / by formula	0.125 / 0.117 N/mm
Initial tension used	7.0 N
Body coils, mass	98, 53 g
Safe force (governed by)	40.9 N (hook torsion)
Safe extension (wire) / spec cap	271 / 265 mm
Surge (fixed-fixed)	24 Hz

Launch (chain spring model)

Item	Value
Stroke, ratio, pitch	241 mm, 1.0:1, 35 deg
Spring extension full / rest	265 / 24 mm
Stored energy (2 springs)	12.1 J
Carriage force at full draw	80 N
Muzzle speed	7.05 m/s
Peak acceleration, stroke time	19 g, 57 ms
Efficiency (dart KE / stored)	57 %
Carriage KE at stop, pad peak	2.00 J, 299 N (19 mm pad)
Spring KE at stop, coil clash	0.84 J, 8.8 m/s

Flight (bead-chain tether + reel model, level ground)

Item	Value
Line path at the shot	taut (path only lengthens)
Range at 0.25 / 0.5 / 0.75 N brake	5.00 / 4.52 / 4.09 m (pitch 38)
Same shot, no tether	5.41 m

Item	Value
Flight time, apex (0.5 N)	0.93 s, 1.37 m
Impact speed, angle (0.5 N)	6.5 m/s, 53 deg down
Ballistics table	6 to 47 deg, best 38.5 deg: 4.52 m

Storage reel payout (LIGHT 0.25 N at full reel)

Item	Value
Reel empty mass, inertia full	34 g, 18.4 kg mm2
Line radius full / core	27.4 / 20.0 mm (brake +37 % at core)
Line pull in the stroke (spin-up)	2.88 N
Reel rpm max	2451 rpm
Overrun after landing	1.02 m in 0.42 s

Capstan and tow

Item	Value
Wraps for tow limit at the capstan (59 N, mu 0.15)	2.6 raw, 4 design
Take-up needed for the tow at the actual wraps (7)	0.08 N
Tow at limit: torque, speed, current	3.31 N m, 0.144 m/s, 5.14 A
Wiper at limiter (5 N m), 12.6 V: torque, current, line pull	5.0 N m, 7.0 A, 78 N (break 667 N)
Wiper motor stall (no limiter), 12.6 V: torque, current, line pull	14.1 N m, 17.0 A, 219 N
Tow sensing at the dart: N per wiper amp above free	dart = 14.2 x (I - I_free) - 1.7 N (capstan side x1.18)

Drivetrain (wiper at 11.1 V with sag)

Item	Value
aim: peak torque, current, time	0.13 N m, 1.6 A, 7 s
cock: peak torque, current, time	2.93 N m, 4.7 A, 3 s
reel: peak torque, current, time	0.38 N m, 1.9 A, 105 s
tow: peak torque, current, time	3.31 N m, 5.1 A, 35 s

To measure (then edit base/spec/params.py and run ./run_all.sh)

1. Spring hang test, each spring: hang 1, 2, 3 kg and measure the length each time (hooks included). python calib.py spring FREE_MM 1:MM 2:MM 3:MM, then SPRING k_n_per_mm and F0_n (mean of the two); the stronger spring decides the anchor holes (MEASURE_LIST B4). Every range number scales with the stored energy.
2. Wiper (MEASURE_LIST A6): free speed and current from a 40 % bench jog in telemetry, then WIPER n0_rpm = jog rpm x 2.5 (never the raw jog rpm) and free_a = amps. The stall cannot be read (jogs trip at 2.5 to 3 A, A7): keep stall_nm and stall_a.
3. Dart mass on a kitchen scale with ballast and tether knot (spec 0.279 kg), then DART mass_kg.
4. Payout brake (MEASURE_LIST G4, hanging weights: a luggage scale cannot read this low): in LAUNCH 20 g on the line must not pull line off and 30 g must (LIGHT 0.25 N); in NEUTRAL the FIRM clutch (5 N) lets line off at about 0.5 kg on the luggage scale.
5. First shots on level ground: python calib.py range pitch:brake:range ..., then LAUNCH v_scale = FACTOR.

13. Troubleshooting

First move for any problem: send `estop`; if anything still moves, pull lead #2's XT60 (section 2, Stopping). Then read the `fault` field in telemetry. The operator page STOP button sends a plain `stop` (wiper off, drum stays); `estop` also moves the drum to NEUTRAL when safe.

Base

Symptom	Likely cause	Fix
No reply from the base on the Jetson	Charge-only USB cable; cable in the UART port; wrong serial device	Data cable in the port marked USB. On the Jetson, `ls -l /dev/grappstar-base` must show a link to ttyACM0 (jetson/README.md section 4)
ESP32 resets when the wiper starts	Ground or USB power dip	Check the ESP32 GND wire to the BTS7960 GND; short USB cable; report BROWNOUT faults
Wiper does nothing, no fault	R_EN / L_EN not wired, 20 A fuse blown	Battery unplugged, check the fuse and the two enable wires
Wiper turns one way only	RPWM or LPWM loose	Re-seat the Dupont jumper
20 A fuse blows	Stall into a hard stop, short on the wiper leads	Find the bind by turning by hand; never fit a bigger fuse
Wiper runs but nothing turns, after a jam	The coupler shear pin broke (it is the torque limiter, about 5 N m, decision D26)	Battery unplugged. Find and clear the jam first. Push the pieces of the old pin out of the coupler with a 3 mm rod, push a spare printed pin (d 4, six in the print pack) in until flush, then jog gently. Print more spares from the pack if you use them up. Never replace it with a metal pin: it protects the printed gears
Fault STALL or OVERCURRENT	Something binds, springs stronger than planned	Battery unplugged, turn the train by hand, find the tight spot; redo the spring measurement
Fault DRUM_TIMEOUT	Drum N20 not turning: no 9 V, board asleep, gear slipping	Meter on DRV8833 VM (9.0 V); check AIN1 / AIN2 / GND; check the board make (MEASURE_LIST D3); check the printed pinion is still pressed on the N20's D shaft (no set screw: a loose one needs a reprint)
Fault DRUM_SENSOR or SWITCH_DISAGREE	A drum contact does not close or stays closed	Battery unplugged, ESP32 on USB only (contacts read on USB power): watch `sw` while turning the drum by hand by its gear rim; bend or re-screw the contact
Fault ANGLE_SENSOR	AS5600 cable out, magnet too far	Re-seat the STEMMA cable; check the magnet cap is fully on the shaft end and the board sits on its carrier pegs (the carrier sets the 2 mm gap: never shim it closer, MEASURE_LIST G13)
`pitch_deg` wrong after a restart	Pitch reference lost	Run the pitch part of the calibration again (section 10)
Drum stuck between positions with the power off	Expected: the worm holds it	Loosen the cradle swing screw, swing the N20 out of mesh, turn the drum by hand by its gear rim, battery unplugged. Never turn the drum by hand while the carriage is drawn (`cock` above 0 or `SW_HOME` open) or while the line is under tension: shifting under load can dry fire the launcher or drop the line. De-cock and slacken the line first.
Base cocked and you need to unload without firing		Send `{"cmd":"cal","what":"decock"}`: the wiper runs backward slowly and the springs unwind under the worm until `SW_HOME` closes. If it fails: `clear`, `mode` LAUNCH, `decock` again; then, if it says `SWITCH_DISAGREE` and the carriage is visibly drawn, `{"cmd":"decock","force":true}`; last resort the manual unload in section 2 (battery out, one spring at a time with a hook tool). Never hand-release the carriage

Rover and video

Symptom	Likely cause	Fix
Jetson turns off under load	Battery low, or voltage dip	Charge the LiPo; check the barrel plug is tight; check `vbat`
No dart video	VTX and RC832 on different channels; RC832 not powered	Set the RC832 to the VTX channel (scan button); check its LED and the RCA lead
Dart video is snow	Range, antenna off, bad channel	Antenna on both ends; try another channel
Kiyo X shows nothing or errors	JPEG library mismatch on JetPack	The software forces YUYV or MJPEG (jetson/README.md); unplug and replug

Dart

Symptom	Likely cause	Fix
Serial Monitor blank	Wrong port, monitor not at 115200, or the board needs a reset	Check Tools > Port, 115200 baud, press the XIAO reset button once (the sketch will not even compile with USB CDC On Boot disabled)
No KEDGE WiFi	XIAO not powered or crashed	Loop plug in? 5 V at the XIAO? Press reset
A motor turns the wrong way	Leads reversed	Loop plug out, swap that motor's two thin input wires at its driver (section 6, "Reverse a motor"); only if just one drill is wrong, swap that drill's motor leads
Current trip during anchoring	Stall on a stone, or the trip set too low	Re-measure stall currents (section 14 F3) and update the tuning
INA219 or BNO085 not found	I2C wiring	`z` key scans; SDA and SCL must not be swapped

14. Measure list

From `docs/build/MEASURE_LIST.md`*:*

What to measure or confirm

Everything nobody could know from here. Before printing the base, do A1 to A5, A8, group B, and H1, H3, H4 (all caliper and hang tests, no electronics needed): the base parts that depend on them should not be printed until those numbers are in `base/spec/params.py` and the base pipeline and the physics have been re-run (see "After you measure" at the end). Tools: caliper, multimeter, kitchen scale, luggage scale, phone camera, tape measure.

Photo rule: for every caliper reading, take a photo with the caliper on the part and the reading visible. Put the photos in `originals/measure_2026-11/` so the numbers can be checked later.

A. Wiper motor (params.py `WIPER`, set `measured=True` when done)

Item	How	Changes	Reprint
A1 Output shaft diameter	Caliper across the round part of the shaft, 3 places, take the smallest	`WIPER.shaft_d`	Wiper shaft coupler (drum parts)
A2 Shaft flats or spline	Flats: caliper across the flats (`shaft_flat`). Spline: count the teeth, caliper the outside and root diameter, photo end-on	`WIPER.shaft_flat` (note a spline in REPORT)	Wiper shaft coupler
A3 Shaft end thread and length	Thread a nut you know (M6, M8) onto it. Caliper the shaft length from the housing face to the tip	`WIPER.shaft_thread`, `WIPER.shaft_len`	Wiper shaft coupler
A4 Mount hole pattern	Lay the motor face down on paper, press so the holes mark it (or trace with a pencil). Then caliper each hole centre from the shaft centre along two square directions and each hole diameter. Photo of the paper with a ruler	`WIPER.mount_holes` as (x, y, dia) from the shaft axis	`frame_motor_adapter` and `frame_can_stay`
A5 Housing size	Caliper the gearbox box (length, width, height) and the motor can (diameter, length), and how far the shaft sits from each edge	`WIPER.body_box`	Frame cheeks and guard if they clash (the audit says)
A6 Free speed and free current	After CALIBRATION.md (a) magnet and (b) current scale (G1). Motor clamped, nothing on the shaft, AS5600 magnet on the shaft. A forward bench jog runs at only 40 percent power (take-up duty, 102 of 255): send `{"cmd":"jog","motor":"wiper","dir":1,"ms":1500}` and paste it again every second for about 10 s. Read `rpm`, `amps` and `vbat` in telemetry while it runs steady. Free speed scales with power: write n0_rpm = measured rpm x 2.5 (= 255 / 102), never the raw jog rpm. Check rpm by eye too: tape flag, turns in 10 s x 6. (Not with the meter in series for this run)	`WIPER.n0_rpm` (scaled), `WIPER.free_a`	Nothing; re-run physics
A7 Stall torque (optional)	The firmware trips jogs at 2.5 to 3 A, far below the 18 A stall, so the true stall current cannot be read and is not needed (the trips protect the printed parts). Optional torque check: 300 mm bar on the shaft, luggage scale anchored at 90 degrees, jog until it trips and read the peak: torque at the trip = kg x 9.81 x 0.30	`WIPER.stall_nm` only as a lower bound	Nothing
A8 Mounting boss and hub details	Caliper the boss round the output shaft (diameter `boss_d`, height `boss_h`), the hub (`hub_d`, `hub_h`), the threaded length (`thread_len`), where the motor can starts from the shaft (`can_x0`), and the bolt head and nut sizes the motor takes (`bolt_*`, M6 assumed: head 10, nut 10 AF x 5). The motor sits gearbox above the main axis (decision D20, `WIPER.orient_deg` 90): check the can clears the frame in that position	`WIPER.boss_d`, `boss_h`, `hub_d`, `hub_h`, `thread_len`, `can_x0`, `bolt_*`	`frame_motor_adapter`, `frame_can_stay`

B. Launch springs (params.py `SPRING`, set `measured=True` when done)

The two springs are extension springs with initial tension: they need a small force before they stretch at all, so one weight is not enough. Use three loads. Every range number scales with the energy the springs store, so this is the most important measurement.

Item	How	Changes	Reprint
B1 Free length L0	Steel rule or tape, hook to hook (inside of the hooks), no load (the spring is longer than a caliper)	`SPRING.free_len`	Nothing

Item	How	Changes	Reprint
B2 Rate k and initial tension F0	Use water bottles in a shopping bag, weighed on the kitchen scale: about 1, 2 and 3 kg (W1, W2, W3 in kg). Hang each from ONE spring and measure hook to hook (L1, L2, L3). Plot or compute: k (N/mm) = (W3 - W1) x 9.81 / (L3 - L1). F0 (N) = W1 x 9.81 - k x (L1 - L0). The three points should lie on a line; if L1 equals L0 (the spring did not move), drop W1 and add a 3.5 kg load (never more: 4 kg is over the 38.4 N rating), but only if L3 - L0 is under 160 mm. Do both springs. Never stretch a spring longer than L0 + 250 mm. Or let the computer fit it: `python3 base/physics/calib.py spring L0 W1:L1 W2:L2 W3:L3` (kg:mm pairs) prints the rate, the initial tension and the force at full cock. Enter the mean of the two springs in params; the stronger spring decides B4	`SPRING.k_n_per_mm`, `SPRING.F0_n` (mean of the two springs)	Nothing unless the stroke changes (physics says); re-run physics and the ballistics table
B3 Wire and outside diameter	Caliper	`SPRING.wire`, `SPRING.od`	Spring anchors only if OD differs by more than 0.5 mm
B4 Which preload hole	Decision D30: the springs are Century C-243, rated 38.4 N maximum safe load. Full cock is 265 mm extension (about 40 N each with the estimated rate; preload 23.8 mm, stroke 240.9 mm). A fallback anchor hole set F, 8.5 mm rearward, gives 256.5 mm at full cock. After B2, `calib.py spring` prints each spring's force at full cock: if the STRONGER spring is over 38.4 N, use the fallback hole set (the springs pull side by side, so each hook carries its own spring's force)	assembly choice (spring anchor holes)	Nothing

C. Parts box: confirm you have these (no new purchases)

Item	How	Changes	Reprint
C1 608 bearings (buy 10)	Look in the parts box. Needed for: the CAD uses 4 (aim crank 2, storage reel 2); the coupons borrow some (fit coupon tile A 3, risk coupon 1) and give them back	If not: the only purchase: search "608ZZ bearing 8x22x7"	
C2 One M8 x 25 bolt and nut (the CAD's bought list), and any M8 bolts or 8 mm steel rod	Look in the parts box. Needed for: an axle through 608s; rod is optional elsewhere	If not: printed PPA-CF axles (tell the coordinator, it is a params switch)	
C3 Worm N20 count: you need 4 (dart kick, deploy, foot, and the base selector drum)	Look in the parts box. Needed for: dart and base	If not: The kedge notes said 2 in hand and one to buy; the August order listed 8 worm N20s. Count them	
C4 MP1584 #2 (the spare from the dart build; confirm it, decision D19)	Look in the parts box. Needed for: drum N20 supply at 9.0 V	If not: No other way to run the DRV8833 from the 3S: tell the coordinator	
C5 DRV8833 #3 (freed from the dart by the new pin map)	Look in the parts box. Needed for: drum N20		
C6 100 kg load cell + HX711 (August order)	Look in the parts box. Needed for: optional line tension reading	If not: Leave it out: the firmware uses wiper current instead	
C7 Resistors: 2 x 10 k, 1 x 1 k, 1 x 680 R, 1 x 100 k, 1 x 22 k	Look in the parts box. Needed for: BTS7960 enable pull-downs, IS divider, battery divider (firmware/base/WIRING.md)	If not: Any value within 20 percent works for the pull-downs; tell fw-base if the dividers differ. Without the battery divider at all, send `{"cmd":"set","key":"vbat_wired","val":false}` after every start-up (it comes back true at boot, and the base then refuses with NO_POWER)	
C8 Spare XT60 plugs, 14 AWG wire, two small inline fuse holders with a 2 A and a 1 A fuse	Look in the parts box. Needed for: XT60 Y harness; fuses for the MP1584 and RC832 taps	If not: Without the small fuses: use 18 AWG for those taps and keep them under 10 cm	

Item	How	Changes	Reprint
C9 Barrel plug 5.5 x 2.5 mm with lead	Look in the parts box. Needed for: Jetson from the LiPo	If not: Run the Jetson from its own supply on the bench; the rover needs one	
C10 Yellow RCA lead	Look in the parts box. Needed for: RC832 AV out to the DIGITNOW		
C11 USB data cable for the DevKitC, STEMMA QT to Dupont cable	Look in the parts box. Needed for: base ESP32, AS5600		
C12 M3 screws (lengths from the base print list), M3 nuts, M3 heat-set inserts	Look in the parts box. Needed for: printed contacts, frame joints		
C13 AS5600 magnet (recommended): any small neodymium disc (6 to 10 mm, 2 to 3 mm thick, mounted edge-on) or a 6 x 2.5 mm diametric disc. The Adafruit board does not include one	Look in the parts box. Needed for: main shaft angle	If not: the firmware has a no-magnet mode (decision D12): print the 6-bump index cam, fit the SW_INDEX contact on GPIO35, switch SW_INDEX on in firmware/base config.h (WIRING.md section 1) and re-flash. It is less precise; a magnet is better	
C14 Launch springs: two only (decision D38)	Nothing to buy. The CAD keeps holes for a second pair, but the drive train is not rated for it (about 5.8 N m per cock, above the 5 N m shear pin and the 6.5 A trip): do not fit four	none	
C15 8 mm steel rod (or M8 bolts) for: reel shaft S, aim worm wheel axle, storage reel axle, the two spring anchor pins (double shear in post and strap); the carriage yoke pegs take 8 mm rod too (printed PPA-CF pegs are the default)	Look in the parts box. Needed for: reel and aim	If not: Printed PPA-CF shafts (the reel worker's fallback); tell the coordinator	
C16 Hardware the CAD assumes: M3 x 4 x 5 heat-set inserts; M3 screws x 6, 8, 10, 12 mm (frame) and M3 x 25 cap screws + nylocs (rover mount); M6 bolts and nuts for the wiper (head 10, nut 10 AF); one M8 nut (drum coupler); cable ties, a hose clamp, glue (magnet). Printed, nothing to find: the reel cup C-clips (the v7 clip design), the aim drag pad and idler clips, the yoke pegs	Look in the parts box. Needed for: frame, drum, launch, reel	If not: Exact counts in the base print pack / assembly steps	
C17 JST SH 4-pin (STEMMA QT) cable 400 mm or longer for the AS5600	Look in the parts box. Needed for: the 200 mm one is too short	If not: Extend with 4 soldered wires and heat shrink	
C18 Carriage stop	Nothing to find: it is a printed PLA flexure with about 19 mm travel (decision D26). TPU pads are an optional upgrade if you get TPU	none	
C19 No PETG needed: the brake rocker prints in PPA-CF with an adjustable FIRM screw	Nothing to find	none	
C20 How many 693ZZ / F693ZZ bearings are left after the dart (the dart uses 5 F693ZZ). The base needs 1 x 693ZZ (the lip roller; D49 has no tether pulleys); the coupons borrow one 693ZZ and one F693ZZ and give them back	Look in the parts box. Needed for: the lip roller on the base	If no 693ZZ is left: Tell the coordinator the count	
C21 Coupon extras, exactly as the coupon sheets list them (base/pack/coupons/COUPON_TEST.md and base/pack/fit_coupon/COUPON.md): today M3 screws x 10, 12, 16 and 20, M3 nuts and a jam nut, M3 heat-set inserts, string, and 2 x M3 x 16 for the swing splitter coupon (its drag pad and idler clip are printed, no O-ring or e-clip)	Look in the parts box. Needed for: the day-one coupons	If not: ask before buying (the only planned purchase is 608s)	

Item	How	Changes	Reprint
C22 Perfboard about 55 x 31 mm (optional)	Look in the parts box. Needed for: nothing essential. The bay has two bosses for one (2 x M3 x 10) to carry the dividers, the pull-downs and the 3V3 and GND buses, but firmware/base/WIRING.md solders the resistors at the boards and shares one GND wire	If not: leave the two bosses empty	

D. Electrical labels to read

Item	How	Changes	Reprint
D1 RC832 input voltage	Read the label or the listing. Most take 7 to 24 V. If it says 12 V only, it still runs from the 3S (12.6 V full) but report it	SYSTEM_ELECTRICAL.md run P8	
D2 Jetson DC input range	Label on the Jetson's own supply and the NVIDIA dev kit page. The 3S gives 9.9 to 12.6 V in use and sags during a wiper stall. If the minimum is above 10 V, charge earlier (3.7 V per cell at rest)	SYSTEM_ELECTRICAL.md run P7	
D3 DRV8833 carrier make	The order shows Pololu 2130 (x3). Confirm the board is marked pololu (pads VIN, GND, BIN, AIN, SLP). Pololu holds SLP high on the board, which is why the base leaves SLP unwired (D9)	TEST_PLAN stage 2 fallback if it is not	
D4 DevKitC board version	Printed on the back (v1.0 or v1.1). The RGB LED pin differs; the firmware avoids both	Nothing	
D5 Video standard	When the dart video shows on the Jetson, note the size (720 x 480 = NTSC, 720 x 576 = PAL)	jetson config	

E. Rover

Item	How	Changes	Reprint
E1 Deck size and mounting holes	Tape measure the free deck area; caliper the hole positions and diameters from one corner	params.py `FRAMEWORK` (rover mount)	Base deck / rover mount
E2 Drive train and whether the Jetson drives it	Tell the coordinator what drives the wheels and what controls it	jetson rover module	Nothing
E3 Kiyo X height and tilt on the rover	Tape measure lens to ground; phone level app on the camera for tilt	jetson camera model (target calibration)	Nothing
E5 Rover mass, wheelbase, centre of mass	Bathroom scale for the mass; tape for wheelbase and track; centre of mass: balance the rover on a broom handle, front to back	physics `recoil.py` ROVER block (recoil check)	Nothing
E4 Base launch height	Tape measure from the ground to the main shaft axis when the base is mounted	physics (ballistics table)	Nothing

F. Dart (from the KEDGE notes, still open)

Item	How	Changes	Reprint
F1 Dart v30 is the one being built	Confirm	print pack choice	
F2 Bought part weights and the finished dart weight	Kitchen scale	`DART.mass_kg`, dart physics	
F3 Stall current of each dart motor at 2S	The dart `i` key while braking the output for 1 s with a printed stick against the auger or pliers on the coupler (never a cloth or fingers), as BENCH_TEST.md says	firmware/dart tuning values (see firmware/dart/README.md)	
F4 OTOTEC BMS size and F693ZZ flange	Caliper (pocket is 40.8 x 8 x 2.1 plus 0.5 each way)	dart geometry if it does not fit	
F5 Nose ballast	Weigh it: 15 g	flight stability	

G. Bench and field numbers (measured during the TEST_PLAN stages)

Item	How	Changes	Reprint
G1 BTS7960 current sense scale	firmware/base/CALIBRATION.md (meter in series at low duty, compare with `amps`)	firmware `set` key (stored on the ESP32)	
G2 Battery voltage reading	Meter on the LiPo vs `vbat` telemetry	firmware `set` key	

Item	How	Changes	Reprint
G3 AS5600 zero and drum positions	firmware/base/CALIBRATION.md	stored on the ESP32	
G4 Storage reel payout drag	A luggage scale cannot read this low: use hanging weights in LAUNCH mode. Tie 40 g (coins in a bag, weighed) to the line end over the edge of the bench: it must NOT pull line off. 50 g must pull line off slowly. That sets the brake at 0.5 N (the design value). Above 1 N costs range, 3 N ruins the flight. Then in NEUTRAL the clutch is FIRM, 5 N (decision D24): with the luggage scale, line pulls off at about 0.5 kg	slip clutch preload (reel parts)	
G5 Muzzle speed	Phone slow-motion video side-on, tape measure laid in the flight line (same distance from the phone as the dart). Count the frames the nose takes to cross 0.5 m; speed m/s = 0.5 x slow-motion fps (240 or 120) / frames	physics model, ballistics table	
G6 Range at three pitches	Tape measure from the ground point under the rail pivot (the main shaft) to where the dart first touched down, at 3 pitches, tether on, brake unchanged	ballistics table check (jetson uses it)	
G9 Line tension per amp (dart-side tension, decision D43)	CALIBRATION.md (f): `wrap`, `mode` REEL, the line END (where the dart would be) tied to a luggage scale on a post, stand beside the line. `{"cmd":"reel","duty":0.3,"until":"stop"}`, send `stop` at about 10, 20 and 30 N (the worm holds between), read `amps` each time; stay under 40 N. Fit the slope and offset. Default law: line N = 13.1 x (amps - free amps) - 2.3. It is a model, so this check matters before any tow	firmware `k_n_per_a`, `tension_offset_n`	
G10 Tow limit (choose)	The limit is the tension at the dart (D43). The firmware ships `limit_n` 50 N and refuses anything above 55 N at the dart (D46). Keep the first field tows at 40 N or less until the wrap cup ring has been checked for wear (D46). Lower it with `{"cmd":"set","key":"limit_n","val":40}`; keep it below the holding force from G7	firmware `limit_n`	
G11 Payout brake trim	If G4 is off: turn the 2 x M3 trim screws through the reel gear and the wall (reel assembly steps), a quarter turn at a time, until 40 g on the line does not pay out and 50 g just pays out (no shims)	reel trim screws	
G12 Contact gaps	Set each contact to the gap its assembly step gives, then check in telemetry `sw` that it opens and closes at the right place: SW_HOME and SW_SEATED screw head to the nut then back off one full turn (0.5 mm); SW_INDEX about 1.5 mm on a low part of the cam; drum code contacts about 1.8 mm with the nub low. The coupon contact (risk coupon 2) is set to 0.9 mm and is only a test piece. One full turn of an M3 screw is 0.5 mm	nothing	
G13 AS5600 air gap	Magnet face to chip: 2.0 mm design, fixed by the carrier (Sensors 4 step); 0.5 to 3 mm still reads. Do not shim the board closer: its connectors would touch the turning cap. Feeler: a stack of paper	nothing	
G14 Wiper coupler shear pin (before the first powered cock, decision D44)	The torque limiter is a printed PLA pin (d 4), printed LYING DOWN, that shears at about 5 N m (3.5 to 6 N m). Break THREE spare pins on a torque arm: pin in the coupler, hub held in a vice, 300 mm bar on the output side, luggage scale at 90 degrees. N m = kg x 9.81 x 0.30 (about 1.2 to 2.0 kg). They must agree within 20 percent (risk coupon 9 is the same test)	pin diameter if outside 3.5 to 6 N m (tell the coordinator)	spare shear pins
G7 Soil holding force	Dart anchored, luggage scale on the tether, pull slowly up to 50 N, note where it moves	reel tension limit	
G8 Rail friction	Before the rail is mounted: lay the rail part on a flat board, dart in the cradle, lift one end of the board slowly until the dart slides; phone level app on the board. Friction = tan(angle)	physics (launch efficiency)	

H. Other parts the CAD assumed

Item	How	Changes	Reprint
H1 Drum N20 output shaft	Caliper the diameter (3 mm assumed) and the D flat depth	drum N20 pinion (drum_NOTES)	`drum` N20 pinion
H2 Drum N20 stall torque at 9 V	50 mm lever on the shaft, pressing on the kitchen scale; jog the drum for at most 1 s. Torque N m = kg x 9.81 x 0.05. It must stall at 0.171 N m or more at 9 V (the drum design's margin; estimate 0.22 N m). Laying the wraps is the hardest job for it, so a weak N20 shows up there first	drum gear ratio if it is much weaker (tell the coordinator)	`drum_drum`, `drum_n20_cradle` if the ratio changes
H3 CSK20PP keyway depths	Caliper the inner keyway (2.8 assumed) and outer keyway (1.5 assumed) depths	capstan tube key rib, capstan block rib (reel_NOTES)	capstan tube and block
H4 BTS7960 module	Height of the heatsink under the board (22 assumed) and the 4 mounting hole spacing (40 x 40 assumed)	`frame_ebay` standoffs	electronics bay
H5 DevKitC USB socket	Micro-USB (official board) or USB-C (clone). The slot fits both: just confirm, and have the right cable	nothing	nothing

After you measure

1. Fill in `docs/build/MEASURE_RESULTS.md` and put the photos in `originals/measure_2026-11/` (create the folder).
2. Easy way: open Claude Code in the repo and ask it to enter MEASURE_RESULTS.md into `base/spec/params.py` (`measured=True` in `WIPER` and `SPRING`), re-run the base pipeline and the physics, regenerate the firmware numbers and rebuild this guide (the commands in step 3).
3. By hand, in Terminal on the Mac:

```
 cd ~/GrappStar
 bash kedge/setup_mac.sh                      # once: builds kedge/.venv (Python and the CAD
libraries)
 # edit base/spec/params.py: the WIPER and SPRING blocks, measured=True
 FORCE=1 bash base/run_all.sh base            # assembly, audit, print pack in base/out/base/pack,
assembly guide
 sh base/physics/run_all.sh                   # physics: ballistics table for the Jetson, SUMMARY,
tests
 kedge/.venv/bin/python firmware/base/tools/gen_params.py   # firmware numbers, then re-flash the
base ESP32
 kedge/.venv/bin/python docs/build/make_guide.py            # rebuild this guide
```

The audit is expected to end NOT CLEAN with only the known items (README.md, "Open items", CAD), so the base run needs `FORCE=1` to build the pack and the assembly guide, and it ends with "Problems: audit NOT CLEAN (forced on)". Check `base/out/base/audit.txt`: if it lists a failure beyond the known items (a new COLLISION, STATIC, CLEARANCE FAIL, ERROR or other line above the `warn` lines), do not print; give the file to the Claude session. The physics run's last line says PASS, or FAIL with the steps that failed (give that output to the Claude session). It rewrites `base/out/ballistics.json`, the table the Jetson aims with: copy it to the Jetson or commit it and `git pull` there. The copy, from `~/GrappStar`: `scp base/out/ballistics.json <user>@GrappStar.local:GrappStar/base/out/`.

Fields params.py does not have yet (wiper boss and hub, N20 shaft, keyway depths, BTS7960 heatsink) go to the CAD owners through MEASURE_RESULTS.md; the Claude session maps them.

4. Reprint only the parts named in the "Reprint" column. If you have not printed them yet, print after step 3.

Numbers still assumed in params.py right now

params.py version: `1.4.0`.

params.py name	State	Who fixes it
WIPER	assumed, not measured yet	MEASURE_LIST
SPRING	assumed, not measured yet	MEASURE_LIST

15. Sources and appendices

This guide is built by `docs/build/make_guide.py` from the files below. A "NOT YET" row is a section that still shows a placeholder; re-run the script after it lands.

Source	Written by	In the repo
`base/spec/BASE_SPEC.md`	coordinator	yes
`base/pack`	cad-infra (print pack)	yes
`base/out`	cad-infra (assembly pipeline)	yes
`base/out/base/guide.pdf`	cad-infra (base assembly guide, final run)	yes
`base/out/base/pack/PRINT_SHEET.md`	cad-infra (base print sheet, final run)	yes
`base/out/base/BOM.md`	cad-infra (bill of materials)	yes
`base/pack/fit_coupon/COUPON.md`	cad-infra (base fit coupon)	yes
`base/pack/coupons/COUPON_TEST.md`	cad-infra (risk coupons)	yes
`kedge/pack_v30/KEDGE_v30_run_sheet.pdf`	dart v30 pack (done)	yes
`firmware/base/WIRING.md`	fw-base	yes
`firmware/base/README.md`	fw-base	yes
`firmware/base/CALIBRATION.md`	fw-base	yes
`firmware/base/BENCH_TEST.md`	fw-base	yes
`firmware/base/PROTOCOL.md`	fw-base	yes
`firmware/dart/WIRING.md`	fw-dart	yes
`firmware/dart/README.md`	fw-dart	yes
`firmware/dart/API.md`	fw-dart	yes
`firmware/dart/BENCH_TEST.md`	fw-dart	yes
`jetson/README.md`	jetson	yes

Appendix attached after this page (when present): the base assembly guide `base/out/base/guide.pdf`. Print separately: the KEDGE v30 dart run sheet `kedge/pack_v30/KEDGE_v30_run_sheet.pdf` (follow section 6 of this guide for the new dart pin map).

Printable one-pagers next to this guide: `docs/build/FIRST_DAY.md`, `Wiring_Cheat_Sheet_Base.pdf`, `Wiring_Cheat_Sheet_Dart.pdf`, `Commands_Cheat_Sheet.pdf`, `MEASURE_RESULTS.pdf` (fill-in).

Other documents in the repo: `kedge/pack_v30/KEDGE_v30_engineering_report.pdf` (dart physics and loads), `base/spec/BASE_SPEC.md` and `base/spec/DECISIONS.md` (base design and the decisions behind it), `firmware/base/PROTOCOL.md` (base serial protocol), `firmware/dart/API.md` (dart HTTP API), `jetson/INTERFACES.md` (Jetson software).

16. Glossary

Word	Meaning here
3S, 2S	Battery cells in series: 3S LiPo = 3 cells, 9.9 to 12.6 V in use; 2S 18350 = 2 cells, 6.0 to 8.4 V
AF	Across flats: the spanner size of a nut or hex (M3 nut 5.5 AF)
AS5600	Magnetic angle sensor: reads a magnet on the end of the main shaft to know the wiper angle
AWG	Wire size: smaller number = thicker wire. 14 AWG for the wiper, 22 to 28 AWG for signals
Backlash	The small free play between two gears in mesh
BASE_SPEC, params.py	The base design description and the file with every shared number (`base/spec/`)
BTS7960	The big motor driver board that runs the wiper motor in both directions
Buck (MP1584)	Small board that turns a higher voltage into a lower one, set with its brass screw
Capstan	A drum the line wraps round several times; friction from the wraps lets it pull hard
Cock / de-cock	Draw the launch springs back / let them back down slowly without firing
Crush rib	A thin printed rib that squashes to hold a bearing firmly in its seat
CSK20PP, sprag	One-way bearing: turns freely one way, locks the other
D9, D12, D27 ...	Numbered design decisions in `base/spec/DECISIONS.md`; they override older notes
Deadman	A move that stops by itself unless the same command is sent again in time (the `jog` command)
Dog clutch, collar	A sliding ring with teeth that locks the main shaft to one gear or another
Drum (selector drum)	The N20-driven barrel with grooves that moves the dog collar and lays the wraps
DRV8833	Small motor driver board for the N20 motors
E-clip	A small spring-steel clip that snaps into a groove on a pin to stop it sliding out
estop	Emergency stop command: wiper off, drum to a safe position
Fly-in	Entering REEL with the wiper creeping forward so the dog teeth find their slots
Fuse	Thin link that melts and cuts the power if the current is far too high
Gray code	The three drum contacts read as a 3-digit code that changes one digit at a time
Heat-set insert	Brass M3 thread pressed into a printed hole with a hot soldering iron
Hole compensation	Bambu Studio setting that makes printed holes slightly bigger or smaller
IS	The BTS7960's current-sense output, read by the ESP32 to know the wiper current
Jam nut	A second nut tightened against the first so a screw cannot turn
JSON line	One command or reply as text in curly brackets, for example `{"cmd":"ping"}`
Module (m1.5, m2)	Gear tooth size: pitch diameter = module x number of teeth
N20	Small 12 mm gearmotor. "Worm" N20s are right-angle and hold their position unpowered
PPA-CF	Strong carbon-filled nylon filament, for loaded gears and keyed shafts
Printed contact	A switch made from an M3 screw head touching an M3 nut
pty	A pretend serial port on the Jetson, used by the fake base for tests
Pull-down / pull-up	A resistor that holds a pin low (or high) when nothing else drives it
Self-tapper	A screw that cuts its own thread into plastic
SHCS	Socket head cap screw (hex key head), ISO 4762
Shear pin	A printed pin that breaks on purpose to protect the gears from the wiper's full torque
Slip clutch	Friction clutch that slips above a set force: the payout brake and the take-up
STEMMA QT	Adafruit's small 4-pin plug and cable (JST SH) for I2C sensors such as the AS5600
Telemetry	The status line the base sends 10 times a second (`tel`)
Tether	The Dyneema line between the base and the dart
VTX	The dart's video transmitter (5.8 GHz). Never power it without its antenna
XT60	The yellow battery plug used on the LiPo and its leads

GrappStar base station build guide

04 November 2026

Subsystems: drum, frame, launch, reel, sensors. Model exported 2026-11-04 13:38.
All sizes in mm. Masses and print times are estimates.

1. Safety

- Wiring always starts with the battery unplugged. Plug it in last, after a check of every connection.
- Keep hands, hair and loose cloth away from the gears, the reel and the launcher while the motor can run.
- Never stand in front of the launcher. Treat it as loaded whenever the spring is cocked.
- Release the spring before you work on the launcher.
- Wear eye protection when you cut, drill or press parts, and when you test a launch.
- Let hot printed parts and the nozzle cool before you touch them. Print PPA-CF with the enclosure closed.
- Charge LiPo batteries on a fireproof surface and never leave them charging alone.

2. First print: test coupons

Print these before any big part. Each coupon uses the same geometry and numbers as the real part. Files: base/pack/coupons (3MF plates, STL, COUPON_TEST.pdf). Also print the fit coupon (python -m tools.coupon OUT) to tune cad/lib/fits.py.

Print these before the big parts. Each coupon is made from the same geometry functions and numbers as the real part, so a pass here is a pass there. Let parts cool 10 minutes before testing. The fit coupon (tolerance tiles A to H, base/pack/fit_coupon/COUPON.md) is printed alongside.

Settings

PLA: 0.4 mm nozzle, 0.2 mm layers, 4 walls, 40 % gyroid infill unless noted, 5 top and 5 bottom layers, hole and XY compensation 0, no supports. PPA-CF: hardened steel nozzle (never brass), filament dried (80 C, 8 h) and fed from a dry box, 0.2 mm layers, 6 walls, 60 % gyroid infill, 6 top and 6 bottom layers, hole and XY compensation 0, no supports. The 3MF files carry geometry only: set the material and these settings in Bambu Studio.

If a coupon fails

To change a number: on the Mac, after bash kedge/setup_mac.sh (from the repo root), cd base and run ../kedge/.venv/bin/python -m tools.risk_coupon (it rewrites base/pack/coupons and base/pack/fit_coupon). Easier: give the failing result to a Claude session and let it change the numbers. Change one number at a time and reprint only that coupon.

Plates

plate material objects
risk_plate_1_PLA.3mf PLA 24
risk_plate_2_PPA-CF.3mf PPA-CF 26

Tests

1. Swing-arm direction splitter

- Parts: rc1_plate, rc1_sleeve, rc1_sun, rc1_arm, rc1_drag_pad, rc1_drag_shim, rc1_idler_clip, rc1_idler, rc1_output, rc1_crank, rc1_stop, rc1_stop_e
- Bought: 4 x M3x16 (two stop fingers); everything else printed
- Do: Push the drag pad root block and the shim into the arm hub pocket (finger in its recess). On the plate: sleeve on the centre pin, sun on its rib, arm on the sleeve above the sun with its pin down through the idler, printed C clip into the pin groove, output on its pin, crank on top, the two stop fingers on their columns (2 x M3x16 each). Turn the crank 10 turns each way.
- Pass: Clockwise seen from above: the arm swings the idler into the output at once and comes to rest on the engaged stop finger, the output turns, no skipped teeth. Brake the output lightly with a finger and keep turning: the arm stays on that finger (the idler is held at full depth, never pulled in deeper) and the crank still turns smoothly, no binding. Anticlockwise: the arm swings out to the other finger within about 20 deg of crank, the idler runs free and the output stays still. No jam at engagement. The pad drags the arm reliably both ways and the clip stays on.
- If it fails, change: launch_aim.py: FIT (pin and bore play), DRAG_INT and DRAG_SHIM_ADJ (drag: or print a thicker shim), clips.py for the C clip, A() alpha_d = alpha_e + 20 (swing out), _stop_post_xy (swung-out stop), _stop_post_e_xy (engaged stop: binding with the output braked means it lets the idler in too deep); launch_gears.py center_distance extra / spur backlash (mesh).

- Made from: launch_aim.sun_solid, arm_solid, idler_solid (m1.5 z30/18/30, face 12, FIT 0.35), launch_gears.spur for the output, launch_aim.A: centre distances 36.10 / 36.10, output at (11.7, 68.4), engaged arm 96.2 deg on the engaged stop (launch_aim._stop_post_e_xy), swung out 116.2 deg on the other stop (launch_aim._stop_post_xy), sleeve rib launch.RIB_W 5.8 / RIB_R 12.6; drag_pad_solid, drag_shim_solid (DRAG_INT 0.45, shim up to +0.25), pin_clip_solid

2. Printed contact switch

- Parts: rc2_switch_base, rc2_switch_lever

- Bought: M3x10 SHCS ISO 4762 + M3 jam nut, M3 nut, 2x M3x10 self tapping (pan or SHCS) through the lever root into the pad, at most M3x11.7, 2 wires, multimeter

- Do: Nut into the pillar trap (drop of glue), lever clamped with 2 M3 self tappers, contact screw head down through the lever, set the head 0.9 mm above the nut (0.5 mm per turn), lock the jam nut. Meter on beep between screw and nut. Press the lever tip 100 times.

- Pass: Measure the tip travel with calipers (depth rod from the lever tip down to the base top, note the reading at rest, press until it beeps, read again): beep within 0.8 to 1.4 mm of tip travel (design 1.14), silent when let go, tip stops on the hard stop without cracking, lever back flat, same after 100 presses.

- If it fails, change: cad/lib/switch.py contact_switch args travel, overtravel, t, b; STRAIN_LIMIT, NUT_PRESS (loose nut), SCREW_LENGTHS.

- Made from: cad.lib.switch.contact_switch(lever_len=36) with the library defaults (travel 0.8, overtravel 0.6, beam t 1.6): gap 0.91, pretravel at the tip 1.14, close force 0.83 N, strain 0.0061

3. Dog clutch and compliant fork

- Parts: rc3_base, rc3_reel_face, rc3_collar, rc3_fork, rc3_dog_post, rc3_rail

- Bought: M3 nut (yoke tab), 2 x M3x12 (SW_DOG screws in the post), 1 x M3 (post clamp), multimeter

- Do: Press the rail into the base. REEL face on the journal (dogs up), collar on the hex (dogs down), fork pads in the collar groove and its sleeve on the rail, dog post clamped on the rail below the fork, SW_DOG screws in the post set 0.5 mm short of the tab nut at full engagement. Push the fork 7 mm down with the dogs tip to tip, then turn the REEL face until they drop in. Turn the face hard both ways, then pull the fork back. Meter on beep between the lower screw and the tab nut.

- Pass: Yoke flexes when tip to tip (no jam, roofs slide off), dogs drop in within a quarter turn, under hand torque the dogs pull in, never walk out, a light pull on the fork releases them, the yoke springs back flat; the meter beeps only with the dogs fully in, never tip to tip.

- If it fails, change: _drum_kin.py DEFAULTS dog_backcut, dog_roof, dog_deg, dog_gap, fork_k; drum.py GEOM fork_t, yoke_r, groove, dog_touch, dogpost_s; fits.py SHAFT_IN_BORE (collar on the hex, face on the journal), HOLE_SLIDE (fork on the rail).

- Made from: drum.dog_teeth (dog_n 4, dog_h 5, dog_deg 36, backcut 4, roof 1.2, r 18..26), drum.fork, drum.dog_post (dog_touch 0.5), drum.rail, drum.hexprism, GEOM journal d33, groove (82.0, 88.0), stroke 7, neutral gap 2.5

4. Face slip clutch with the flexure spring

- Parts: rc4_hub_plate, rc4_friction_disc, rc4_pressure_plate, rc4_flexure, rc4_adjuster_plate

- Bought: 3 x M3x20 SHCS, 3 x M3 nut, string, coins in a small bag, kitchen scale

- Do: Stack on the key stub: hub plate (counterbores down), friction disc, pressure plate (pads up), flexure (beam down, tabs on the screws), adjuster plate (traps up); 3 screws from below, nuts on top. Run the nuts down until the flexure just touches (zero), then turn all 3 nuts the same, 1/4 turn (0.125 mm) at a time. Clamp the stub level in a vice, tie a string to a disc lug (r 28.5 mm), lay it over the rim and hang a bag of coins weighed on a kitchen scale. Add or take coins to find the hold and slip weights at each setting.

- Pass: At each setting the lower weight must not pull the disc round and the higher one must (smooth steady slip, no judder); same result 3 times; the flexure never bottoms before 1 turn. Bands: 1/4 turn: 35 g holds, 60 g slips (design 48 g, 0.013 N m); 1/2 turn: 70 g holds, 120 g slips (design 95 g, 0.027 N m); 3/4 turn: 105 g holds, 180 g slips (design 143 g, 0.040 N m); 1 turn: 140 g holds, 240 g slips (design 190 g, 0.053 N m).

- If it fails, change: cad/lib/clutch.py: slip_clutch args flex_t, flex_gap, flex_ro, r_in, r_out, n_grooves; MU_PLA_PLA, E_PLA, SIGMA_FLEX_PLA_SUSTAINED (creep: if the torque falls overnight).

- Made from: cad.lib.clutch.slip_clutch() defaults (reel does not use it: its clutch is the spool face plus the PETG rocker): shaft d20 key 6, friction r 18.5..27, mu 0.3, flexure t 0.8 gap 0.5, 0.106 N m per mm, travel 0.50; on the shared keyed stub of coupon 7

5. Bearing seats

- Parts: rc5_seats_608_693, rc5_csk20_seat

- Bought: 608ZZ, F693ZZ, 693ZZ, CSK20PP

- Do: Press each bearing in square with a flat block or a vice (tile left to right: 608, F693ZZ, 693ZZ); CSK20PP into the round boss with its outer keyway on the printed key.

- Pass: Firm press, seated flat on the lip, stays in when tapped upside down, spins as freely as before; the CSK20PP key slides in without force, it still freewheels one way and locks the other.

- If it fails, change: fits.py BRG_RIB_INTERF, BRG_RIB_BORE (all seats); bearings.py CSK20_KEY_RCLEAR, CSK20_OKEY_W (key), KEY_CLEAR in fits.py. If the 608 is off, print tile A of the fit coupon (base/pack/fit_coupon, COUPON.md).

- Made from: cad.lib.bearings.seat_608(bottom lip, lip_d 14) as reel, seat_f693(), seat_693(), seat_csk20(key_deg 90) as the reel capstan block (wall 4); fits BRG_RIB_INTERF 0.15, BRG_RIB_BORE 0.3, CSK20_KEY_RCLEAR 0.2

6. M3 holes

- Parts: rc6_m3_tile

- Bought: M3x4x5 heat-set insert, M3 nut, 2 x M3x10

- Do: Left to right: heat-set the insert (220 C), press the nut in, push a screw through, self-tap a screw in.

- Pass: Insert flush and does not spin when the screw is tightened; nut presses in and holds; screw drops through the clearance hole; the tap hole cuts a thread and holds a hard pull without splitting.

- If it fails, change: fits.py M3_INSERT_D, M3_NUT_AF, M3_CLEAR_D, M3_TAP_D (tiles C, D, E of the fit coupon in base/pack/fit_coupon bracket them).

- Made from: cad.lib.fasteners.m3_insert (d 4.0), m3_nut_trap (AF 5.8), m3_clear (d 3.4), m3_tap (d 2.6)

7. Key rib and spigot

- Parts: rc7_key_stub, rc7_key_bore, rc7_spigot, rc7_bushing

- Bought: CSK20PP from coupon 5 (inner keyway)

- Do: Slide the key stub into the keyed tile and into the CSK20PP inner race (the real interface). Push the spigot into the sleeve slice and turn it.

- Pass: Stub slides in by hand, key in the keyway, no visible rock. Play: clamp the tile, tape a ruler to the stub end, mark the ruler 60 mm from the axis, rock the stub both ways: the mark moves under 1 mm (about 1 deg). The spigot turns freely; rocked at its free end it moves under 0.5 mm.

- If it fails, change: fits.py KEY_CLEAR, SHAFT_IN_BORE; shafts.py KEY20_H, SHAFT_PRINT_UNDERSIZE; launch.py SPIGOT_BORE (now 12.35); drum.py GEOM spig_d.

- Made from: cad.lib.shafts.keyed_shaft(d 20, key_w 6, key_h 2.6) and keyed_bore (clear 0.3); drum.main_shaft (spig_d 12); launch.sleeve_solid (SPIGOT_BORE 12.35)

8. Gear meshes m1.25, m1.5, m2, m2.5

- Parts: rc8_m125_plate, rc8_m125_seg, rc8_m125_pinion, rc8_m15_plate, rc8_m15_z42, rc8_m15_z24, rc8_m2_plate, rc8_m2_z50, rc8_m2_z22, rc8_m25_plate, rc8_m25_big, rc8_m25_drive

- Bought: none

- Do: Each pair on its plate (label = module), flat faces down; turn or rock through every tooth.

- Pass: Smooth, no tight spot, a little play when rocked; m2.5: the tall tooth and the cut-down last tooth pass without catching, then the mesh lets go at the gap.

- If it fails, change: fits.py GEAR_BACKLASH, GEAR_CENTER_EXTRA (m1.25, m1.5, m2); launch_gears.py center_distance extra, spur backlash, sector_gear_2d tall_factor (m2.5); _drum_kin.py pin_x (m1.25 pinion).

- Made from: m1.25: gears.spur_gear z60 sector / z10 shift 0.4 tip_short 0.08 (drum.tip_short_for), cd 44.33 (drum.a_seg); m1.5: gears.spur_gear z42 sector / z24, cd 49.60; m2: gears.spur_gear z50 sector (drum GEOM out_m, out_z) / z22 (reel Z_CAPG), cd 72.10; m2.5: launch_gears.sector_gear_2d z40 (n_present 27, last 6 + gap) / launch_gears.spur z26, cd 82.60 (launch L); coupon face 6, fits GEAR_BACKLASH 0.15, GEAR_CENTER_EXTRA 0.1

9. Shear-pin torque limiter

- Parts: rc9_hub_lever, rc9_outer_lever, rc9_shear_pin x3

- Bought: string or a strap, luggage scale

- Do: Print the pins lying down on their flat, as exported (D44: shear plane across the layers; never stand them up). Hub slice into the outer slice, pin through both holes (head outside). Clamp the hub lever, hang the luggage scale on the outer lever hole (r 60 mm) at right angles and pull slowly until the pin shears. Break all 3 pins. Torque = kg x 9.81 x 0.06 N m.

- Pass: Each pin shears cleanly between 5.9 and 10.2 kg on the scale (3.5 to 6 N m, drum_NOTES; design 4.9 N m = 8.3 kg), the 3 pins within 20 percent of each other, both slices undamaged (holes still round); a fresh pin goes in by hand.

- If it fails, change: drum.py GEOM shear_d (torque goes with d squared), hub_od (radius), shear_s; fits for the hole: drum.coupler shear hole (shear_d + 0.1) and shear_pin (shear_d - 0.04). Spec D8 / D26 limit 5 N m.

- Made from: drum.coupler_hub and drum.coupler (sliced s 4.5..20), drum.shear_pin: d4 PLA in single shear at r 13 (shear_s 9); 4.9 N m at 30 MPa

10. AS5600 magnet cap (bench test)

- Parts: rc10_magnet_cap

- Bought: magnet: not included with the board, see MEASURE_LIST C13 (any small neodymium disc edge-on, or 6 x 2.5 diametric), M3x6, superglue

- Do: Glue the magnet into the end pocket (flush, any way round), push the cap on the shaft end (flat to the flat), M3x6 onto the flat. It fits the d12 main shaft spigot of the base (for the bare-motor bench test, tape the magnet to the wiper shaft end instead). AS5600 chip on the shaft axis, 1 to 2 mm from the magnet face.

- Pass: Cap does not wobble or slip on the shaft; the AS5600 angle follows the shaft smoothly through full turns (no jumps), magnet status OK (not too weak, not too strong) at 1 to 2 mm.

- If it fails, change: drum.py GEOM cap_s, spig_d; magnet_holder pocket d6.2 x 2.6 (magnet loose or tight); fits.py HOLE_SLIDE (cap on the spigot).

- Made from: drum.magnet_holder (spig_d 12 with its flat, cap_s (195.0, 201.5), magnet pocket d6.2 x 2.6), printed as drum prints it

Parts

part material qty print bbox mm what
rc1_plate PLA 1 110.1 x 82.2 x 33.1 Splitter base: centre pin d8, output pin d8, two stop finger columns (2 x M3 tap each)
rc1_sleeve PPA-CF 1 26.0 x 26.0 x 29.3 Sleeve stub d20 with the integral key rib (launch.RIB_W, RIB_R) under the sun and the crank
rc1_sun PPA-CF 1 47.8 x 48.0 x 12.0 launch_aim.sun_solid: sun m1.5 z30 as built
rc1_arm PPA-CF 1 61.4 x 41.0 x 19.3 launch_aim.arm_solid: swing arm as built (drag pad pocket, d8 pin, clip groove)
rc1_drag_pad PLA 1 24.6 x 8.7 x 4.0 launch_aim.drag_pad_solid: printed flexure drag pad as built
rc1_drag_shim PLA 1 9.4 x 3.0 x 4.0 launch_aim.drag_shim_solid: nominal preload shim as built
rc1_idler_clip PLA 1 10.8 x 10.0 x 1.5 launch_aim.pin_clip_solid: printed C clip for the d8 idler pin
rc1_idler PPA-CF 1 30.0 x 29.7 x 12.0 launch_aim.idler_solid: idler m1.5 z18 as built
rc1_output PPA-CF 1 47.8 x 48.0 x 12.0 Output m1.5 z30 (launch_gears.spur, face 12) on a d8 pin
rc1_crank PPA-CF 1 48.2 x 30.3 x 20.0 Hand crank, keyed on the sleeve rib like the sun
rc1_stop PLA 1 47.1 x 10.0 x 6.4 Stop finger: rounded end at the as-built swung-out stop post position
rc1_stop_e PLA 1 64.4 x 10.0 x 6.4 Engaged stop finger: rounded end at the as-built engaged stop post position (launch_aim._stop_post_e_xy)
rc2_switch_base PLA 1 54.0 x 35.0 x 11.5 contact_switch base: pad, nut pillar, hard stop, wire channel
rc2_switch_lever PLA 1 50.0 x 5.0 x 8.0 contact_switch lever, printed on its side (Y up)
rc3_base PLA 1 40.0 x 91.5 x 97.5 Base: d30 journal (REEL sleeve) and hex 22 AF (main shaft), rail hole (press)
rc3_reel_face PPA-CF 1 61.6 x 54.0 x 9.5 REEL gear dog face (drum.dog_teeth, phase 45) with a handle
rc3_collar PPA-CF 1 60.0 x 60.0 x 18.0 Collar lower half: drum.dog_teeth facing down, fork groove, hex bore

rc3_fork PLA 1 87.1 x 94.1 x 83.0 drum.fork as built: compliant yoke, rail sleeve, follower pin, M3 nut in the tab
rc3_dog_post PLA 1 20.0 x 75.9 x 41.7 drum.dog_post as built: SW_DOG_L / SW_DOG_R screw bosses, clamps on the rail
rc3_rail PPA-CF 1 110.0 x 10.0 x 9.0 drum.rail profile d10 with the print flat, cut to 110 long
rc4_hub_plate PLA 1 54.0 x 54.0 x 6.0 clutch.slip_clutch hub_plate (library defaults)
rc4_friction_disc PLA 1 57.0 x 55.0 x 3.0 clutch.slip_clutch friction_disc (library defaults)
rc4_pressure_plate PLA 1 54.0 x 54.0 x 3.5 clutch.slip_clutch pressure_plate (library defaults)
rc4_flexure PLA 1 50.0 x 50.0 x 1.7 clutch.slip_clutch flexure (library defaults)
rc4_adjuster_plate PLA 1 54.0 x 54.0 x 4.0 clutch.slip_clutch adjuster_plate (library defaults)
rc5_seats_608_693 PLA 1 60.0 x 28.0 x 9.6 608 (reel spool seat), F693ZZ and 693ZZ crush-rib seats
rc5_csk20_seat PLA 1 55.3 x 55.3 x 16.0 CSK20PP seat with the printed key (reel capstan block, key_deg 90)
rc6_m3_tile PLA 1 54.0 x 18.0 x 8.6 M3 heat-set hole, nut trap, clearance and tap holes (fits.py values)
rc7_key_stub PPA-CF 1 21.2 x 20.0 x 31.0 shafts.keyed_shaft d20 key 6 (reel capstan tube numbers), 31 long, also the coupon 4 clutch shaft
rc7_key_bore PPA-CF 1 36.0 x 36.0 x 8.6 shafts.keyed_bore d20 key 6, 8 long (keyed hubs on the sleeve rib are PPA-CF)
rc7_spigot PPA-CF 1 28.0 x 25.4 x 22.0 drum.main_shaft slice: hex 8 long + d12 spigot 20 long, on a hex flat
rc7_bushing PPA-CF 1 32.0 x 32.0 x 16.0 launch.sleeve_solid slice 16 long: spigot bore d12.35
rc8_m125_plate PPA-CF 1 62.3 x 18.0 x 11.0 Pins at 44.33
rc8_m125_seg PLA 1 38.1 x 46.7 x 6.0 drum segment gear m1.25 z60 sector
rc8_m125_pinion PPA-CF 1 15.8 x 15.2 x 6.0 drum wrap pinion m1.25 z10 x0.4 tip_short 0.08
rc8_m15_plate PPA-CF 1 67.6 x 18.0 x 11.0 Pins at 49.60
rc8_m15_z42 PPA-CF 1 41.0 x 29.6 x 6.0 drum wrap idler output m1.5 z42 sector
rc8_m15_z24 PPA-CF 1 38.9 x 38.9 x 6.0 reel cup ring gear m1.5 z24
rc8_m2_plate PPA-CF 1 90.1 x 18.0 x 11.0 Pins at 72.10
rc8_m2_z50 PPA-CF 1 61.0 x 39.3 x 6.0 drum REEL output m2 z50 sector
rc8_m2_z22 PPA-CF 1 48.0 x 47.7 x 6.0 reel capstan gear m2 z22
rc8_m25_plate PPA-CF 1 100.6 x 18.0 x 11.0 Pins at 82.60
rc8_m25_big PPA-CF 1 61.5 x 49.0 x 6.0 launch big gear m2.5 z40 sector: teeth 21..26 (tall, relieved)
rc8_m25_drive PPA-CF 1 42.5 x 44.0 x 6.0 launch drive gear m2.5 z26 sector
rc9_hub_lever PPA-CF 1 79.0 x 26.0 x 15.5 drum.coupler_hub slice (shear pin hole) with a lever, string hole at r 60
rc9_outer_lever PPA-CF 1 86.0 x 40.0 x 9.5 drum.coupler slice (shear pin hole) with a lever, string hole at r 60
rc9_shear_pin PLA 3 12.1 x 5.0 x 4.1 drum.shear_pin as built (d4 PLA), printed LYING on its 0.4 mm flat as drum prints it (D44): the shear plane runs across the layers
rc10_magnet_cap PLA 1 20.0 x 20.0 x 6.5 drum.magnet_holder as built: AS5600 magnet cap on the d12 spigot end

3. Parts list

Printed parts

part	subsystem	material	qty	mass g	notes
drum_bracket_b	drum	PLA	1	25.6	Drum bracket B: axle, guide rod, wrap idler journal; foot screwed to the reel head spine
drum_dog_post	drum	PLA	1	7.1	Dog contact post on the shift rail: SW_DOG_R and SW_DOG_L screws (collar fully engaged)
drum_drum_a	drum	PLA	1	103.2	Drum A: N20 gear 120T m0.8, groove 1 barrel (follower), hex spigot
drum_drum_b	drum	PLA	1	82.4	Drum B: brake cam (spool clutch lever), Gray code bands A B C, wrap segment gear + locking arc
drum_follower	drum	PLA	1	6.2	Follower slider: pin in groove 1, slides along the guide rod; the link takes its motion to the bridge
drum_fork	drum	PLA	1	19.9	Compliant shift fork: thin yoke is the spring (about 1.5 N/mm), lug for the bell crank pin, contact tab
drum_index_block	drum	PLA	1	33.2	SW_INDEX block: column on the motor plate front face (2 x M3 x 10 into the frame inserts), M3 nut contact
drum_index_lever	drum	PLA	1	0.9	SW_INDEX lever on the 6-bump coupler cam (D12 fallback encoder)
drum_link_eye_1	drum	PLA	1	3.0	Link eye pressed on the rod end, M3 screw through the follower lug
drum_link_eye_2	drum	PLA	1	3.0	Link eye pressed on the rod end, M3 screw through the bell crank input arm
drum_magnet_cap	drum	PLA	1	1.9	AS5600 magnet cap on the spigot end (6 x 2.5 diametric or edge-on disc), M3 x 6 onto the flat

part	subsystem	material	qty	mass g	notes
drum_n20_cradle	drum	PLA	1	12.1	N20 cradle on the reel head front plate: pivot + swing screw (hand fallback), SW_N20 contact
drum_n20_strap	drum	PLA	1	2.7	End plate closing the N20 cradle, 2 x M3 x 8
drum_shear_pin	drum	PLA	6	0.2	Torque limiter shear pin d4 (about 5 N m), prints lying; print 6, 5 are spares
drum_sw_lever_0	drum	PLA	1	0.9	Code switch lever A: flexure, round nub, M3 x 6 contact screw
drum_sw_lever_1	drum	PLA	1	0.9	Code switch lever B: flexure, round nub, M3 x 6 contact screw
drum_sw_lever_2	drum	PLA	1	0.9	Code switch lever C: flexure, round nub, M3 x 6 contact screw
drum_switch_block	drum	PLA	1	18.3	Code switch block on top of the drum: 3 lever feet, 3 M3 nut contacts (SW_DRUM_A, B, C)
frame_cable_anchor_1	frame	PLA	1	4.0	cable-tie anchor (strain relief), 1 x M3 x 12 into the deck
frame_cable_anchor_2	frame	PLA	1	4.0	cable-tie anchor (strain relief), 1 x M3 x 12 into the deck
frame_camera_mount	frame	PLA	1	166.5	Kiyo X mount: its own clip grips the 16 mm edge plate (or a 1/4-20 screw), 4 x M3 x 10 into the deck
frame_can_stay	frame	PLA	1	28.4	can stay: post on the motor plate (2 x M3 x 12 into plate inserts), cable tie or hose clamp round the can through the cradle slots; reprint with the adapter
frame_cheek_r	frame	PLA	1	205.2	motor plate (y -10..0, front face = the spec's y 0): window for the puck adapter, 8 rim holes (M3 x 12 from the front into the puck lip), bridge to the motor leg (2 x M3 x 12), can stay and hood inserts; carved round the other subsystems
frame_deck_1	frame	PLA	1	233.6	deck: M3 inserts for every frame foot; 10 x M3 rover holes (it is its own drill template)
frame_deck_2	frame	PLA	1	304.3	deck: M3 inserts for every frame foot; 12 x M3 rover holes (it is its own drill template)
frame_deck_3	frame	PLA	1	231.4	deck: M3 inserts for every frame foot; 9 x M3 rover holes (it is its own drill template)
frame_deck_4	frame	PLA	1	301.6	deck: M3 inserts for every frame foot; 14 x M3 rover holes (it is its own drill template)
frame_ebay	frame	PLA	1	175.3	electronics bay: ESP32-S3 cradle (USB slot), BTS7960 on standoffs with vents, DRV8833, MP1584, HX711 pockets, fuse cradle, wire exits with tie anchors
frame_ebay_esp_clamp	frame	PLA	1	1.6	ESP32 hold-down over the module end, 2 x M3 x 8 into the cradle posts
frame_ebay_lid	frame	PLA	1	71.4	bay lid: grille over the BTS7960, 4 x M3 into the corner bosses
frame_guard_front	frame	PLA	1	56.8	front gear guard between the drum gears and the bay, M3 x 10 into the deck
frame_guard_hood	frame	PLA	1	14.1	gear hood over the reel and bus gears, M3 x 10 through its lugs into the motor plate
frame_motor_leg	frame	PLA	1	117.3	motor support leg (prints upright): 4 x M3 x 10 into the deck, 2 x M3 x 12 along y into the bridge behind the motor plate
frame_rc832_holder	frame	PLA	1	22.8	RC832 holder: receiver stands antenna-end up in the slot, leads out low, 2 x M3 x 10
launch_aim_cap	launch	PLA	1	5.2	Worm shaft saddle cap, 4 x M3x30 through the legs into 13 deep pilots in the saddles
launch_aim_crankpin_clip	launch	PLA	1	0.1	Printed C clip on the crank pin, inboard of the conrod

part	subsystem	material	qty	mass g	notes
launch_aim_drag_pad	launch	PLA	1	0.4	Flexure drag pad: cantilever finger t 2, tip bump 0.45 into the sleeve d20 (strain 0.43 %, about 2.3 N), root block pushed axially into the arm hub pocket
launch_aim_drag_shim	launch	PLA	1	0.1	Preload shim behind the drag pad root block (nominal; print up to 0.25 thicker for more drag, strain stays under 0.7 %)
launch_aim_housing	launch	PLA	1	185.4	Aim housing on the deck (4 x M3 x 10 from the top through its counterbored foot flanges into the deck inserts): sleeve end bushing, countershaft bushings, worm saddles, M8 wheel bolt (head on the outer boss, nut in the inner face hex pocket), engaged and swung-out swing arm stop posts
launch_aim_idler_clip	launch	PLA	1	0.1	Printed C clip on the arm pin, holds the idler
launch_aim_lugpin_clip	launch	PLA	1	0.1	Printed C clip on the lug pin, outboard of the conrod
launch_carriage	launch	PLA	1	59.6	Carriage: rack, shoes, dart cradle, pusher with tether hole, spring arms and pegs, SW feet
launch_carriage_yoke	launch	PLA	1	23.5	Bolted yoke: both spring arms (hit the stop, d8.1 holes for the 8 mm steel rod spring pegs, D5 holes at y +-56 empty), cross bar, keys, SW_SEATED block, rigid SW_HOME foot. Prints on the peg boss bottoms and key plates (w 62)
launch_drive_washer	launch	PLA	1	0.2	Spacer washer: drive gear hub to the left cheek
launch_gear_clip	launch	PLA	1	0.3	C clip on the big gear pin groove
launch_gear_washer	launch	PLA	1	0.5	Spacer washer: big gear hub to the left cheek
launch_guard_l_a	launch	PLA	1	65.0	Spring guard left rear (end wall, top cover) (D33): 2 mm side plate outside both spring pairs, 4 wall brackets (M3x12 self tap into the rail wall); fit by moving it inward
launch_guard_l_b1	launch	PLA	1	4.5	Spring guard left mid front of the stop frame (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall); fit by moving it inward
launch_guard_l_b2	launch	PLA	1	17.9	Spring guard left mid under the stop spring (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall) and a foot on the front plate (M3x10 self tap down); fit by moving it inward
launch_guard_l_c	launch	PLA	1	4.8	Spring guard left front (top cover) (D33): 2 mm side plate outside both spring pairs, 2 bosses on the front plate edge (M3x12 self tap along y); fit by moving it inward
launch_guard_r_a	launch	PLA	1	65.0	Spring guard right rear (end wall, top cover) (D33): 2 mm side plate outside both spring pairs, 4 wall brackets (M3x12 self tap into the rail wall); fit by moving it inward
launch_guard_r_b1	launch	PLA	1	4.5	Spring guard right mid front of the stop frame (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall); fit by moving it inward
launch_guard_r_b2	launch	PLA	1	21.8	Spring guard right mid under the stop spring (D33): 2 mm side plate outside both spring pairs, 2 wall brackets (M3x12 self tap into the rail wall) and a foot on the front plate (M3x10 self tap down); fit by moving it inward
launch_guard_r_c	launch	PLA	1	4.8	Spring guard right front (top cover) (D33): 2 mm side plate outside both spring pairs, 2 bosses on the front plate edge (M3x12 self tap along y); fit by moving it inward
launch_muzzle_roller	launch	PLA	1	0.5	Lip roller at the carriage front: the tether bends down over it in tow and retrieval (groove r 6 for the 1 mm line), 693ZZ inside

part	subsystem	material	qty	mass g	notes
launch_muzzle_washer_l	launch	PLA	1	0.0	Lip roller thrust washer (0.9 thick, flange on the cheek side) between the bearing inner race and the cheek: put both in the roller bore, then drop roller and washers into the fork slot
launch_muzzle_washer_r	launch	PLA	1	0.0	Lip roller thrust washer (0.9 thick, flange on the cheek side) between the bearing inner race and the cheek: put both in the roller bore, then drop roller and washers into the fork slot
launch_rail_front	launch	PLA	1	137.7	Rail front piece: shoe track end, spring anchor posts with straps, muzzle tie
launch_rail_mid	launch	PLA	1	178.7	Rail mid piece: shoe track, cheeks on the sleeve, big gear pin bosses, lever seats, SW_GAP bracket pilots
launch_rail_rear	launch	PLA	1	115.9	Rail rear piece: shoe track, walls, flange at the rear end for the reel head (D49)
launch_rail_splice_l	launch	PLA	1	0.3	Top splice strap over the front rail joint (4 x M3 x 10 self tap)
launch_rail_splice_r	launch	PLA	1	0.3	Top splice strap over the front rail joint (4 x M3 x 10 self tap)
launch_seat_clip	launch	PLA	1	0.1	Head cap on the plunger pin (rests on the yoke block)
launch_stop_frame_l	launch	PLA	1	32.7	Stop frame (print outboard face down): outboard preload ledge, anchor seat, rear foot (2 x M3x12 into the rail wall), front foot (2 x M3x16 into the rail front plate, hex key through the d6.5 holes in the front wall). Fit before the spring guards and the main springs
launch_stop_frame_r	launch	PLA	1	32.7	Stop frame (print outboard face down): outboard preload ledge, anchor seat, rear foot (2 x M3x12 into the rail wall), front foot (2 x M3x16 into the rail front plate, hex key through the d6.5 holes in the front wall). Fit before the spring guards and the main springs
launch_stop_spring_l	launch	PLA	1	81.6	Carriage stop flexure: 16 loops (31 tapered leaves t 3.15/1.0 x 14), print upside down on the flat w 83.5 face, 100% infill / all perimeters; preload 6.6 mm by 2 x M3x20 draw screws through the frame front wall
launch_stop_spring_r	launch	PLA	1	81.6	Carriage stop flexure: 16 loops (31 tapered leaves t 3.15/1.0 x 14), print upside down on the flat w 83.5 face, 100% infill / all perimeters; preload 6.6 mm by 2 x M3x20 draw screws through the frame front wall
launch_sw_home	launch	PLA	1	0.8	SW_HOME printed flex lever on the right rail lip, at rest (prints on its side): beam 0.8 thick, contact gap 0.50 (one screw turn), pad pretravel 0.70 + overtravel 0.25, peak strain 0.10 % (limit 0.8 %)
launch_sw_seated	launch	PLA	1	0.8	SW_SEATED printed flex lever on the left rail lip, at rest (prints on its side): beam 0.8 thick, contact gap 0.50 (one screw turn), pad pretravel 0.70 + overtravel 0.25, peak strain 0.10 % (limit 0.8 %)
reel_bridge	reel	PLA	1	118.5	Rail bridge: plate inboard of the main-axis stack with countershafts C and T, the c1' stub, pillars to the u-shaft bushings, cad-drum's shift rail; box arm bolted to rail cheek A. Print on its inboard face
reel_cup	reel	PLA	1	12.7	Wrap cup (Harry v7, D49 axial eyelet at r 26): the line runs straight through the eyelet. Print open end down
reel_cup_clip	reel	PLA	1	0.2	Harry's v7 C-clip on the cup shaft neck in front of the cup
reel_head_front	reel	PLA	1	108.4	Head front plate: bolts the reel head to the rail rear flange, u-shaft T rear bushing, link window. Print flat
reel_head_mid	reel	PLA	1	19.3	Head mid plate: spool shaft inboard bushing. Print flat

part	subsystem	material	qty	mass g	notes
reel_head_p1	reel	PLA	1	55.0	Head plate P1: capstan tube, u-shaft C and G_b bushings, clutch lever post, spool shaft leg. Print flat
reel_head_p1b	reel	PLA	1	10.7	Head plate P1b: capstan tube rear bushing, SW_ARM. Print flat
reel_head_p2	reel	PLA	1	94.1	Head plate P2: cup shaft and wrap idler rear journals. Print flat
reel_head_spine	reel	PLA	1	104.2	Head spine: inboard wall from the front plate to P2; the drum brackets and plates screw to it. Print flat
reel_shroud	reel	PLA	1	7.6	Spool shroud: keeps loose loops on the spool, open at the top front where the line leaves
reel_spool	reel	PLA	1	30.8	Spool d40 core, d62 flanges, 20 wide: 20 m of 1 mm Dyneema, turns free on two 608s. Print flange down
reel_sw_base	reel	PLA	1	7.0	SW_ARM contact base on P2's front face
reel_sw_lever	reel	PLA	1	1.3	SW_ARM lever: rides the cup gear's raised track (contact closed), drops into its home notch (open)
sensors_as5600_carrier	sensors	PLA	1	12.7	AS5600 carrier: plate with 4 pegs through the board holes, column screwed to the aim housing spine
sensors_as5600_strip_bot	sensors	PLA	1	1.0	AS5600 clamp strip on the board edge over the pegs, M3 from the strip front into a nut in the carrier back
sensors_as5600_strip_top	sensors	PLA	1	1.0	AS5600 clamp strip on the board edge over the pegs, M3 from the strip front into a nut in the carrier back
sensors_sw_gap	sensors	PLA	1	9.9	SW_GAP printed contact base (cad.lib.switch, coupon 2 geometry): with its arm screwed to the right rail cheek outer face below the big gear, closed by the big gear cam bump (gear in its gap)
sensors_sw_gap_lever	sensors	PLA	1	1.4	SW_GAP lever, t 1.4 mm, print on its side so layers run along the beam
sensors_sw_rail_low	sensors	PLA	1	7.3	SW_RAIL_LOW printed contact base (cad.lib.switch, coupon 2 geometry): on its bracket on the aim housing foot, closed by the crank cam at the LOW dead point
sensors_sw_rail_low_bracket	sensors	PLA	1	17.8	SW_RAIL_LOW bracket on the aim housing foot extension: an M3x10 into the extension and its own deck screw, an M3x16 from the top through the counterbored foot and the extension into a deck insert
sensors_sw_rail_low_lever	sensors	PLA	1	1.3	SW_RAIL_LOW lever, t 1.4 mm, print on its side so layers run along the beam
drum_axle	drum	PPA-CF	1	8.9	Fixed drum axle d10, pressed into the reel head front plate and bracket B
drum_collar	drum	PPA-CF	1	44.0	Dog collar: hex bore, 4 back-cut roofed dogs each face, fork groove (bridges over the lower dogs)
drum_coupler	drum	PPA-CF	1	89.3	Coupler: 5 N m shear pin limiter (D8), bus gear 50T m1.5 (D25), SW_INDEX cam, REEL journal, hex socket; prints hub end down, tree support under the bus gear ring
drum_coupler_hub	drum	PPA-CF	1	14.6	Coupler hub on the wiper shaft (bore parametric: WIPER), M8 nut inside, shear pin hole
drum_crank	drum	PPA-CF	1	6.9	Bell crank 1:1 on the bridge shelf: turns the link's motion along the rail into the fork's shift
drum_dogring_launch	drum	PPA-CF	1	19.0	LAUNCH dog ring: 3 slots for the launch sleeve lugs, 3 x M3 from the sleeve side
drum_gear_reel	drum	PPA-CF	1	95.9	REEL output gear m2 50T face 12 (D20), free on its sleeve, dog face toward the collar; meshes C1 on the rail bridge (D49)

part	subsystem	material	qty	mass g	notes
drum_idler	drum	PPA-CF	1	8.7	Wrap idler shaft: pinion 10T m1.25 (locked at home by the drum lock arc), lock cam, journals in bracket B and P2
drum_idler_gear	drum	PPA-CF	1	35.8	Wrap idler output gear 49T m1.5 (D bore) to the cad-reel cup gear 28T: 7 cup turns
drum_n20_pinion	drum	PPA-CF	1	0.5	N20 pinion 10T m0.8, D bore pressed on the N20 shaft
drum_reel_washer	drum	PPA-CF	1	0.9	REEL gear retaining washer on the coupler journal end, 2 x M3 x 6
drum_rod	drum	PPA-CF	1	8.9	Follower guide rod d10, pressed into the reel head front plate and bracket B
drum_shaft	drum	PPA-CF	1	45.2	Main shaft: hex 22 AF, d12 spigot into the launch sleeve; prints on a hex flat (support under the spigot)
drum_shift_rail	drum	PPA-CF	1	8.2	Shift rail d10 on the rail bridge (pressed into the bridge plate boss)
frame_motor_adapter	frame	PPA-CF	1	302.8	puck adapter: slotted M6 holes (+-4 mm) for the motor bosses, rear lip with 8 M3 inserts for the rim screws, 2 inserts for cad-drum's SW_INDEX block; reprint when the motor is measured
launch_aim_arm	launch	PPA-CF	1	6.6	Swing arm on the sleeve, dragged by the printed PLA flexure pad in its hub; printed idler pin with a printed C clip groove
launch_aim_bevel30	launch	PPA-CF	1	8.8	Bevel m1.5 z30 on the countershaft D flat, drives the z15 2:1
launch_aim_conrod	launch	PPA-CF	1	14.2	Conrod 156 between pin centres (PPA-CF), printed C clips on both pins
launch_aim_crank	launch	PPA-CF	1	35.3	Worm wheel m2 z20 face 16 = crank gear, crank arm r30 and d8 crank pin in one part, two 608 inside
launch_aim_idler	launch	PPA-CF	1	7.5	Splitter idler m1.5 z18 face 12 on the arm pin
launch_aim_lug	launch	PPA-CF	1	16.9	Lug bracket (PPA-CF), 2 x M3 into the rail wall, d8 lug pin at (u 115.2, w 0)
launch_aim_output	launch	PPA-CF	1	26.2	Splitter output m1.5 z30 face 12 with its d8 countershaft (D flat)
launch_aim_sun	launch	PPA-CF	1	18.3	Splitter sun m1.5 z30 face 12, keyed on the launch sleeve
launch_aim_worm	launch	PPA-CF	1	21.3	Worm m2 pd24 single start (lead angle 4.8 deg, self-locking) + d8 shaft + bevel z15
launch_big_gear	launch	PPA-CF	1	177.6	Big gear m2.5: sector layer 27 of 40 teeth (rack, v10 release teeth) + full 40T layer (drive), SW_GAP cam bump, on the rail pin
launch_drive_gear	launch	PPA-CF	1	29.9	Drive gear m2.5 26T, sprag #1 inside on an integral key rib
launch_gear_pin	launch	PPA-CF	1	5.4	Snap pin d10 for the big gear, C clip; prints lying on its flat
launch_seat_plunger	launch	PPA-CF	1	1.0	SW_SEATED plunger: foot pad + stem with a 45 deg chamfer, clip groove on top
launch_seat_pushrod	launch	PPA-CF	1	0.2	SW_SEATED pushrod d3.2 with the stem slot
launch_sleeve	launch	PPA-CF	1	49.3	Launch sleeve d20: dog ring flange, integral key ribs (sprag #1, splitter sun), journals
launch_yoke_peg_l	launch	PPA-CF	1	1.2	Spring peg on the yoke arm (printed PPA-CF lying down, or 8 mm steel rod 20 long): pressed into the blind d8.1 hole from below up to its end (a drop of CA if it is loose), the spring hook sits on the 6 mm below the boss (held at w 60 by the spring line); ends deburred

part	subsystem	material	qty	mass g	notes
launch_yoke_peg_r	launch	PPA-CF	1	1.2	Spring peg on the yoke arm (printed PPA-CF lying down, or 8 mm steel rod 20 long): pressed into the blind d8.1 hole from below up to its end (a drop of CA if it is loose), the spring hook sits on the 6 mm below the boss (held at w 60 by the spring line); ends deburred
reel_b1g	reel	PPA-CF	1	2.6	Bevel B1 gear m1.5 24T (2:1 up to u-shaft C)
reel_b1p	reel	PPA-CF	1	1.2	Bevel B1 pinion m1.5 12T on the front end of u-shaft C
reel_b2g	reel	PPA-CF	1	0.9	Bevel B2 m1 20T on countershaft T
reel_b2p	reel	PPA-CF	1	1.3	Bevel B2 m1 20T on the front end of u-shaft T
reel_b3a	reel	PPA-CF	1	1.3	Bevel B3 m1 20T on the rear end of u-shaft T
reel_b3b	reel	PPA-CF	1	1.4	Bevel B3 m1 20T on the spool shaft
reel_c	reel	PPA-CF	1	17.4	Capstan tube C (PPA-CF d20): key rib in the drum, G_c on its flat, journals in P1 and P1b. Print standing on its rear end
reel_c1	reel	PPA-CF	1	93.7	C1 m2 49T: meshes the REEL output gear 50T across the pivot
reel_c1p	reel	PPA-CF	1	27.8	c1' m1.5 46T, free on its stub: driven by the bus gear 50T, drives the swing idler
reel_cs_c	reel	PPA-CF	1	11.4	Countershaft C (PPA-CF): sprag #2, C1 and the B1 bevel. Print standing
reel_cs_t	reel	PPA-CF	1	2.8	Countershaft T (PPA-CF): T1 and the B2 bevel
reel_d	reel	PPA-CF	1	7.6	Clutch disk D on the spool shaft flat: its friction ring presses the spool's outboard face. Print ring up
reel_drum	reel	PPA-CF	1	45.0	Capstan drum dia 40 x 40 (Harry v7), keyed on the tube. Print flange down
reel_ga	reel	PPA-CF	1	9.6	G_a m1.5 26T on the rear end of u-shaft C
reel_gb	reel	PPA-CF	1	15.1	G_b m1.5 32T idler on its stub in P1
reel_gc	reel	PPA-CF	1	5.3	Capstan gear G_c m1.5 23T on the tube flat
reel_gi	reel	PPA-CF	1	11.4	Swing idler m1.5 30T
reel_gk	reel	PPA-CF	1	15.0	Cup gear G_k m1.5 28T: driven by the drum's wrap idler 49T (7 cup turns per wrap); the raised track on its rear face holds SW_ARM closed, the notch at home lets it open. Print track face down
reel_k	reel	PPA-CF	1	10.2	Cup shaft K (PPA-CF d10): runs inside the tube, cup in front, G_k behind P1b, journal in P2. Print standing on its rear end
reel_lever	reel	PPA-CF	1	6.9	Clutch lever (PPA-CF): the drum's brake cam presses its upper arm; the lower arm is the clutch spring leaf, its FIRM screw squeezes the spool. Print flat
reel_s	reel	PPA-CF	1	5.7	Spool shaft S (PPA-CF d8): B3 bevel and the clutch disk on its flats, journals in the mid plate and the P1 leg. Print lying on its 1 mm flat
reel_sprag_hub	reel	PPA-CF	1	12.2	Sprag #2 hub on the bridge: holds the CSK20PP capstan backstop
reel_stub	reel	PPA-CF	1	1.1	c1' stub axle (PPA-CF d8, pressed into the bridge)
reel_swing_arm	reel	PPA-CF	1	3.2	Swing arm on c1's hub: carries the idler into mesh with T1 on forward wiper turns
reel_t	reel	PPA-CF	1	1.1	Thrust ring: the lever's FIRM screw pushes it onto the inboard 608 inner race (clutch squeeze)
reel_t1	reel	PPA-CF	1	4.6	T1 m1.5 20T: take-up at 2.5 x wiper

part	subsystem	material	qty	mass g	notes
reel_uc	reel	PPA-CF	1	23.7	u-shaft C (PPA-CF d10, 244 long): along the rail from the B1 bevel to G_a. Print standing (244 of the 260 mm height); an 8 mm steel rod with printed sleeves is the stiff fallback
reel_ut	reel	PPA-CF	1	6.5	u-shaft T (PPA-CF d8): along the rail from the B2 bevel to the B3 bevel. Print standing

Printed mass: PLA 4011 g (1344.9 m of filament); PPA-CF 1456 g (484.4 m of filament).

Bought parts

item	type	count	used by
608ZZ 8x22x7	bearings	4	launch_aim_brg_1, launch_aim_brg_2, reel_brg_1, reel_brg_2
693ZZ 3x8x4	bearings	1	launch_muzzle_roller_brg
Adafruit AS5600 STEMMA QT PID 6357	bearings	1	sensors_as5600
CSK20PP 20x47x14 sprag #1	bearings	1	launch_sprag
CSK20PP one-way bearing 20x47x14	bearings	1	reel_sprag
M3 nut	fasteners	6	launch_sw_seated_nut, launch_sw_home_nut, sensors_as5600_strip_top, sensors_as5600_strip_bot, sensors_sw_rail_low_nut, sensors_sw_gap_nut
M3 nut (jam)	fasteners	2	sensors_sw_rail_low_screw, sensors_sw_gap_screw
M3 nyloc nut	fasteners	1	launch_muzzle_roller_axle
M3x10 self tap	fasteners	2	launch_guard_l_b2, launch_guard_r_b2
M3x10 self tap (lever clamp)	fasteners	6	reel_sw_lever, sensors_sw_rail_low_lever, sensors_sw_gap_lever
M3x10 SHCS	fasteners	4	drum_index_block, sensors_sw_rail_low_screw, sensors_sw_gap_screw
M3x10 SHCS (base to P2)	fasteners	4	reel_sw_base
M3x10 SHCS (foot into the housing)	fasteners	1	sensors_sw_rail_low_bracket
M3x10 SHCS + 2 M3 nuts	fasteners	1	reel_sw_hw
M3x10 SHCS + M3 nut	fasteners	1	reel_firm_screw
M3x12 self tap	fasteners	17	launch_guard_l_a, launch_guard_l_b1, launch_guard_l_b2, launch_guard_l_c, launch_guard_r_a, launch_guard_r_b1, launch_guard_r_b2, launch_guard_r_c
M3x12 SHCS	fasteners	4	reel_head_mid, sensors_as5600_strip_top, sensors_as5600_strip_bot
M3x12 SHCS (arm into the right cheek)	fasteners	2	sensors_sw_gap
M3x12 SHCS (posts)	fasteners	2	reel_head_p1b
M3x12 SHCS (spine)	fasteners	7	reel_head_front, reel_head_p1, reel_head_p2
M3x12 SHCS (through the spine)	fasteners	4	drum_bracket_b, drum_switch_block
M3x16 SHCS	fasteners	5	launch_muzzle_roller_axle, reel_sprag_hub
M3x16 SHCS (arm to cheek A)	fasteners	2	reel_bridge
M3x16 SHCS (deck screw)	fasteners	1	sensors_sw_rail_low_bracket
M3x16 SHCS (into the housing spine)	fasteners	2	sensors_as5600_carrier
M3x16 SHCS (into the rail flange inserts)	fasteners	4	reel_head_front
M3x16 SHCS (pivot, tapped into the shelf)	fasteners	1	drum_crank
M3x16 SHCS + M3 nut	fasteners	1	reel_light_screw
M3x16 SHCS + nyloc nut	fasteners	2	drum_link_eye_1, drum_link_eye_2

item	type	count	used by
M3x16 SHCS + washer	fasteners	1	reel_lever_pin
M3x6 SHCS	fasteners	2	launch_sw_seated_screw, launch_sw_home_screw
M3x8 self tap (base to bracket)	fasteners	4	sensors_sw_rail_low
M3x8 SHCS (tab to the P1 leg)	fasteners	1	reel_shroud
M8 hex nut ISO 4032	fasteners	1	launch_aim_nut
M8x25 hex bolt ISO 4017	fasteners	1	launch_aim_bolt
STEMMA QT JST SH 4 pin cable	fasteners	1	sensors_as5600
(part of the wiper motor)	other	1	frame_wiper_shaft
12 V car wiper motor (self-locking worm), 2-wire rebuild	other	1	frame_wiper_motor
8 mm steel rod, 26 mm long	other	1	reel_gb_axle
8 mm steel rod, cut to length (see note), ends deburred	other	2	launch_anchor_pin_l, launch_anchor_pin_r
8 mm steel rod, cut to the length on the print sheet	other	1	drum_link
BTS7960 43 A module (IBT-2)	other	1	frame_pcb_bts7960
Century C-243 (5/8 x 6-1/2 in x 0.054 in wire) extension spring	other	2	launch_spring_l_hook_a, launch_spring_r_hook_a
Dyneema braid 1 mm (LINE)	other	21	launch_line_a0, launch_line_a1, launch_line_a2, launch_line_a3, launch_line_a4, launch_line_a5, launch_line_a6, launch_line_a7, launch_line_a8, launch_line_a9, launch_line_a10, launch_line_a11, launch_line_a12, launch_line_a13, launch_line_a14, launch_line_a15, launch_line_a16, launch_line_a17, launch_line_a18, launch_line_a19, launch_line_a20
Espressif ESP32-S3-DevKitC-1-N8	other	1	frame_pcb_esp32
HX711 load cell amplifier (optional)	other	1	frame_pcb_hx711
inline ATO fuse holder + 20 A fuse	other	1	frame_fuse_20a
KEDGE v30 dart (reference, built separately)	other	1	launch_dart
launch_line_p (bought, no bought_ref)	other	1	launch_line_p
launch_spring_l_0 (bought, no bought_ref)	other	1	launch_spring_l_0
launch_spring_l_1 (bought, no bought_ref)	other	1	launch_spring_l_1
launch_spring_l_2 (bought, no bought_ref)	other	1	launch_spring_l_2
launch_spring_l_3 (bought, no bought_ref)	other	1	launch_spring_l_3
launch_spring_l_4 (bought, no bought_ref)	other	1	launch_spring_l_4
launch_spring_l_hook_c (bought, no bought_ref)	other	1	launch_spring_l_hook_c
launch_spring_r_0 (bought, no bought_ref)	other	1	launch_spring_r_0
launch_spring_r_1 (bought, no bought_ref)	other	1	launch_spring_r_1
launch_spring_r_2 (bought, no bought_ref)	other	1	launch_spring_r_2

item	type	count	used by
launch_spring_r_3 (bought, no bought_ref)	other	1	launch_spring_r_3
launch_spring_r_4 (bought, no bought_ref)	other	1	launch_spring_r_4
launch_spring_r_hook_c (bought, no bought_ref)	other	1	launch_spring_r_hook_c
MP1584EN buck module, set to 9.0 V	other	1	frame_pcb_mp1584
N20 right-angle self-locking worm gearmotor 12 V 27 rpm (have)	other	1	drum_n20
perfboard 55 x 31 (dividers, pull-downs, 3V3 and GND buses), 2 x M3 x 10 on wall bosses	other	1	frame_perfboard
Pololu DRV8833 2130 (headers off)	other	1	frame_pcb_drv8833
Razer Kiyo X webcam	other	1	frame_kiyo_x
RC832 5.8 GHz receiver (80 x 65 x 15)	other	1	frame_rc832
sensors_as5600_col_screws (bought, no bought_ref)	other	1	sensors_as5600_col_screws
sensors_as5600_nut_bot (bought, no bought_ref)	other	1	sensors_as5600_nut_bot
sensors_as5600_nut_top (bought, no bought_ref)	other	1	sensors_as5600_nut_top
sensors_as5600_screw_bot (bought, no bought_ref)	other	1	sensors_as5600_screw_bot
sensors_as5600_screw_top (bought, no bought_ref)	other	1	sensors_as5600_screw_top
sensors_sw_gap_jam (bought, no bought_ref)	other	1	sensors_sw_gap_jam
sensors_sw_rail_low_jam (bought, no bought_ref)	other	1	sensors_sw_rail_low_jam

4. Print list

150 printed parts on 19 plates. The 3MF plates and STL files are in the print pack.

part	material	qty	print size mm	supports
frame_deck_2	PLA	1	222 x 212 x 8	no
frame_deck_4	PLA	1	214 x 212 x 8	no
frame_deck_1	PLA	1	222 x 169 x 8	no
frame_deck_3	PLA	1	214 x 169 x 8	no
frame_cheek_r	PLA	1	215 x 220 x 30	yes, tree supports from the plate (1084 mm2 over 45 deg)
frame_camera_mount	PLA	1	58 x 72 x 236	yes, tree supports from the plate (672 mm2 over 45 deg)
launch_aim_housing	PLA	1	176 x 59 x 183	yes, tree supports from the plate (761 mm2 over 45 deg)
launch_rail_mid	PLA	1	92 x 68 x 170	yes, tree supports from the plate (2356 mm2 over 45 deg)
frame_motor_leg	PLA	1	49 x 69 x 187	yes, tree supports from the plate (222 mm2 over 45 deg)
launch_rail_front	PLA	1	132 x 128 x 48	yes, tree supports from the plate (7890 mm2 over 45 deg)

part	material	qty	print size mm	supports
frame_ebay	PLA	1	124 x 124 x 58	yes, tree supports from the plate (879 mm2 over 45 deg)
drum_drum_a	PLA	1	97 x 97 x 43	yes, tree supports from the plate (719 mm2 over 45 deg)
launch_rail_rear	PLA	1	151 x 68 x 40	yes, tree supports from the plate (3676 mm2 over 45 deg)
reel_bridge	PLA	1	220 x 192 x 88	yes, tree supports from the plate (686 mm2 over 45 deg)
reel_head_front	PLA	1	188 x 186 x 6	no
reel_head_spine	PLA	1	131 x 188 x 6	no
drum_drum_b	PLA	1	77 x 77 x 41	yes, tree supports from the plate (3126 mm2 over 45 deg)
reel_head_p2	PLA	1	104 x 182 x 6	no
launch_stop_spring_l	PLA	1	155 x 52 x 14	no
launch_stop_spring_r	PLA	1	155 x 52 x 14	no
frame_ebay_lid	PLA	1	124 x 124 x 7	no
reel_head_p1	PLA	1	100 x 177 x 37	no
launch_guard_l_a	PLA	1	250 x 49 x 40	no
launch_guard_r_a	PLA	1	250 x 49 x 40	no
launch_carriage	PLA	1	205 x 240 x 43	yes, tree supports from the plate (6453 mm2 over 45 deg)
frame_guard_front	PLA	1	155 x 104 x 16	no
drum_index_block	PLA	1	25 x 68 x 40	no
frame_can_stay	PLA	1	55 x 55 x 71	yes, tree supports from the plate (906 mm2 over 45 deg)
reel_spool	PLA	1	62 x 62 x 22	yes, tree supports from the plate (1791 mm2 over 45 deg)
launch_stop_frame_l	PLA	1	175 x 53 x 55	yes, tree supports from the plate (541 mm2 over 45 deg)
launch_stop_frame_r	PLA	1	175 x 53 x 55	yes, tree supports from the plate (541 mm2 over 45 deg)
drum_bracket_b	PLA	1	76 x 109 x 6	no
drum_switch_block	PLA	1	26 x 66 x 21	yes, tree supports from the plate (216 mm2 over 45 deg)
launch_carriage_yoke	PLA	1	43 x 127 x 28	yes, tree supports from the plate (1406 mm2 over 45 deg)
frame_rc832_holder	PLA	1	97 x 22 x 40	no
reel_head_mid	PLA	1	54 x 58 x 6	yes, tree supports from the plate (143 mm2 over 45 deg)
launch_guard_r_b2	PLA	1	145 x 32 x 40	no
drum_fork	PLA	1	86 x 68 x 51	yes, tree supports from the plate (1227 mm2 over 45 deg)
sensors_sw_rail_low_bracket	PLA	1	31 x 48 x 74	no
launch_guard_l_b2	PLA	1	145 x 32 x 40	no
sensors_as5600_carrier	PLA	1	34 x 68 x 11	no
frame_guard_hood	PLA	1	122 x 56 x 14	no
reel_cup	PLA	1	48 x 53 x 18	yes, tree supports from the plate (1210 mm2 over 45 deg)

part	material	qty	print size mm	supports
drum_n20_cradle	PLA	1	54 x 32 x 20	yes, tree supports from the plate (114 mm2 over 45 deg)
reel_head_p1b	PLA	1	38 x 44 x 16	no
sensors_sw_gap	PLA	1	63 x 43 x 12	no
reel_shroud	PLA	1	70 x 66 x 30	no
sensors_sw_rail_low	PLA	1	48 x 35 x 12	no
drum_dog_post	PLA	1	22 x 65 x 48	no
reel_sw_base	PLA	1	46 x 35 x 12	no
drum_follower	PLA	1	26 x 39 x 17	yes, tree supports from the plate (260 mm2 over 45 deg)
launch_aim_cap	PLA	1	46 x 14 x 19	no
launch_guard_l_c	PLA	1	32 x 54 x 26	no
launch_guard_r_c	PLA	1	32 x 54 x 26	no
launch_guard_l_b1	PLA	1	12 x 49 x 40	no
launch_guard_r_b1	PLA	1	12 x 49 x 40	no
frame_cable_anchor_1	PLA	1	16 x 24 x 10	yes, tree supports from the plate (141 mm2 over 45 deg)
frame_cable_anchor_2	PLA	1	24 x 16 x 10	yes, tree supports from the plate (141 mm2 over 45 deg)
drum_link_eye_1	PLA	1	13 x 13 x 23	no
drum_link_eye_2	PLA	1	13 x 13 x 23	no
drum_n20_strap	PLA	1	54 x 18 x 3	no
drum_magnet_cap	PLA	1	20 x 20 x 6	no
frame_ebay_esp_clamp	PLA	1	43 x 8 x 4	no
sensors_sw_gap_lever	PLA	1	40 x 9 x 8	no
sensors_sw_rail_low_lever	PLA	1	44 x 5 x 8	no
reel_sw_lever	PLA	1	42 x 5 x 8	no
sensors_as5600_strip_top	PLA	1	25 x 12 x 3	no
sensors_as5600_strip_bot	PLA	1	25 x 12 x 3	no
drum_index_lever	PLA	1	13 x 48 x 6	no
drum_sw_lever_0	PLA	1	14 x 48 x 6	no
drum_sw_lever_1	PLA	1	14 x 48 x 6	no
drum_sw_lever_2	PLA	1	14 x 48 x 6	no
launch_sw_seated	PLA	1	42 x 8 x 6	no
launch_sw_home	PLA	1	42 x 8 x 6	no
launch_muzzle_roller	PLA	1	14 x 14 x 5	no
launch_gear_washer	PLA	1	18 x 18 x 2	no
launch_aim_drag_pad	PLA	1	9 x 24 x 4	no
launch_gear_clip	PLA	1	17 x 18 x 1	no
launch_rail_splice_l	PLA	1	22 x 5 x 3	no
launch_rail_splice_r	PLA	1	22 x 5 x 3	no
launch_drive_washer	PLA	1	32 x 32 x 1	no
drum_shear_pin	PLA	6	12 x 5 x 4	no
reel_cup_clip	PLA	1	10 x 10 x 2	no

part	material	qty	print size mm	supports
launch_aim_drag_shim	PLA	1	9 x 7 x 4	no
launch_aim_idler_clip	PLA	1	11 x 10 x 1	no
launch_aim_crankpin_clip	PLA	1	11 x 10 x 1	no
launch_aim_lugpin_clip	PLA	1	11 x 10 x 1	no
launch_seat_clip	PLA	1	6 x 6 x 2	no
launch_muzzle_washer_l	PLA	1	6 x 6 x 1	no
launch_muzzle_washer_r	PLA	1	6 x 6 x 1	no
frame_motor_adapter	PPA-CF	1	194 x 194 x 16	yes, tree supports from the plate (5484 mm2 over 45 deg)
launch_big_gear	PPA-CF	1	105 x 105 x 27	yes, tree supports from the plate (639 mm2 over 45 deg)
drum_gear_reel	PPA-CF	1	104 x 104 x 19	no
reel_c1	PPA-CF	1	102 x 102 x 12	no
drum_coupler	PPA-CF	1	78 x 78 x 62	yes, tree supports from the plate (1998 mm2 over 45 deg)
drum_shaft	PPA-CF	1	170 x 25 x 22	yes, tree supports from the plate (922 mm2 over 45 deg)
reel_drum	PPA-CF	1	46 x 46 x 43	yes, tree supports from the plate (251 mm2 over 45 deg)
launch_sleeve	PPA-CF	1	64 x 64 x 85	yes, tree supports from the plate (2907 mm2 over 45 deg)
drum_collar	PPA-CF	1	60 x 60 x 26	yes, tree supports from the plate (2694 mm2 over 45 deg)
launch_aim_crank	PPA-CF	1	62 x 60 x 38	yes, tree supports from the plate (985 mm2 over 45 deg)
drum_idler_gear	PPA-CF	1	76 x 76 x 7	no
launch_drive_gear	PPA-CF	1	70 x 70 x 29	yes, tree supports from the plate (2457 mm2 over 45 deg)
reel_c1p	PPA-CF	1	72 x 72 x 10	no
launch_aim_output	PPA-CF	1	48 x 48 x 60	no
reel_uc	PPA-CF	1	10 x 10 x 244	no
launch_aim_worm	PPA-CF	1	28 x 28 x 54	yes, tree supports from the plate (2779 mm2 over 45 deg)
drum_dogring_launch	PPA-CF	1	62 x 62 x 10	yes, tree supports from the plate (123 mm2 over 45 deg)
launch_aim_sun	PPA-CF	1	48 x 48 x 12	no
reel_c	PPA-CF	1	23 x 20 x 58	no
launch_aim_lug	PPA-CF	1	30 x 58 x 28	yes, tree supports from the plate (995 mm2 over 45 deg)
reel_gb	PPA-CF	1	51 x 51 x 7	no
reel_gk	PPA-CF	1	45 x 45 x 12	yes, tree supports from the plate (1031 mm2 over 45 deg)
drum_coupler_hub	PPA-CF	1	26 x 26 x 32	no
launch_aim_conrod	PPA-CF	1	173 x 17 x 6	no
reel_cs_c	PPA-CF	1	23 x 20 x 40	yes, tree supports from the plate (182 mm2 over 45 deg)
reel_sprag_hub	PPA-CF	1	56 x 56 x 14	no

part	material	qty	print size mm	supports
reel_gi	PPA-CF	1	48 x 48 x 6	no
reel_k	PPA-CF	1	10 x 10 x 109	no
reel_ga	PPA-CF	1	42 x 42 x 7	no
drum_axle	PPA-CF	1	95 x 10 x 9	no
drum_rod	PPA-CF	1	95 x 10 x 9	no
launch_aim_bevel30	PPA-CF	1	53 x 52 x 10	yes, tree supports from the plate (590 mm2 over 45 deg)
drum_idler	PPA-CF	1	22 x 19 x 54	yes, tree supports from the plate (331 mm2 over 45 deg)
drum_shift_rail	PPA-CF	1	88 x 10 x 9	no
reel_d	PPA-CF	1	53 x 53 x 3	yes, tree supports from the plate (1050 mm2 over 45 deg)
launch_aim_idler	PPA-CF	1	30 x 30 x 12	no
drum_crank	PPA-CF	1	56 x 58 x 10	no
reel_lever	PPA-CF	1	67 x 66 x 8	no
launch_aim_arm	PPA-CF	1	64 x 41 x 19	no
reel_ut	PPA-CF	1	8 x 8 x 104	no
reel_s	PPA-CF	1	99 x 8 x 7	no
launch_gear_pin	PPA-CF	1	53 x 16 x 12	no
reel_gc	PPA-CF	1	37 x 37 x 7	no
reel_t1	PPA-CF	1	33 x 33 x 6	no
reel_swing_arm	PPA-CF	1	73 x 19 x 10	no
reel_cs_t	PPA-CF	1	12 x 12 x 24	no
reel_b1g	PPA-CF	1	39 x 39 x 6	yes, tree supports from the plate (679 mm2 over 45 deg)
reel_b3b	PPA-CF	1	22 x 22 x 9	no
reel_b2p	PPA-CF	1	22 x 22 x 8	yes, tree supports from the plate (149 mm2 over 45 deg)
reel_b3a	PPA-CF	1	22 x 22 x 8	no
launch_yoke_peg_l	PPA-CF	1	20 x 8 x 7	no
launch_yoke_peg_r	PPA-CF	1	20 x 8 x 7	no
reel_b1p	PPA-CF	1	21 x 21 x 11	no
reel_stub	PPA-CF	1	20 x 8 x 7	no
reel_t	PPA-CF	1	18 x 18 x 6	no
launch_seat_plunger	PPA-CF	1	30 x 15 x 20	no
drum_reel_washer	PPA-CF	1	35 x 35 x 1	no
reel_b2g	PPA-CF	1	22 x 22 x 7	yes, tree supports from the plate (149 mm2 over 45 deg)
drum_n20_pinion	PPA-CF	1	10 x 9 x 8	no
launch_seat_pushrod	PPA-CF	1	16 x 5 x 4	no

plate_1_PLA.3mf: drum_link_eye_1, drum_link_eye_2, drum_n20_cradle, launch_carriage, launch_guard_l_c, launch_guard_r_c, sensors_sw_rail_low.

plate_2_PLA.3mf: drum_index_block, drum_magnet_cap, drum_switch_block, frame_cheek_r, frame_rc832_holder.

plate_3_PLA.3mf: drum_index_lever, drum_n20_strap, drum_sw_lever_2, frame_cable_anchor_1, frame_deck_2, launch_aim_cap.

plate_4_PLA.3mf: drum_follower, frame_cable_anchor_2, frame_deck_4, launch_aim_drag_pad, sensors_as5600_strip_bot, sensors_as5600_strip_top.

plate_5_PLA.3mf: drum_dog_post, reel_bridge, reel_cup, reel_head_p1b, reel_sw_base, sensors_sw_rail_low_bracket.

plate_6_PLA.3mf: drum_shear_pin__c1, drum_shear_pin__c2, drum_shear_pin__c3, drum_shear_pin__c4, drum_shear_pin__c5, frame_deck_1, launch_aim_crankpin_clip, launch_aim_lugpin_clip, launch_gear_clip, launch_gear_washer, launch_guard_r_b1, launch_rail_rear, launch_sw_home, launch_sw_seated, reel_cup_clip, reel_shroud.

plate_7_PLA.3mf: frame_deck_3, launch_guard_l_b2, launch_guard_r_b2, launch_rail_mid.

plate_8_PLA.3mf: drum_shear_pin__c6, frame_motor_leg, launch_aim_idler_clip, launch_guard_l_b1, launch_muzzle_washer_l, launch_muzzle_washer_r, launch_stop_frame_l, launch_stop_spring_l, reel_head_front.

plate_9_PLA.3mf: launch_stop_frame_r, reel_head_mid, reel_head_p2, reel_head_spine.

plate_10_PLA.3mf: launch_aim_housing, launch_carriage_yoke, launch_rail_front, reel_head_p1, reel_spool, sensors_sw_gap_lever.

plate_11_PLA.3mf: drum_drum_a, drum_fork, drum_sw_lever_0, drum_sw_lever_1, frame_ebay, frame_ebay_esp_clamp, frame_guard_front, launch_aim_drag_shim, launch_drive_washer, launch_muzzle_roller, launch_seat_clip, sensors_as5600_carrier.

plate_12_PLA.3mf: drum_bracket_b, frame_camera_mount, frame_can_stay, frame_ebay_lid, frame_guard_hood, launch_rail_splice_l, launch_rail_splice_r, launch_stop_spring_r, reel_sw_lever, sensors_sw_gap, sensors_sw_rail_low_lever.

plate_13_PLA.3mf: drum_drum_b.

plate_14_PLA.3mf: launch_guard_l_a.

plate_15_PLA.3mf: launch_guard_r_a.

plate_16_PPA-CF.3mf: drum_n20_pinion, drum_reel_washer, drum_rod, frame_motor_adapter, launch_aim_arm, reel_b1g, reel_b1p, reel_b2g, reel_b3a, reel_drum, reel_ga, reel_gc, reel_gk.

plate_17_PPA-CF.3mf: drum_coupler, drum_coupler_hub, drum_gear_reel, drum_idler, drum_shift_rail, launch_aim_bevel30, launch_big_gear, launch_gear_pin, launch_seat_pushrod, reel_b2p, reel_b3b, reel_c, reel_c1, reel_cs_c, reel_d, reel_gb, reel_s, reel_ut.

plate_18_PPA-CF.3mf: drum_axle, drum_crank, drum_dogring_launch, drum_idler_gear, drum_shaft, launch_aim_crank, launch_aim_idler, launch_aim_lug, launch_aim_worm, launch_drive_gear, launch_seat_plunger, launch_sleeve, launch_yoke_peg_l, launch_yoke_peg_r, reel_c1p, reel_cs_t, reel_k, reel_lever, reel_sprag_hub, reel_stub, reel_t1, reel_uc.

plate_19_PPA-CF.3mf: drum_collar, launch_aim_conrod, launch_aim_output, launch_aim_sun, reel_gi, reel_swing_arm, reel_t.

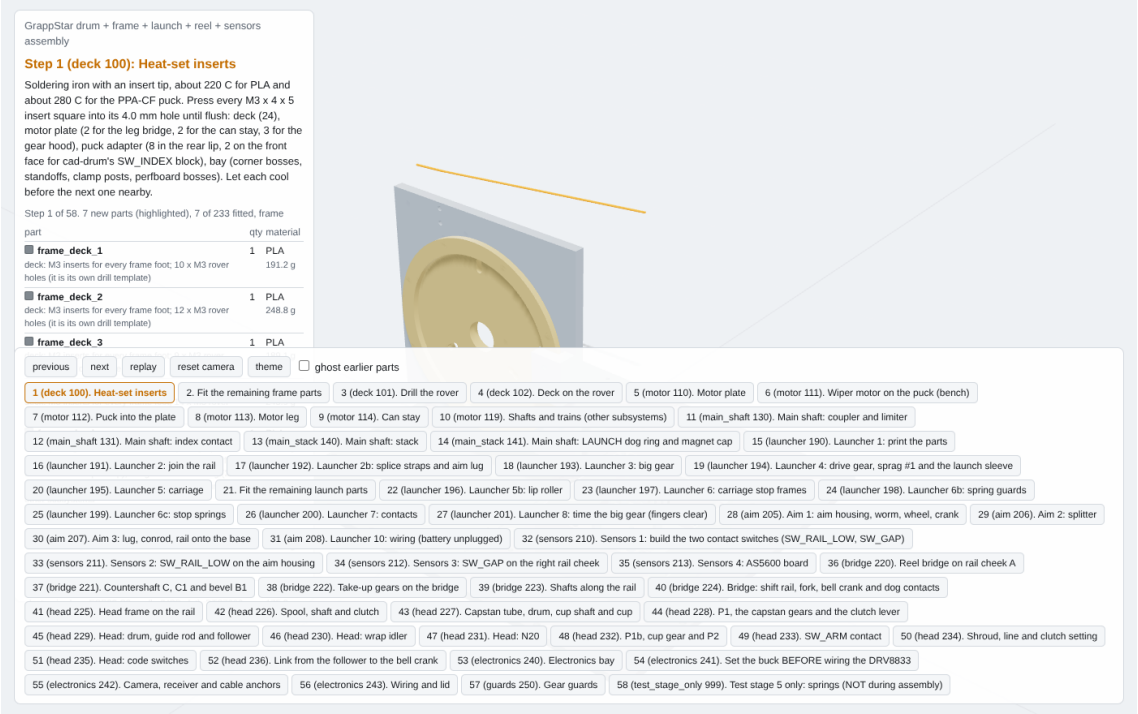
Print settings are in PRINT_SHEET.pdf. Use a hardened nozzle for PPA-CF and dry it first.

5. Assembly steps

Step 1 (deck 100). Heat-set inserts

Battery unplugged. Soldering iron with an insert tip, about 220 C for PLA and about 280 C for the PPA-CF puck. Press every M3 x 4 x 5 insert square into its 4.0 mm hole until flush: deck (24), motor plate (2 for the leg bridge, 2 for the can stay, 3 for the gear hood), puck adapter (8 in the rear lip, 2 on the front face for cad-drum's SW_INDEX block), bay (corner bosses, standoffs, clamp posts, perfboard bosses). Let each cool before the next one nearby.

Parts: frame_deck_1, frame_deck_2, frame_deck_3, frame_deck_4, frame_cheek_r, frame_motor_adapter, frame_ebay.



Step 2. Fit the remaining frame parts

Parts without an explicit assembly step: frame_perfboard.

Parts: frame_perfboard.

Step 3 (deck 101). Drill the rover

Lay the deck on the rover top plate where it will live (launch direction forward). Mark through the 45 perimeter holes with a 3 mm drill or a scriber (or tape down the 1:1 PDF template from cad/parts/_frame_tests/drill_template.py), take the deck off, drill 3.4 mm. Deburr.

Also handle (fitted earlier): frame_deck_1, frame_deck_2, frame_deck_3, frame_deck_4.

Step 4 (deck 102). Deck on the rover

Slide the tiles together on the dovetails (from above). M3 x 25 SHCS down through every perimeter hole, washer and nyloc under the rover plate. Snug, not crushed. Do this before the other subsystems put their feet down: some rover bolts sit under their feet.

Also handle (fitted earlier): frame_deck_1, frame_deck_2, frame_deck_3, frame_deck_4.

Step 5 (motor 110). Motor plate

Hold the motor plate upright (front face, the side without the bridge, towards +y) with its bridge against the top of the leg. 2 x M3 x 12 along y through the leg's long counterbores into the bridge inserts.

Also handle (fitted earlier): frame_cheek_r.

Step 6 (motor 111). Wiper motor on the puck (bench)

Battery unplugged. Put the M6 bolts through the puck slots from the front (heads sit in the slot counterbores), into the motor bosses (threaded bosses, or a nut behind each boss ear). Slide the motor so its shaft is centred in the bore (a 10 mm drill shank in the coupler bore helps), then tighten.

Parts: frame_wiper_motor, frame_wiper_shaft.
Also handle (fitted earlier): frame_motor_adapter.

Step 7 (motor 112). Puck into the plate

Feed the puck into the plate window from behind (the motor side) until its lip sits on the plate's rear face. From the front, M3 x 12 through the plate into the lip inserts (8 of the 8 rim holes have both plate and lip; use those).

Also handle (fitted earlier): frame_motor_adapter.

Step 8 (motor 113). Motor leg

Stand the leg on the deck in the corner between the reel and the bay, foot along y. 4 x M3 x 10 down through the foot into the deck inserts.

Parts: frame_motor_leg.

Step 9 (motor 114). Can stay

Post against the plate's rear face, 2 x M3 x 12 into the plate inserts. Cable tie or hose clamp round the motor can through the two cradle slots, firm but not crushing.

Parts: frame_can_stay.

Step 10 (motor 119). Shafts and trains (other subsystems)

Battery unplugged. Now cad-drum (coupler on the wiper shaft, SW_INDEX block on this plate's front face inserts: drum step "Main shaft: index contact"), cad-launch (rail and aim housing) and then the rail bridge and reel head (cad-reel, cad-drum: D49, they ride the rail) go in. Follow their steps, then come back. Leave a flex loop for the head's wires across the pivot (about 10 conductors, 41 deg of pitch).

Step 11 (main_shaft 130). Main shaft: coupler and limiter

Push the coupler hub onto the wiper output shaft (flats in line) until it stops on the end of the flats; its lower end goes into the puck bore, just clear of the motor's output boss. Drop the M8 nut into the hub pocket onto the thread (it sits about 8 mm down the pocket) and tighten it with a 13 mm socket (1/4 inch drive). Slide the outer coupler over the hub, line up the holes, push a shear pin in (head outside, its flat facing the outboard end) and put a piece of tape round the coupler over the head. Print the pins lying on their flat (D44: shear across the layers). Keep the 5 spare pins with the base. A broken pin stub comes out with an M3 screw turned into it. Then slide the REEL gear onto the coupler journal (plain face first) and screw the retaining washer to the journal end (2 x M3 x 6): the gear must spin free.

Parts: drum_coupler_hub, drum_coupler, drum_shear_pin, drum_gear_reel, drum_reel_washer.

Step 12 (main_shaft 131). Main shaft: index contact

Battery unplugged. Press an M3 nut into the index block pocket, screw the index lever foot to the block (M3 x 12), thread an M3 x 6 into the lever boss, head toward the nut. Screw the block's flange to the motor plate front face with two M3 x 10 into the frame inserts, nub on the coupler cam (D49: SW_INDEX stays on the frame, never on the rail). Set the gap so the screw head touches the nut near the top of a bump (about 1.5 mm on a low part). Battery unplugged: nut to GND, screw to SW_INDEX.

Parts: drum_index_block, drum_index_lever.

Step 13 (main_stack 140). Main shaft: stack

Slide the collar onto the main shaft from the spigot end and push the shaft about 42 mm into the coupler socket. The shaft is held endwise later by the launch sleeve and the magnet cap.

Parts: drum_shaft, drum_collar.

Step 14 (main_stack 141). Main shaft: LAUNCH dog ring and magnet cap

Fit the LAUNCH dog ring on the launch sleeve flange (sleeve lugs in the ring slots, 3 x M3 x 10 from the sleeve side into the ring) before the launch sleeve goes on the spigot (launcher steps). After the launcher: glue the AS5600 magnet into the cap, push the cap on the spigot end (flat to flat) and fit the M3 x 6 onto the flat.

Parts: drum_dogring_launch, drum_magnet_cap.

Step 15 (launcher 190). Launcher 1: print the parts

All PLA unless named. PPA-CF: big gear, drive gear, launch sleeve, big gear pin, SW_SEATED plunger and pushrod, and the aim gears (sun, idler, output, bevel, worm, crank). Use each part's print orientation from the print sheet: the carriage body lies on its side across the bed diagonal; the stop springs print upside down at 100 percent infill (all perimeters); rail pieces lie on their largest flat face. 0.2 mm layers, 4 walls on every loaded part. Hardware (launcher and aim train): M3 screws 2 x 6, 3 x 8, 14 x 10, 23 x 12, 5 x 16, 12 x 20, 4 x 30, plus 4 x 10 for the aim housing to the deck (Aim 3); 10 M3 nuts, 1 M3 nyloc; M8 x 25 bolt and M8 nut; 2 x 608; 1 x 693ZZ (lip roller); 4 M3 heat-set inserts (rail rear flange for the reel head, D49); 2 x 8 mm steel rod pins (cut to the length on the print sheet) for the spring anchors; sprag #1 (CSK20PP); the two springs.

Step 16 (launcher 191). Launcher 2: join the rail

Lay the rear, mid and front rail pieces in a line on a flat table. Push each joint together and fit 2 x M3 x 20 with nuts through each flange pair (8 in total). Tighten evenly; file any step inside the track where the pieces meet so the carriage shoes slide over it.

Parts: launch_rail_rear, launch_rail_mid, launch_rail_front.

Step 17 (launcher 192). Launcher 2b: splice straps and aim lug

Screw a splice strap on each wall top across the front joint (4 x M3 x 10 self tap each). Bolt the aim lug to the left rail wall now (2 x M3 x 12 self tap): the spring guards cover its screws later.

Parts: launch_rail_splice_l, launch_rail_splice_r, launch_aim_lug.

Step 18 (launcher 193). Launcher 3: big gear

Hold the big gear between the two cheeks of the mid piece, the toothed sector towards the drum side (right) and the full ring of teeth towards the left, with the spacer washer between the gear and the left cheek. Push the big gear pin through cheek, gear, washer, cheek from the right and snap the C clip into its groove. The gear must spin freely with almost no end play.

Parts: launch_big_gear, launch_gear_pin, launch_gear_clip, launch_gear_washer.

Step 19 (launcher 194). Launcher 4: drive gear, sprag #1 and the launch sleeve

First bolt cad-drum's LAUNCH dog ring to the sleeve flange: lugs in the ring slots, 3 x M3 x 8 from the flange side into the tapped ring (the heads are out of reach once the sleeve is in the cheeks). Slide sprag #1 into the drive gear so its outer keyway takes the rib inside the gear. Check the direction: look at the gear from its collar side (the side that goes to the right, towards the motor), hold the gear and turn the inner race with a finger. It must turn freely counter-clockwise (BACKWARD) and lock clockwise (FORWARD: clockwise seen from the motor side is the cock direction). If it is the other way round, push the sprag out, turn it over and slide it in other face first. Hold the drive gear between the cheeks (collar side right, then the spacer washer on its left), meshing the big gear's full ring of teeth. Turn the cheek bores so their slots point down, then push the launch sleeve in from the motor side, outboard end first, ribs lined up with the slots and the sprag keyway, until the long (15 mm) rib sits inside the sprag.

Parts: launch_drive_gear, launch_sprag, launch_sleeve, launch_drive_washer.

Step 20 (launcher 195). Launcher 5: carriage

Turn the big gear so its toothless gap faces the rack. Slide the carriage BODY into the track from the FRONT end (the rear flange blocks the rear), rack teeth down, until its rear face reaches the stop position. Yoke: push the SW_SEATED pushrod into the block bore from the rear, collar first, then push the plunger's pin up through the block from below (pad under the left arm) and press the head cap on the pin top. Lower the yoke onto the carriage rear so its key plates drop into the slots in the body webs, and fit 1 x M3 x 10 per side through the yoke into the webs. Press a printed spring peg up into each inner (y 38) peg hole from below, flat to the rear, to the end of the hole.

Parts: launch_carriage, launch_carriage_yoke, launch_seat_plunger, launch_seat_pushrod, launch_seat_clip, launch_yoke_peg_l, launch_yoke_peg_r.

Step 21. Fit the remaining launch parts

Parts without an explicit assembly step: launch_dart.

Parts: launch_dart.

Step 22 (launcher 196). Launcher 5b: lip roller

Press a 693ZZ into the lip roller, put a printed thrust washer each side in its bore (hub to the bearing), drop roller and washers into the fork at the carriage front and fit the M3 x 16 and a nyloc nut; it must spin freely. The tether runs over it in tow and retrieval, and the dart's tail slides in over it through the funnel onto the cradle.

Parts: launch_muzzle_roller, launch_muzzle_roller_brg, launch_muzzle_roller_axle, launch_muzzle_washer_l, launch_muzzle_washer_r.

Step 23 (launcher 197). Launcher 6: carriage stop frames

For each side, before any spring: slide the stop frame onto the rail from the outboard side (it does not drop straight down past the inboard preload ledge). 2 x M3 x 12 through the rear foot into the rail wall, and 2 x M3 x 16 down through the front foot into the rail front plate, driven through the access holes in the frame's front wall.

Parts: launch_stop_frame_l, launch_stop_frame_r.

Step 24 (launcher 198). Launcher 6b: spring guards

Screw the spring guard plates to the rail walls and anchor posts (M3 self tap into the pilots, see the print sheet), four per side: a (rear, with the end wall), b1 and b2 (either side of the stop frame's rear leg) and c (front, 2 x M3 x 12 into the front plate edge). Slide each one in from the side, under the stop frame's outboard bar. Wall brackets: M3 x 12 self tap, head and hex key through the hole in the plate; b2 foot: M3 x 10 down. Never cock with a guard off.

Parts: launch_guard_l_a, launch_guard_l_b1, launch_guard_l_b2, launch_guard_l_c, launch_guard_r_a, launch_guard_r_b1, launch_guard_r_b2, launch_guard_r_c.

Step 25 (launcher 199). Launcher 6c: stop springs

Put each stop spring on its ledges inside its frame, rear bar towards the yoke arm, and pull it onto the frame's front wall with the 2 x M3 x 20 draw screws (heads on the front wall), a turn each in turn, until it sits on the wall (6.6 mm of preload). The rear bar must rest on both ledges.

Parts: launch_stop_spring_l, launch_stop_spring_r.

Step 26 (launcher 200). Launcher 7: contacts

Battery unplugged. SW_HOME (right lip) and SW_SEATED (left lip): drop an M3 nut into the slot in the rail wall with a stripped wire end under it. Put a ring lug (or a stripped wire end wrapped once) on an M3 x 6 under its head and screw it up into the lever's boss, head down, until the tip is flush with the boss top (the pad beside it must stay the highest point), then screw the lever root to the lip with one M3 x 10 (self tap). Set the gap with the carriage pushed back at least 10 mm (lever free): turn the screw down until the head just touches the nut (meter beeps), then back it off ONE full turn (0.5 mm gap; a dab of nail varnish on the thread holds it). Carriage on its stop: the right foot presses the pad about 0.95 mm and SW_HOME must read closed; 2 mm back it must read open (if not, a quarter turn at a time). SW_SEATED closes only with a dart on the sled (no springs yet: hold the carriage on its stop by hand to check). SW_GAP, SW_RAIL_LOW and the AS5600 carrier are fitted in the Sensors steps (cad-sensors).

Parts: launch_sw_home, launch_sw_home_screw, launch_sw_home_nut, launch_sw_seated, launch_sw_seated_screw, launch_sw_seated_nut.

Step 27 (launcher 201). Launcher 8: time the big gear (fingers clear)

No springs fitted yet. Hold the carriage on its stop by hand and turn the drive gear FORWARD until the first tooth after the toothless gap just touches the rearmost rack tooth: this is HOME (SW_GAP is set to it in the Sensors steps). Turning on pulls the carriage back smoothly; at full cock the cut last tooth lets go (push the carriage home by hand). Hold the drive gear from the side and never put a hand in front of the carriage.

Step 28 (aim 205). Aim 1: aim housing, worm, wheel, crank

Build the aim housing on the bench (it goes onto the base later, after the rail). The AS5600 carrier and SW_RAIL_LOW are cad-sensors parts (Sensors steps), screwed into the pilots on the housing. Push the countershaft in from the inboard side through the inner wall bushing, through the big bevel (D bore, slid on from the outboard end of the shaft) and into the outer wall bushing. Drop the M8 nut into the hex pocket on the inner face of the outer wall, fit the two 608 bearings into the crank (worm wheel) hub, hold the crank in place and push the M8 x 25 bolt in from outboard through the wall, the nut and both bearings; tighten so the head and nut clamp the wall. Lay the worm shaft with the small bevel in its saddles and screw the cap on (4 x M3 x 30 self tap). The housing screws to the deck later (Aim 3), from the top: 4 x M3 x 10 down through the counterbored holes in its foot flanges (two behind the spine, one in front of the outer wall, one on the inboard tab) into the deck's heat-set inserts. The SW_RAIL_LOW bracket on the foot extension has its own deck screw (Sensors 2).

Parts: launch_aim_housing, launch_aim_cap, launch_aim_worm, launch_aim_crank, launch_aim_bolt, launch_aim_brg_1, launch_aim_brg_2, launch_aim_output, launch_aim_bevel30, launch_aim_nut.

Step 29 (aim 206). Aim 2: splitter

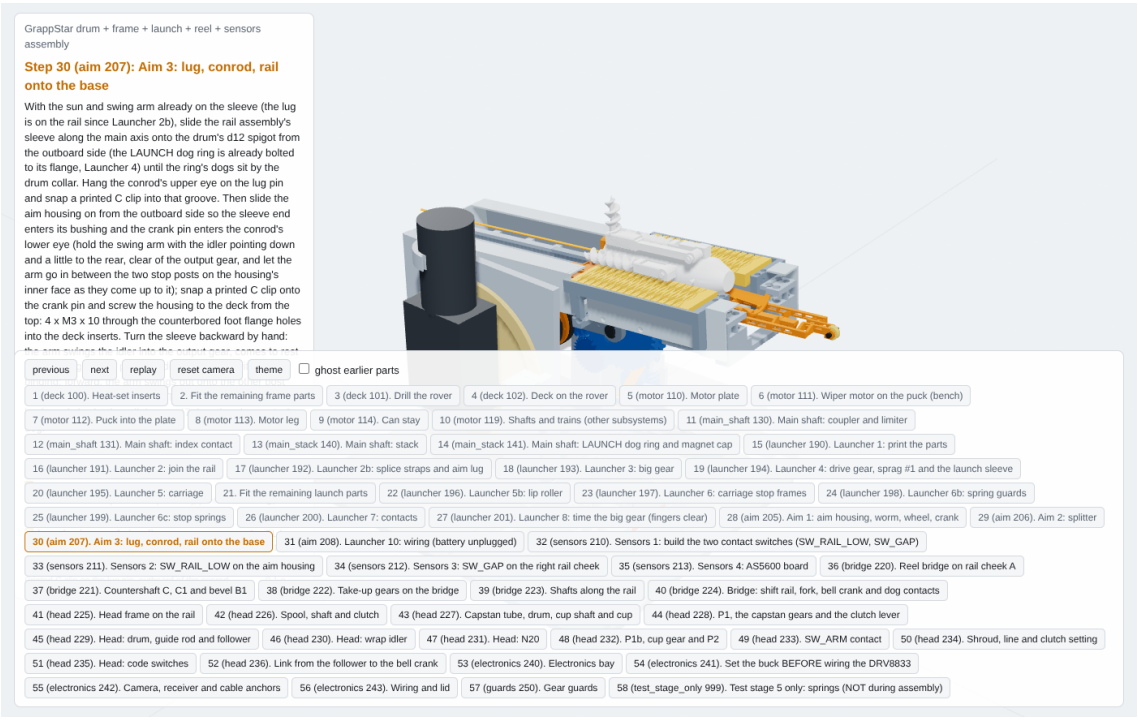
Slide the sun onto the sleeve's outer rib. On the bench: fit the idler on the swing arm pin and snap the printed C clip into the pin groove, and push the printed drag pad into the arm hub pocket with the shim behind its root block (a thicker shim gives more drag). Then slide the arm onto the sleeve next to the sun (there is no room to fit the idler once the arm is on).

Parts: launch_aim_sun, launch_aim_arm, launch_aim_idler, launch_aim_drag_pad, launch_aim_drag_shim, launch_aim_idler_clip.

Step 30 (aim 207). Aim 3: lug, conrod, rail onto the base

With the sun and swing arm already on the sleeve (the lug is on the rail since Launcher 2b), slide the rail assembly's sleeve along the main axis onto the drum's d12 spigot from the outboard side (the LAUNCH dog ring is already bolted to its flange, Launcher 4) until the ring's dogs sit by the drum collar. Hang the conrod's upper eye on the lug pin and snap a printed C clip into that groove. Then slide the aim housing on from the outboard side so the sleeve end enters its bushing and the crank pin enters the conrod's lower eye (hold the swing arm with the idler pointing down and a little to the rear, clear of the output gear, and let the arm go in between the two stop posts on the housing's inner face as they come up to it); snap a printed C clip onto the crank pin and screw the housing to the deck from the top: 4 x M3 x 10 through the counterbored foot flange holes into the deck inserts. Turn the sleeve backward by hand: the arm swings the idler into the output gear, comes to rest on its engaged stop post and drives the gear without binding; forward, the arm swings out onto the other post and the output gear stays still. Then turn the wiper backward slowly: the rail rises to 47 deg, then lowers to 6 deg.

Parts: launch_aim_conrod, launch_aim_crankpin_clip, launch_aim_lugpin_clip.



Step 31 (aim 208). Launcher 10: wiring (battery unplugged)

Battery unplugged. Run the SW_HOME, SW_SEATED and SW_GAP wires along the rail to the pivot and leave a 150 mm loop at the main axis before they go to the frame. No wire goes onto the carriage. Each contact: one wire to its ESP32 pin (SW_HOME 21, SW_GAP 39, SW_RAIL_LOW 40, SW_SEATED 41, as params.SWITCHES), the other to GND.

Step 32 (sensors 210). Sensors 1: build the two contact switches (SW_RAIL_LOW, SW_GAP)

Battery unplugged. For each switch: press an M3 nut into the hex trap on the base pillar (a drop of glue), solder a wire to the nut through the side slot and lay it in the groove. Screw the lever onto the pad with 2 x M3x10 self tapping screws. Screw an M3x10 through the lever boss from the top, head down, with a ring lug and the second wire under a jam nut on top. Turn the screw until the head touches the nut, then back it off 1.9 turns (gap about 0.9 mm, 0.5 mm per turn) and lock the jam nut.

Parts: sensors_sw_rail_low, sensors_sw_rail_low_lever, sensors_sw_rail_low_screw, sensors_sw_rail_low_jam, sensors_sw_rail_low_nut.

Step 33 (sensors 211). Sensors 2: SW_RAIL_LOW on the aim housing

Screw the switch base to its bracket (4 x M3x8 self tapping), lever pointing to the crank. Set the bracket foot on the housing foot extension in front of the crank. Screw it down with an M3x10 in the hole nearest the crank (self taps into the housing) and an M3x16 in the counterbored hole at the front end of the foot: down through the foot and the foot extension into the deck insert under it (the bracket's own deck screw). Turn the crank by hand (wiper backwards) to the lowest rail position: the cam lobe presses the lever tip and the meter shows the switch closed. Turn the crank about 20 deg either side: the switch must open.

Parts: sensors_sw_rail_low_bracket.

Step 34 (sensors 212). Sensors 3: SW_GAP on the right rail cheek

Build SW_GAP as in Sensors 1 (its base has its own arm). Hold the arm tab on the outer face of the right rail cheek, below the big gear, and screw it on with 2 x M3x12 (heads in the counterbores). With the big gear at rest (toothless gap at the rack) the cam bump presses the lever: switch closed. Turn the gear 20 deg either way: switch open.

Parts: sensors_sw_gap, sensors_sw_gap_lever, sensors_sw_gap_screw, sensors_sw_gap_jam, sensors_sw_gap_nut.

Step 35 (sensors 213). Sensors 4: AS5600 board

Battery unplugged. Do this after the magnet cap is on the main shaft end (drum steps): the board sits 2 mm in front of it. Battery unplugged. Plug the STEMMA QT cable into the board's connector on the +x end (the end away from the housing spine). Push the board onto the 4 pegs of the carrier, chip facing out. Drop an M3 nut into each hex pocket in the carrier back, lay the strips over the board edges (pegs through their holes) and screw each from the strip front with an M3x12 into its nut. Screw the carrier column to the aim housing spine with 2 x M3x16. Check by eye: the chip sits on the shaft axis, about 2.0 mm from the magnet face, and the cap turns free.

Parts: sensors_as5600, sensors_as5600_carrier, sensors_as5600_strip_top, sensors_as5600_strip_bot, sensors_as5600_nut_top, sensors_as5600_nut_bot, sensors_as5600_screw_top, sensors_as5600_screw_bot, sensors_as5600_col_screws.

Step 36 (bridge 220). Reel bridge on rail cheek A

Press the 8 mm c1' stub into its hole in the bridge plate. Press the CSK20PP into the sprag hub (key slot on the printed rib; seen from the outboard side the inner ring must turn freely counter-clockwise and lock clockwise) and screw the hub to the plate's outboard face with four M3 x 16. Hold the bridge inboard of the main shaft stack, below the REEL gear, and bolt its box arm to the inner face of rail cheek A with two M3 x 16 (heads reached through the two holes in the arm).

Parts: reel_bridge, reel_stub, reel_sprag_hub, reel_sprag.

Step 37 (bridge 221). Countershaft C, C1 and bevel B1

Slide countershaft C into the sprag from the outboard side, key rib in the keyway, until its collar meets the plate. Slide C1 (49 teeth) onto the D-flat so it meshes the REEL gear, then the 24-tooth bevel, teeth outboard. Turn the REEL gear by hand: C1 turns one way only (the sprag holds the other).

Parts: reel_cs_c, reel_c1, reel_b1g.

Step 38 (bridge 222). Take-up gears on the bridge

Slide c1' (46 teeth) onto the stub, hub outboard, meshing the bus gear. Clip the swing arm's split ring over c1's hub and put the 30-tooth swing idler on the arm pin. Push countershaft T into its bushing and slide T1 (20 teeth) and the B2 bevel onto its flat. Turn the wiper forward by hand: the idler swings into T1; backward it swings out.

Parts: reel_c1p, reel_swing_arm, reel_gi, reel_cs_t, reel_t1, reel_b2g.

Step 39 (bridge 223). Shafts along the rail

Slide the 12-tooth B1 pinion onto the front flat of u-shaft C and feed the shaft rearward through the bridge block until the pinion meshes the B1 bevel. Slide the B2 pinion onto the front flat of u-shaft T and feed it rearward through the bridge pillar until the pinion meshes B2. Both shafts reach back to the reel head.

Parts: reel_b1p, reel_uc, reel_b2p, reel_ut.

Step 40 (bridge 224). Bridge: shift rail, fork, bell crank and dog contacts

Battery unplugged. Press the shift rail into the bridge plate boss (flat facing the rail). Press an M3 nut into the hex hole of the fork's tab. Thread an M3 x 8 into each dog post boss, heads toward each other. Slide the fork sleeve onto the rail with the yoke under the collar and drop the yoke pads into the collar groove, then slide the dog post onto the rail end and tighten its clamp screw (M3 x 8, tapped) on the rail flat. Put the bell crank on the shelf with its pin up in the fork lug's slot and fit the M3 x 16 pivot screw (tapped into the shelf, the crank must swing). Battery unplugged: nut wire to GND, REEL side screw to SW_DOG_R, LAUNCH side screw to SW_DOG_L; the wires run in the flex loop to the frame.

Parts: drum_shift_rail, drum_fork, drum_dog_post, drum_crank.

Step 41 (head 225). Head frame on the rail

Screw the front plate to the spine (three M3 x 12 through the spine into the plate). Slide the front plate's bushing over the rear end of u-shaft T and bolt the plate to the rail's rear flange with four M3 x 16 into the flange inserts. Screw the mid plate to the front plate (M3 x 12).

Parts: reel_head_front, reel_head_spine, reel_head_mid.

Step 42 (head 226). Spool, shaft and clutch

Press a 608 into each end of the spool. Push the line end through the small hole near the inboard flange, knot it inside (figure eight) and wind 20 m on (line over the top, leaving toward the rail). Slide the clutch disk onto the outboard flat of the spool shaft (friction ring inboard), then from the other end the spool until its outboard face meets the disk, then the thrust ring, boss toward the spool. Feed the inboard end through the mid plate bushing and slide the B3 bevel onto its flat. Slide the B3 bevel for u-shaft T onto the shaft's rear flat; the two bevels mesh.

Parts: reel_spool, reel_brg_1, reel_brg_2, reel_s, reel_d, reel_t, reel_b3a, reel_b3b.

Step 43 (head 227). Capstan tube, drum, cup shaft and cup

Slide the capstan drum onto the front end of the capstan tube, key slot on the rib. Push the cup shaft through the tube from the front (neck end in front), slide the cup onto it (flat in the bore onto the flat, eyelet lug at the bottom, in line with the spool) and push Harry's small C-clip onto the neck.

Parts: reel_c, reel_drum, reel_k, reel_cup, reel_cup_clip.

Step 44 (head 228). P1, the capstan gears and the clutch lever

Press the G_b axle (8 mm rod, 26 long) into P1. Slide P1 over the capstan tube, the rear end of u-shaft C and the spool shaft end (its leg), and screw it to the spine (two M3 x 12) and to the mid plate. Put G_c (23 teeth) on the tube flat, G_a (26) on u-shaft C's flat and G_b (32) on its axle. Hang the clutch lever on the P1 post with the M3 x 16 pivot screw (it must swing), its upper arm under the drum's brake cam, the FIRM screw toward the thrust ring. Screw the LIGHT stop screw into P1 under the upper arm.

Parts: reel_head_p1, reel_gb_axle, reel_gc, reel_ga, reel_gb, reel_lever, reel_lever_pin, reel_firm_screw, reel_light_screw.

Step 45 (head 229). Head: drum, guide rod and follower

Press the drum axle and the follower guide rod into the reel head front plate (flats as on the print sheet). Slide drum A onto the axle, gear first, then the follower onto the rod with its pin in drum A's groove, then drum B (hex socket over the spigot; it fits one way only). Press bracket B onto the axle and rod ends with the wrap idler shaft in its journal and screw the bracket's foot to the spine (2 x M3 x 12 from outside the spine).

Parts: drum_axle, drum_rod, drum_drum_a, drum_follower, drum_drum_b, drum_bracket_b.

Step 46 (head 230). Head: wrap idler

Push the 49-tooth output gear onto the idler shaft's flat (D bore). Feed the shaft into bracket B from the front so the small pinion sits at the drum's segment teeth and the lock cam over the lock ring; its rear journal goes into P2 when P2 goes on. Turn the drum to LAUNCH: the flat-sided lock cam must sit in the drum lock ring and the idler can no longer turn by hand.

Parts: drum_idler, drum_idler_gear.

Step 47 (head 231). Head: N20

Battery unplugged. Press the pinion onto the N20 output shaft (flat to flat) leaving 1.5 mm of shaft below it (that end runs in the front plate slot). Slide the N20 into the cradle from the open end, fit the end plate (2 x M3 x 8). Press an M3 nut into the pocket in the front plate's rear face, wire along its groove. Screw the cradle to the front plate: pivot screw M3 x 12 through the round hole, swing screw M3 x 12 through the arc slot at its full-mesh end, both tapped. Turn the SW_N20 screw (M3 x 8, tapped in the cradle plate) in until its tip just touches the nut. Strap the N20 body to the cradle with a cable tie (it rides the rail and sees every shot). Battery unplugged: nut to GND, screw to SW_N20.

Parts: drum_n20_pinion, drum_n20, drum_n20_cradle, drum_n20_strap.

Step 48 (head 232). P1b, cup gear and P2

Fit P1b over the rear end of the capstan tube and screw it to the two P1 posts (M3 x 12). Slide the cup gear G_k (28 teeth, raised track forward) onto the cup shaft flat so it meshes the drum's wrap idler output with the cup at home (eyelet at the bottom). Screw P2 to the spine over the cup shaft and wrap idler ends (two M3 x 12). The cup must turn freely both ways.

Parts: reel_head_p1b, reel_gk, reel_head_p2.

Step 49 (head 233). SW_ARM contact

Battery unplugged. Build the printed contact (see the contact switch page): nut pressed in the base trap, lever screwed onto the pad with 2 x M3 x 10 self tapping screws, M3 x 10 contact screw with its jam nut in the lever. Screw the base to P1b's rear face with four M3 x 10. SW_ARM is closed while the cup is away from home and opens at home: the lever rides the raised track on the front of the cup gear and drops into its notch at home. Cup at home: turn the contact screw until its head just touches the nut, back it off 1.9 turns (gap about 0.9 mm) and lock the jam nut. Check with the meter: about 15 degrees of cup turn either way closes it; at home it must be open.

Parts: reel_sw_base, reel_sw_lever, reel_sw_hw.

Step 50 (head 234). Shroud, line and clutch setting

Battery unplugged. Put the shroud round the spool, open side at the top toward the rail, and screw its tab to the P1 leg (M3 x 8). Lead the line from the top of the spool straight forward under the capstan drum and through the cup eyelet from behind, then forward through the pusher hole to the dart's tail cap. Wind 20 m through the eyelet once on the bench before the first shot (D49: the pack mounds in the middle). Set LIGHT with the drum at LAUNCH: turn the LIGHT stop screw until a 25 g weight hung on the line end just pulls line off slowly (0.25 N). Set FIRM with the drum at NEUTRAL: turn the FIRM screw until the line slips at 5.0 N on the luggage scale, then lock its nut.

Parts: reel_shroud, launch_line_p, launch_line_a0, launch_line_a1, launch_line_a2, launch_line_a3, launch_line_a4, launch_line_a5, launch_line_a6, launch_line_a7, launch_line_a8, launch_line_a9, launch_line_a10, launch_line_a11, launch_line_a12, launch_line_a13, launch_line_a14, launch_line_a15, launch_line_a16, launch_line_a17, launch_line_a18, launch_line_a19, launch_line_a20.

Step 51 (head 235). Head: code switches

Battery unplugged. Press an M3 nut into each of the 3 pockets of the switch block and screw its foot to the spine (2 x M3 x 12 from outside the spine). Screw each lever foot to the block (M3 x 12), nub on its band. Thread an M3 x 6 into each lever boss, head toward the nut, so the head touches the nut near the top of a bump (gap about 1.8 mm with the nub low). Battery unplugged: nuts to GND, screws to SW_DRUM_A (nearest the drum gear), SW_DRUM_B, SW_DRUM_C.

Parts: drum_switch_block, drum_sw_lever_0, drum_sw_lever_1, drum_sw_lever_2.

Step 52 (head 236). Link from the follower to the bell crank

Cut the 8 mm steel rod to the length on the print sheet, deburr the ends and press a printed eye onto each end, both eye holes parallel. Pass the rod through the window in the head front plate, screw the top eye to the follower lug and the bottom eye to the bell crank's input arm (M3 x 16 and nyloc nut each, snug: the eyes must swing). Turn the drum by hand from LAUNCH to REEL: the follower slides 14 mm, the crank turns and the collar moves from the LAUNCH dogs to the REEL dogs.

Parts: drum_link, drum_link_eye_1, drum_link_eye_2.

Step 53 (electronics 240). Electronics bay

Battery unplugged. Bay on the deck, 4 x M3 x 10 through the tabs. Boards in: BTS7960 on its 4 standoffs (M3 x 6), heatsink down; ESP32 DevKitC pins down into the cradle, USB end into the wall slot, clamp with 2 x M3 x 8; DRV8833, MP1584 and HX711 into their pockets with a cable tie through the floor slots or a drop of hot glue. Fuse holder in its cradle with a cable tie.

Parts: frame_ebay_lid, frame_ebay_esp_clamp, frame_pcb_esp32, frame_pcb_bts7960, frame_pcb_mp1584, frame_pcb_drv8833, frame_pcb_hx711, frame_fuse_20a.
Also handle (fitted earlier): frame_ebay.

Step 54 (electronics 241). Set the buck BEFORE wiring the DRV8833

Battery unplugged. Wire only the buck input to the fuse output and ground. Plug the battery in, meter on the buck output, turn the trimpot until it reads 9.0 V. Unplug the battery. Only then wire buck output to DRV8833 VIN (the DRV8833 dies above 10.8 V).

Also handle (fitted earlier): frame_pcb_mp1584, frame_pcb_drv8833.

Step 55 (electronics 242). Camera, receiver and cable anchors

Battery unplugged. Camera column on the deck, 4 x M3 x 10 into the inserts. Clip the Kiyo X onto the 16 mm edge plate like onto a monitor (or use a 1/4-20 tripod screw up through the plate hole); it looks forward along the launch line. RC832: stand it antenna-end up in its holder slot, leads out of the low notch, holder 2 x M3 x 10. Cable anchors 1 x M3 x 12 each; tie the camera USB, the receiver AV lead and the capture dongle lead to them so no plug carries the cable weight.

Parts: frame_camera_mount, frame_rc832_holder, frame_rc832, frame_kiyo_x, frame_cable_anchor_1, frame_cable_anchor_2.

Step 56 (electronics 243). Wiring and lid

Battery unplugged. Follow firmware/base/WIRING.md. Every wire leaving the bay goes through its wall exit and is cable-tied to the anchor beside it (strain relief). 14 AWG for battery, fuse, BTS7960 and motor. Lid on, 4 x M3 x 6. USB cable from the ESP32 native USB port to the Jetson.

Also handle (fitted earlier): frame_ebay_lid.

Step 57 (guards 250). Gear guards

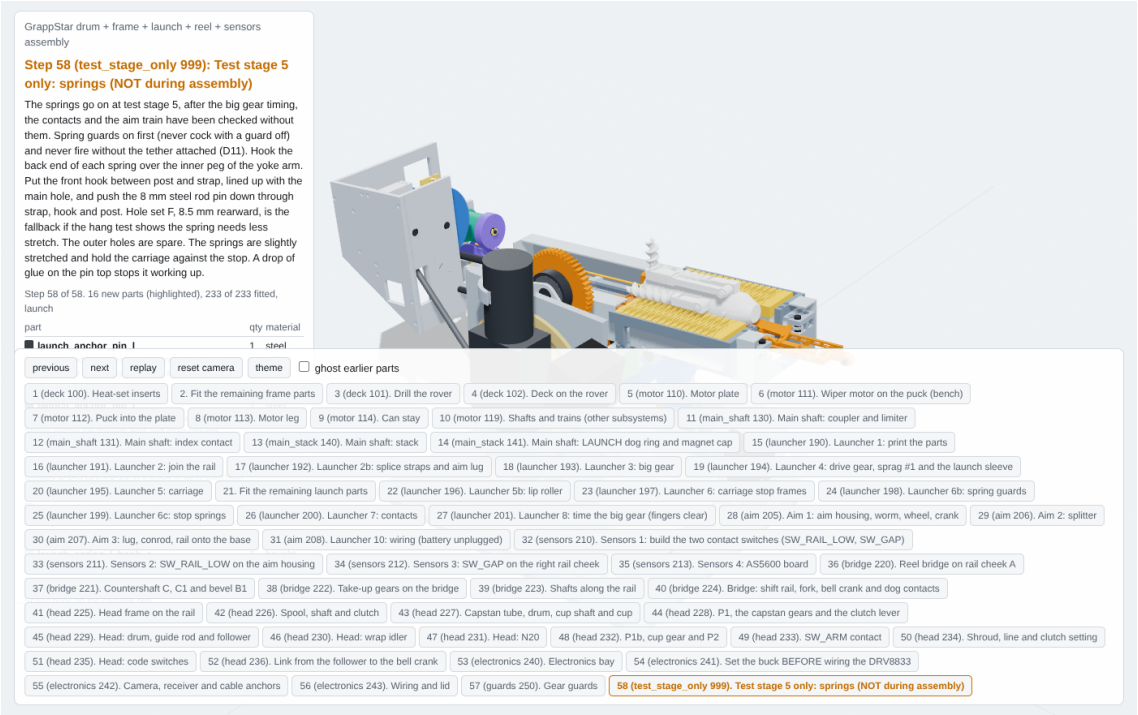
After every train has been turned by hand: the hood goes over the gears in front of the motor plate, M3 x 10 through each lug into the plate inserts; the front guard stands on the deck between the drum gears and the bay, M3 x 10 into the deck inserts. Turn the wiper by hand once: nothing rubs.

Parts: frame_guard_hood, frame_guard_front.

Step 58 (test_stage_only 999). Test stage 5 only: springs (NOT during assembly)

The springs go on at test stage 5, after the big gear timing, the contacts and the aim train have been checked without them. Spring guards on first (never cock with a guard off) and never fire without the tether attached (D11). Hook the back end of each spring over the inner peg of the yoke arm. Put the front hook between post and strap, lined up with the main hole, and push the 8 mm steel rod pin down through strap, hook and post. Hole set F, 8.5 mm rearward, is the fallback if the hang test shows the spring needs less stretch. The outer holes are spare. The springs are slightly stretched and hold the carriage against the stop. A drop of glue on the pin top stops it working up.

Parts: launch_anchor_pin_l, launch_anchor_pin_r, launch_spring_l_hook_a, launch_spring_r_hook_a, launch_spring_l_hook_c, launch_spring_r_hook_c, launch_spring_l_0, launch_spring_l_1, launch_spring_l_2, launch_spring_l_3, launch_spring_l_4, launch_spring_r_0, launch_spring_r_1, launch_spring_r_2, launch_spring_r_3, launch_spring_r_4.



6. Audit summary

Result: NOT CLEAN

233 parts checked over 46676 states.

Collision drum_coupler with drum_index_lever: 24.517 mm3 in 67 states.

Collision launch_aim_sun with launch_aim_idler: 55.565 mm3 in 12 states.

Allowed contacts: 274.